Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**DESIGN CALCULATION** **3640 LOW GRADE CONDITIONING, LEACHING AND ADSORPTION** **DESIGN OF MISCELLANEOUS FOUNDATION**
Document Number:	**MD-216-3640-EG-CV-CAL-1326**
Revision:	**D01**
Revision Date:	**23.11.2023**
Status:	**AS BUILT**

Outotec Document Number:	900707	GAU01	HL13	CED01	36402	OU100682596	2
L&T Document Number:	O19045	450	C	K	24138	002	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	21.05.2020	FIRST ISSUE	MMJ	AVG	SJM	
C01	20.06.2020	ISSUED FOR CONSTRUCTION	MMJ	AVG	SJM	
D01	23.11.2023	AS BUILT	MMJ	AVG	SJM	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The purpose of this document is to analysis and design of **LOW GRADE CONDITIONING, LEACHING AND ADSORPTION MISCELLANEOUS FOUNDATION**. In this document design & analysis of foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Load data for foundation design taken from document no. **MD-216-3640-EG-ST-CAL-2286, MD-216-3640-EG-ME-GAD-1038** and from respective superstructure Staad model.

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

LOAD 1 = DEAD LOAD (DL)

LOAD 2 = LIVE LOAD (LL)

LOAD 3 = OPERATING FLUID LOAD (FL)

LOAD 4 = EQP DL (EQDL)

LOAD 5 = EQP LL (EQLL)

LOAD 6 = WIND LOAD (+X) WL(+X)

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LOAD 7 = WIND LOAD (-X) WL(-X)

LOAD 8 = WIND LOAD (+Z) WL(+Z)

LOAD 9 = WIND LOAD (-Z) WL(-Z)

LOAD 10 = SEISMIC LOAD SL(+X)

LOAD 11 = SEISMIC LOAD SL(-X)

LOAD 12 = SEISMIC LOAD SL(+Z)

LOAD 13 = SEISMIC LOAD SL(-Z)

LOAD 14 = TEMPERATURE LOAD (TL)

LOAD 15 = PIPE LATERAL LOAD (PL)

4.3 Load combination

A. The following load combination are considered for member strength calculation.

101 1.4(DL+EQDL+FL)

102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)

103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)

104 1.4(DL+EQDL+FL+PL)

105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)

106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)

107 1.2(DL+EQDL)+0.8WL(+X)

108 1.2(DL+EQDL)+0.8WL(-X)

109 1.2(DL+EQDL)+0.8WL(+Z)

110 1.2(DL+EQDL)+0.8WL(-Z)

111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)

112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)

113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)

114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)

115 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL

116 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL

117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL

118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL

119 0.9(DL+EQDL)+1.6WL(+X)

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120 0.9(DL+EQDL)+1.6WL(-X)

121 0.9(DL+EQDL)+1.6WL(+Z)

122 0.9(DL+EQDL)+1.6WL(-Z)

123 0.9(DL+EQDL)+SL(+X)

124 0.9(DL+EQDL)+SL(-X)

125 0.9(DL+EQDL)+SL(+Z)

126 0.9(DL+EQDL)+SL(-Z)

B) The following load combination are considered for base pressure, sliding & overturning check.

201 DL+EQDL+FL

202 DL+EQDL+FL+LL+EQLL+TL

203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL

204 DL+EQDL+FL+PL

205 DL+EQDL+FL+PL+LL+EQLL+TL

206 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL

207 DL+EQDL+FL+WL(+X)

208 DL+EQDL+FL+WL(-X)

209 DL+EQDL+FL+WL(+Z)

210 DL+EQDL+FL+WL(-Z)

211 DL+EQDL+FL+0.7SL(+X)

212 DL+EQDL+FL+0.7SL(-X)

213 DL+EQDL+FL+0.7SL(+Z)

214 DL+EQDL+FL+0.7SL(-Z)

215 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)

216 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)

217 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)

218 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)

219 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)

220 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)

221 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)

222 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)

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223 0.6(DL+EQDL)+WL(+X)

224 0.6(DL+EQDL)+WL(-X)

225 0.6(DL+EQDL)+WL(+Z)

226 0.6(DL+EQDL)+WL(-Z)

227 0.6(DL+EQDL)+0.7SL(+X)

228 0.6(DL+EQDL)+0.7SL(-X)

229 0.6(DL+EQDL)+0.7SL(+Z)

230 0.6(DL+EQDL)+0.7SL(-Z)

4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m2 at 1.2m depth below EGL (RL 912.200) refer Table 5 - BH-27 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 40500 kN/m^2/m at 1.2m depth below EGL (RL 912.200) refer Table 5 - BH-27 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5.1 DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-X	Along-Z
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	6.00m	3.00m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.40	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

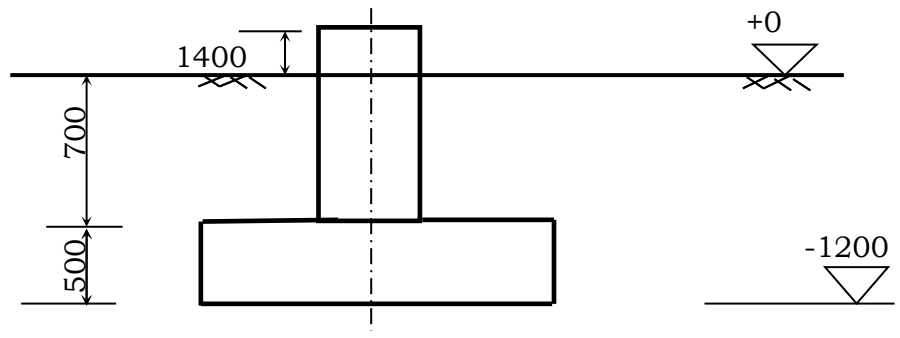
Calculation of Weight of Foundation :

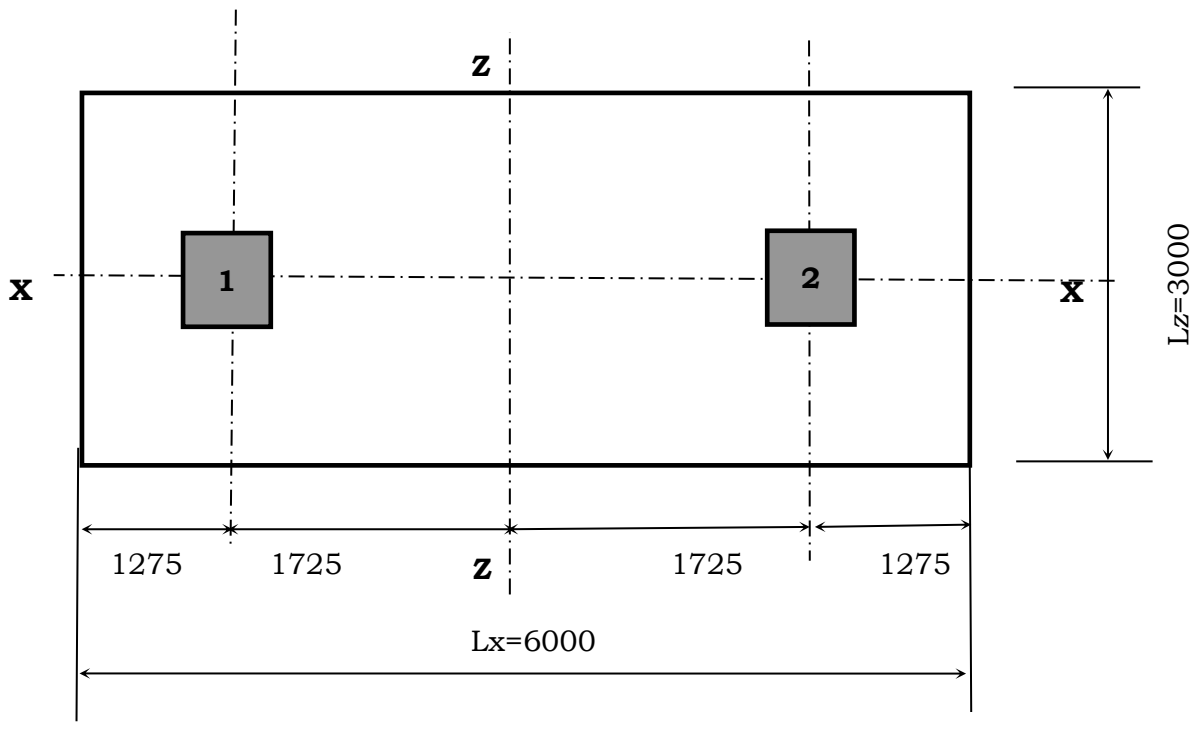
Weight of raft = (6*3*0.5*25) = 225.00KN

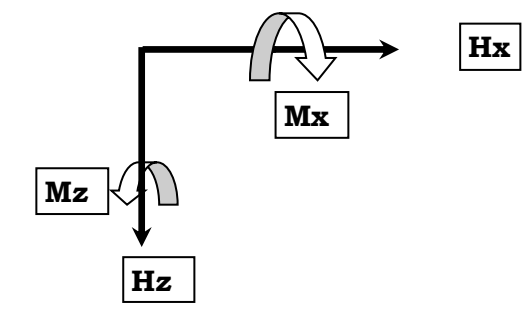
Weight of soil = ((6*3)-2*(0.5*0.5))*0.7*18 = 220.50KN

Total = 445.50KN

Submerged wt. = 445.5-6*3*1.2*10 = 229.50KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	59.0	4.0	-10.4	-7.0	-18.2	141.0	3.0	-7.8	-20.0	-52.0
202 DL+EQDL+FL+LL+EQLL+TL	171.0	34.0	-88.4	-40.0	-104.0	288.0	-17.0	44.2	-72.0	-187.2
203 DL+EQDL+FL+0.75(LL+EQLL)+	143.0	26.5	-68.9	-31.8	-82.6	251.3	-12.0	31.2	-59.0	-153.4
204 DL+EQDL+FL+WL(+X)	-76.0	-28.0	72.8	-4.0	-10.4	238.0	-19.0	49.4	-27.0	-70.2
205 DL+EQDL+FL+WL(-X)	194.0	36.0	-93.6	-10.0	-26.0	44.0	25.0	-65.0	-13.0	-33.8
206 DL+EQDL+FL+WL(+Z)	92.0	2.0	-5.2	-22.0	-57.2	177.0	1.0	-2.6	-44.0	-114.4
207 DL+EQDL+FL+WL(-Z)	26.0	6.0	-15.6	8.0	20.8	105.0	5.0	-13.0	4.0	10.4
208 DL+EQDL+FL+0.7SL(+X)	5.8	-7.9	20.5	-5.6	-14.6	180.9	-6.8	17.7	-24.2	-62.9
209 DL+EQDL+FL+0.7SL(-X)	112.2	15.9	-41.3	-8.4	-21.8	101.1	12.8	-33.3	-15.8	-41.1
210 DL+EQDL+FL+0.7SL(+Z)	66.7	3.3	-8.6	-10.5	-27.3	155.7	2.3	-6.0	-30.5	-79.3
211 DL+EQDL+FL+0.7SL(-Z)	51.3	4.7	-12.2	-3.5	-9.1	126.3	3.7	-9.6	-9.5	-24.7
212 DL+EQDL+FL+0.75WL(+X)+0.7	34.3	-17.0	44.2	-11.5	-29.9	327.0	-11.3	29.3	-40.3	-104.7
213 DL+EQDL+FL+0.75WL(-X)+0.7	236.8	31.0	-80.6	-16.0	-41.6	181.5	21.8	-56.6	-29.8	-77.4
214 DL+EQDL+FL+0.75WL(+Z)+0.7	160.3	5.5	-14.3	-25.0	-65.0	281.3	3.8	-9.8	-53.0	-137.8
215 DL+EQDL+FL+0.75WL(-Z)+0.7	110.8	8.5	-22.1	-2.5	-6.5	227.3	6.8	-17.6	-17.0	-44.2
216 DL+EQDL+FL+0.525SL(+X)+0.	95.6	-1.9	5.0	-12.7	-33.0	284.2	-2.1	5.5	-38.2	-99.2
217 DL+EQDL+FL+0.525SL(-X)+0.7	175.4	15.9	-41.4	-14.8	-38.5	224.3	12.6	-32.8	-31.9	-82.8
218 DL+EQDL+FL+0.525SL(+Z)+0.	141.3	6.5	-16.8	-16.4	-42.6	265.3	4.7	-12.3	-42.9	-111.5
219 DL+EQDL+FL+0.525SL(-Z)+0.7	129.7	7.5	-19.6	-11.1	-28.9	243.2	5.8	-15.0	-27.1	-70.5
220 0.6(DL+EQDL)+WL(+X)	-102.6	-30.2	78.5	-0.6	-1.6	147.4	-20.8	54.1	-14.2	-36.9
221 0.6(DL+EQDL)+WL(-X)	167.4	33.8	-87.9	-6.6	-17.2	-46.6	23.2	-60.3	-0.2	-0.5
222 0.6(DL+EQDL)+WL(+Z)	65.4	-0.2	0.5	-18.6	-48.4	86.4	-0.8	2.1	-31.2	-81.1
223 0.6(DL+EQDL)+WL(-Z)	-0.6	3.8	-9.9	11.4	29.6	14.4	3.2	-8.3	16.8	43.7
224 0.6(DL+EQDL)+0.7SL(+X)	-20.8	-10.1	26.3	-2.2	-5.7	90.3	-8.6	22.4	-11.4	-29.6
225 0.6(DL+EQDL)+0.7SL(-X)	85.6	13.7	-35.6	-5.0	-13.0	10.5	11.0	-28.6	-3.0	-7.8
226 0.6(DL+EQDL)+0.7SL(+Z)	40.1	1.1	-2.9	-7.1	-18.5	65.1	0.5	-1.3	-17.7	-46.0
227 0.6(DL+EQDL)+0.7SL(-Z)	24.7	2.5	-6.5	-0.1	-0.3	35.7	1.9	-4.9	3.3	8.6
228 DL+EQTL	57.0	3.0	-7.8	-6.0	-15.6	92.0	2.0	-5.2	-14.0	-36.4

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL+EQDL+FL	645.5	7.00	123.3	-27.0	-70.2	15.7	13.8	36.9	9.6
202 DL+EQDL+FL+LL+EQLL+TL	904.5	17.00	157.6	-112.0	-291.2	17.2	4.7	21.3	3.2
203 DL+EQDL+FL+0.75(LL+EQLL)+	839.8	14.50	149.1	-90.8	-236.0	16.9	5.3	23.2	3.7
204 DL+EQDL+FL+WL(+X)	607.5	-47.00	663.9	-31.0	-80.6	2.7	11.3	5.2	7.8
205 DL+EQDL+FL+WL(-X)	683.5	61.00	-417.4	-23.0	-59.8	4.9	17.1	4.5	11.9
206 DL+EQDL+FL+WL(+Z)	714.5	3.00	138.8	-66.0	-171.6	15.4	6.2	95.3	4.3
207 DL+EQDL+FL+WL(-Z)	576.5	11.00	107.7	12.0	31.2	16.1	27.7	21.0	19.2
208 DL+EQDL+FL+0.7SL(+X)	632.2	-14.70	340.2	-29.8	-77.5	5.6	12.2	17.2	8.5
209 DL+EQDL+FL+0.7SL(-X)	658.8	28.70	-93.7	-24.2	-62.9	21.1	15.7	9.2	10.9
210 DL+EQDL+FL+0.7SL(+Z)	667.9	5.60	138.9	-41.0	-106.6	14.4	9.4	47.7	6.5
211 DL+EQDL+FL+0.7SL(-Z)	623.1	8.40	107.6	-13.0	-33.8	17.4	27.7	29.7	19.2
212 DL+EQDL+FL+0.75WL(+X)+0.7	806.8	-28.30	578.4	-51.8	-134.6	4.2	9.0	11.4	6.2
213 DL+EQDL+FL+0.75WL(-X)+0.7	863.8	52.80	-232.6	-45.8	-119.0	11.1	10.9	6.5	7.5
214 DL+EQDL+FL+0.75WL(+Z)+0.7	887.1	9.30	184.6	-78.0	-202.8	14.4	6.6	38.2	4.5
215 DL+EQDL+FL+0.75WL(-Z)+0.7	783.6	15.30	161.3	-19.5	-50.7	14.6	23.2	20.5	16.1
216 DL+EQDL+FL+0.525SL(+X)+0.	825.3	-4.00	335.8	-50.9	-132.2	7.4	9.4	82.5	6.5
217 DL+EQDL+FL+0.525SL(-X)+0.7	845.2	28.50	10.2	-46.7	-121.3	249.8	10.5	11.9	7.2
218 DL+EQDL+FL+0.525SL(+Z)+0.	852.1	11.20	184.8	-59.3	-154.1	13.8	8.3	30.4	5.7
219 DL+EQDL+FL+0.525SL(-Z)+0.7	818.4	13.30	161.2	-38.2	-99.4	15.2	12.4	24.6	8.6
220 0.6(DL+EQDL)+WL(+X)	490.3	-51.00	563.9	-14.8	-38.5	2.6	19.1	3.8	13.3
221 0.6(DL+EQDL)+WL(-X)	566.3	57.00	-517.4	-6.8	-17.7	3.3	48.0	4.0	33.3
222 0.6(DL+EQDL)+WL(+Z)	597.3	-1.00	38.8	-49.8	-129.5	46.2	6.9	238.9	4.8
223 0.6(DL+EQDL)+WL(-Z)	459.3	7.00	7.7	28.2	73.3	179.5	9.4	26.2	6.5
224 0.6(DL+EQDL)+0.7SL(+X)	515.0	-18.70	240.3	-13.6	-35.3	6.4	21.9	11.0	15.1
225 0.6(DL+EQDL)+0.7SL(-X)	541.6	24.70	-193.7	-8.0	-20.8	8.4	39.1	8.8	27.1
226 0.6(DL+EQDL)+0.7SL(+Z)	550.7	1.60	38.9	-24.8	-64.5	42.4	12.8	137.7	8.9
227 0.6(DL+EQDL)+0.7SL(-Z)	505.9	4.40	7.6	3.2	8.3	200.4	91.4	46.0	63.2
228 DL+EQTL	594.5	5.00	47.4	-20.0	-52.0	37.6	17.1	47.6	11.9
					Min.=	**2.6**	**4.7**	**3.8**	**3.2**

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	161	201	0	36.29	0
Min Px	161	201	0	36.29	0
Max Py	700	202	0	99.80	0
Min Py	165	204	0	0.00	0
Max Pz	161	201	0	36.29	0
Min Pz	161	201	0	36.29	0

Gross Maximum Base pressure, p_{max} = 99.8 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.7 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 462.6 KN/m^2
> 99.8 KN/m^2
Hence OK

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Design of foundation raft F1

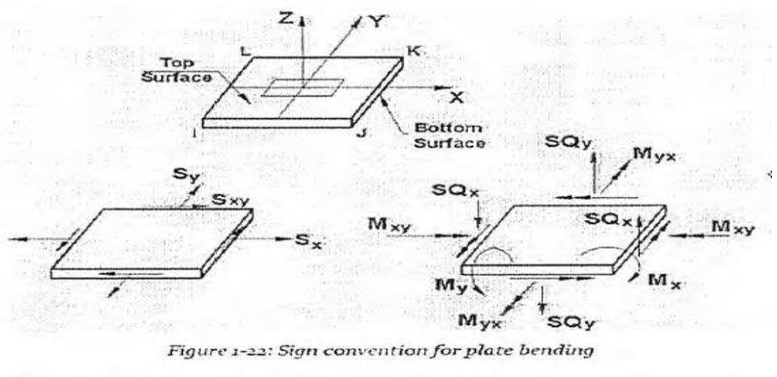
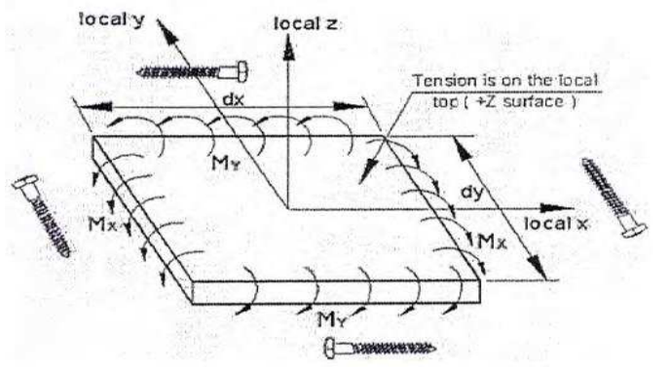


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	491	0.454	-0.337	0	0	0	-50	-81.276	2.745	-52.749	-84.021	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Min Qx	403	-0.839	0.031	0	0	0	-137.6	-91.04	0	-137.552	-91.04	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max Qy	418	-0.002	0.827	0	0	0	-103.1	-120.86	15.937	-119.057	-136.797	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Min Qy	76	0.153	-0.677	0	0	0	-62.9	-113.811	2.612	-65.515	-116.423	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max Sx	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Min Sx	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Max Sy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Min Sy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Max Sxy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Min Sxy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Max Mx	389	0.186	0.086	0	0	0	58.964	7.988	7.461	66.425	15.449	116 0.9(DL+EQDL)+1.6WL(+X)
Min Mx	403	-0.839	0.031	0	0	0	-137.6	-91.04	17.558	-155.11	-108.598	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max My	225	0.033	-0.102	0	0	0	-22.48	42.693	-27.441	-49.924	70.134	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Min My	418	-0.002	0.827	0	0	0	-103.1	-120.86	15.937	-119.057	-136.797	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max Mxy	213	0.036	-0.049	0	0	0	18.709	-8.645	25.863	44.572	-34.508	109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
Min Mxy	531	-0.102	0.111	0	0	0	1.032	5.186	-41.545	42.577	46.731	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	539	0.257	0.177	0	0	0	-44.77	-44.888	19.865	-64.63	-64.753	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Min Qx	402	-0.472	0.012	0	0	0	-69.37	-82.311	0	-69.371	-82.311	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max Qy	432	0.002	0.449	0	0	0	-96.57	-53.119	9.762	-106.329	-62.881	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Min Qy	72	0.044	-0.373	0	0	0	-51.43	-52.527	-5.082	-56.508	-57.609	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max Sx	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Min Sx	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Max Sy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Min Sy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Max Sxy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Min Sxy	1015	0.006	0.006	0	0	0	-1.118	-1.086	0.671	-1.789	-1.757	101 1.4(DL+EQDL+FL)
Max Mx	135	-0.011	-0.021	0	0	0	46.587	-6.62	-4.714	51.301	-11.334	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Min Mx	432	0.002	0.449	0	0	0	-96.57	-53.119	9.762	-106.329	-62.881	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max My	715	0.139	0.035	0	0	0	16.282	26.197	-2.246	18.528	28.443	116 0.9(DL+EQDL)+1.6WL(+X)
Min My	402	-0.472	0.012	0	0	0	-69.37	-82.311	11.4	-80.771	-93.711	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max Mxy	374	-0.192	-0.243	0	0	0	-60.19	-37.628	23.806	-83.991	-61.434	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Min Mxy	209	-0.108	0.123	0	0	0	14.437	-23.994	-24.557	38.994	-48.551	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

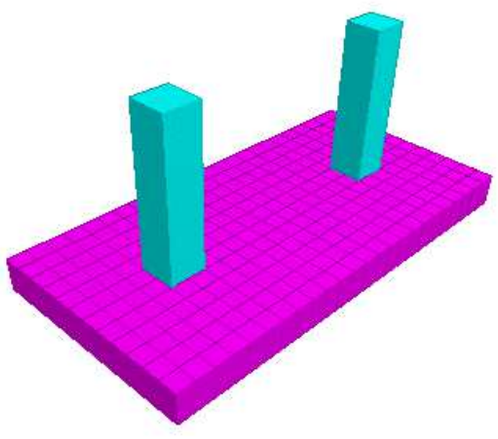
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
116 0.9(D	66.43	top face	50	442	66.425	0.45	0.0011	480.8	795.6	16	252.82	150	1341	18.91	23.64	0.053	tension control
103 1.2(D	-155.11	bottom face	50	442	155.11	1.06	0.0026	1134.6		16	177.28	150	1341	18.91	23.64	0.053	tension control
109 1.2(D	44.57	top face	50	442	44.572	0.30	0.0007	321.8		16	252.82	150	1341	18.91	23.64	0.053	tension control
103 1.2(D	42.58	top face	50	442	42.577	0.29	0.0007	307.3		16	252.82	150	1341	18.91	23.64	0.053	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.257	1	128.5	64.63	0.003	0.442	438.419	441.182	758.323	438.42	1623	> 128.5	ok	328.8	φVc>Vu	not required
	Min Qx	0.472	1	236	69.371			452.665			441.18	1625	> 236	ok	330.9	φVc>Vu	not required

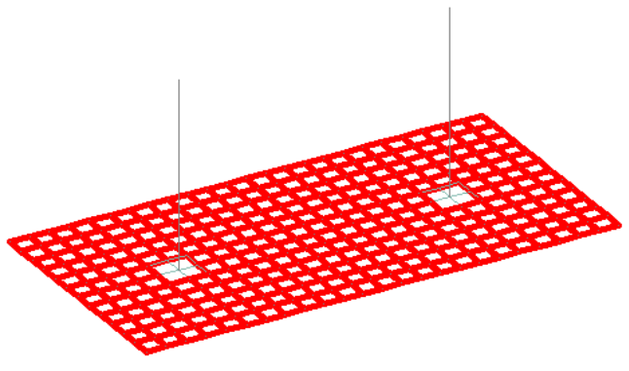
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103 1.2(D	70.13	top face	50	442	70.134	0.48	0.0011	507.8	795.6	16	252.82	150	1341	18.91	23.64	0.053	tension control
103 1.2(D	-136.80	bottom face	50	442	136.8	0.93	0.0023	998.4		16	201.46	150	1341	18.91	23.64	0.053	tension control
109 1.2(D	-34.51	bottom face	50	442	34.508	0.24	0.0006	248.8		16	252.82	150	1341	18.91	23.64	0.053	tension control
103 1.2(D	46.73	top face	50	442	46.731	0.32	0.0008	337.5		16	252.82	150	1341	18.91	23.64	0.053	tension control

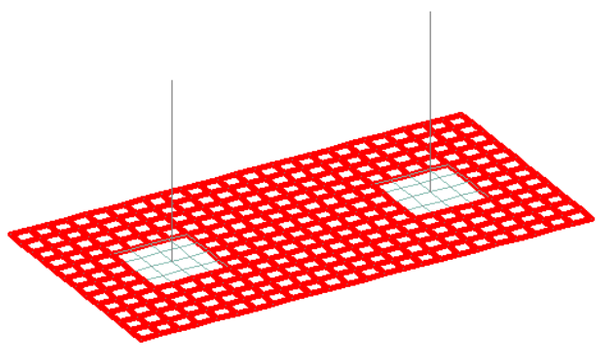
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	103 1.2(D	0.449	1	224.5	62.881	0.003	0.442	454.36	441.182	fc'758.3	441.18	460	> 224.5	ok	330.9	φVc>Vu	not required
	103 1.2(D	0.373	1	186.5	57.609			451.006			441.18	460	> 186.5	ok	330.9	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	16ϕT	**150**	c/c
At Top	16ϕT	**150**	c/c
Along Z Direction :			
At Bottom	16ϕT	**150**	c/c
At Top	16ϕT	**150**	c/c

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5.2 DESIGN OF CALCULATION OF FOOTING F2

STAAD Node - 162

	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	3.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.20 m	
U/S level Base Plate :	1.40 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

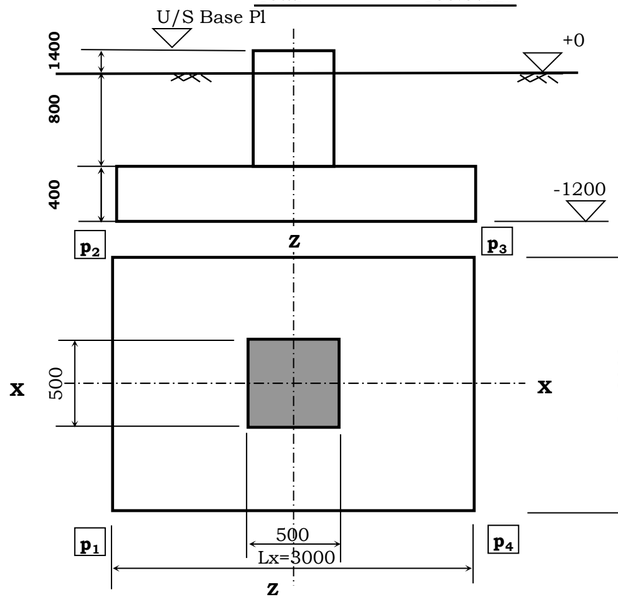
Yx 162

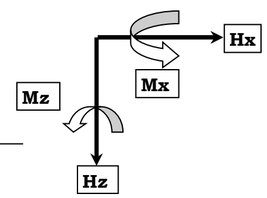
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(3*2*0.4*24)+(0.5*0.5*2.2*24) =	70.80KN
Weight of soil =	((3*2)-(0.5*0.5))*0.8*18 =	82.80KN
	Total =	153.60KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 332mm = 0.332m

Clear projection along Z direction = 0.75m

Clear projection along X direction = 1.25m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING F2

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+EQDL+FL	152.3	21.0	-54.6	-3.0	-7.8	305.90	21.00	-54.60	-3.00	-7.80
202	DL+EQDL+FL+LL+EQLL+TL	221.3	59.0	-153.4	-6.0	-15.6	374.90	59.00	-153.40	-6.00	-15.60
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	204.0	49.5	-128.7	-5.3	-13.7	357.60	49.50	-128.70	-5.30	-13.70
204	DL+EQDL+FL+WL(+X)	185.3	14.0	-36.4	-4.0	-10.4	338.90	14.00	-36.40	-4.00	-10.40
205	DL+EQDL+FL+WL(-X)	119.3	28.0	-72.8	-2.0	-5.2	272.90	28.00	-72.80	-2.00	-5.20
206	DL+EQDL+FL+WL(+Z)	178.3	23.0	-59.8	-5.0	-13.0	331.90	23.00	-59.80	-5.00	-13.00
207	DL+EQDL+FL+WL(-Z)	126.3	19.0	-49.4	-1.0	-2.6	279.90	19.00	-49.40	-1.00	-2.60
208	DL+EQDL+FL+0.7SL(+X)	167.7	19.6	-51.0	-3.7	-9.6	321.30	19.60	-51.00	-3.70	-9.60
209	DL+EQDL+FL+0.7SL(-X)	136.9	22.4	-58.2	-2.3	-6.0	290.50	22.40	-58.20	-2.30	-6.00
210	DL+EQDL+FL+0.7SL(+Z)	166.3	23.1	-60.1	-3.7	-9.6	319.90	23.10	-60.10	-3.70	-9.60
211	DL+EQDL+FL+0.7SL(-Z)	138.3	18.9	-49.1	-2.3	-6.0	291.90	18.90	-49.10	-2.30	-6.00
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQ	230.3	21.8	-56.6	-5.3	-13.7	383.90	21.80	-56.60	-5.30	-13.70
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQ	180.8	32.3	-83.9	-3.8	-9.8	334.40	32.30	-83.90	-3.80	-9.80
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQ	225.0	28.5	-74.1	-6.0	-15.6	378.60	28.50	-74.10	-6.00	-15.60
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQ	186.0	25.5	-66.3	-3.0	-7.8	339.60	25.50	-66.30	-3.00	-7.80
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+E	217.1	26.0	-67.5	-5.0	-13.1	370.70	26.00	-67.50	-5.00	-13.10
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+E	194.0	28.1	-72.9	-4.0	-10.3	347.60	28.10	-72.90	-4.00	-10.30
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+E	216.0	28.6	-74.3	-5.0	-13.1	369.60	28.60	-74.30	-5.00	-13.10
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+E	195.0	25.4	-66.1	-4.0	-10.3	348.60	25.40	-66.10	-4.00	-10.30
220	0.6(DL+EQDL)+WL(+X)	86.6	-0.4	1.0	-2.2	-5.7	240.20	-0.40	1.00	-2.20	-5.70
221	0.6(DL+EQDL)+WL(-X)	20.6	13.6	-35.4	-0.2	-0.5	174.20	13.60	-35.40	-0.20	-0.50
222	0.6(DL+EQDL)+WL(+Z)	79.6	8.6	-22.4	-3.2	-8.3	233.20	8.60	-22.40	-3.20	-8.30
223	0.6(DL+EQDL)+WL(-Z)	27.6	4.6	-12.0	0.8	2.1	181.20	4.60	-12.00	0.80	2.10
224	0.6(DL+EQDL)+0.7SL(+X)	69.0	5.2	-13.5	-1.9	-4.9	222.60	5.20	-13.50	-1.90	-4.90
225	0.6(DL+EQDL)+0.7SL(-X)	38.2	8.0	-20.8	-0.5	-1.3	191.80	8.00	-20.80	-0.50	-1.30
226	0.6(DL+EQDL)+0.7SL(+Z)	67.6	8.7	-22.6	-1.9	-4.9	221.20	8.70	-22.60	-1.90	-4.90
227	0.6(DL+EQDL)+0.7SL(-Z)	39.6	4.5	-11.7	-0.5	-1.3	193.20	4.50	-11.70	-0.50	-1.30
228	DL+EQTL	122.3	16.0	-41.6	-2.0	-5.2	275.90	16.00	-41.60	-2.00	-5.20
Ultimate Conditions											
101	1.4(DL+EQDL+FL)	213.2	29.4	-76.4	-4.2	-10.9	366.80	29.40	-76.40	-4.20	-10.90
102	1.2(DL+EQDL+FL)+1.6(LL+EQLL)	296.3	38.0	-98.8	-6.8	-17.7	449.90	38.00	-98.80	-6.80	-17.70
103	1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)	293.9	74.0	-192.4	-8.0	-20.8	447.50	74.00	-192.40	-8.00	-20.80
104	1.2(DL+EQDL)+0.8WL(+X)	133.5	7.6	-19.8	-3.2	-8.3	287.10	7.60	-19.80	-3.20	-8.30
105	1.2(DL+EQDL)+0.8WL(-X)	80.7	18.8	-48.9	-1.6	-4.2	234.30	18.80	-48.90	-1.60	-4.20
106	1.2(DL+EQDL)+0.8WL(+Z)	127.9	14.8	-38.5	-4.0	-10.4	281.50	14.80	-38.50	-4.00	-10.40
107	1.2(DL+EQDL)+0.8WL(-Z)	86.3	11.6	-30.2	-0.8	-2.1	239.90	11.60	-30.20	-0.80	-2.10
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQL	195.4	6.0	-15.6	-5.0	-13.0	349.00	6.00	-15.60	-5.00	-13.00
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQL	89.8	28.4	-73.8	-1.8	-4.7	243.40	28.40	-73.80	-1.80	-4.70
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQL	184.2	20.4	-53.0	-6.6	-17.2	337.80	20.40	-53.00	-6.60	-17.20
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQL	101.0	14.0	-36.4	-0.2	-0.5	254.60	14.00	-36.40	-0.20	-0.50
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+	196.1	20.2	0.0	-4.9	0.0	349.70	20.20	52.52	-4.90	-12.74
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+	152.1	24.2	0.0	-2.9	0.0	305.70	24.20	62.92	-2.90	-7.54
114	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+	194.1	25.2	0.0	-4.9	0.0	347.70	25.20	65.52	-4.90	-12.74
115	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+	154.1	19.2	0.0	-2.9	0.0	307.70	19.20	49.92	-2.90	-7.54
116	0.9(DL+EQDL)+1.6WL(+X)	133.1	-1.3	0.0	-3.4	0.0	286.70	-1.30	-3.38	-3.40	-8.84
117	0.9(DL+EQDL)+1.6WL(-X)	27.5	21.1	0.0	-0.2	0.0	181.10	21.10	54.86	-0.20	-0.52
118	0.9(DL+EQDL)+1.6WL(+Z)	121.9	13.1	0.0	-5.0	0.0	275.50	13.10	34.06	-5.00	-13.00
119	0.9(DL+EQDL)+1.6WL(-Z)	38.7	6.7	0.0	1.4	0.0	192.30	6.70	17.42	1.40	3.64
120	0.9(DL+EQDL)+SL(+X)	102.3	7.9	0.0	-2.8	0.0	255.90	7.90	20.54	-2.80	-7.28
121	0.9(DL+EQDL)+SL(-X)	58.3	11.9	0.0	-0.8	0.0	211.90	11.90	30.94	-0.80	-2.08
122	0.9(DL+EQDL)+SL(+Z)	100.3	12.9	0.0	-2.8	0.0	253.90	12.90	33.54	-2.80	-7.28
123	0.9(DL+EQDL)+SL(-Z)	60.3	6.9	0.0	-0.8	0.0	213.90	6.90	17.94	-0.80	-2.08
124	1.2DL+EQTL	132.5	17.0	0.0	-2.2	0.0	286.10	17.00	44.20	-2.20	-5.72

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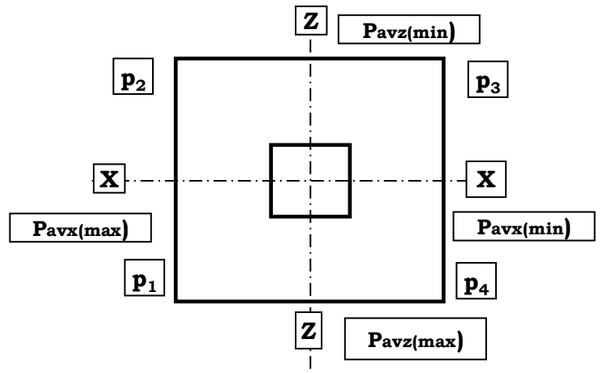
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	DL+EQDL+FL	305.9	21.0	-54.6	-3.0	-7.8	8.4	39.2	5.8	40.8
202	DL+EQDL+FL+LL+EQLL+TL	374.9	59.0	-153.4	-6.0	-15.6	3.7	24.0	2.5	25.0
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	357.6	49.5	-128.7	-5.3	-13.7	4.2	26.1	2.9	27.0
204	DL+EQDL+FL+WL(+X)	338.9	14.0	-36.4	-4.0	-10.4	14.0	32.6	9.7	33.9
205	DL+EQDL+FL+WL(-X)	272.9	28.0	-72.8	-2.0	-5.2	5.6	52.5	3.9	54.6
206	DL+EQDL+FL+WL(+Z)	331.9	23.0	-59.8	-5.0	-13.0	8.3	25.5	5.8	26.6
207	DL+EQDL+FL+WL(-Z)	279.9	19.0	-49.4	-1.0	-2.6	8.5	107.7	5.9	112.0
208	DL+EQDL+FL+0.7SL(+X)	321.3	19.6	-51.0	-3.7	-9.6	9.5	33.5	6.6	34.7
209	DL+EQDL+FL+0.7SL(-X)	290.5	22.4	-58.2	-2.3	-6.0	7.5	48.4	5.2	50.5
210	DL+EQDL+FL+0.7SL(+Z)	319.9	23.1	-60.1	-3.7	-9.6	8.0	33.3	5.5	34.6
211	DL+EQDL+FL+0.7SL(-Z)	291.9	18.9	-49.1	-2.3	-6.0	8.9	48.7	6.2	50.8
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)	383.9	21.8	-56.6	-5.3	-13.7	10.2	28.0	7.0	29.0
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)	334.4	32.3	-83.9	-3.8	-9.8	6.0	34.1	4.1	35.2
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)	378.6	28.5	-74.1	-6.0	-15.6	7.7	24.3	5.3	25.2
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)	339.6	25.5	-66.3	-3.0	-7.8	7.7	43.5	5.3	45.3
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL	370.7	26.0	-67.5	-5.0	-13.1	8.2	28.3	5.7	29.7
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)	347.6	28.1	-72.9	-4.0	-10.3	7.2	33.7	4.9	34.8
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL	369.6	28.6	-74.3	-5.0	-13.1	7.5	28.2	5.2	29.6
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)	348.6	25.4	-66.1	-4.0	-10.3	7.9	33.8	5.5	34.9
220	0.6(DL+EQDL)+WL(+X)	240.2	-0.4	1.0	-2.2	-5.7	360.3	42.1	240.2	43.7
221	0.6(DL+EQDL)+WL(-X)	174.2	13.6	-35.4	-0.2	-0.5	7.4	348.4	5.1	348.4
222	0.6(DL+EQDL)+WL(+Z)	233.2	8.6	-22.4	-3.2	-8.3	15.6	28.1	10.8	29.2
223	0.6(DL+EQDL)+WL(-Z)	181.2	4.6	-12.0	0.8	2.1	22.7	86.3	15.8	90.6
224	0.6(DL+EQDL)+0.7SL(+X)	222.6	5.2	-13.5	-1.9	-4.9	24.7	45.4	17.1	46.9
225	0.6(DL+EQDL)+0.7SL(-X)	191.8	8.0	-20.8	-0.5	-1.3	13.8	147.5	9.6	153.4
226	0.6(DL+EQDL)+0.7SL(+Z)	221.2	8.7	-22.6	-1.9	-4.9	14.7	45.1	10.2	46.6
227	0.6(DL+EQDL)+0.7SL(-Z)	193.2	4.5	-11.7	-0.5	-1.3	24.8	148.6	17.2	154.6
228	DL+EQTL	275.9	16.0	-41.6	-2.0	-5.2	9.9	53.1	6.9	55.2
						Min.=	**3.7**	**24.0**	**2.5**	**25.0**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F2



Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 471.6 kN/m² For WL/SL case = 584.1 kN/m²

Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	Base Pressure (kN/m²) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	DL+EQDL+FL	305.90	-54.60	-7.80	50.98	18.20	3.90	73.08	65.28	36.68	28.88	51.48	69.18	32.78	54.88	47.08	O.K
202	DL+EQDL+FL+LL+EQLL+TL	374.90	-153.40	-15.60	62.48	51.13	7.80	121.42	105.82	19.15	3.55	99.82	113.62	11.35	70.29	54.69	O.K
203	DL+EQDL+FL+0.75(LL+EQLL)+0	357.60	-128.70	-13.70	59.60	42.90	6.85	109.35	95.65	23.55	9.85	87.75	102.50	16.70	66.45	52.75	O.K
204	DL+EQDL+FL+WL(+X)	338.90	-36.40	-10.40	56.48	12.13	5.20	73.82	63.42	49.55	39.15	52.22	68.62	44.35	61.69	51.29	O.K
205	DL+EQDL+FL+WL(-X)	272.90	-72.80	-5.20	45.48	24.27	2.60	72.35	67.15	23.82	18.62	50.75	69.75	21.22	48.09	42.89	O.K
206	DL+EQDL+FL+WL(+Z)	331.90	-59.80	-13.00	55.32	19.93	6.50	81.75	68.75	41.88	28.88	60.15	75.25	35.38	61.82	48.82	O.K
207	DL+EQDL+FL+WL(-Z)	279.90	-49.40	-2.60	46.65	16.47	1.30	64.42	61.82	31.48	28.88	42.82	63.12	30.18	47.95	45.35	O.K
208	DL+EQDL+FL+0.7SL(+X)	321.30	-51.00	-9.60	53.55	17.00	4.80	75.35	65.75	41.35	31.75	53.75	70.55	36.55	58.35	48.75	O.K
209	DL+EQDL+FL+0.7SL(-X)	290.50	-58.20	-6.00	48.42	19.40	3.00	70.82	64.82	32.02	26.02	49.22	67.82	29.02	51.42	45.42	O.K
210	DL+EQDL+FL+0.7SL(+Z)	319.90	-60.10	-9.60	53.32	20.03	4.80	78.15	68.55	38.08	28.48	56.55	73.35	33.28	58.12	48.52	O.K
211	DL+EQDL+FL+0.7SL(-Z)	291.90	-49.10	-6.00	48.65	16.37	3.00	68.02	62.02	35.28	29.28	46.42	65.02	32.28	51.65	45.65	O.K
212	DL+EQDL+FL+0.75WL(+X)+0.75	383.90	-56.60	-13.70	63.98	18.87	6.85	89.70	76.00	51.97	38.27	68.10	82.85	45.12	70.84	57.14	O.K
213	DL+EQDL+FL+0.75WL(-X)+0.75(	334.40	-83.90	-9.80	55.73	27.97	4.90	88.60	78.80	32.67	22.87	67.00	83.70	27.77	60.64	50.84	O.K
214	DL+EQDL+FL+0.75WL(+Z)+0.75	378.60	-74.10	-15.60	63.10	24.70	7.80	95.60	80.00	46.20	30.60	74.00	87.80	38.40	70.90	55.30	O.K

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215 DL+EQDL+FL+0.75WL(-Z)+0.75(	339.60	-66.30	-7.80	56.60	22.10	3.90	82.60	74.80	38.40	30.60	61.00	78.70	34.50	60.50	52.70	O.K
216 DL+EQDL+FL+0.525SL(+X)+0.75	370.70	-67.50	-13.10	61.78	22.50	6.55	90.83	77.73	45.83	32.73	69.23	84.28	39.28	68.33	55.23	O.K
217 DL+EQDL+FL+0.525SL(-X)+0.75	347.60	-72.90	-10.30	57.93	24.30	5.15	87.38	77.08	38.78	28.48	65.78	82.23	33.63	63.08	52.78	O.K
218 DL+EQDL+FL+0.525SL(+Z)+0.75	369.60	-74.30	-13.10	61.60	24.77	6.55	92.92	79.82	43.38	30.28	71.32	86.37	36.83	68.15	55.05	O.K
219 DL+EQDL+FL+0.525SL(-Z)+0.75	348.60	-66.10	-10.30	58.10	22.03	5.15	85.28	74.98	41.22	30.92	63.68	80.13	36.07	63.25	52.95	O.K
220 0.6(DL+EQDL)+WL(+X)	240.20	1.00	-5.70	40.03	0.33	2.85	43.22	37.52	42.55	36.85	21.62	40.37	39.70	42.89	37.19	O.K
221 0.6(DL+EQDL)+WL(-X)	174.20	-35.40	-0.50	29.03	11.80	0.25	41.08	40.58	17.48	16.98	19.48	40.83	17.23	29.28	28.78	O.K
222 0.6(DL+EQDL)+WL(+Z)	233.20	-22.40	-8.30	38.87	7.47	4.15	50.48	42.18	35.55	27.25	28.88	46.33	31.40	43.02	34.72	O.K
223 0.6(DL+EQDL)+WL(-Z)	181.20	-12.00	2.10	30.20	4.00	1.05	35.25	33.15	27.25	25.15	13.65	34.20	26.20	31.25	29.15	O.K
224 0.6(DL+EQDL)+0.7SL(+X)	222.60	-13.50	-4.90	37.10	4.50	2.45	44.05	39.15	35.05	30.15	22.45	41.60	32.60	39.55	34.65	O.K
225 0.6(DL+EQDL)+0.7SL(-X)	191.80	-20.80	-1.30	31.97	6.93	0.65	39.55	38.25	25.68	24.38	17.95	38.90	25.03	32.62	31.32	O.K
226 0.6(DL+EQDL)+0.7SL(+Z)	221.20	-22.60	-4.90	36.87	7.53	2.45	46.85	41.95	31.78	26.88	25.25	44.40	29.33	39.32	34.42	O.K
227 0.6(DL+EQDL)+0.7SL(-Z)	193.20	-11.70	-1.30	32.20	3.90	0.65	36.75	35.45	28.95	27.65	15.15	36.10	28.30	32.85	31.55	O.K
228 DL+EQTL	275.90	-41.60	-5.20	45.98	13.87	2.60	62.45	57.25	34.72	29.52	40.85	59.85	32.12	48.59	43.39	O.K

Ultimate Conditions																
101 1.4(DL+EQDL+FL)	366.80	-76.40	-10.90	61.13	25.47	5.45	92.05	81.15	41.12	30.22	70.45	86.60	35.67	66.59	55.69	
102 1.2(DL+EQDL+FL)+1.6(LL+EQLL	449.90	-98.80	-17.70	74.98	32.93	8.85	116.77	99.07	50.90	33.20	95.17	107.92	42.05	83.84	66.14	
103 1.2(DL+EQDL+FL+TL)+1.6(LL+E	447.50	-192.40	-20.80	74.58	64.13	10.40	149.12	128.32	20.85	0.05	127.52	138.72	10.45	84.99	64.19	
104 1.2(DL+EQDL)+0.8WL(+X)	287.10	-19.80	-8.30	47.85	6.60	4.15	58.60	50.30	45.40	37.10	37.00	54.45	41.25	52.00	43.70	
105 1.2(DL+EQDL)+0.8WL(-X)	234.30	-48.90	-4.20	39.05	16.30	2.10	57.45	53.25	24.85	20.65	35.85	55.35	22.75	41.15	36.95	
106 1.2(DL+EQDL)+0.8WL(+Z)	281.50	-38.50	-10.40	46.92	12.83	5.20	64.95	54.55	39.28	28.88	43.35	59.75	34.08	52.12	41.72	
107 1.2(DL+EQDL)+0.8WL(-Z)	239.90	-30.20	-2.10	39.98	10.07	1.05	51.10	49.00	30.97	28.87	29.50	50.05	29.92	41.04	38.94	
108 1.2(DL+EQDL)+1.6WL(+X)+0.5(L	349.00	-15.60	-13.00	58.17	5.20	6.50	69.87	56.87	59.47	46.47	48.27	63.37	52.97	64.67	51.67	

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109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LI	243.40	-73.80	-4.70	40.57	24.60	2.35	67.52	62.82	18.32	13.62	45.92	65.17	15.97	42.92	38.22	
110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(L	337.80	-53.00	-17.20	56.30	17.67	8.60	82.57	65.37	47.23	30.03	60.97	73.97	38.63	64.90	47.70	
111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LI	254.60	-36.40	-0.50	42.43	12.13	0.25	54.82	54.32	30.55	30.05	33.22	54.57	30.30	42.69	42.19	
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+E	349.70	52.52	-12.74	58.28	17.51	6.37	82.16	69.42	47.15	34.41	60.56	75.79	40.78	64.66	51.92	
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+E	305.70	62.92	-7.54	50.95	20.97	3.77	75.69	68.15	33.75	26.21	54.09	71.92	29.98	54.72	47.18	
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+E	347.70	65.52	-12.74	57.95	21.84	6.37	86.16	73.42	42.48	29.74	64.56	79.79	36.11	64.32	51.58	
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+E	307.70	49.92	-7.54	51.28	16.64	3.77	71.69	64.15	38.41	30.87	50.09	67.92	34.64	55.05	47.51	
116 0.9(DL+EQDL)+1.6WL(+X)	286.70	-3.38	-8.84	47.78	1.13	4.42	53.33	44.49	51.08	42.24	31.73	48.91	46.66	52.21	43.37	
117 0.9(DL+EQDL)+1.6WL(-X)	181.10	54.86	-0.52	30.18	18.29	0.26	48.73	48.21	12.16	11.64	27.13	48.47	11.90	30.45	29.93	
118 0.9(DL+EQDL)+1.6WL(+Z)	275.50	34.06	-13.00	45.92	11.35	6.50	63.77	50.77	41.06	28.06	42.17	57.27	34.56	52.42	39.42	
119 0.9(DL+EQDL)+1.6WL(-Z)	192.30	17.42	3.64	32.05	5.81	1.82	39.68	36.04	28.06	24.42	18.08	37.86	26.24	33.87	30.23	
120 0.9(DL+EQDL)+SL(+X)	255.90	20.54	-7.28	42.65	6.85	3.64	53.14	45.86	39.44	32.16	31.54	49.50	35.80	46.29	39.01	
121 0.9(DL+EQDL)+SL(-X)	211.90	30.94	-2.08	35.32	10.31	1.04	46.67	44.59	26.04	23.96	25.07	45.63	25.00	36.36	34.28	
122 0.9(DL+EQDL)+SL(+Z)	253.90	33.54	-7.28	42.32	11.18	3.64	57.14	49.86	34.78	27.50	35.54	53.50	31.14	45.96	38.68	
123 0.9(DL+EQDL)+SL(-Z)	213.90	17.94	-2.08	35.65	5.98	1.04	42.67	40.59	30.71	28.63	21.07	41.63	29.67	36.69	34.61	
124 1.2DL+EQTL	286.10	44.20	-5.72	47.68	14.73	2.86	65.28	59.56	35.81	30.09	43.68	62.42	32.95	50.55	44.83	
Max :	450	66	4													
Min :	174	-192	-21													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 51.48 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 61.48 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 99.82 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 109.82 kN/m²

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Sample calculation of values of base pressure:

Load case : 202 DL+EQDL+FL+LL+EQLL+TL

Maximum pressure, p_1 (pmax) = 62.48+51.13+7.80= 121.42 kN/m^2

p_2= 62.48+51.13-7.80= 105.82 kN/m^2

p_3= 62.48-(51.13+7.80)= 3.55 kN/m^2

p_4= 62.48-51.13+7.80= 19.15 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 121.42-1.2*18= 99.82 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (121.42+105.82)/2= 113.62 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (3.55+19.15)/2= 11.35 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (121.42+19.15)/2= 70.28 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (3.55+105.82)/2= 54.68 kN/m^2

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DESIGN OF ISOLATED FOOTING F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.332)+(0.5+2*0.5*0.332))*2

= 3.328 m

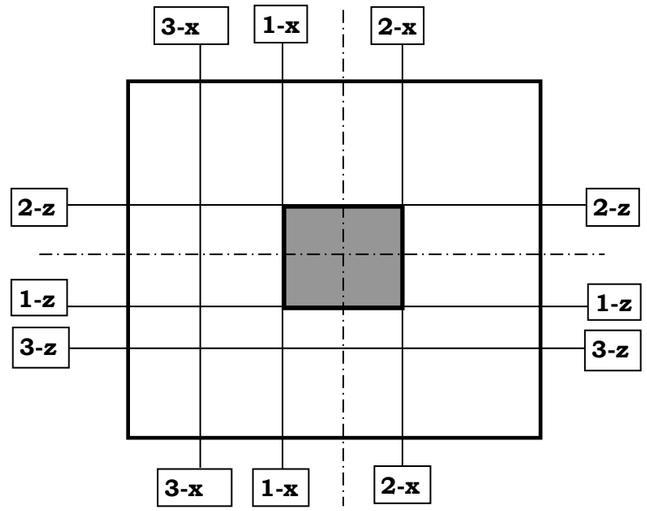
Punching Area= ((0.5+2*0.5*0.332)*(0.5+2*0.5*0.332))

= 0.69 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+0.8*18)

= 24.00 kN/m^2

Area of footing, A_f= 6.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+EQDL+FL)	65.38	56.89	71.22	43.38	14.64	49.75	197.18
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	80.47	69.50	88.03	58.42	21.25	67.02	270.72
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	85.27	63.90	99.98	75.71	3.33	86.39	268.60
104	1.2(DL+EQDL)+0.8WL(+X)	48.95	46.75	50.46	22.36	14.91	25.78	126.64
105	1.2(DL+EQDL)+0.8WL(-X)	41.77	36.33	45.50	20.95	2.56	23.94	79.92
106	1.2(DL+EQDL)+0.8WL(+Z)	49.05	44.78	52.00	25.14	10.66	28.88	121.69
107	1.2(DL+EQDL)+0.8WL(-Z)	41.66	38.31	43.97	18.17	6.81	20.85	84.87
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(	59.04	57.30	60.23	29.63	23.76	34.25	181.43

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109 1.2(DL+EQDL)+1.6WL(-X)+0.5(	44.67	36.47	50.31	26.83	-0.93	30.57	87.97
110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(	59.25	53.36	63.30	35.20	15.26	40.44	171.51
111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(	44.46	40.41	47.24	21.25	7.56	24.38	97.88
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+	61.20	55.37	65.22	36.66	16.91	42.13	182.04
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+	54.45	47.46	59.25	32.89	9.22	37.68	143.10
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+	61.59	54.31	66.60	38.85	14.20	44.57	180.27
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+	54.05	48.51	57.87	30.70	11.92	35.24	144.87
116 0.9(DL+EQDL)+1.6WL(+X)	47.97	47.60	48.23	19.22	17.95	22.26	126.29
117 0.9(DL+EQDL)+1.6WL(-X)	33.23	27.14	37.43	15.15	-5.49	17.17	32.83
118 0.9(DL+EQDL)+1.6WL(+Z)	47.81	44.02	50.41	23.53	10.71	27.04	116.38
119 0.9(DL+EQDL)+1.6WL(-Z)	33.02	31.08	34.35	9.57	3.01	10.97	42.75
120 0.9(DL+EQDL)+SL(+X)	43.79	41.51	45.36	18.44	10.71	21.23	99.03
121 0.9(DL+EQDL)+SL(-X)	37.03	33.60	39.40	14.66	3.02	16.77	60.09
122 0.9(DL+EQDL)+SL(+Z)	44.18	40.46	46.75	20.62	8.00	23.67	97.26
123 0.9(DL+EQDL)+SL(-Z)	36.65	34.65	38.02	12.48	5.73	14.34	61.86
124 1.2DL+EQTL	50.14	45.23	53.52	26.82	10.19	30.78	125.76

Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1x} = 86.6-(86.6-35.67)*1.25/3 = 65.38 kN/m^2 Overhang = 1.25 m

p_{2x} = 35.67+(86.6-35.67)*1.25/3 = 56.89 kN/m^2 Effective depth = 0.344 m

p_{3x} = 86.6-(86.6-35.67)*(1.25-0.344)/3 = 71.22 kN/m^2

BM_{1x} = (65.38-24)*1.25*1.25/2+0.5*(86.6-65.38)*1.25*1.25*0.67 = 43.38 kN-m/m

BM_{2x} = (56.89-24)*1.25*1.25/2-0.5*(-35.67+56.89)*1.25*1.25*0.67 = 14.64 kN-m/m

SF_{3x} = (71.22-24)*(1.25-0.344)+0.5*(86.6-71.22)*(1.25-0.344) = 49.75 kN/m

$S.F_{(punching)}$ = (6-0.69)*(61.13-24) = 197.18 kN

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	75.71	0.64	0.85	0.002	708.97	619.20	12ϕT	159.52	300	150	753.98
At top	-5.49	-0.05	0.06	0.000	50.67	309.60	10ϕT	253.68	300	250	314.16

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 86.39 kN/m

Shear stress , τ_v= (86.39*1000)/(1000*344)

= 0.25 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 270.72 kN

Strength of Concrete in shear Vc1 = 3386.46 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2295.63 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2257.64 kN **Table 22.6.5.2_ (c)**

Design Vc= 2257.64 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (270.72*1000)/(3328*344)

= 0.24 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+EQDL+FL)	62.50	59.78	64.31	11.60	9.29	17.33
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	77.20	72.78	80.14	16.21	12.47	24.24
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	77.19	71.99	80.64	16.43	12.03	24.59
104	1.2(DL+EQDL)+0.8WL(+X)	48.89	46.81	50.27	7.59	5.83	11.34
105	1.2(DL+EQDL)+0.8WL(-X)	39.58	38.53	40.27	4.68	3.79	6.99
106	1.2(DL+EQDL)+0.8WL(+Z)	48.22	45.62	49.95	7.55	5.35	11.30
107	1.2(DL+EQDL)+0.8WL(-Z)	40.25	39.73	40.60	4.72	4.27	7.03
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(	59.80	56.55	61.95	10.99	8.23	16.43
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(	41.16	39.98	41.94	5.16	4.16	7.70
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(	58.45	54.15	61.31	10.90	7.26	16.34
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(	42.50	42.38	42.59	5.24	5.13	7.79
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+	59.88	56.70	62.00	10.99	8.30	16.44
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+	51.89	50.01	53.14	8.38	6.78	12.51
114	1.2(DL+EQDL)+SL(+Z)+0.5(LL+	59.54	56.36	61.66	10.90	8.20	16.30

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115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+I	52.22	50.34	53.47	8.47	6.87	12.65
116 0.9(DL+EQDL)+1.6WL(+X)	48.90	46.69	50.36	7.63	5.76	11.41
117 0.9(DL+EQDL)+1.6WL(-X)	30.26	30.13	30.34	1.80	1.69	2.67
118 0.9(DL+EQDL)+1.6WL(+Z)	47.55	44.30	49.70	7.54	4.79	11.31
119 0.9(DL+EQDL)+1.6WL(-Z)	32.51	31.60	33.11	2.65	1.88	3.97
120 0.9(DL+EQDL)+SL(+X)	43.56	41.74	44.77	6.02	4.47	9.00
121 0.9(DL+EQDL)+SL(-X)	35.58	36.17	35.93	3.40	3.09	5.08
122 0.9(DL+EQDL)+SL(+Z)	43.23	36.49	44.44	5.92	2.58	8.86
123 0.9(DL+EQDL)+SL(-Z)	35.91	39.85	36.26	3.50	5.13	5.21
124 1.2DL+EQTL	48.41	28.02	49.35	7.27	-4.15	10.85

Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1z} = 66.59-(66.59-55.69)*0.75/2= 62.50 kN/m²

p_{2z} = 55.69+(66.59-55.69)*0.75/2= 59.78 kN/m²

p_{3z} = 66.59-(66.59-55.69)*(0.75-0.332)/2= 64.31 kN/m²

Overhang = 0.75 m

Effective depth = 0.332 m

BM_{1z} = (62.50-24)*0.75*0.75/2+0.5*(66.59-62.50)*0.75*0.75*0.67 = 11.60 kN-m/m

BM_{2z} = (59.78-24)*0.75*0.75/2-0.5*(-55.69+59.78)*0.75*0.75*0.67 = 9.29 kN-m/m

p_{1z} = 52.12-(52.12-41.72)*0.75/2= 48.22 kN/m²

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.	Ast (mm^2)
At bottom	16.43	0.149	0.199	0.0005	163.33	597.60	12ϕT	189.25	300	150	754.0
At top	-4.15	-0.038	-0.050	-0.0001	-0.40	298.80	10ϕT	262.85	250	250	314.2

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm²

Shear at a distance 'd' from face of column, Vu= 24.59 kN/m

Shear stress , τ_v= (24.59*1000)/(1000*332)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	**150**	c/c
	At Top	10ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	12ϕT	**150**	c/c
	At Top	10ϕT	**250**	c/c

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5.3 DESIGN OF CALCULATION OF FOOTING F3

STAAD Node - 161

	Along-X	Along-Z
Size of column :	0.65m	0.65m
Size of fdn. raft :	3.00m	3.00m
Thickness of raft :	0.60 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.20 m	
U/S level Base Plate :	1.40 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

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SYMBOLS AND NOTATIONS

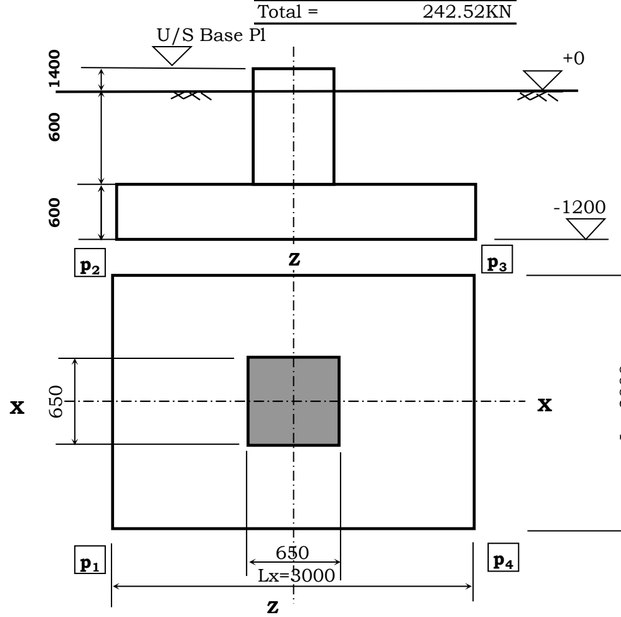
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(3*3*0.6*24)+(0.65*0.65*2*24) =	149.88KN
Weight of soil =	((3*3)-(0.65*0.65))*0.6*18 =	92.64KN
	Total =	242.52KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

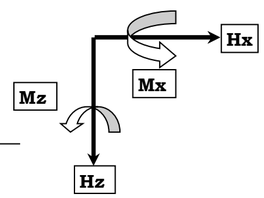


Effective depth of slab along X direction = 542mm = 0.542m

Effective depth of slab along Z direction = 526mm = 0.526m

Clear projection along Z direction = 1.175m

Clear projection along X direction = 1.175m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING F3

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+EQDL+FL	307.5	-20.0	52.0	93.0	241.8	550.02	-20.00	52.00	93.00	241.80
202	DL+EQDL+FL+LL+EQLL+TL	547.5	-74.0	192.4	177.0	530.4	790.02	-74.00	192.40	177.00	530.40
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	487.5	-60.5	157.3	170.3	458.3	730.02	-60.50	157.30	170.30	458.30
204	DL+EQDL+FL+WL(+X)	408.5	-62.0	161.2	109.0	283.4	651.02	-62.00	161.20	109.00	283.40
205	DL+EQDL+FL+WL(-X)	206.5	22.0	-57.2	77.0	200.2	449.02	22.00	-57.20	77.00	200.20
206	DL+EQDL+FL+WL(+Z)	279.5	-16.0	41.6	75.0	195.0	522.02	-16.00	41.60	75.00	195.00
207	DL+EQDL+FL+WL(-Z)	335.5	-24.0	62.4	111.0	288.6	578.02	-24.00	62.40	111.00	288.60
208	DL+EQDL+FL+0.7SL(+X)	356.5	-36.8	95.7	101.4	263.6	599.02	-36.80	95.70	101.40	263.60
209	DL+EQDL+FL+0.7SL(-X)	258.5	-3.2	8.3	84.6	220.0	501.02	-3.20	8.30	84.60	220.00
210	DL+EQDL+FL+0.7SL(+Z)	280.2	-17.9	46.5	77.6	201.8	522.72	-17.90	46.50	77.60	201.80
211	DL+EQDL+FL+0.7SL(-Z)	334.8	-22.1	57.5	108.4	281.8	577.32	-22.10	57.50	108.40	281.80
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQ	558.0	-67.3	174.9	149.3	388.1	800.52	-67.30	174.90	149.30	388.10
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQ	406.5	-4.3	11.1	125.3	325.7	649.02	-4.30	11.10	125.30	325.70
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQ	461.2	-32.8	85.2	123.8	321.8	703.72	-32.80	85.20	123.80	321.80
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQ	503.2	-38.8	100.8	150.8	392.0	745.72	-38.80	100.80	150.80	392.00
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+E	519.0	-48.4	125.7	143.6	373.2	761.52	-48.40	125.70	143.60	373.20
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQ	445.5	-23.2	60.2	131.0	340.5	688.02	-23.20	60.20	131.00	340.50
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+E	461.7	-34.2	88.9	125.7	326.8	704.22	-34.20	88.90	125.70	326.80
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQ	502.7	-37.3	97.0	148.8	386.9	745.22	-37.30	97.00	148.80	386.90
220	0.6(DL+EQDL)+WL(+X)	193.7	-48.6	126.4	42.4	110.2	436.22	-48.60	126.40	42.40	110.20
221	0.6(DL+EQDL)+WL(-X)	-8.3	35.4	-92.0	10.4	27.0	234.22	35.40	-92.00	10.40	27.00
222	0.6(DL+EQDL)+WL(+Z)	64.7	-2.6	6.8	8.4	21.8	307.22	-2.60	6.80	8.40	21.80
223	0.6(DL+EQDL)+WL(-Z)	120.7	-10.6	27.6	44.4	115.4	363.22	-10.60	27.60	44.40	115.40
224	0.6(DL+EQDL)+0.7SL(+X)	141.7	-23.4	60.8	34.8	90.5	384.22	-23.40	60.80	34.80	90.50
225	0.6(DL+EQDL)+0.7SL(-X)	43.7	10.2	-26.5	18.0	46.8	286.22	10.20	-26.50	18.00	46.80
226	0.6(DL+EQDL)+0.7SL(+Z)	65.4	-4.5	11.7	11.0	28.6	307.92	-4.50	11.70	11.00	28.60
227	0.6(DL+EQDL)+0.7SL(-Z)	120.0	-8.7	22.6	41.8	108.7	362.52	-8.70	22.60	41.80	108.70
228	DL+EQTL	150.5	-11.0	28.6	51.0	132.6	393.02	-11.00	28.60	51.00	132.60
Ultimate Conditions											
101	1.4(DL+EQDL+FL)	430.4	-28.0	72.8	130.2	338.5	672.92	-28.00	72.80	130.20	338.50
102	1.2(DL+EQDL+FL)+1.6(LL+EQLL)	741.8	-57.6	149.8	206.0	535.6	984.32	-57.60	149.80	206.00	535.60
103	1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)	750.2	-97.2	252.7	268.4	697.8	992.72	-97.20	252.70	268.40	697.80
104	1.2(DL+EQDL)+0.8WL(+X)	266.2	-46.8	121.7	65.6	170.6	508.72	-46.80	121.70	65.60	170.60
105	1.2(DL+EQDL)+0.8WL(-X)	104.6	20.4	-53.0	40.0	104.0	347.12	20.40	-53.00	40.00	104.00
106	1.2(DL+EQDL)+0.8WL(+Z)	163.0	-10.0	26.0	38.4	99.8	405.52	-10.00	26.00	38.40	99.80
107	1.2(DL+EQDL)+0.8WL(-Z)	207.8	-16.4	42.6	67.2	174.7	450.32	-16.40	42.60	67.20	174.70
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQL	463.5	-90.9	236.3	107.9	280.5	706.02	-90.90	236.30	107.90	280.50
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQL	140.3	43.5	-113.1	56.7	147.4	382.82	43.50	-113.10	56.70	147.40
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQL	257.1	-17.3	45.0	53.5	139.1	499.62	-17.30	45.00	53.50	139.10
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQL	346.7	-30.1	78.3	111.1	288.9	589.22	-30.10	78.30	111.10	288.90
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+	448.4	-52.2	0.0	118.8	0.0	690.92	-52.20	-135.72	118.80	308.88
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+(	308.4	-4.2	0.0	94.8	0.0	550.92	-4.20	-10.92	94.80	246.48
114	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+(	339.4	-25.2	0.0	84.8	0.0	581.92	-25.20	-65.52	84.80	220.48
115	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+(	417.4	-31.2	0.0	128.8	0.0	659.92	-31.20	-81.12	128.80	334.88
116	0.9(DL+EQDL)+1.6WL(+X)	300.6	-77.1	0.0	65.2	0.0	543.12	-77.10	-200.46	65.20	169.52
117	0.9(DL+EQDL)+1.6WL(-X)	-22.6	57.3	0.0	14.0	0.0	219.92	57.30	148.98	14.00	36.40
118	0.9(DL+EQDL)+1.6WL(+Z)	94.2	-3.5	0.0	10.8	0.0	336.72	-3.50	-9.10	10.80	28.08
119	0.9(DL+EQDL)+1.6WL(-Z)	183.8	-16.3	0.0	68.4	0.0	426.32	-16.30	-42.38	68.40	177.84
120	0.9(DL+EQDL)+SL(+X)	209.0	-33.9	0.0	51.6	0.0	451.52	-33.90	-88.14	51.60	134.16
121	0.9(DL+EQDL)+SL(-X)	69.0	14.1	0.0	27.6	0.0	311.52	14.10	36.66	27.60	71.76
122	0.9(DL+EQDL)+SL(+Z)	100.0	-6.9	0.0	17.6	0.0	342.52	-6.90	-17.94	17.60	45.76
123	0.9(DL+EQDL)+SL(-Z)	178.0	-12.9	0.0	61.6	0.0	420.52	-12.90	-33.54	61.60	160.16
124	1.2DL+EQTL	168.0	-12.4	0.0	54.8	0.0	410.52	-12.40	-32.24	54.80	142.48

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Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	DL+EQDL+FL	550.0	-20.0	52.0	93.0	241.8	15.9	3.4	11.0	2.4
202	DL+EQDL+FL+LL+EQLL+TL	790.0	-74.0	192.4	177.0	530.4	6.2	2.2	4.3	1.8
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	730.0	-60.5	157.3	170.3	458.3	7.0	2.4	4.8	1.7
204	DL+EQDL+FL+WL(+X)	651.0	-62.0	161.2	109.0	283.4	6.1	3.4	4.2	2.4
205	DL+EQDL+FL+WL(-X)	449.0	22.0	-57.2	77.0	200.2	11.8	3.4	8.2	2.3
206	DL+EQDL+FL+WL(+Z)	522.0	-16.0	41.6	75.0	195.0	18.8	4.0	13.1	2.8
207	DL+EQDL+FL+WL(-Z)	578.0	-24.0	62.4	111.0	288.6	13.9	3.0	9.6	2.1
208	DL+EQDL+FL+0.7SL(+X)	599.0	-36.8	95.7	101.4	263.6	9.4	3.4	6.5	2.4
209	DL+EQDL+FL+0.7SL(-X)	501.0	-3.2	8.3	84.6	220.0	90.5	3.4	62.6	2.4
210	DL+EQDL+FL+0.7SL(+Z)	522.7	-17.9	46.5	77.6	201.8	16.9	3.9	11.7	2.7
211	DL+EQDL+FL+0.7SL(-Z)	577.3	-22.1	57.5	108.4	281.8	15.1	3.1	10.4	2.1
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)	800.5	-67.3	174.9	149.3	388.1	6.9	3.1	4.8	2.1
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)	649.0	-4.3	11.1	125.3	325.7	87.7	3.0	60.4	2.1
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)	703.7	-32.8	85.2	123.8	321.8	12.4	3.3	8.6	2.3
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)	745.7	-38.8	100.8	150.8	392.0	11.1	2.9	7.7	2.0
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL	761.5	-48.4	125.7	143.6	373.2	9.1	3.1	6.3	2.1
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)	688.0	-23.2	60.2	131.0	340.5	17.1	3.0	11.9	2.1
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL	704.2	-34.2	88.9	125.7	326.8	11.9	3.2	8.2	2.2
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)	745.2	-37.3	97.0	148.8	386.9	11.5	2.9	8.0	2.0
220	0.6(DL+EQDL)+WL(+X)	436.2	-48.6	126.4	42.4	110.2	5.2	5.9	3.6	4.1
221	0.6(DL+EQDL)+WL(-X)	234.2	35.4	-92.0	10.4	27.0	3.8	13.0	2.6	9.0
222	0.6(DL+EQDL)+WL(+Z)	307.2	-2.6	6.8	8.4	21.8	67.8	21.1	47.3	14.6
223	0.6(DL+EQDL)+WL(-Z)	363.2	-10.6	27.6	44.4	115.4	19.7	4.7	13.7	3.3
224	0.6(DL+EQDL)+0.7SL(+X)	384.2	-23.4	60.8	34.8	90.5	9.5	6.4	6.6	4.4
225	0.6(DL+EQDL)+0.7SL(-X)	286.2	10.2	-26.5	18.0	46.8	16.2	9.2	11.2	6.4
226	0.6(DL+EQDL)+0.7SL(+Z)	307.9	-4.5	11.7	11.0	28.6	39.5	16.1	27.4	11.2
227	0.6(DL+EQDL)+0.7SL(-Z)	362.5	-8.7	22.6	41.8	108.7	24.1	5.0	16.7	3.5
228	DL+EQTL	393.0	-11.0	28.6	51.0	132.6	20.6	4.4	14.3	3.1
						Min.=	**3.8**	**2.2**	**2.6**	**1.7**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F3

Load Combination & Calculation of Base pressure

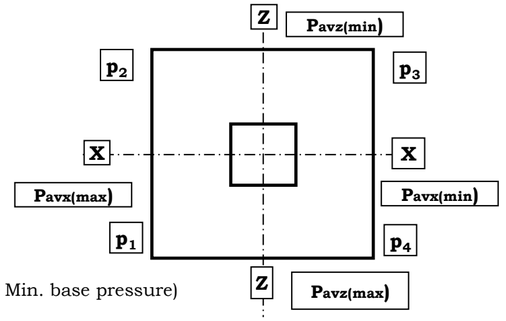
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 471.6 kN/m^2 For WL/SL case = 584.1 kN/m^2



*Max. Base pressure(pmax) modification done using Teng's chart (For negative Min. base pressure)

Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)^2.Lz	6.Mx/Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_1 (p_{max}) - modified using Teng's chart*	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	DL+EQDL+FL	550.02	52.00	241.80	61.11	11.56	53.73	126.40	126.52	18.94	103.29	0.00	104.92	72.73	51.65	114.91	9.47	O.K
202	DL+EQDL+FL+LL+EQLL+TL	790.02	192.40	530.40	87.78	42.76	117.87	248.40	269.89	12.67	162.89	0.00	248.29	141.28	81.45	216.39	6.34	O.K
203	DL+EQDL+FL+0.75(LL+EQLL)+0	730.02	157.30	458.30	81.11	34.96	101.84	217.91	230.64	14.22	148.00	0.00	209.04	122.43	74.00	189.32	7.11	O.K
204	DL+EQDL+FL+WL(+X)	651.02	161.20	283.40	72.34	35.82	62.98	171.14	173.16	45.18	99.49	0.00	151.56	109.17	49.75	136.33	22.59	O.K
205	DL+EQDL+FL+WL(-X)	449.02	-57.20	200.20	49.89	12.71	44.49	107.09	107.37	18.11	81.67	0.00	85.77	62.74	40.84	94.52	9.06	O.K
206	DL+EQDL+FL+WL(+Z)	522.02	41.60	195.00	58.00	9.24	43.33	110.58	110.58	23.91	92.09	5.42	88.98	67.25	48.76	101.34	14.67	O.K
207	DL+EQDL+FL+WL(-Z)	578.02	62.40	288.60	64.22	13.87	64.13	142.22	143.11	13.96	114.49	0.00	121.51	78.54	57.25	128.80	6.98	O.K
208	DL+EQDL+FL+0.7SL(+X)	599.02	95.70	263.60	66.56	21.27	58.58	146.40	147.07	29.25	103.87	0.00	125.47	88.16	51.94	125.47	14.63	O.K
209	DL+EQDL+FL+0.7SL(-X)	501.02	8.30	220.00	55.67	1.84	48.89	106.40	106.40	8.62	102.71	4.94	84.80	57.51	53.83	104.56	6.78	O.K
210	DL+EQDL+FL+0.7SL(+Z)	522.72	46.50	201.80	58.08	10.33	44.84	113.26	113.26	23.57	92.59	2.90	91.66	68.42	47.75	102.93	13.24	O.K
211	DL+EQDL+FL+0.7SL(-Z)	577.32	57.50	281.80	64.15	12.78	62.62	139.55	140.11	14.30	113.99	0.00	118.51	77.21	57.00	127.05	7.15	O.K
212	DL+EQDL+FL+0.75WL(+X)+0.75	800.52	174.90	388.10	88.95	38.87	86.24	214.06	217.24	41.57	136.32	0.00	195.64	129.41	68.16	176.78	20.79	O.K
213	DL+EQDL+FL+0.75WL(-X)+0.75(	649.02	11.10	325.70	72.11	2.47	72.38	146.96	147.03	2.20	142.02	0.00	125.43	74.62	71.01	144.53	1.10	O.K
214	DL+EQDL+FL+0.75WL(+Z)+0.75	703.72	85.20	321.80	78.19	18.93	71.51	168.64	169.14	25.61	130.77	0.00	147.54	97.38	65.39	149.96	12.81	O.K

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215 DL+EQDL+FL+0.75WL(-Z)+0.75(	745.72	100.80	392.00	82.86	22.40	87.11	192.37	194.83	18.15	147.57	0.00	173.23	106.49	73.79	171.20	9.08	O.K
216 DL+EQDL+FL+0.525SL(+X)+0.75	761.52	125.70	373.20	84.61	27.93	82.93	195.48	197.35	29.61	139.61	0.00	175.75	113.48	69.81	168.48	14.81	O.K
217 DL+EQDL+FL+0.525SL(-X)+0.75	688.02	60.20	340.50	76.45	13.38	75.67	165.49	166.12	14.16	138.74	0.00	144.52	90.14	69.37	152.43	7.08	O.K
218 DL+EQDL+FL+0.525SL(+Z)+0.75	704.22	88.90	326.80	78.25	19.76	72.62	170.62	171.28	25.38	131.11	0.00	149.68	98.33	65.56	151.20	12.69	O.K
219 DL+EQDL+FL+0.525SL(-Z)+0.75	745.22	97.00	386.90	82.80	21.56	85.98	190.34	192.48	18.38	147.22	0.00	170.88	105.43	73.61	169.85	9.19	O.K
220 0.6(DL+EQDL)+WL(+X)	436.22	126.40	110.20	48.47	28.09	24.49	101.05	101.13	52.07	44.87	0.00	79.53	76.60	22.44	73.00	26.04	O.K
221 0.6(DL+EQDL)+WL(-X)	234.22	-92.00	27.00	26.02	20.44	6.00	52.47	52.50	40.47	11.58	0.00	30.90	46.49	5.79	32.04	20.24	O.K
222 0.6(DL+EQDL)+WL(+Z)	307.22	6.80	21.80	34.14	1.51	4.84	40.49	40.49	30.80	37.47	27.78	18.89	35.65	32.63	38.98	29.29	O.K
223 0.6(DL+EQDL)+WL(-Z)	363.22	27.60	115.40	40.36	6.13	25.64	72.14	72.14	20.85	59.87	8.58	50.54	46.50	34.23	66.01	14.72	O.K
224 0.6(DL+EQDL)+0.7SL(+X)	384.22	60.80	90.50	42.69	13.51	20.11	76.31	76.31	36.09	49.29	9.07	54.71	56.20	29.18	62.80	22.58	O.K
225 0.6(DL+EQDL)+0.7SL(-X)	286.22	-26.50	46.80	31.80	5.89	10.40	48.09	48.09	27.29	36.31	15.51	26.49	37.69	25.91	42.20	21.40	O.K
226 0.6(DL+EQDL)+0.7SL(+Z)	307.92	11.70	28.60	34.21	2.60	6.36	43.17	43.17	30.46	37.97	25.26	21.57	36.82	31.62	40.57	27.86	O.K
227 0.6(DL+EQDL)+0.7SL(-Z)	362.52	22.60	108.70	40.28	5.02	24.16	69.46	69.46	21.15	59.41	11.10	47.86	45.31	35.26	64.44	16.13	O.K
228 DL+EQTL	393.02	28.60	132.60	43.67	6.36	29.47	79.49	79.49	20.56	66.78	7.85	57.89	50.03	37.32	73.14	14.21	O.K

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_1 (p_{max}) - modified using Teng's chart*	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
Ultimate Conditions																		
101	1.4(DL+EQDL+FL)	672.92	72.80	338.50	74.77	16.18	75.22	166.17	167.29	15.72	133.81	0.00	145.69	91.50	66.91	150.55	7.86	
102	1.2(DL+EQDL+FL)+1.6(LL+EQLL	984.32	149.80	535.60	109.37	33.29	119.02	261.68	266.44	23.64	195.10	0.00	244.84	145.04	97.55	230.77	11.82	
103	1.2(DL+EQDL+FL+TL)+1.6(LL+E	992.72	252.70	697.80	110.30	56.16	155.07	321.52	356.65	11.39	209.21	0.00	335.05	184.02	104.61	282.93	5.70	
104	1.2(DL+EQDL)+0.8WL(+X)	508.72	121.70	170.60	56.52	27.04	37.91	121.48	121.81	45.66	67.39	0.00	100.21	83.73	33.70	94.60	22.83	
105	1.2(DL+EQDL)+0.8WL(-X)	347.12	-53.00	104.00	38.57	11.78	23.11	73.46	73.46	27.24	49.90	3.68	51.86	50.35	26.79	61.68	15.46	
106	1.2(DL+EQDL)+0.8WL(+Z)	405.52	26.00	99.80	45.06	5.78	22.18	73.01	73.01	28.66	61.46	17.10	51.41	50.84	39.28	67.24	22.88	
107	1.2(DL+EQDL)+0.8WL(-Z)	450.32	42.60	174.70	50.04	9.47	38.82	98.32	98.32	20.68	79.39	1.75	76.72	59.50	40.57	88.86	11.22	
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(L	706.02	236.30	280.50	78.45	52.51	62.33	193.29	197.02	68.62	88.27	0.00	175.42	132.82	44.14	142.65	34.31	

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109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LI	382.82	-113.10	147.40	42.54	25.13	32.76	100.42	101.58	34.91	50.16	0.00	79.98	68.25	25.08	75.87	17.46	
110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(L	499.62	45.00	139.10	55.51	10.00	30.91	96.42	96.42	34.60	76.42	0.00	74.82	65.51	38.21	86.42	17.30	
111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LI	589.22	78.30	288.90	65.47	17.40	64.20	147.07	148.11	18.67	112.27	0.00	126.51	83.39	56.14	130.19	9.34	
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+E	690.92	-135.72	308.88	76.77	30.16	68.64	175.57	176.99	38.29	115.25	0.00	155.39	107.64	57.63	146.12	19.15	
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+E	550.92	-10.92	246.48	61.21	2.43	54.77	118.41	118.41	8.87	113.56	0.00	96.81	63.64	56.78	115.99	4.44	
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+E	581.92	-65.52	220.48	64.66	14.56	49.00	128.21	128.21	30.22	99.09	0.00	106.61	79.22	49.55	113.65	15.11	
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+E	659.92	-81.12	334.88	73.32	18.03	74.42	165.77	167.23	16.93	129.72	0.00	145.63	92.08	64.86	148.47	8.47	
116 0.9(DL+EQDL)+1.6WL(+X)	543.12	-200.46	169.52	60.35	44.55	37.67	142.56	144.22	67.22	53.47	0.00	122.62	105.72	26.74	98.85	33.61	
117 0.9(DL+EQDL)+1.6WL(-X)	219.92	148.98	36.40	24.44	33.11	8.09	65.63	69.99	49.45	-0.58	0.00	48.39	59.72	-0.29	34.70	24.73	
118 0.9(DL+EQDL)+1.6WL(+Z)	336.72	-9.10	28.08	37.41	2.02	6.24	45.68	45.68	33.20	41.63	29.15	24.08	39.44	35.39	43.65	31.18	
119 0.9(DL+EQDL)+1.6WL(-Z)	426.32	-42.38	177.84	47.37	9.42	39.52	96.31	96.32	17.27	77.47	0.00	74.72	56.80	38.74	86.90	8.64	
120 0.9(DL+EQDL)+SL(+X)	451.52	-88.14	134.16	50.17	19.59	29.81	99.57	99.57	39.94	60.40	0.77	77.97	69.75	30.59	79.98	20.36	
121 0.9(DL+EQDL)+SL(-X)	311.52	36.66	71.76	34.61	8.15	15.95	58.71	58.72	26.81	42.41	0.00	37.12	42.77	21.21	50.57	13.41	
122 0.9(DL+EQDL)+SL(+Z)	342.52	-17.94	45.76	38.06	3.99	10.17	52.21	52.21	31.88	44.24	23.90	30.61	42.05	34.07	48.23	27.89	
123 0.9(DL+EQDL)+SL(-Z)	420.52	-33.54	160.16	46.72	7.45	35.59	89.77	89.77	18.59	74.86	3.68	68.17	54.18	39.27	82.32	11.14	
124 1.2DL+EQTL	410.52	-32.24	142.48	45.61	7.16	31.66	84.44	84.44	21.12	70.11	6.79	62.84	52.78	38.45	77.28	13.96	
Max :	993	253	698														
Min :	220	-200	22														

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 104.92 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 114.92 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 248.29 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 258.29 kN/m^2

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Sample calculation of values of base pressure:

Load case : 202 DL+EQDL+FL+LL+EQLL+TL

Maximum pressure, p_1 (pmax) = 87.78+42.76+117.87= 248.40 kN/m^2

p_2= 87.78+42.76-117.87= 12.67 kN/m^2

p_3= 87.78-(42.76+117.87)= -72.84 kN/m^2

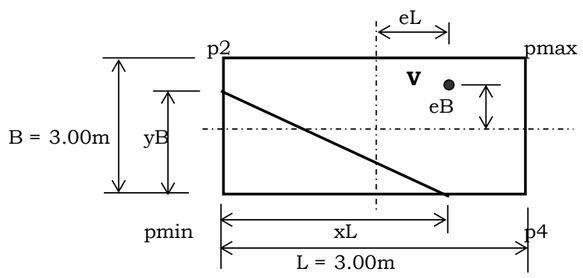
p_4= 87.78-42.76+117.87= 162.89 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 248.40-1.2*18= 226.80 kN/m^2

Sample calculation of values of base pressure modification:

Load case : 202 DL+EQDL+FL+LL+EQLL+TL

CASE - II **of Teng's chart**



Maximum average Pressure along X, $p_{avx(max)}$ = (248.40+12.67)/2= 130.54 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+162.89)/2= 45.02 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (248.40+162.89)/2= 205.65 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+12.67)/2= -30.09 kN/m^2

A	=	9	m2
K	=	3.07457	
x	=	0.8191	
y	=	0.33505	
V	=	790.02	kN
pmax = KV/A =(3.07x790.02/9)	=	269.886	kN/m2
xL =(0.82x3)	=	2.45731	m
yB =(0.34x3)	=	1.00515	m
Aeff = A - 1/2*(xL)*(yB)	=	7.76501	m2
% Area of contact = Aeff /A * 100 %	=	86.2779	%

eL =	MZ/ V	=	(192.4/790.02)	0.24354 m
eB =	MX / V	=	(530.4/790.02)	0.67138 m
eL/L	=	(0.24/3)	=	0.08118
eB/B	=	(0.67/3)	=	0.22379

From Teng's chart

eL/L	eB/B	K value
0.08118	0.22379	3.07457

From Teng's chart

eL/L	eB/B	x	y
0.08118	0.22379	0.8191	0.335051

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DESIGN OF ISOLATED FOOTING F3

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

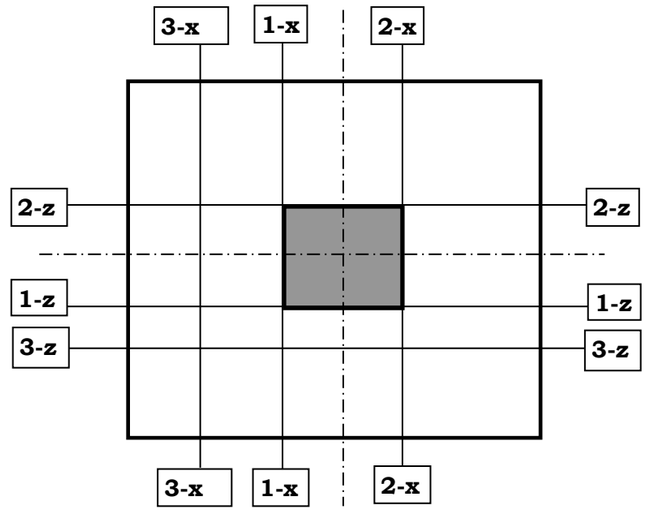
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.65+2*0.5*0.526)+(0.65+2*0.5*0.526))*2
= 4.704 m

Punching Area= ((0.65+2*0.5*0.526)*(0.65+2*0.5*0.526))
= 1.38 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.6*24+0.6*18)
= 25.20 kN/m^2

Area of footing, A_f= 9.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+EQDL+FL)	81.87	76.54	86.31	43.55	31.01	40.33	377.71
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	126.44	116.15	135.02	78.45	54.22	72.69	641.37
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	152.92	135.71	167.26	102.48	61.97	95.23	648.48
104	1.2(DL+EQDL)+0.8WL(+X)	64.13	53.30	73.17	35.90	10.38	33.71	238.69
105	1.2(DL+EQDL)+0.8WL(-X)	41.12	36.02	45.38	15.24	3.22	14.35	101.87
106	1.2(DL+EQDL)+0.8WL(+Z)	46.31	43.81	48.40	16.66	10.76	15.46	151.32
107	1.2(DL+EQDL)+0.8WL(-Z)	52.09	47.98	55.51	21.97	12.32	20.45	189.25
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(	98.09	78.87	114.11	66.30	21.07	62.20	405.74
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(	51.34	41.99	59.14	25.83	3.81	24.37	132.10
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(	54.82	48.90	59.75	25.37	11.44	23.69	230.99
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(	72.72	66.81	77.64	37.71	23.81	35.01	306.85
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+	88.05	77.22	97.09	52.40	26.89	48.84	392.95
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+	60.95	59.47	62.19	25.92	22.42	23.87	274.42

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114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+	67.60	61.17	72.96	34.62	19.48	32.21	300.67
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+I	81.42	75.52	86.34	43.71	29.83	40.52	366.71
116 0.9(DL+EQDL)+1.6WL(+X)	74.79	57.67	89.06	48.47	8.18	45.69	267.82
117 0.9(DL+EQDL)+1.6WL(-X)	36.22	23.21	47.06	18.42	-12.19	17.84	-5.83
118 0.9(DL+EQDL)+1.6WL(+Z)	37.85	36.98	38.59	9.47	7.40	8.74	93.07
119 0.9(DL+EQDL)+1.6WL(-Z)	49.73	45.81	52.99	20.19	10.97	18.80	168.93
120 0.9(DL+EQDL)+SL(+X)	54.41	45.93	61.49	27.22	7.25	25.58	190.26
121 0.9(DL+EQDL)+SL(-X)	34.33	29.65	38.22	10.19	-0.81	9.68	71.73
122 0.9(DL+EQDL)+SL(+Z)	38.92	37.20	40.37	10.91	6.84	10.13	97.98
123 0.9(DL+EQDL)+SL(-Z)	48.34	45.11	51.03	18.66	11.06	17.35	164.02
124 1.2DL+EQTL	47.17	44.06	49.76	17.75	10.44	16.50	155.55

Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1x} = 91.5-(91.5-66.91)*1.175/3 = 81.87 kN/m^2 Overhang = 1.175 m

p_{2x} = 66.91+(91.5-66.91)*1.175/3 = 76.54 kN/m^2 Effective depth = 0.542 m

p_{3x} = 91.5-(91.5-66.91)*(1.175-0.542)/3 = 86.31 kN/m^2

BM_{1x} = (81.87-25.2)*1.175*1.175/2+0.5*(91.5-81.87)*1.175*1.175*0.67 = 43.55 kN-m/m

BM_{2x} = (76.54-25.2)*1.175*1.175/2-0.5*(-66.91+76.54)*1.175*1.175*0.67 = 31.01 kN-m/m

SF_{3x} = (86.31-25.2)*(1.175-0.542)+0.5*(91.5-86.31)*(1.175-0.542) = 40.33 kN/m

$S.F_{(punching)}$ = (9-1.38)*(74.77-25.2) = 377.71 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	102.48	0.35	0.47	0.001	605.01	975.60	16ϕT	206.09	300	175	1148.93
At top	-12.19	-0.04	0.06	0.000	71.45	487.80	12ϕT	231.85	300	200	565.49

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 95.23 kN/m

Shear stress , τ_v= (95.23*1000)/(1000*542)

= 0.18 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 648.48 kN

Strength of Concrete in shear Vc1 = 7541.72 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 5410.46 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 5027.82 kN **Table 22.6.5.2_ (c)**

Design Vc= 5027.82 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (648.48*1000)/(4704*542)

= 0.25 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M$_{1z}$ kNm	B.M$_{2z}$ kNm	S.F$_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+EQDL+FL)	94.66	63.75	119.68	73.80	0.76	71.34
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	145.01	97.58	183.40	122.37	10.30	118.04
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	174.35	114.28	222.96	153.18	11.27	147.81

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104 1.2(DL+EQDL)+0.8WL(+X)	66.49	50.94	79.07	41.50	4.77	40.00
105 1.2(DL+EQDL)+0.8WL(-X)	43.58	33.56	51.68	21.06	-2.60	20.43
106 1.2(DL+EQDL)+0.8WL(+Z)	49.87	40.25	57.64	25.06	2.36	24.17
107 1.2(DL+EQDL)+0.8WL(-Z)	58.45	41.63	72.06	37.02	-2.72	35.86
108 1.2(DL+EQDL)+1.6WL(+X)+0.5(	100.22	76.74	119.21	71.41	15.96	68.62
109 1.2(DL+EQDL)+1.6WL(-X)+0.5(	52.99	40.34	63.23	29.77	-0.13	28.78
110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(	59.35	44.37	71.47	36.09	0.71	34.88
111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(	82.86	56.67	104.05	61.69	-0.17	59.65
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+	96.39	68.88	118.65	72.14	7.15	69.56
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+	72.30	48.13	91.86	52.72	-4.38	51.09
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+	75.06	53.70	92.33	52.27	1.83	50.49
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+	93.64	63.30	118.18	72.60	0.94	70.17
116 0.9(DL+EQDL)+1.6WL(+X)	73.30	59.16	84.74	45.02	11.63	43.22
117 0.9(DL+EQDL)+1.6WL(-X)	30.80	28.63	32.54	5.67	0.57	5.47
118 0.9(DL+EQDL)+1.6WL(+Z)	38.77	36.06	40.95	11.62	5.24	11.10
119 0.9(DL+EQDL)+1.6WL(-Z)	56.25	39.29	69.97	35.61	-4.45	34.55
120 0.9(DL+EQDL)+SL(+X)	56.63	43.71	67.08	32.50	1.98	31.37
121 0.9(DL+EQDL)+SL(-X)	36.02	24.05	42.53	14.20	0.97	13.86
122 0.9(DL+EQDL)+SL(+Z)	40.26	42.21	43.83	14.08	-0.32	13.52
123 0.9(DL+EQDL)+SL(-Z)	54.44	35.42	66.92	33.08	-2.75	32.07
124 1.2DL+EQTL	52.48	8.49	63.58	30.30	-15.46	29.35

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Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1z} = 150.55-(150.55-7.86)*1.175/3= 94.66 kN/m^2

p_{2z} = 7.86+(150.55-7.86)*1.175/3= 63.75 kN/m^2

p_{3z} = 150.55-(150.55-7.86)*(1.175-0.526)/3= 119.68 kN/m^2

Overhang = 1.175 m

Effective depth = 0.526 m

BM_{1z} = (94.66-25.2)*1.175*1.175/2+0.5*(150.55-94.66)*1.175*1.175*0.67 = 73.80 kN-m/m

BM_{2z} = (63.75-25.2)*1.175*1.175/2-0.5*(-7.86+63.75)*1.175*1.175*0.67 = 0.76 kN-m/m

SF_{3z} = (119.68-25.2)*(1.175-0.526)+0.5*(150.55-119.68)*(1.175-0.526) = 71.34 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	153.18	0.554	0.738	0.0018	964.73	946.80	16ϕT	208.41	300	175	1148.9
At top	-15.46	-0.056	-0.075	-0.0002	-0.93	473.40	12ϕT	238.90	250	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 147.81 kN/m

Shear stress , τ_v= (147.81*1000)/(1000*526)

= 0.28 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	**175**	c/c
	At Top	12ϕT	**200**	c/c
Along Z Direction :				
	At Bottom	16ϕT	**175**	c/c
	At Top	12ϕT	**200**	c/c

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5.4 DESIGN OF CALCULATION OF FOOTING F4

STAAD Node - 162

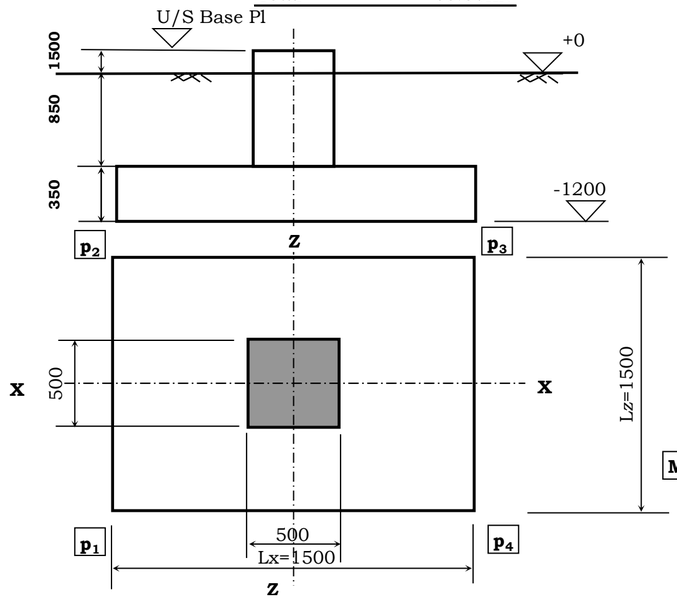
	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.20 m	
U/S level Base Plate :	1.50 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

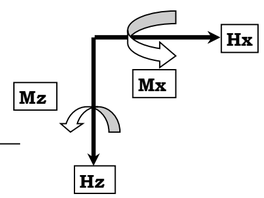
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.5*1.5*0.35*24)+(0.5*0.5*2.35*24) = 33.00KN
Weight of soil = ((1.5*1.5)-(0.5*0.5))*0.85*18 = 30.60KN
Total = 63.60KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.5m

Clear projection along X direction = 0.5m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING F4

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+EQDL+FL	207.9	0.0	0.0	0.0	0.0	271.50	0.00	0.00	0.00	0.00
202	DL+EQDL+FL+LL+EQLL+TL	207.9	0.0	0.0	0.0	0.0	271.50	0.00	0.00	0.00	0.00
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	207.9	0.0	0.0	0.0	0.0	271.50	0.00	0.00	0.00	0.00
204	DL+EQDL+FL+WL(+X)	207.9	-5.3	14.3	0.0	0.0	271.50	-5.30	14.30	0.00	0.00
205	DL+EQDL+FL+WL(-X)	207.9	5.3	-14.3	0.0	0.0	271.50	5.30	-14.30	0.00	0.00
206	DL+EQDL+FL+WL(+Z)	207.9	0.0	0.0	-5.3	-14.3	271.50	0.00	0.00	-5.30	-14.30
207	DL+EQDL+FL+WL(-Z)	207.9	0.0	0.0	5.3	14.3	271.50	0.00	0.00	5.30	14.30
208	DL+EQDL+FL+0.7SL(+X)	207.9	-7.0	18.9	0.0	0.0	271.50	-7.00	18.90	0.00	0.00
209	DL+EQDL+FL+0.7SL(-X)	207.9	7.0	-18.9	0.0	0.0	271.50	7.00	-18.90	0.00	0.00
210	DL+EQDL+FL+0.7SL(+Z)	207.9	0.0	0.0	-7.0	-18.9	271.50	0.00	0.00	-7.00	-18.90
211	DL+EQDL+FL+0.7SL(-Z)	207.9	0.0	0.0	7.0	18.9	271.50	0.00	0.00	7.00	18.90
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+E	207.9	-4.0	10.7	0.0	0.0	271.50	-4.00	10.70	0.00	0.00
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EC	207.9	4.0	-10.7	0.0	0.0	271.50	4.00	-10.70	0.00	0.00
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EC	207.9	0.0	0.0	-4.0	-10.7	271.50	0.00	0.00	-4.00	-10.70
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQ	207.9	0.0	0.0	4.0	10.7	271.50	0.00	0.00	4.00	10.70
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+E	207.9	-5.3	14.2	0.0	0.0	271.50	-5.30	14.20	0.00	0.00
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+E	207.9	5.3	-14.2	0.0	0.0	271.50	5.30	-14.20	0.00	0.00
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+E	207.9	0.0	0.0	-5.3	-14.2	271.50	0.00	0.00	-5.30	-14.20
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+E	207.9	0.0	0.0	5.3	14.2	271.50	0.00	0.00	5.30	14.20
220	0.6(DL+EQDL)+WL(+X)	22.7	-5.3	14.3	0.0	0.0	86.30	-5.30	14.30	0.00	0.00
221	0.6(DL+EQDL)+WL(-X)	22.7	5.3	-14.3	0.0	0.0	86.30	5.30	-14.30	0.00	0.00
222	0.6(DL+EQDL)+WL(+Z)	22.7	0.0	0.0	-5.3	-14.3	86.30	0.00	0.00	-5.30	-14.30
223	0.6(DL+EQDL)+WL(-Z)	22.7	0.0	0.0	5.3	14.3	86.30	0.00	0.00	5.30	14.30
224	0.6(DL+EQDL)+0.7SL(+X)	22.7	-7.0	18.9	0.0	0.0	86.30	-7.00	18.90	0.00	0.00
225	0.6(DL+EQDL)+0.7SL(-X)	22.7	7.0	-18.9	0.0	0.0	86.30	7.00	-18.90	0.00	0.00
226	0.6(DL+EQDL)+0.7SL(+Z)	22.7	0.0	0.0	-7.0	-18.9	86.30	0.00	0.00	-7.00	-18.90
227	0.6(DL+EQDL)+0.7SL(-Z)	22.7	0.0	0.0	7.0	18.9	86.30	0.00	0.00	7.00	18.90
228	DL+EQTL	207.9	0.0	0.0	0.0	0.0	271.50	0.00	0.00	0.00	0.00
Ultimate Conditions											
101	1.4(DL+EQDL+FL)	291.0	0.0	0.0	0.0	0.0	354.60	0.00	0.00	0.00	0.00
102	1.2(DL+EQDL+FL)+1.6(LL+EQLL)	249.5	0.0	0.0	0.0	0.0	313.10	0.00	0.00	0.00	0.00
103	1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)	249.5	0.0	0.0	0.0	0.0	313.10	0.00	0.00	0.00	0.00
104	1.2(DL+EQDL)+0.8WL(+X)	45.5	-4.2	11.4	0.0	0.0	109.10	-4.20	11.40	0.00	0.00
105	1.2(DL+EQDL)+0.8WL(-X)	45.5	4.2	-11.4	0.0	0.0	109.10	4.20	-11.40	0.00	0.00
106	1.2(DL+EQDL)+0.8WL(+Z)	45.5	0.0	0.0	-4.2	-11.4	109.10	0.00	0.00	-4.20	-11.40
107	1.2(DL+EQDL)+0.8WL(-Z)	45.5	0.0	0.0	4.2	11.4	109.10	0.00	0.00	4.20	11.40
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQL	45.5	-8.5	22.9	0.0	0.0	109.10	-8.50	22.90	0.00	0.00
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQL	45.5	8.5	-22.9	0.0	0.0	109.10	8.50	-22.90	0.00	0.00
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQL	45.5	0.0	0.0	-8.5	-22.9	109.10	0.00	0.00	-8.50	-22.90
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQL	45.5	0.0	0.0	8.5	22.9	109.10	0.00	0.00	8.50	22.90
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+(	130.5	-10.0	0.0	0.0	0.0	194.10	-10.00	-27.00	0.00	0.00
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+(	130.5	10.0	0.0	0.0	0.0	194.10	10.00	27.00	0.00	0.00
114	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+(	130.5	0.0	0.0	-10.0	0.0	194.10	0.00	0.00	-10.00	-27.00
115	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+(	130.5	0.0	0.0	10.0	0.0	194.10	0.00	0.00	10.00	27.00
116	0.9(DL+EQDL)+1.6WL(+X)	34.1	-8.5	0.0	0.0	0.0	97.70	-8.50	-22.95	0.00	0.00
117	0.9(DL+EQDL)+1.6WL(-X)	34.1	8.5	0.0	0.0	0.0	97.70	8.50	22.95	0.00	0.00
118	0.9(DL+EQDL)+1.6WL(+Z)	34.1	0.0	0.0	-8.5	0.0	97.70	0.00	0.00	-8.50	-22.95
119	0.9(DL+EQDL)+1.6WL(-Z)	34.1	0.0	0.0	8.5	0.0	97.70	0.00	0.00	8.50	22.95
120	0.9(DL+EQDL)+SL(+X)	34.1	-10.0	0.0	0.0	0.0	97.70	-10.00	-27.00	0.00	0.00
121	0.9(DL+EQDL)+SL(-X)	34.1	10.0	0.0	0.0	0.0	97.70	10.00	27.00	0.00	0.00
122	0.9(DL+EQDL)+SL(+Z)	34.1	0.0	0.0	-10.0	0.0	97.70	0.00	0.00	-10.00	-27.00
123	0.9(DL+EQDL)+SL(-Z)	34.1	0.0	0.0	10.0	0.0	97.70	0.00	0.00	10.00	27.00
124	1.2DL+EQTL	211.3	0.0	0.0	0.0	0.0	274.90	0.00	0.00	0.00	0.00

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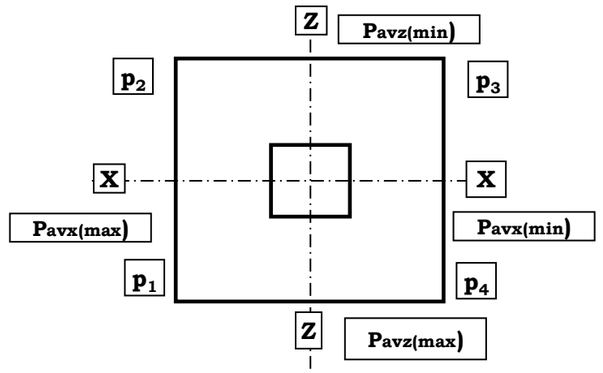
Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	DL+EQDL+FL	271.5	0.0	0.0	0.0	0.0	NA	NA	NA	NA
202	DL+EQDL+FL+LL+EQLL+TL	271.5	0.0	0.0	0.0	0.0	NA	NA	NA	NA
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	271.5	0.0	0.0	0.0	0.0	NA	NA	NA	NA
204	DL+EQDL+FL+WL(+X)	271.5	-5.3	14.3	0.0	0.0	14.2	NA	20.5	NA
205	DL+EQDL+FL+WL(-X)	271.5	5.3	-14.3	0.0	0.0	14.2	NA	20.5	NA
206	DL+EQDL+FL+WL(+Z)	271.5	0.0	0.0	-5.3	-14.3	NA	14.2	NA	20.5
207	DL+EQDL+FL+WL(-Z)	271.5	0.0	0.0	5.3	14.3	NA	14.2	NA	20.5
208	DL+EQDL+FL+0.7SL(+X)	271.5	-7.0	18.9	0.0	0.0	10.8	NA	15.5	NA
209	DL+EQDL+FL+0.7SL(-X)	271.5	7.0	-18.9	0.0	0.0	10.8	NA	15.5	NA
210	DL+EQDL+FL+0.7SL(+Z)	271.5	0.0	0.0	-7.0	-18.9	NA	10.8	NA	15.5
211	DL+EQDL+FL+0.7SL(-Z)	271.5	0.0	0.0	7.0	18.9	NA	10.8	NA	15.5
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)	271.5	-4.0	10.7	0.0	0.0	19.0	NA	27.2	NA
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)	271.5	4.0	-10.7	0.0	0.0	19.0	NA	27.2	NA
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)	271.5	0.0	0.0	-4.0	-10.7	NA	19.0	NA	27.2
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)	271.5	0.0	0.0	4.0	10.7	NA	19.0	NA	27.2
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL	271.5	-5.3	14.2	0.0	0.0	14.3	NA	20.5	NA
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)	271.5	5.3	-14.2	0.0	0.0	14.3	NA	20.5	NA
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL	271.5	0.0	0.0	-5.3	-14.2	NA	14.3	NA	20.5
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)	271.5	0.0	0.0	5.3	14.2	NA	14.3	NA	20.5
220	0.6(DL+EQDL)+WL(+X)	86.3	-5.3	14.3	0.0	0.0	4.5	NA	6.5	NA
221	0.6(DL+EQDL)+WL(-X)	86.3	5.3	-14.3	0.0	0.0	4.5	NA	6.5	NA
222	0.6(DL+EQDL)+WL(+Z)	86.3	0.0	0.0	-5.3	-14.3	NA	4.5	NA	6.5
223	0.6(DL+EQDL)+WL(-Z)	86.3	0.0	0.0	5.3	14.3	NA	4.5	NA	6.5
224	0.6(DL+EQDL)+0.7SL(+X)	86.3	-7.0	18.9	0.0	0.0	3.4	NA	4.9	NA
225	0.6(DL+EQDL)+0.7SL(-X)	86.3	7.0	-18.9	0.0	0.0	3.4	NA	4.9	NA
226	0.6(DL+EQDL)+0.7SL(+Z)	86.3	0.0	0.0	-7.0	-18.9	NA	3.4	NA	4.9
227	0.6(DL+EQDL)+0.7SL(-Z)	86.3	0.0	0.0	7.0	18.9	NA	3.4	NA	4.9
228	DL+EQTL	271.5	0.0	0.0	0.0	0.0	NA	NA	NA	NA
						Min.=	**3.4**	**3.4**	**4.9**	**4.9**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F4



Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 471.6 kN/m² For WL/SL case = 584.1 kN/m²

Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/Lx.(Lz)²	Base Pressure (kN/m²) p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	Remarks
201	DL+EQDL+FL	271.50	0.00	0.00	120.67	0.00	0.00	120.67	120.67	120.67	120.67	99.07	120.67	120.67	120.67	120.67	O.K
202	DL+EQDL+FL+LL+EQLL+TL	271.50	0.00	0.00	120.67	0.00	0.00	120.67	120.67	120.67	120.67	99.07	120.67	120.67	120.67	120.67	O.K
203	DL+EQDL+FL+0.75(LL+EQLL)+0	271.50	0.00	0.00	120.67	0.00	0.00	120.67	120.67	120.67	120.67	99.07	120.67	120.67	120.67	120.67	O.K
204	DL+EQDL+FL+WL(+X)	271.50	14.30	0.00	120.67	25.42	0.00	146.09	146.09	95.24	95.24	124.49	146.09	95.24	120.67	120.67	O.K
205	DL+EQDL+FL+WL(-X)	271.50	-14.30	0.00	120.67	25.42	0.00	146.09	146.09	95.24	95.24	124.49	146.09	95.24	120.67	120.67	O.K
206	DL+EQDL+FL+WL(+Z)	271.50	0.00	-14.30	120.67	0.00	25.42	146.09	95.24	146.09	95.24	124.49	120.67	120.67	146.09	95.24	O.K
207	DL+EQDL+FL+WL(-Z)	271.50	0.00	14.30	120.67	0.00	25.42	146.09	95.24	146.09	95.24	124.49	120.67	120.67	146.09	95.24	O.K
208	DL+EQDL+FL+0.7SL(+X)	271.50	18.90	0.00	120.67	33.60	0.00	154.27	154.27	87.07	87.07	132.67	154.27	87.07	120.67	120.67	O.K
209	DL+EQDL+FL+0.7SL(-X)	271.50	-18.90	0.00	120.67	33.60	0.00	154.27	154.27	87.07	87.07	132.67	154.27	87.07	120.67	120.67	O.K
210	DL+EQDL+FL+0.7SL(+Z)	271.50	0.00	-18.90	120.67	0.00	33.60	154.27	87.07	154.27	87.07	132.67	120.67	120.67	154.27	87.07	O.K
211	DL+EQDL+FL+0.7SL(-Z)	271.50	0.00	18.90	120.67	0.00	33.60	154.27	87.07	154.27	87.07	132.67	120.67	120.67	154.27	87.07	O.K
212	DL+EQDL+FL+0.75WL(+X)+0.75	271.50	10.70	0.00	120.67	19.02	0.00	139.69	139.69	101.64	101.64	118.09	139.69	101.64	120.67	120.67	O.K
213	DL+EQDL+FL+0.75WL(-X)+0.75(	271.50	-10.70	0.00	120.67	19.02	0.00	139.69	139.69	101.64	101.64	118.09	139.69	101.64	120.67	120.67	O.K
214	DL+EQDL+FL+0.75WL(+Z)+0.75	271.50	0.00	-10.70	120.67	0.00	19.02	139.69	101.64	139.69	101.64	118.09	120.67	120.67	139.69	101.64	O.K

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215 DL+EQDL+FL+0.75WL(-Z)+0.75(	271.50	0.00	10.70	120.67	0.00	19.02	139.69	101.64	139.69	101.64	118.09	120.67	120.67	139.69	101.64	O.K
216 DL+EQDL+FL+0.525SL(+X)+0.75	271.50	14.20	0.00	120.67	25.24	0.00	145.91	145.91	95.42	95.42	124.31	145.91	95.42	120.67	120.67	O.K
217 DL+EQDL+FL+0.525SL(-X)+0.75	271.50	-14.20	0.00	120.67	25.24	0.00	145.91	145.91	95.42	95.42	124.31	145.91	95.42	120.67	120.67	O.K
218 DL+EQDL+FL+0.525SL(+Z)+0.75	271.50	0.00	-14.20	120.67	0.00	25.24	145.91	95.42	145.91	95.42	124.31	120.67	120.67	145.91	95.42	O.K
219 DL+EQDL+FL+0.525SL(-Z)+0.75	271.50	0.00	14.20	120.67	0.00	25.24	145.91	95.42	145.91	95.42	124.31	120.67	120.67	145.91	95.42	O.K
220 0.6(DL+EQDL)+WL(+X)	86.30	14.30	0.00	38.36	25.42	0.00	63.78	63.78	12.93	12.93	42.18	63.78	12.93	38.36	38.36	O.K
221 0.6(DL+EQDL)+WL(-X)	86.30	-14.30	0.00	38.36	25.42	0.00	63.78	63.78	12.93	12.93	42.18	63.78	12.93	38.36	38.36	O.K
222 0.6(DL+EQDL)+WL(+Z)	86.30	0.00	-14.30	38.36	0.00	25.42	63.78	12.93	63.78	12.93	42.18	38.36	38.36	63.78	12.93	O.K
223 0.6(DL+EQDL)+WL(-Z)	86.30	0.00	14.30	38.36	0.00	25.42	63.78	12.93	63.78	12.93	42.18	38.36	38.36	63.78	12.93	O.K
224 0.6(DL+EQDL)+0.7SL(+X)	86.30	18.90	0.00	38.36	33.60	0.00	71.96	71.96	4.76	4.76	50.36	71.96	4.76	38.36	38.36	O.K
225 0.6(DL+EQDL)+0.7SL(-X)	86.30	-18.90	0.00	38.36	33.60	0.00	71.96	71.96	4.76	4.76	50.36	71.96	4.76	38.36	38.36	O.K
226 0.6(DL+EQDL)+0.7SL(+Z)	86.30	0.00	-18.90	38.36	0.00	33.60	71.96	4.76	71.96	4.76	50.36	38.36	38.36	71.96	4.76	O.K
227 0.6(DL+EQDL)+0.7SL(-Z)	86.30	0.00	18.90	38.36	0.00	33.60	71.96	4.76	71.96	4.76	50.36	38.36	38.36	71.96	4.76	O.K
228 DL+EQTL	271.50	0.00	0.00	120.67	0.00	0.00	120.67	120.67	120.67	120.67	99.07	120.67	120.67	120.67	120.67	O.K

Ultimate Conditions																
101 1.4(DL+EQDL+FL)	354.60	0.00	0.00	157.60	0.00	0.00	157.60	157.60	157.60	157.60	136.00	157.60	157.60	157.60	157.60	
102 1.2(DL+EQDL+FL)+1.6(LL+EQLL	313.10	0.00	0.00	139.16	0.00	0.00	139.16	139.16	139.16	139.16	117.56	139.16	139.16	139.16	139.16	
103 1.2(DL+EQDL+FL+TL)+1.6(LL+E	313.10	0.00	0.00	139.16	0.00	0.00	139.16	139.16	139.16	139.16	117.56	139.16	139.16	139.16	139.16	
104 1.2(DL+EQDL)+0.8WL(+X)	109.10	11.40	0.00	48.49	20.27	0.00	68.76	68.76	28.22	28.22	47.16	68.76	28.22	48.49	48.49	
105 1.2(DL+EQDL)+0.8WL(-X)	109.10	-11.40	0.00	48.49	20.27	0.00	68.76	68.76	28.22	28.22	47.16	68.76	28.22	48.49	48.49	
106 1.2(DL+EQDL)+0.8WL(+Z)	109.10	0.00	-11.40	48.49	0.00	20.27	68.76	28.22	68.76	28.22	47.16	48.49	48.49	68.76	28.22	
107 1.2(DL+EQDL)+0.8WL(-Z)	109.10	0.00	11.40	48.49	0.00	20.27	68.76	28.22	68.76	28.22	47.16	48.49	48.49	68.76	28.22	
108 1.2(DL+EQDL)+1.6WL(+X)+0.5(L	109.10	22.90	0.00	48.49	40.71	0.00	89.20	89.20	7.78	7.78	67.60	89.20	7.78	48.49	48.49	

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109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LI	109.10	-22.90	0.00	48.49	40.71	0.00	89.20	89.20	7.78	7.78	67.60	89.20	7.78	48.49	48.49	
110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(L	109.10	0.00	-22.90	48.49	0.00	40.71	89.20	7.78	89.20	7.78	67.60	48.49	48.49	89.20	7.78	
111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LI	109.10	0.00	22.90	48.49	0.00	40.71	89.20	7.78	89.20	7.78	67.60	48.49	48.49	89.20	7.78	
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+E	194.10	-27.00	0.00	86.27	48.00	0.00	134.27	134.27	38.27	38.27	112.67	134.27	38.27	86.27	86.27	
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+E	194.10	27.00	0.00	86.27	48.00	0.00	134.27	134.27	38.27	38.27	112.67	134.27	38.27	86.27	86.27	
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+E	194.10	0.00	-27.00	86.27	0.00	48.00	134.27	38.27	134.27	38.27	112.67	86.27	86.27	134.27	38.27	
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+E	194.10	0.00	27.00	86.27	0.00	48.00	134.27	38.27	134.27	38.27	112.67	86.27	86.27	134.27	38.27	
116 0.9(DL+EQDL)+1.6WL(+X)	97.70	-22.95	0.00	43.42	40.80	0.00	84.22	84.22	2.62	2.62	62.62	84.22	2.62	43.42	43.42	
117 0.9(DL+EQDL)+1.6WL(-X)	97.70	22.95	0.00	43.42	40.80	0.00	84.22	84.22	2.62	2.62	62.62	84.22	2.62	43.42	43.42	
118 0.9(DL+EQDL)+1.6WL(+Z)	97.70	0.00	-22.95	43.42	0.00	40.80	84.22	2.62	84.22	2.62	62.62	43.42	43.42	84.22	2.62	
119 0.9(DL+EQDL)+1.6WL(-Z)	97.70	0.00	22.95	43.42	0.00	40.80	84.22	2.62	84.22	2.62	62.62	43.42	43.42	84.22	2.62	
120 0.9(DL+EQDL)+SL(+X)	97.70	-27.00	0.00	43.42	48.00	0.00	91.42	91.42	-4.58	-4.58	69.82	91.42	-4.58	43.42	43.42	
121 0.9(DL+EQDL)+SL(-X)	97.70	27.00	0.00	43.42	48.00	0.00	91.42	91.42	-4.58	-4.58	69.82	91.42	-4.58	43.42	43.42	
122 0.9(DL+EQDL)+SL(+Z)	97.70	0.00	-27.00	43.42	0.00	48.00	91.42	-4.58	91.42	-4.58	69.82	43.42	43.42	91.42	-4.58	
123 0.9(DL+EQDL)+SL(-Z)	97.70	0.00	27.00	43.42	0.00	48.00	91.42	-4.58	91.42	-4.58	69.82	43.42	43.42	91.42	-4.58	
124 1.2DL+EQTL	274.90	0.00	0.00	122.18	0.00	0.00	122.18	122.18	122.18	122.18	100.58	122.18	122.18	122.18	122.18	
Max :	355	27	27													
Min :	86	-27	-27													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 99.07 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 109.07 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 132.67 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 142.67 kN/m^2

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Sample calculation of values of base pressure:

Load case : 202 DL+EQDL+FL+LL+EQLL+TL

Maximum pressure, p_1 (pmax) = 120.67+0.00+0.00= 120.67 kN/m^2

p_2= 120.67+0.00-0.00= 120.67 kN/m^2

p_3= 120.67-(0.00+0.00)= 120.67 kN/m^2

p_4= 120.67-0.00+0.00= 120.67 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 120.67-1.2*18= 99.07 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (120.67+120.67)/2= 120.67 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (120.67+120.67)/2= 120.67 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (120.67+120.67)/2= 120.67 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (120.67+120.67)/2= 120.67 kN/m^2

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DESIGN OF ISOLATED FOOTING F4

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

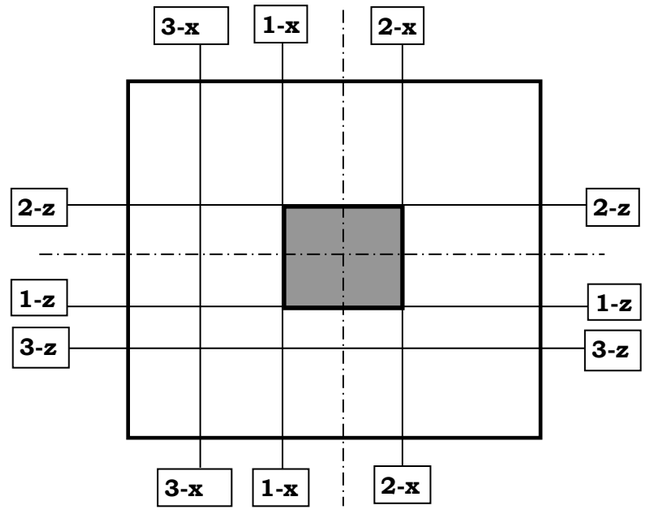
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.282)+(0.5+2*0.5*0.282))*2
= 3.128 m

Punching Area= ((0.5+2*0.5*0.282)*(0.5+2*0.5*0.282))
= 0.61 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+0.85*18)
= 23.70 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+EQDL+FL)	157.60	157.60	157.60	16.74	16.74	27.58	219.60
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	139.16	139.16	139.16	14.43	14.43	23.78	189.35
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	139.16	139.16	139.16	14.43	14.43	23.78	189.35
104	1.2(DL+EQDL)+0.8WL(+X)	55.25	41.73	63.19	5.07	1.13	8.71	40.65
105	1.2(DL+EQDL)+0.8WL(-X)	55.25	41.73	63.19	5.07	1.13	8.71	40.65
106	1.2(DL+EQDL)+0.8WL(+Z)	48.49	48.49	48.49	3.10	3.10	5.11	40.65
107	1.2(DL+EQDL)+0.8WL(-Z)	48.49	48.49	48.49	3.10	3.10	5.11	40.65
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(	62.06	34.92	78.02	7.06	-0.86	12.34	40.65
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(	62.06	34.92	78.02	7.06	-0.86	12.34	40.65
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(	48.49	48.49	48.49	3.10	3.10	5.11	40.65

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111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(I	48.49	48.49	48.49	3.10	3.10	5.11	40.65
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+	102.27	70.27	121.09	12.49	3.15	21.42	102.61
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+I	102.27	70.27	121.09	12.49	3.15	21.42	102.61
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+	86.27	86.27	86.27	7.82	7.82	12.89	102.61
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+I	86.27	86.27	86.27	7.82	7.82	12.89	102.61
116 0.9(DL+EQDL)+1.6WL(+X)	57.02	29.82	73.01	6.43	-1.50	11.31	32.34
117 0.9(DL+EQDL)+1.6WL(-X)	57.02	29.82	73.01	6.43	-1.50	11.31	32.34
118 0.9(DL+EQDL)+1.6WL(+Z)	43.42	43.42	43.42	2.47	2.47	4.06	32.34
119 0.9(DL+EQDL)+1.6WL(-Z)	43.42	43.42	43.42	2.47	2.47	4.06	32.34
120 0.9(DL+EQDL)+SL(+X)	59.42	27.42	78.24	7.13	-2.20	12.59	32.34
121 0.9(DL+EQDL)+SL(-X)	59.42	27.42	78.24	7.13	-2.20	12.59	32.34
122 0.9(DL+EQDL)+SL(+Z)	43.42	43.42	43.42	2.47	2.47	4.06	32.34
123 0.9(DL+EQDL)+SL(-Z)	43.42	43.42	43.42	2.47	2.47	4.06	32.34
124 1.2DL+EQTL	122.18	122.18	122.18	12.31	12.31	20.29	161.50

Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1x} = 157.6-(157.6-157.6)*0.5/1.5 = 157.60 kN/m^2 Overhang = 0.5 m

p_{2x} = 157.6+(157.6-157.6)*0.5/1.5 = 157.60 kN/m^2 Effective depth = 0.294 m

p_{3x} = 157.6-(157.6-157.6)*(0.5-0.294)/1.5 = 157.60 kN/m^2

BM_{1x} = (157.60-23.7)*0.5*0.5/2+0.5*(157.6-157.60)*0.5*0.5*0.67 = 16.74 kN-m/m

BM_{2x} = (157.60-23.7)*0.5*0.5/2-0.5*(-157.6+157.60)*0.5*0.5*0.67 = 16.74 kN-m/m

SF_{3x} = (157.60-23.7)*(0.5-0.294)+0.5*(157.6-157.60)*(0.5-0.294) = 27.58 kN/m

$S.F_{(punching)}$ = (2.25-0.61)*(157.60-23.7) = 219.60 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	16.74	0.19	0.26	0.001	181.52	529.20	12ϕT	213.71	300	200	565.49

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 27.58 kN/m

Shear stress , τ_v= (27.58*1000)/(1000*294)

= 0.09 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 219.60 kN

Strength of Concrete in shear Vc1 = 2720.31 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1759.04 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1813.54 kN **Table 22.6.5.2_ (c)**

Design Vc= 1759.04 kN

Allowable shear stress, (ϕ Vc)/bd = 1.43 N/mm^2

Design punching shear stress, τ = (219.6*1000)/(3128*294)

= 0.24 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

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Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+EQDL+FL)	157.60	157.60	157.60	16.74	16.74	29.19
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	139.16	139.16	139.16	14.43	14.43	25.17
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	139.16	139.16	139.16	14.43	14.43	25.17
104	1.2(DL+EQDL)+0.8WL(+X)	48.49	48.49	48.49	3.10	3.10	5.40
105	1.2(DL+EQDL)+0.8WL(-X)	48.49	48.49	48.49	3.10	3.10	5.40
106	1.2(DL+EQDL)+0.8WL(+Z)	55.25	41.73	62.87	5.08	1.12	9.18
107	1.2(DL+EQDL)+0.8WL(-Z)	55.25	41.73	62.87	5.08	1.12	9.18
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(	48.49	48.49	48.49	3.10	3.10	5.40
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(	48.49	48.49	48.49	3.10	3.10	5.40
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(	62.06	34.92	77.37	7.07	-0.87	12.99
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(	62.06	34.92	77.37	7.07	-0.87	12.99
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+	86.27	86.27	86.27	7.82	7.82	13.64
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+	86.27	86.27	86.27	7.82	7.82	13.64
114	1.2(DL+EQDL)+SL(+Z)+0.5(LL+	102.27	70.27	120.32	12.50	3.14	22.58
115	1.2(DL+EQDL)+SL(-Z)+0.5(LL+	102.27	70.27	120.32	12.50	3.14	22.58
116	0.9(DL+EQDL)+1.6WL(+X)	43.42	43.42	43.42	2.47	2.47	4.30
117	0.9(DL+EQDL)+1.6WL(-X)	43.42	43.42	43.42	2.47	2.47	4.30
118	0.9(DL+EQDL)+1.6WL(+Z)	57.02	29.82	72.36	6.44	-1.51	11.90
119	0.9(DL+EQDL)+1.6WL(-Z)	57.02	29.82	72.36	6.44	-1.51	11.90
120	0.9(DL+EQDL)+SL(+X)	43.42	43.42	43.42	2.47	2.47	4.30
121	0.9(DL+EQDL)+SL(-X)	43.42	52.93	43.42	2.47	-0.38	4.30
122	0.9(DL+EQDL)+SL(+Z)	59.42	20.93	77.47	7.15	-1.70	13.24
123	0.9(DL+EQDL)+SL(-Z)	59.42	37.17	77.47	7.15	8.68	13.24
124	1.2DL+EQTL	122.18	81.45	122.18	12.31	0.40	21.47

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Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1z} = 157.6-(157.6-157.6)*0.5/1.5= 157.60 kN/m^2

p_{2z} = 157.6+(157.6-157.6)*0.5/1.5= 157.60 kN/m^2

p_{3z} = 157.6-(157.6-157.6)*(0.5-0.282)/1.5= 157.60 kN/m^2

Overhang = 0.5 m

Effective depth = 0.282 m

BM_{1z} = (157.60-23.7)*0.5*0.5/2+0.5*(157.6-157.60)*0.5*0.5*0.67 = 16.74 kN-m/m

BM_{2z} = (157.60-23.7)*0.5*0.5/2-0.5*(-157.6+157.60)*0.5*0.5*0.67 = 16.74 kN-m/m

SF_{3z} = (157.60-23.7)*(0.5-0.282)+0.5*(157.6-157.60)*(0.5-0.282) = 29.19 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	16.74	0.210	0.281	0.0007	197.37	507.60	12ϕT	222.81	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 29.19 kN/m

Shear stress , τ_v= (29.19*1000)/(1000*282)

= 0.10 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	**12ϕT**	**200**	c/c
Along Z Direction :				
	At Bottom	**12ϕT**	**200**	c/c

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5.5 DESIGN CALCULATION OF FOOTING F5

Size of Columns at node	Along-Z	Along-X
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	4.50m	1.50m
Thickness of raft:	0.35	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.50	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

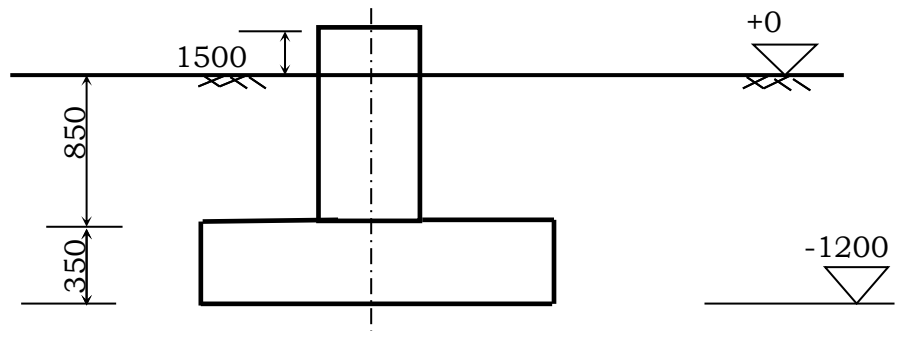
Calculation of Weight of Foundation :

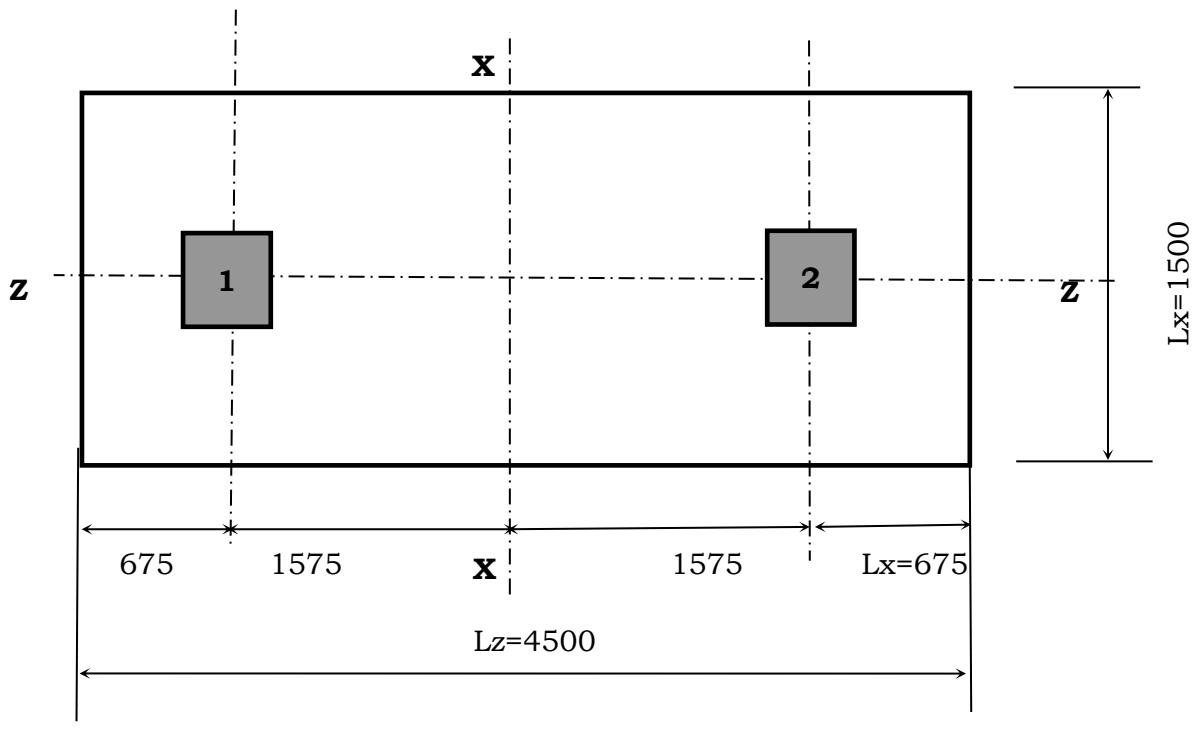
Weight of raft = (4.5*1.5*0.35*25) = 59.06KN

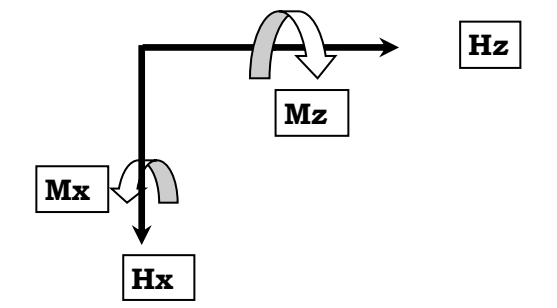
Weight of soil = ((4.5*1.5)-2*(0.5*0.5))*0.85*18 = 95.63KN

Total = 154.69KN

Submerged wt. = 154.69-4.5*1.5*1.2*10 = 73.69KN







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F5

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	207.9	0.0	0.0	0.0	0.0	207.9	0.0	0.0	0.0	0.0
202 DL+EQDL+FL+LL+EQLL+TL	207.9	0.0	0.0	0.0	0.0	207.9	0.0	0.0	0.0	0.0
203 DL+EQDL+FL+0.75(LL+EQLL)+	207.9	0.0	0.0	0.0	0.0	207.9	0.0	0.0	0.0	0.0
204 DL+EQDL+FL+WL(+X)	207.9	-5.3	14.3	0.0	0.0	207.9	-5.3	14.3	0.0	0.0
205 DL+EQDL+FL+WL(-X)	207.9	5.3	-14.3	0.0	0.0	207.9	5.3	-14.3	0.0	0.0
206 DL+EQDL+FL+WL(+Z)	207.9	0.0	0.0	-5.3	-14.3	207.9	0.0	0.0	-5.3	-14.3
207 DL+EQDL+FL+WL(-Z)	207.9	0.0	0.0	5.3	14.3	207.9	0.0	0.0	5.3	14.3
208 DL+EQDL+FL+0.7SL(+X)	207.9	-7.0	18.9	0.0	0.0	207.9	-7.0	18.9	0.0	0.0
209 DL+EQDL+FL+0.7SL(-X)	207.9	7.0	-18.9	0.0	0.0	207.9	7.0	-18.9	0.0	0.0
210 DL+EQDL+FL+0.7SL(+Z)	207.9	0.0	0.0	-7.0	-18.9	207.9	0.0	0.0	-7.0	-18.9
211 DL+EQDL+FL+0.7SL(-Z)	207.9	0.0	0.0	7.0	18.9	207.9	0.0	0.0	7.0	18.9
212 DL+EQDL+FL+0.75WL(+X)+0.7	207.9	-4.0	10.7	0.0	0.0	207.9	-4.0	10.7	0.0	0.0
213 DL+EQDL+FL+0.75WL(-X)+0.7	207.9	4.0	-10.7	0.0	0.0	207.9	4.0	-10.7	0.0	0.0
214 DL+EQDL+FL+0.75WL(+Z)+0.7	207.9	0.0	0.0	-4.0	-10.7	207.9	0.0	0.0	-4.0	-10.7
215 DL+EQDL+FL+0.75WL(-Z)+0.7	207.9	0.0	0.0	4.0	10.7	207.9	0.0	0.0	4.0	10.7
216 DL+EQDL+FL+0.525SL(+X)+0.	207.9	-5.3	14.2	0.0	0.0	207.9	-5.3	14.2	0.0	0.0
217 DL+EQDL+FL+0.525SL(-X)+0.7	207.9	5.3	-14.2	0.0	0.0	207.9	5.3	-14.2	0.0	0.0
218 DL+EQDL+FL+0.525SL(+Z)+0.	207.9	0.0	0.0	-5.3	-14.2	207.9	0.0	0.0	-5.3	-14.2
219 DL+EQDL+FL+0.525SL(-Z)+0.7	207.9	0.0	0.0	5.3	14.2	207.9	0.0	0.0	5.3	14.2
220 0.6(DL+EQDL)+WL(+X)	22.7	-5.3	14.3	0.0	0.0	22.7	-5.3	14.3	0.0	0.0
221 0.6(DL+EQDL)+WL(-X)	22.7	5.3	-14.3	0.0	0.0	22.7	5.3	-14.3	0.0	0.0
222 0.6(DL+EQDL)+WL(+Z)	22.7	0.0	0.0	-5.3	-14.3	22.7	0.0	0.0	-5.3	-14.3
223 0.6(DL+EQDL)+WL(-Z)	22.7	0.0	0.0	5.3	14.3	22.7	0.0	0.0	5.3	14.3
224 0.6(DL+EQDL)+0.7SL(+X)	22.7	-7.0	18.9	0.0	0.0	22.7	-7.0	18.9	0.0	0.0
225 0.6(DL+EQDL)+0.7SL(-X)	22.7	7.0	-18.9	0.0	0.0	22.7	7.0	-18.9	0.0	0.0
226 0.6(DL+EQDL)+0.7SL(+Z)	22.7	0.0	0.0	-7.0	-18.9	22.7	0.0	0.0	-7.0	-18.9
227 0.6(DL+EQDL)+0.7SL(-Z)	22.7	0.0	0.0	7.0	18.9	22.7	0.0	0.0	7.0	18.9
228 DL+EQTL	207.9	0.0	0.0	0.0	0.0	207.9	0.0	0.0	0.0	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL+EQDL+FL	570.5	0.00	0.0	0.0	0.0	NA	NA	NA	NA
202 DL+EQDL+FL+LL+EQLL+TL	570.5	0.00	0.0	0.0	0.0	NA	NA	NA	NA
203 DL+EQDL+FL+0.75(LL+EQLL)+	570.5	0.00	0.0	0.0	0.0	NA	NA	NA	NA
204 DL+EQDL+FL+WL(+X)	570.5	0.00	0.0	-10.6	28.6	15.0	NA	21.5	NA
205 DL+EQDL+FL+WL(-X)	570.5	0.00	0.0	10.6	-28.6	15.0	NA	21.5	NA
206 DL+EQDL+FL+WL(+Z)	570.5	-10.60	-28.6	0.0	0.0	NA	44.9	NA	21.5
207 DL+EQDL+FL+WL(-Z)	570.5	10.60	28.6	0.0	0.0	NA	44.9	NA	21.5
208 DL+EQDL+FL+0.7SL(+X)	570.5	0.00	0.0	-14.0	37.8	11.3	NA	16.3	NA
209 DL+EQDL+FL+0.7SL(-X)	570.5	0.00	0.0	14.0	-37.8	11.3	NA	16.3	NA
210 DL+EQDL+FL+0.7SL(+Z)	570.5	-14.00	-37.8	0.0	0.0	NA	34.0	NA	16.3
211 DL+EQDL+FL+0.7SL(-Z)	570.5	14.00	37.8	0.0	0.0	NA	34.0	NA	16.3
212 DL+EQDL+FL+0.75WL(+X)+0.7	570.5	0.00	0.0	-8.0	21.4	20.0	NA	28.5	NA
213 DL+EQDL+FL+0.75WL(-X)+0.7	570.5	0.00	0.0	8.0	-21.4	20.0	NA	28.5	NA
214 DL+EQDL+FL+0.75WL(+Z)+0.7	570.5	-8.00	-21.4	0.0	0.0	NA	60.0	NA	28.5
215 DL+EQDL+FL+0.75WL(-Z)+0.7	570.5	8.00	21.4	0.0	0.0	NA	60.0	NA	28.5
216 DL+EQDL+FL+0.525SL(+X)+0.	570.5	0.00	0.0	-10.6	28.4	15.1	NA	21.5	NA
217 DL+EQDL+FL+0.525SL(-X)+0.7	570.5	0.00	0.0	10.6	-28.4	15.1	NA	21.5	NA
218 DL+EQDL+FL+0.525SL(+Z)+0.	570.5	-10.60	-28.4	0.0	0.0	NA	45.2	NA	21.5
219 DL+EQDL+FL+0.525SL(-Z)+0.7	570.5	10.60	28.4	0.0	0.0	NA	45.2	NA	21.5
220 0.6(DL+EQDL)+WL(+X)	200.1	0.00	0.0	-10.6	28.6	5.2	NA	7.6	NA
221 0.6(DL+EQDL)+WL(-X)	200.1	0.00	0.0	10.6	-28.6	5.2	NA	7.6	NA
222 0.6(DL+EQDL)+WL(+Z)	200.1	-10.60	-28.6	0.0	0.0	NA	15.7	NA	7.6
223 0.6(DL+EQDL)+WL(-Z)	200.1	10.60	28.6	0.0	0.0	NA	15.7	NA	7.6
224 0.6(DL+EQDL)+0.7SL(+X)	200.1	0.00	0.0	-14.0	37.8	4.0	NA	5.7	NA
225 0.6(DL+EQDL)+0.7SL(-X)	200.1	0.00	0.0	14.0	-37.8	4.0	NA	5.7	NA
226 0.6(DL+EQDL)+0.7SL(+Z)	200.1	-14.00	-37.8	0.0	0.0	NA	11.9	NA	5.7
227 0.6(DL+EQDL)+0.7SL(-Z)	200.1	14.00	37.8	0.0	0.0	NA	11.9	NA	5.7
228 DL+EQTL	570.5	0.00	0.0	0.0	0.0	NA	NA	NA	NA
Min.=						**4.0**	**11.9**	**5.7**	**5.7**

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Base pressure check of footing F5

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	162	201	0	105.74	0
Min Px	162	201	0	105.74	0
Max Py	591	208	0	141.38	0
Min Py	648	224	0	0.00	0
Max Pz	162	201	0	105.74	0
Min Pz	162	201	0	105.74	0

Gross Maximum Base pressure, p_{max} = 141.4 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.85 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 465.3 KN/m^2
> 141.4 KN/m^2
Hence OK

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Design of foundation raft F5

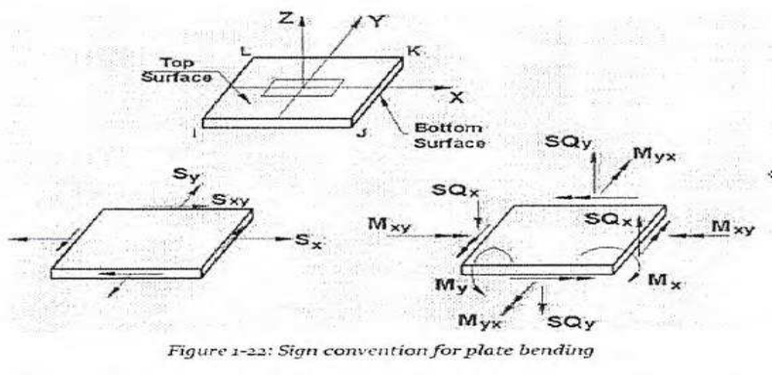
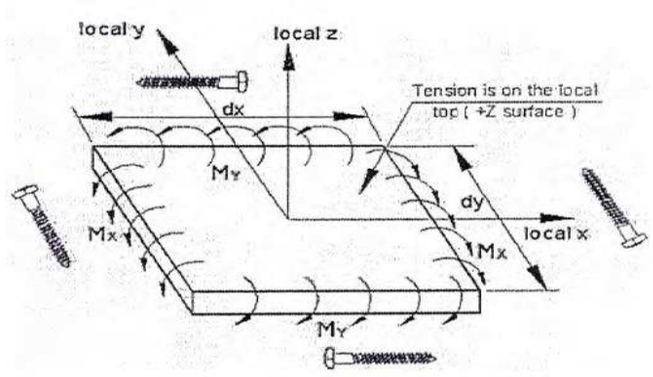


Figure 1-22: Sign convention for plate bending



Sign convention :

Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :

Mx : BM along X direction, +ve means tension on +Z surface of plate

My : BM along Y direction, +ve means tension on +Z surface of plate

For -ve Mx & My it is vice-virsa

Shear force :

SQx : Shear at face perpendicular to X axis

SQy : Shear at face perpendicular to Y axis

Axial force :

+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS : FDN_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	504	0.253	0.057	0	0	0	-11.91	-8.708	-2.133	-14.04	-10.841	101 1.4(DL+EQDL+FL)
Min Qx	493	-0.253	0.057	0	0	0	-11.91	-8.706	0	-11.906	-8.706	101 1.4(DL+EQDL+FL)
Max Qy	602	0.037	0.378	0	0	0	-22.96	25.867	4.071	-27.028	29.938	101 1.4(DL+EQDL+FL)
Min Qy	646	0.036	-0.378	0	0	0	-22.95	25.887	-4.068	-27.017	29.955	101 1.4(DL+EQDL+FL)
Max Sx	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Min Sx	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Max Sy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Min Sy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Max Sxy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Min Sxy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Max Mx	521	-0.044	-0.028	0	0	0	4.602	1.958	-1.982	6.584	3.94	121 0.9(DL+EQDL)+SL(-X)
Min Mx	517	-0.056	0.111	0	0	0	-32.59	-1.832	-2.316	-34.908	-4.148	101 1.4(DL+EQDL+FL)
Max My	626	0	-0.02	0	0	0	-0.422	78.356	-0.04	-0.462	78.396	101 1.4(DL+EQDL+FL)
Min My	602	0.029	0.106	0	0	0	-5.324	-13.081	0.115	-5.439	-13.196	123 0.9(DL+EQDL)+SL(-Z)
Max Mxy	592	-0.025	-0.06	0	0	0	-1.067	12.38	7.392	-8.459	19.772	112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
Min Mxy	647	0.025	-0.06	0	0	0	-1.067	12.38	-7.392	-8.459	19.772	113 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL

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PLATE CENTER STRESS : FDN_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	505	0.075	0.045	0	0	0	-1.854	-1.679	-0.34	-2.194	-2.019	101 1.4(DL+EQDL+FL)
Min Qx	491	-0.075	0.045	0	0	0	-1.854	-1.677	0	-1.854	-1.677	101 1.4(DL+EQDL+FL)
Max Qy	606	0.025	0.245	0	0	0	-11.97	47.291	1.912	-13.88	49.203	101 1.4(DL+EQDL+FL)
Min Qy	642	0.025	-0.245	0	0	0	-11.97	47.297	-1.912	-13.88	49.209	101 1.4(DL+EQDL+FL)
Max Sx	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Min Sx	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Max Sy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Min Sy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Max Sxy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Min Sxy	652	0.066	-0.125	0	0	0	-1.351	7.142	-1.818	-3.169	8.96	101 1.4(DL+EQDL+FL)
Max Mx	505	-0.018	-0.008	0	0	0	0.92	2.36	1.491	2.411	3.851	121 0.9(DL+EQDL)+SL(-X)
Min Mx	544	-0.025	0.245	0	0	0	-11.97	47.291	-1.914	-13.882	49.205	101 1.4(DL+EQDL+FL)
Max My	626	0	-0.02	0	0	0	-0.422	78.356	-0.04	-0.462	78.396	101 1.4(DL+EQDL+FL)
Min My	605	0.012	0.012	0	0	0	-0.041	-8.837	0.041	-0.082	-8.878	123 0.9(DL+EQDL)+SL(-Z)
Max Mxy	587	-0.016	-0.075	0	0	0	-2.445	19.664	5.497	-7.942	25.161	112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
Min Mxy	542	-0.016	0.075	0	0	0	-2.445	19.661	-5.497	-7.942	25.158	112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		350	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
121 0.9(D	6.58	top face	50	294	6.584	0.10	0.0002	71.2	529.2	12	213.8	200	565.8	7.978	9.972	0.085	tension control
101 1.4(D	-34.91	bottom face	50	294	34.908	0.54	0.0013	380.4		12	213.8	200	565.8	7.978	9.972	0.085	tension control
112 1.2(D	-8.46	bottom face	50	294	8.459	0.13	0.0003	91.5		12	213.8	200	565.8	7.978	9.972	0.085	tension control
113 1.2(D	-8.46	bottom face	50	294	8.459	0.13	0.0003	91.5		12	213.8	200	565.8	7.978	9.972	0.085	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.075	1	26.25	2.194	0.002	0.294	312.126	287.911	504.405	287.91	1077	> 26.25	ok	215.9	φVc>Vu	not required
	Min Qx	0.075	1	26.25	1.854			318.331			287.91	1077	> 26.25	ok	215.9	φVc>Vu	not required

Reinforcement Along Z:

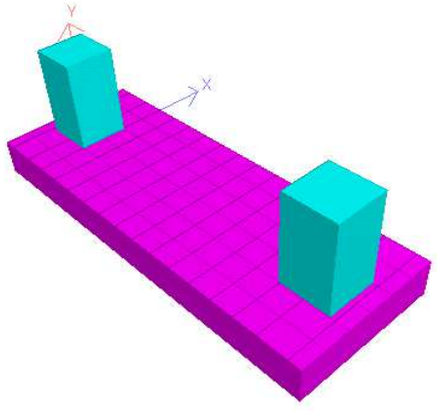
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
101 1.4(D	78.40	top face	50	294	78.396	1.21	0.0029	864.5	529.2	12	130.88	125	905.2	12.76	15.95	0.052	tension control
123 0.9(D	-13.20	bottom face	50	294	13.196	0.20	0.0005	143.0		12	213.8	200	565.8	7.978	9.972	0.085	tension control
112 1.2(D	19.77	top face	50	294	19.772	0.30	0.0007	214.6		12	213.8	200	565.8	7.978	9.972	0.085	tension control
113 1.2(D	19.77	top face	50	294	19.772	0.30	0.0007	214.6		12	213.8	200	565.8	7.978	9.972	0.085	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	101 1.4(D	0.245	1	85.75	49.203	0.003	0.294	283.221	287.911	fc'504.4	283.22	299	> 85.75	ok	212.4	φVc>Vu	not required
	101 1.4(D	0.245	1	85.75	49.209			283.22			283.22	299	> 85.75	ok	212.4	φVc>Vu	not required

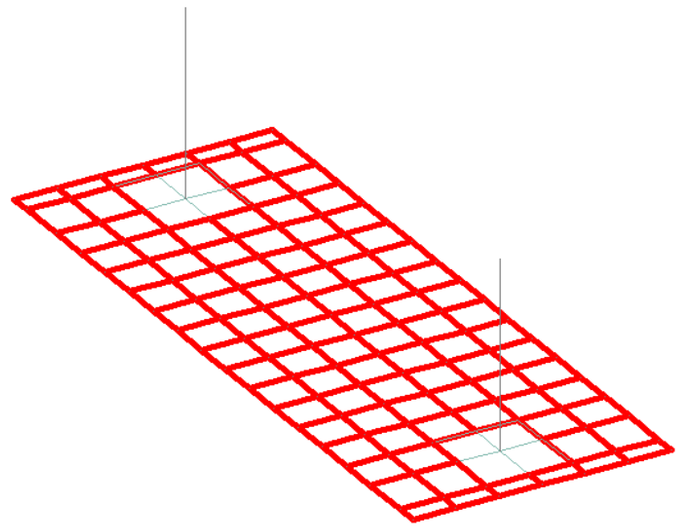
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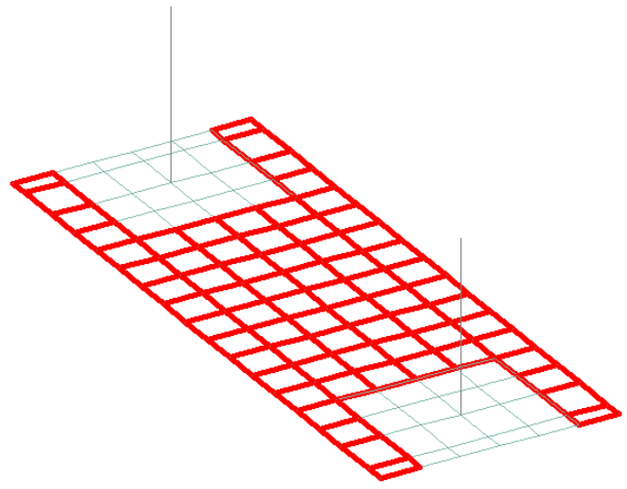
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3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	**200**	c/c
At Top	12ϕT	**200**	c/c
Along Z Direction :			
At Bottom	12ϕT	**200**	c/c
At Top	12ϕT	**125**	c/c

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5.6 DESIGN OF CALCULATION OF FOOTING F6

STAAD Node - 164

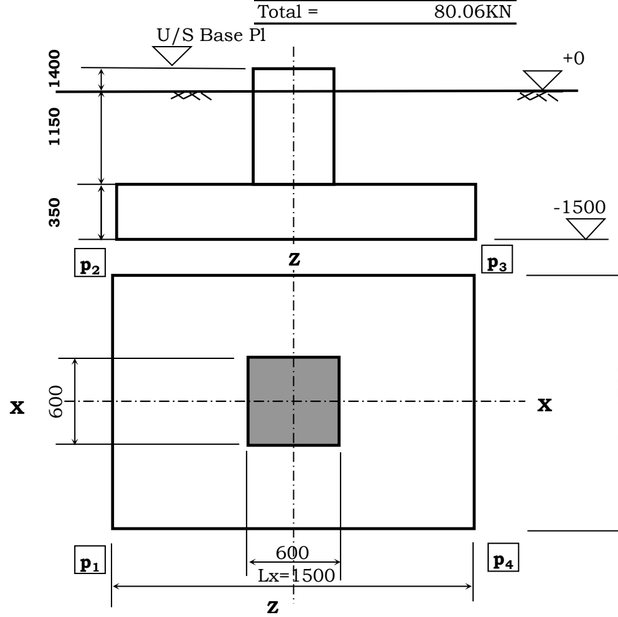
	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	1.40 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

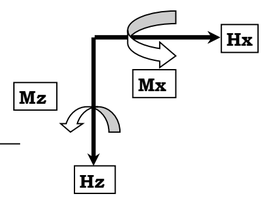
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.5*1.5*0.35*24)+(0.6*0.6*2.55*24) = 40.93KN
Weight of soil = ((1.5*1.5)-(0.6*0.6))*1.15*18 = 39.12KN
Total = 80.06KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.45m

Clear projection along X direction = 0.45m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING F6

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+EQDL+FL	141.1	0.0	0.0	0.0	0.0	221.16	0.00	0.00	0.00	0.00
202	DL+EQDL+FL+LL+EQLL+TL	141.1	0.0	0.0	0.0	0.0	221.16	0.00	0.00	0.00	0.00
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	141.1	0.0	0.0	0.0	0.0	221.16	0.00	0.00	0.00	0.00
204	DL+EQDL+FL+WL(+X)	141.1	-6.0	17.4	0.0	0.0	221.16	-6.00	17.40	0.00	0.00
205	DL+EQDL+FL+WL(-X)	141.1	6.0	-17.4	0.0	0.0	221.16	6.00	-17.40	0.00	0.00
206	DL+EQDL+FL+WL(+Z)	141.1	0.0	0.0	-6.0	-17.4	221.16	0.00	0.00	-6.00	-17.40
207	DL+EQDL+FL+WL(-Z)	141.1	0.0	0.0	6.0	17.4	221.16	0.00	0.00	6.00	17.40
208	DL+EQDL+FL+0.7SL(+X)	141.1	-2.1	6.1	0.0	0.0	221.16	-2.10	6.10	0.00	0.00
209	DL+EQDL+FL+0.7SL(-X)	141.1	2.1	-6.1	0.0	0.0	221.16	2.10	-6.10	0.00	0.00
210	DL+EQDL+FL+0.7SL(+Z)	141.1	0.0	0.0	-2.1	-6.1	221.16	0.00	0.00	-2.10	-6.10
211	DL+EQDL+FL+0.7SL(-Z)	141.1	0.0	0.0	2.1	6.1	221.16	0.00	0.00	2.10	6.10
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQ	141.1	-4.5	13.1	0.0	0.0	221.16	-4.50	13.10	0.00	0.00
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQ	141.1	4.5	-13.1	0.0	0.0	221.16	4.50	-13.10	0.00	0.00
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQ	141.1	0.0	0.0	-4.5	-13.1	221.16	0.00	0.00	-4.50	-13.10
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQ	141.1	0.0	0.0	4.5	13.1	221.16	0.00	0.00	4.50	13.10
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+E	141.1	-1.6	4.6	0.0	0.0	221.16	-1.60	4.60	0.00	0.00
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+E	141.1	1.6	-4.6	0.0	0.0	221.16	1.60	-4.60	0.00	0.00
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+E	141.1	0.0	0.0	-1.6	-4.6	221.16	0.00	0.00	-1.60	-4.60
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+E	141.1	0.0	0.0	1.6	4.6	221.16	0.00	0.00	1.60	4.60
220	0.6(DL+EQDL)+WL(+X)	37.3	-6.0	17.4	0.0	0.0	117.36	-6.00	17.40	0.00	0.00
221	0.6(DL+EQDL)+WL(-X)	37.3	6.0	-17.4	0.0	0.0	117.36	6.00	-17.40	0.00	0.00
222	0.6(DL+EQDL)+WL(+Z)	37.3	0.0	0.0	-6.0	-17.4	117.36	0.00	0.00	-6.00	-17.40
223	0.6(DL+EQDL)+WL(-Z)	37.3	0.0	0.0	6.0	17.4	117.36	0.00	0.00	6.00	17.40
224	0.6(DL+EQDL)+0.7SL(+X)	37.3	-2.1	6.1	0.0	0.0	117.36	-2.10	6.10	0.00	0.00
225	0.6(DL+EQDL)+0.7SL(-X)	37.3	2.1	-6.1	0.0	0.0	117.36	2.10	-6.10	0.00	0.00
226	0.6(DL+EQDL)+0.7SL(+Z)	37.3	0.0	0.0	-2.1	-6.1	117.36	0.00	0.00	-2.10	-6.10
227	0.6(DL+EQDL)+0.7SL(-Z)	37.3	0.0	0.0	2.1	6.1	117.36	0.00	0.00	2.10	6.10
228	DL+EQTL	141.1	0.0	0.0	0.0	0.0	221.16	0.00	0.00	0.00	0.00
Ultimate Conditions											
101	1.4(DL+EQDL+FL)	197.5	0.0	0.0	0.0	0.0	277.56	0.00	0.00	0.00	0.00
102	1.2(DL+EQDL+FL)+1.6(LL+EQLL)	169.3	0.0	0.0	0.0	0.0	249.36	0.00	0.00	0.00	0.00
103	1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)	169.3	0.0	0.0	0.0	0.0	249.36	0.00	0.00	0.00	0.00
104	1.2(DL+EQDL)+0.8WL(+X)	74.5	-4.8	13.9	0.0	0.0	154.56	-4.80	13.90	0.00	0.00
105	1.2(DL+EQDL)+0.8WL(-X)	74.5	4.8	-13.9	0.0	0.0	154.56	4.80	-13.90	0.00	0.00
106	1.2(DL+EQDL)+0.8WL(+Z)	74.5	0.0	0.0	-4.8	-13.9	154.56	0.00	0.00	-4.80	-13.90
107	1.2(DL+EQDL)+0.8WL(-Z)	74.5	0.0	0.0	4.8	13.9	154.56	0.00	0.00	4.80	13.90
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQL	74.5	-9.6	27.8	0.0	0.0	154.56	-9.60	27.80	0.00	0.00
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQL	74.5	9.6	-27.8	0.0	0.0	154.56	9.60	-27.80	0.00	0.00
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQL	74.5	0.0	0.0	-9.6	-27.8	154.56	0.00	0.00	-9.60	-27.80
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQL	74.5	0.0	0.0	9.6	27.8	154.56	0.00	0.00	9.60	27.80
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+	114.0	-3.0	0.0	0.0	0.0	194.06	-3.00	-8.70	0.00	0.00
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+	114.0	3.0	0.0	0.0	0.0	194.06	3.00	8.70	0.00	0.00
114	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+	114.0	0.0	0.0	-3.0	0.0	194.06	0.00	0.00	-3.00	-8.70
115	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+	114.0	0.0	0.0	3.0	0.0	194.06	0.00	0.00	3.00	8.70
116	0.9(DL+EQDL)+1.6WL(+X)	55.9	-9.6	0.0	0.0	0.0	135.96	-9.60	-27.84	0.00	0.00
117	0.9(DL+EQDL)+1.6WL(-X)	55.9	9.6	0.0	0.0	0.0	135.96	9.60	27.84	0.00	0.00
118	0.9(DL+EQDL)+1.6WL(+Z)	55.9	0.0	0.0	-9.6	0.0	135.96	0.00	0.00	-9.60	-27.84
119	0.9(DL+EQDL)+1.6WL(-Z)	55.9	0.0	0.0	9.6	0.0	135.96	0.00	0.00	9.60	27.84
120	0.9(DL+EQDL)+SL(+X)	55.9	-3.0	0.0	0.0	0.0	135.96	-3.00	-8.70	0.00	0.00
121	0.9(DL+EQDL)+SL(-X)	55.9	3.0	0.0	0.0	0.0	135.96	3.00	8.70	0.00	0.00
122	0.9(DL+EQDL)+SL(+Z)	55.9	0.0	0.0	-3.0	0.0	135.96	0.00	0.00	-3.00	-8.70
123	0.9(DL+EQDL)+SL(-Z)	55.9	0.0	0.0	3.0	0.0	135.96	0.00	0.00	3.00	8.70
124	1.2DL+EQTL	146.3	0.0	0.0	0.0	0.0	226.36	0.00	0.00	0.00	0.00

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Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	DL+EQDL+FL	221.2	0.0	0.0	0.0	0.0	NA	NA	NA	NA
202	DL+EQDL+FL+LL+EQLL+TL	221.2	0.0	0.0	0.0	0.0	NA	NA	NA	NA
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	221.2	0.0	0.0	0.0	0.0	NA	NA	NA	NA
204	DL+EQDL+FL+WL(+X)	221.2	-6.0	17.4	0.0	0.0	9.5	NA	14.7	NA
205	DL+EQDL+FL+WL(-X)	221.2	6.0	-17.4	0.0	0.0	9.5	NA	14.7	NA
206	DL+EQDL+FL+WL(+Z)	221.2	0.0	0.0	-6.0	-17.4	NA	9.5	NA	14.7
207	DL+EQDL+FL+WL(-Z)	221.2	0.0	0.0	6.0	17.4	NA	9.5	NA	14.7
208	DL+EQDL+FL+0.7SL(+X)	221.2	-2.1	6.1	0.0	0.0	27.2	NA	42.1	NA
209	DL+EQDL+FL+0.7SL(-X)	221.2	2.1	-6.1	0.0	0.0	27.2	NA	42.1	NA
210	DL+EQDL+FL+0.7SL(+Z)	221.2	0.0	0.0	-2.1	-6.1	NA	27.2	NA	42.1
211	DL+EQDL+FL+0.7SL(-Z)	221.2	0.0	0.0	2.1	6.1	NA	27.2	NA	42.1
212	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)	221.2	-4.5	13.1	0.0	0.0	12.7	NA	19.7	NA
213	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)	221.2	4.5	-13.1	0.0	0.0	12.7	NA	19.7	NA
214	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)	221.2	0.0	0.0	-4.5	-13.1	NA	12.7	NA	19.7
215	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)	221.2	0.0	0.0	4.5	13.1	NA	12.7	NA	19.7
216	DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL	221.2	-1.6	4.6	0.0	0.0	36.1	NA	55.3	NA
217	DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)	221.2	1.6	-4.6	0.0	0.0	36.1	NA	55.3	NA
218	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL	221.2	0.0	0.0	-1.6	-4.6	NA	36.1	NA	55.3
219	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)	221.2	0.0	0.0	1.6	4.6	NA	36.1	NA	55.3
220	0.6(DL+EQDL)+WL(+X)	117.4	-6.0	17.4	0.0	0.0	5.1	NA	7.8	NA
221	0.6(DL+EQDL)+WL(-X)	117.4	6.0	-17.4	0.0	0.0	5.1	NA	7.8	NA
222	0.6(DL+EQDL)+WL(+Z)	117.4	0.0	0.0	-6.0	-17.4	NA	5.1	NA	7.8
223	0.6(DL+EQDL)+WL(-Z)	117.4	0.0	0.0	6.0	17.4	NA	5.1	NA	7.8
224	0.6(DL+EQDL)+0.7SL(+X)	117.4	-2.1	6.1	0.0	0.0	14.4	NA	22.4	NA
225	0.6(DL+EQDL)+0.7SL(-X)	117.4	2.1	-6.1	0.0	0.0	14.4	NA	22.4	NA
226	0.6(DL+EQDL)+0.7SL(+Z)	117.4	0.0	0.0	-2.1	-6.1	NA	14.4	NA	22.4
227	0.6(DL+EQDL)+0.7SL(-Z)	117.4	0.0	0.0	2.1	6.1	NA	14.4	NA	22.4
228	DL+EQTL	221.2	0.0	0.0	0.0	0.0	NA	NA	NA	NA
						Min.=	**5.1**	**5.1**	**7.8**	**7.8**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F6

Load Combination & Calculation of Base pressure

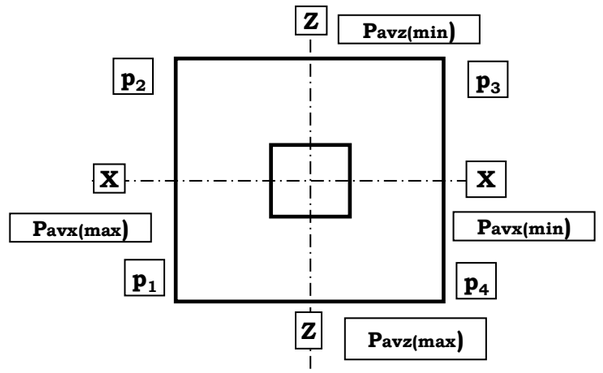
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	DL+EQDL+FL	221.16	0.00	0.00	98.29	0.00	0.00	98.29	98.29	98.29	98.29	71.29	98.29	98.29	98.29	98.29	O.K
202	DL+EQDL+FL+LL+EQLL+TL	221.16	0.00	0.00	98.29	0.00	0.00	98.29	98.29	98.29	98.29	71.29	98.29	98.29	98.29	98.29	O.K
203	DL+EQDL+FL+0.75(LL+EQLL)+0	221.16	0.00	0.00	98.29	0.00	0.00	98.29	98.29	98.29	98.29	71.29	98.29	98.29	98.29	98.29	O.K
204	DL+EQDL+FL+WL(+X)	221.16	17.40	0.00	98.29	30.93	0.00	129.23	129.23	67.36	67.36	102.23	129.23	67.36	98.30	98.30	O.K
205	DL+EQDL+FL+WL(-X)	221.16	-17.40	0.00	98.29	30.93	0.00	129.23	129.23	67.36	67.36	102.23	129.23	67.36	98.30	98.30	O.K
206	DL+EQDL+FL+WL(+Z)	221.16	0.00	-17.40	98.29	0.00	30.93	129.23	67.36	129.23	67.36	102.23	98.30	98.30	129.23	67.36	O.K
207	DL+EQDL+FL+WL(-Z)	221.16	0.00	17.40	98.29	0.00	30.93	129.23	67.36	129.23	67.36	102.23	98.30	98.30	129.23	67.36	O.K
208	DL+EQDL+FL+0.7SL(+X)	221.16	6.10	0.00	98.29	10.84	0.00	109.14	109.14	87.45	87.45	82.14	109.14	87.45	98.30	98.30	O.K
209	DL+EQDL+FL+0.7SL(-X)	221.16	-6.10	0.00	98.29	10.84	0.00	109.14	109.14	87.45	87.45	82.14	109.14	87.45	98.30	98.30	O.K
210	DL+EQDL+FL+0.7SL(+Z)	221.16	0.00	-6.10	98.29	0.00	10.84	109.14	87.45	109.14	87.45	82.14	98.30	98.30	109.14	87.45	O.K
211	DL+EQDL+FL+0.7SL(-Z)	221.16	0.00	6.10	98.29	0.00	10.84	109.14	87.45	109.14	87.45	82.14	98.30	98.30	109.14	87.45	O.K
212	DL+EQDL+FL+0.75WL(+X)+0.75	221.16	13.10	0.00	98.29	23.29	0.00	121.58	121.58	75.00	75.00	94.58	121.58	75.00	98.29	98.29	O.K
213	DL+EQDL+FL+0.75WL(-X)+0.75(	221.16	-13.10	0.00	98.29	23.29	0.00	121.58	121.58	75.00	75.00	94.58	121.58	75.00	98.29	98.29	O.K
214	DL+EQDL+FL+0.75WL(+Z)+0.75	221.16	0.00	-13.10	98.29	0.00	23.29	121.58	75.00	121.58	75.00	94.58	98.29	98.29	121.58	75.00	O.K

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215 DL+EQDL+FL+0.75WL(-Z)+0.75(	221.16	0.00	13.10	98.29	0.00	23.29	121.58	75.00	121.58	75.00	94.58	98.29	98.29	121.58	75.00	O.K
216 DL+EQDL+FL+0.525SL(+X)+0.75	221.16	4.60	0.00	98.29	8.18	0.00	106.47	106.47	90.12	90.12	79.47	106.47	90.12	98.30	98.30	O.K
217 DL+EQDL+FL+0.525SL(-X)+0.75	221.16	-4.60	0.00	98.29	8.18	0.00	106.47	106.47	90.12	90.12	79.47	106.47	90.12	98.30	98.30	O.K
218 DL+EQDL+FL+0.525SL(+Z)+0.75	221.16	0.00	-4.60	98.29	0.00	8.18	106.47	90.12	106.47	90.12	79.47	98.30	98.30	106.47	90.12	O.K
219 DL+EQDL+FL+0.525SL(-Z)+0.75	221.16	0.00	4.60	98.29	0.00	8.18	106.47	90.12	106.47	90.12	79.47	98.30	98.30	106.47	90.12	O.K
220 0.6(DL+EQDL)+WL(+X)	117.36	17.40	0.00	52.16	30.93	0.00	83.09	83.09	21.23	21.23	56.09	83.09	21.23	52.16	52.16	O.K
221 0.6(DL+EQDL)+WL(-X)	117.36	-17.40	0.00	52.16	30.93	0.00	83.09	83.09	21.23	21.23	56.09	83.09	21.23	52.16	52.16	O.K
222 0.6(DL+EQDL)+WL(+Z)	117.36	0.00	-17.40	52.16	0.00	30.93	83.09	21.23	83.09	21.23	56.09	52.16	52.16	83.09	21.23	O.K
223 0.6(DL+EQDL)+WL(-Z)	117.36	0.00	17.40	52.16	0.00	30.93	83.09	21.23	83.09	21.23	56.09	52.16	52.16	83.09	21.23	O.K
224 0.6(DL+EQDL)+0.7SL(+X)	117.36	6.10	0.00	52.16	10.84	0.00	63.00	63.00	41.32	41.32	36.00	63.00	41.32	52.16	52.16	O.K
225 0.6(DL+EQDL)+0.7SL(-X)	117.36	-6.10	0.00	52.16	10.84	0.00	63.00	63.00	41.32	41.32	36.00	63.00	41.32	52.16	52.16	O.K
226 0.6(DL+EQDL)+0.7SL(+Z)	117.36	0.00	-6.10	52.16	0.00	10.84	63.00	41.32	63.00	41.32	36.00	52.16	52.16	63.00	41.32	O.K
227 0.6(DL+EQDL)+0.7SL(-Z)	117.36	0.00	6.10	52.16	0.00	10.84	63.00	41.32	63.00	41.32	36.00	52.16	52.16	63.00	41.32	O.K
228 DL+EQTL	221.16	0.00	0.00	98.29	0.00	0.00	98.29	98.29	98.29	98.29	71.29	98.29	98.29	98.29	98.29	O.K

Ultimate Conditions																
101 1.4(DL+EQDL+FL)	277.56	0.00	0.00	123.36	0.00	0.00	123.36	123.36	123.36	123.36	96.36	123.36	123.36	123.36	123.36	
102 1.2(DL+EQDL+FL)+1.6(LL+EQLL	249.36	0.00	0.00	110.83	0.00	0.00	110.83	110.83	110.83	110.83	83.83	110.83	110.83	110.83	110.83	
103 1.2(DL+EQDL+FL+TL)+1.6(LL+E	249.36	0.00	0.00	110.83	0.00	0.00	110.83	110.83	110.83	110.83	83.83	110.83	110.83	110.83	110.83	
104 1.2(DL+EQDL)+0.8WL(+X)	154.56	13.90	0.00	68.69	24.71	0.00	93.40	93.40	43.98	43.98	66.40	93.40	43.98	68.69	68.69	
105 1.2(DL+EQDL)+0.8WL(-X)	154.56	-13.90	0.00	68.69	24.71	0.00	93.40	93.40	43.98	43.98	66.40	93.40	43.98	68.69	68.69	
106 1.2(DL+EQDL)+0.8WL(+Z)	154.56	0.00	-13.90	68.69	0.00	24.71	93.40	43.98	93.40	43.98	66.40	68.69	68.69	93.40	43.98	
107 1.2(DL+EQDL)+0.8WL(-Z)	154.56	0.00	13.90	68.69	0.00	24.71	93.40	43.98	93.40	43.98	66.40	68.69	68.69	93.40	43.98	
108 1.2(DL+EQDL)+1.6WL(+X)+0.5(L	154.56	27.80	0.00	68.69	49.42	0.00	118.12	118.12	19.27	19.27	91.12	118.12	19.27	68.70	68.70	

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109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LI	154.56	-27.80	0.00	68.69	49.42	0.00	118.12	118.12	19.27	19.27	91.12	118.12	19.27	68.70	68.70	
110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(L	154.56	0.00	-27.80	68.69	0.00	49.42	118.12	19.27	118.12	19.27	91.12	68.70	68.70	118.12	19.27	
111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LI	154.56	0.00	27.80	68.69	0.00	49.42	118.12	19.27	118.12	19.27	91.12	68.70	68.70	118.12	19.27	
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+E	194.06	-8.70	0.00	86.25	15.47	0.00	101.72	101.72	70.78	70.78	74.72	101.72	70.78	86.25	86.25	
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+E	194.06	8.70	0.00	86.25	15.47	0.00	101.72	101.72	70.78	70.78	74.72	101.72	70.78	86.25	86.25	
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+E	194.06	0.00	-8.70	86.25	0.00	15.47	101.72	70.78	101.72	70.78	74.72	86.25	86.25	101.72	70.78	
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+E	194.06	0.00	8.70	86.25	0.00	15.47	101.72	70.78	101.72	70.78	74.72	86.25	86.25	101.72	70.78	
116 0.9(DL+EQDL)+1.6WL(+X)	135.96	-27.84	0.00	60.43	49.49	0.00	109.92	109.92	10.93	10.93	82.92	109.92	10.93	60.43	60.43	
117 0.9(DL+EQDL)+1.6WL(-X)	135.96	27.84	0.00	60.43	49.49	0.00	109.92	109.92	10.93	10.93	82.92	109.92	10.93	60.43	60.43	
118 0.9(DL+EQDL)+1.6WL(+Z)	135.96	0.00	-27.84	60.43	0.00	49.49	109.92	10.93	109.92	10.93	82.92	60.43	60.43	109.92	10.93	
119 0.9(DL+EQDL)+1.6WL(-Z)	135.96	0.00	27.84	60.43	0.00	49.49	109.92	10.93	109.92	10.93	82.92	60.43	60.43	109.92	10.93	
120 0.9(DL+EQDL)+SL(+X)	135.96	-8.70	0.00	60.43	15.47	0.00	75.89	75.89	44.96	44.96	48.89	75.89	44.96	60.43	60.43	
121 0.9(DL+EQDL)+SL(-X)	135.96	8.70	0.00	60.43	15.47	0.00	75.89	75.89	44.96	44.96	48.89	75.89	44.96	60.43	60.43	
122 0.9(DL+EQDL)+SL(+Z)	135.96	0.00	-8.70	60.43	0.00	15.47	75.89	44.96	75.89	44.96	48.89	60.43	60.43	75.89	44.96	
123 0.9(DL+EQDL)+SL(-Z)	135.96	0.00	8.70	60.43	0.00	15.47	75.89	44.96	75.89	44.96	48.89	60.43	60.43	75.89	44.96	
124 1.2DL+EQTL	226.36	0.00	0.00	100.60	0.00	0.00	100.60	100.60	100.60	100.60	73.60	100.60	100.60	100.60	100.60	
Max :	278	28	28													
Min :	117	-28	-28													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 71.29 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 81.29 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 102.23 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 112.23 kN/m²

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Sample calculation of values of base pressure:

Load case : 202 DL+EQDL+FL+LL+EQLL+TL

Maximum pressure, p_1 (pmax) = 98.29+0.00+0.00= 98.29 kN/m^2

p_2= 98.29+0.00-0.00= 98.29 kN/m^2

p_3= 98.29-(0.00+0.00)= 98.29 kN/m^2

p_4= 98.29-0.00+0.00= 98.29 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 98.29-1.5*18= 71.29 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (98.29+98.29)/2= 98.29 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (98.29+98.29)/2= 98.29 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (98.29+98.29)/2= 98.29 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (98.29+98.29)/2= 98.29 kN/m^2

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DESIGN OF ISOLATED FOOTING F6

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

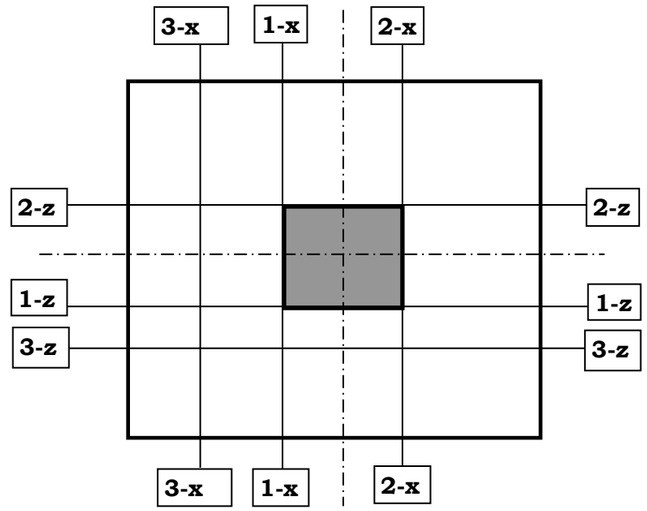
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.282)+(0.6+2*0.5*0.282))*2
= 3.528 m

Punching Area= ((0.6+2*0.5*0.282)*(0.6+2*0.5*0.282))
= 0.78 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+1.15*18)
= 29.10 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+EQDL+FL)	123.36	123.36	123.36	9.54	9.54	14.70	138.56
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	110.83	110.83	110.83	8.28	8.28	12.75	120.14
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	110.83	110.83	110.83	8.28	8.28	12.75	120.14
104	1.2(DL+EQDL)+0.8WL(+X)	78.57	58.81	88.26	6.01	2.01	9.63	58.20
105	1.2(DL+EQDL)+0.8WL(-X)	78.57	58.81	88.26	6.01	2.01	9.63	58.20
106	1.2(DL+EQDL)+0.8WL(+Z)	68.69	68.69	68.69	4.01	4.01	6.18	58.20
107	1.2(DL+EQDL)+0.8WL(-Z)	68.69	68.69	68.69	4.01	4.01	6.18	58.20
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(	88.47	48.93	107.84	8.01	0.01	13.09	58.20

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109 1.2(DL+EQDL)+1.6WL(-X)+0.5(	88.47	48.93	107.84	8.01	0.01	13.09	58.20
110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(	68.70	68.70	68.70	4.01	4.01	6.18	58.20
111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(	68.70	68.70	68.70	4.01	4.01	6.18	58.20
112 1.2(DL+EQDL)+SL(+X)+0.5(LL+	92.44	80.06	98.50	7.04	4.53	11.08	84.01
113 1.2(DL+EQDL)+SL(-X)+0.5(LL+	92.44	80.06	98.50	7.04	4.53	11.08	84.01
114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+	86.25	86.25	86.25	5.79	5.79	8.92	84.01
115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+	86.25	86.25	86.25	5.79	5.79	8.92	84.01
116 0.9(DL+EQDL)+1.6WL(+X)	80.22	40.63	99.63	7.18	-0.84	11.80	46.05
117 0.9(DL+EQDL)+1.6WL(-X)	80.22	40.63	99.63	7.18	-0.84	11.80	46.05
118 0.9(DL+EQDL)+1.6WL(+Z)	60.43	60.43	60.43	3.17	3.17	4.89	46.05
119 0.9(DL+EQDL)+1.6WL(-Z)	60.43	60.43	60.43	3.17	3.17	4.89	46.05
120 0.9(DL+EQDL)+SL(+X)	66.61	54.24	72.67	4.42	1.92	7.05	46.05
121 0.9(DL+EQDL)+SL(-X)	66.61	54.24	72.67	4.42	1.92	7.05	46.05
122 0.9(DL+EQDL)+SL(+Z)	60.43	60.43	60.43	3.17	3.17	4.89	46.05
123 0.9(DL+EQDL)+SL(-Z)	60.43	60.43	60.43	3.17	3.17	4.89	46.05
124 1.2DL+EQTL	100.60	100.60	100.60	7.24	7.24	11.15	105.11

Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1x} = 123.36-(123.36-123.36)*0.45/1.5 = 123.36 kN/m^2 Overhang = 0.45 m

p_{2x} = 123.36+(123.36-123.36)*0.45/1.5 = 123.36 kN/m^2 Effective depth = 0.294 m

p_{3x} = 123.36-(123.36-123.36)*(0.45-0.294)/1.5 = 123.36 kN/m^2

BM_{1x} = (123.36-29.1)*0.45*0.45/2+0.5*(123.36-123.36)*0.45*0.45*0.67 = 9.54 kN-m/m

BM_{2x} = (123.36-29.1)*0.45*0.45/2-0.5*(-123.36+123.36)*0.45*0.45*0.67 = 9.54 kN-m/m

SF_{3x} = (123.36-29.1)*(0.45-0.294)+0.5*(123.36-123.36)*(0.45-0.294) = 14.70 kN/m

$S.F_{(punching)}$ = (2.25-0.78)*(123.36-29.1) = 138.56 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	9.54	0.11	0.15	0.000	103.31	529.20	12ϕT	213.71	300	200	565.49

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 14.70 kN/m

Shear stress , τ_v= (14.7*1000)/(1000*294)

= 0.05 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 138.56 kN

Strength of Concrete in shear Vc1 = 3068.17 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1874.99 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2045.45 kN **Table 22.6.5.2_ (c)**

Design Vc= 1874.99 kN

Allowable shear stress, (ϕ Vc)/bd = 1.36 N/mm^2

Design punching shear stress, τ = (138.56*1000)/(3528*294)

= 0.13 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

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Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+EQDL+FL)	123.36	123.36	123.36	9.54	9.54	15.84
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	110.83	110.83	110.83	8.28	8.28	13.73
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	110.83	110.83	110.83	8.28	8.28	13.73
104	1.2(DL+EQDL)+0.8WL(+X)	68.69	68.69	68.69	4.01	4.01	6.65
105	1.2(DL+EQDL)+0.8WL(-X)	68.69	68.69	68.69	4.01	4.01	6.65
106	1.2(DL+EQDL)+0.8WL(+Z)	78.57	58.81	87.86	6.02	2.00	10.34
107	1.2(DL+EQDL)+0.8WL(-Z)	78.57	58.81	87.86	6.02	2.00	10.34
108	1.2(DL+EQDL)+1.6WL(+X)+0.5(	68.70	68.70	68.70	4.01	4.01	6.65
109	1.2(DL+EQDL)+1.6WL(-X)+0.5(	68.70	68.70	68.70	4.01	4.01	6.65
110	1.2(DL+EQDL)+1.6WL(+Z)+0.5(	88.47	48.93	107.05	8.02	0.00	14.03
111	1.2(DL+EQDL)+1.6WL(-Z)+0.5(	88.47	48.93	107.05	8.02	0.00	14.03
112	1.2(DL+EQDL)+SL(+X)+0.5(LL+	86.25	86.25	86.25	5.79	5.79	9.60
113	1.2(DL+EQDL)+SL(-X)+0.5(LL+	86.25	86.25	86.25	5.79	5.79	9.60
114	1.2(DL+EQDL)+SL(+Z)+0.5(LL+	92.44	80.06	98.25	7.04	4.53	11.91
115	1.2(DL+EQDL)+SL(-Z)+0.5(LL+	92.44	80.06	98.25	7.04	4.53	11.91
116	0.9(DL+EQDL)+1.6WL(+X)	60.43	60.43	60.43	3.17	3.17	5.26
117	0.9(DL+EQDL)+1.6WL(-X)	60.43	60.43	60.43	3.17	3.17	5.26
118	0.9(DL+EQDL)+1.6WL(+Z)	80.22	40.63	98.83	7.19	-0.85	12.65
119	0.9(DL+EQDL)+1.6WL(-Z)	80.22	40.63	98.83	7.19	-0.85	12.65
120	0.9(DL+EQDL)+SL(+X)	60.43	60.43	60.43	3.17	3.17	5.26
121	0.9(DL+EQDL)+SL(-X)	60.43	61.20	60.43	3.17	1.90	5.26
122	0.9(DL+EQDL)+SL(+Z)	66.61	50.37	72.43	4.43	1.54	7.57
123	0.9(DL+EQDL)+SL(-Z)	66.61	60.96	72.43	4.43	5.76	7.57
124	1.2DL+EQTL	100.60	70.42	100.60	7.24	-0.59	12.01

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Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1z} = 123.36-(123.36-123.36)*0.45/1.5= 123.36 kN/m^2

p_{2z} = 123.36+(123.36-123.36)*0.45/1.5= 123.36 kN/m^2

p_{3z} = 123.36-(123.36-123.36)*(0.45-0.282)/1.5= 123.36 kN/m^2

Overhang = 0.45 m

Effective depth = 0.282 m

BM_{1z} = (123.36-29.1)*0.45*0.45/2+0.5*(123.36-123.36)*0.45*0.45*0.67 = 9.54 kN-m/m

BM_{2z} = (123.36-29.1)*0.45*0.45/2-0.5*(-123.36+123.36)*0.45*0.45*0.67 = 9.54 kN-m/m

SF_{3z} = (123.36-29.1)*(0.45-0.282)+0.5*(123.36-123.36)*(0.45-0.282) = 15.84 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	9.54	0.120	0.160	0.0004	112.31	507.60	12ϕT	222.81	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 15.84 kN/m

Shear stress , τ_v= (15.84*1000)/(1000*282)

= 0.06 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	**12ϕT**	**200**	c/c
Along Z Direction :				
	At Bottom	**12ϕT**	**200**	c/c

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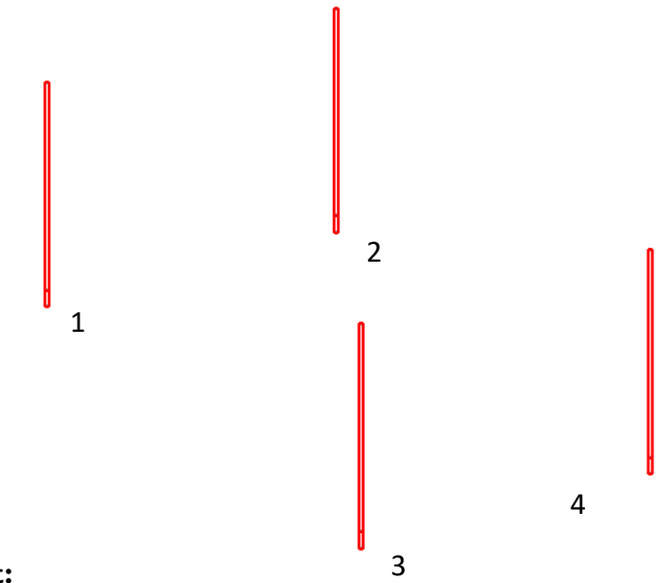
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5.7 DESIGN OF FOOTING F7

(STAAD Node -1 2 3 4)

Size of Columns		Along-X	Along-Z
Node	1	0.55m	0.55m
Node	2	0.55m	0.55m
Node	3	0.55m	0.55m
Node	4	0.55m	0.55m
Size of fdn. raft:		7.70m	6.95m
Thickness of raft:		0.75 m	
Original Gr. Level:		0.00 m	
Finished Gr. Level:		0.00 m	
U/S of B/Pl Above GL		1.40 m	
U/S level of fdn:		-1.20 m	
Unit weight of conc:		25 kN/m^3	
Unit weight of soil:		18 kN/m^3	



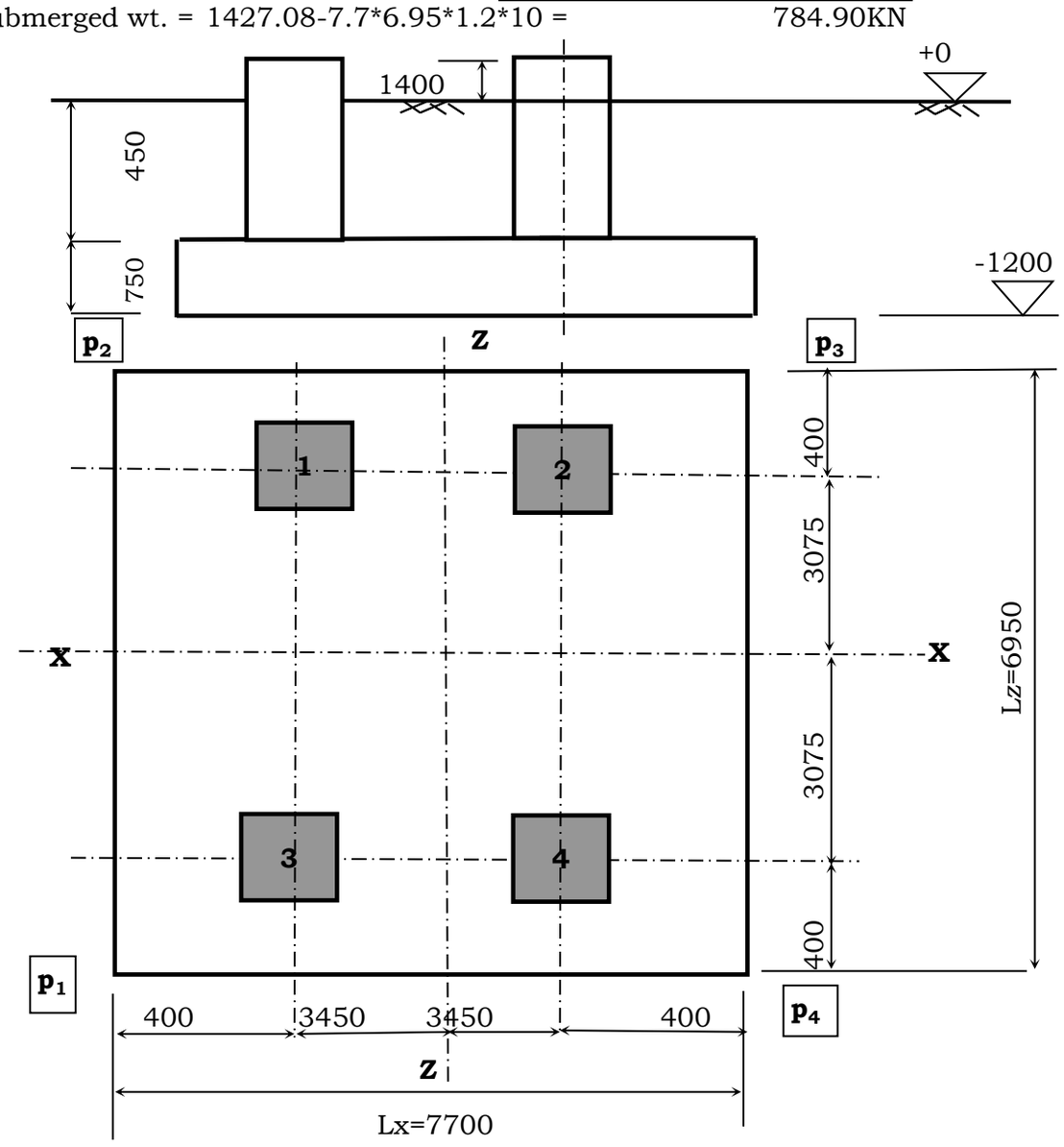
Calculation of Weight of Foundation :

Weight of raft = (7.7*6.95*0.75*25) = 1003.41KN

Weight of soil = ((7.7*6.95)-3*((0.55*0.55)+(0.55*0.55))* 423.67KN

Total = 1427.08KN

Submerged wt. = 1427.08-7.7*6.95*1.2*10 = 784.90KN

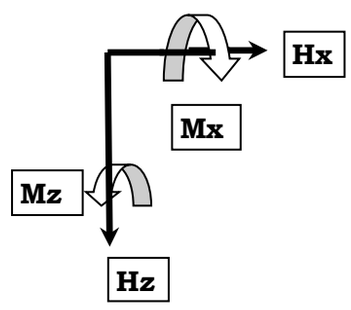


Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	25mm
Dia. of bars provd. along X direction at top =	25mm
Dia. of bars provided along Z direction at bottom=	25mm
Dia. of bars provided along Z direction at top=	25mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 675mm = 0.675m

Effective depth of slab along Z direction = 687.5mm = 0.6875m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F7 :

Node	1					2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	237.9	19.6	-51	0.6	1.6	250.1	-18.4	47.8	20.6	53.6
202 DL+EQDL+FL+LL+EQLL+TL	366.3	60.4	-157	14.1	36.7	527.8	-54.8	142.5	70.7	183.8
203 DL+EQDL+FL+0.75(LL+EQLL)+	334.2	50.2	-130.5	10.7	27.9	458.3	-45.7	118.8	58.2	151.3
204 DL+EQDL+FL+PL	138.2	3.6	-9.4	-6.2	-16.1	255.4	-28.5	74.1	15.2	39.5
205 DL+EQDL+FL+PL+LL+EQLL+TL	266.6	44.4	-115.4	7.3	19	533.1	-64.9	168.7	65.3	169.8
206 DL+EQDL+FL+PL+0.75(LL+EQL	234.5	34.2	-88.9	3.9	10.2	463.6	-55.8	145.1	52.8	137.2
207 DL+EQDL+FL+WL(+X)	123.1	-8.2	21.3	0.2	0.5	365.1	-47.5	123.5	30.4	79
208 DL+EQDL+FL+WL(-X)	352.7	47.4	-123.2	1	2.6	135.1	10.7	-27.8	10.8	28.1
209 DL+EQDL+FL+WL(+Z)	96.2	9.1	-23.7	-35.8	-93.1	80.4	-9.2	23.9	-18.7	-48.6
210 DL+EQDL+FL+WL(-Z)	379.6	30.1	-78.3	37	96.2	419.8	-27.6	71.8	59.9	155.7
211 DL+EQDL+FL+0.7SL(+X)	26.6	-32.8	85.4	17.5	45.4	445.4	-70.3	182.7	21.7	56.5
212 DL+EQDL+FL+0.7SL(-X)	449.1	72	-187.3	-16.3	-42.3	54.8	33.5	-87	19.5	50.6
213 DL+EQDL+FL+0.7SL(+Z)	51.2	6.3	-16.4	-38.7	-100.7	60.1	-6.2	16	-5.3	-13.8
214 DL+EQDL+FL+0.7SL(-Z)	424.5	32.9	-85.5	39.9	103.8	440	-30.7	79.7	46.5	120.9
215 DL+EQDL+FL+0.75WL(+X)+0.7	246.3	12.8	-33.2	1.8	4.7	545.1	-51.2	133.1	52.9	137.6
216 DL+EQDL+FL+0.75WL(-X)+0.75	418.5	54.5	-141.6	2.4	6.2	372.6	-7.5	19.6	38.2	99.4
217 DL+EQDL+FL+0.75WL(+Z)+0.7	226.2	25.8	-67	-25.2	-65.5	331.6	-22.5	58.4	16.1	41.9
218 DL+EQDL+FL+0.75WL(-Z)+0.75	438.7	41.5	-107.9	29.4	76.4	586.1	-36.3	94.3	75.1	195.1
219 DL+EQDL+FL+0.525SL(+X)+0.7	174	-5.7	14.8	14.8	38.4	605.3	-68.3	177.5	46.4	120.7
220 DL+EQDL+FL+0.525SL(-X)+0.7	490.9	72.9	-189.7	-10.6	-27.4	312.4	9.6	-24.8	44.7	116.3
221 DL+EQDL+FL+0.525SL(+Z)+0.7	192.5	23.7	-61.5	-27.4	-71.3	316.4	-20.2	52.4	26.2	68
222 DL+EQDL+FL+0.525SL(-Z)+0.7	472.4	43.6	-113.4	31.6	82.2	601.3	-38.5	100.2	65	169
223 0.6(DL+EQDL)+WL(+X)	-7	-19.1	49.7	0	-0.1	226.6	-37.1	96.4	19.3	50.1
224 0.6(DL+EQDL)+WL(-X)	222.6	36.5	-94.9	0.8	2	-3.4	21.1	-54.9	-0.3	-0.8
225 0.6(DL+EQDL)+WL(+Z)	-33.9	-1.8	4.7	-36	-93.7	-58.1	1.2	-3.2	-29.8	-77.5
226 0.6(DL+EQDL)+WL(-Z)	249.5	19.2	-49.9	36.8	95.6	281.3	-17.2	44.7	48.8	126.8
227 0.6(DL+EQDL)+0.7SL(+X)	-103.5	-43.7	113.7	17.2	44.8	306.9	-59.9	155.6	10.6	27.6
228 0.6(DL+EQDL)+0.7SL(-X)	319.1	61.1	-158.9	-16.5	-42.9	-83.7	43.9	-114.1	8.4	21.7
229 0.6(DL+EQDL)+0.7SL(+Z)	-78.8	-4.6	12	-39	-101.3	-78.4	4.3	-11.1	-16.4	-42.7
230 0.6(DL+EQDL)+0.7SL(-Z)	294.4	22	-57.2	39.7	103.2	301.6	-20.2	52.6	35.4	92

Node	3					4				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	110.6	8.2	-21.3	0.5	1.3	171.9	-9.4	24.4	-21.8	-56.7
202 DL+EQDL+FL+LL+EQLL+TL	281.2	41.8	-108.7	-9.3	-24.2	491.8	-47.2	122.7	-75.5	-196.3
203 DL+EQDL+FL+0.75(LL+EQLL)+	238.5	33.4	-86.8	-6.9	-17.8	411.8	-37.8	98.2	-62.1	-161.4
204 DL+EQDL+FL+PL	125.4	8.9	-23.1	-5.7	-14.8	252.9	-15.8	41.1	-35.2	-91.5
205 DL+EQDL+FL+PL+LL+EQLL+TL	296	42.5	-110.5	-15.5	-40.3	572.8	-53.6	139.4	-88.9	-231.1
206 DL+EQDL+FL+PL+0.75(LL+EQL	253.3	34.1	-88.7	-13.1	-33.9	492.8	-44.2	114.8	-75.5	-196.2
207 DL+EQDL+FL+WL(+X)	6.9	-19	49.4	0.2	0.5	275.5	-37.4	97.2	-31	-80.6
208 DL+EQDL+FL+WL(-X)	214.3	35.4	-92	0.8	2.1	68.3	18.6	-48.4	-12.6	-32.8
209 DL+EQDL+FL+WL(+Z)	252.7	19.6	-51	-32.5	-84.5	341.6	-20	52	-61.2	-159.1
210 DL+EQDL+FL+WL(-Z)	-31.5	-3.2	8.3	33.5	87.1	2.2	1.2	-3.1	17.6	45.8
211 DL+EQDL+FL+0.7SL(+X)	-50.9	-2.9	7.4	15.8	41.2	340.3	-20.7	53.9	-55.3	-143.7
212 DL+EQDL+FL+0.7SL(-X)	272.1	19.3	-50.1	-14.8	-38.6	3.4	1.9	-5	11.7	30.3
213 DL+EQDL+FL+0.7SL(+Z)	315	23	-59.7	-35.7	-92.8	361.8	-23.6	61.4	-47.8	-124.2
214 DL+EQDL+FL+0.7SL(-Z)	-93.8	-6.6	17.1	36.7	95.4	-18.1	4.8	-12.5	4.2	10.8
215 DL+EQDL+FL+0.75WL(+X)+0.7	161.8	-1.2	3.1	1.6	4	489	-44.5	115.7	-56.4	-146.6
216 DL+EQDL+FL+0.75WL(-X)+0.75	317.3	39.6	-103	2	5.2	333.6	-2.5	6.5	-42.6	-110.7
217 DL+EQDL+FL+0.75WL(+Z)+0.7	346.1	27.8	-72.2	-23	-59.7	538.5	-31.5	81.8	-79	-205.5
218 DL+EQDL+FL+0.75WL(-Z)+0.75	133	10.7	-27.8	26.5	69	284	-15.6	40.4	-19.9	-51.8
219 DL+EQDL+FL+0.525SL(+X)+0.7	118.4	10.9	-28.4	13.3	34.5	537.6	-32	83.2	-74.6	-193.9
220 DL+EQDL+FL+0.525SL(-X)+0.7	360.7	27.5	-71.6	-9.7	-25.3	284.9	-15	39	-24.4	-63.4
221 DL+EQDL+FL+0.525SL(+Z)+0.7	392.9	30.3	-78.8	-25.4	-66	553.7	-34.2	88.8	-69	-179.3
222 DL+EQDL+FL+0.525SL(-Z)+0.7	86.3	8.1	-21.2	28.9	75.2	268.8	-12.8	33.4	-30	-78
223 0.6(DL+EQDL)+WL(+X)	-41.7	-23.1	60.1	0	0	187.9	-32.8	85.3	-19.3	-50.3
224 0.6(DL+EQDL)+WL(-X)	165.7	31.3	-81.3	0.6	1.6	-19.3	23.2	-60.3	-0.9	-2.4
225 0.6(DL+EQDL)+WL(+Z)	204.1	15.5	-40.2	-32.7	-85	254	-15.4	40	-49.5	-128.8
226 0.6(DL+EQDL)+WL(-Z)	-80.1	-7.3	19	33.3	86.6	-85.4	5.8	-15.1	29.3	76.1
227 0.6(DL+EQDL)+0.7SL(+X)	-99.5	-7	18.1	15.6	40.6	252.7	-16.1	42	-43.6	-113.4
228 0.6(DL+EQDL)+0.7SL(-X)	223.4	15.1	-39.4	-15	-39.1	-84.1	6.5	-17	23.3	60.6
229 0.6(DL+EQDL)+0.7SL(+Z)	266.4	18.9	-49	-35.9	-93.3	274.3	-19	49.4	-36.1	-93.9
230 0.6(DL+EQDL)+0.7SL(-Z)	-142.4	-10.7	27.8	36.5	94.9	-105.7	9.4	-24.5	15.8	41.2

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L/C	Load at top of foundation					Load at bottom of foundation				
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	770.5	0.0	253.5	-0.1	631.7	2197.6	0.0	253.5	-0.1	631.7
202 DL+EQDL+FL+LL+EQLL+TL	1667.1	0.2	1283.2	0.0	372.4	3094.2	0.2	1283.2	0.0	372.4
203 DL+EQDL+FL+0.75(LL+EQLL)+	1442.8	0.1	1025.7	-0.1	437.3	2869.9	0.1	1025.7	-0.1	437.3
204 DL+EQDL+FL+PL	771.9	-31.8	926.9	-31.9	-35.9	2199.0	-31.8	926.9	-31.9	-35.9
205 DL+EQDL+FL+PL+LL+EQLL+TI	1668.5	-31.6	1956.6	-31.8	-295.1	3095.6	-31.6	1956.6	-31.8	-295.1
206 DL+EQDL+FL+PL+0.75(LL+EQI	1444.2	-31.7	1699.0	-31.9	-230.3	2871.3	-31.7	1699.0	-31.9	-230.3
207 DL+EQDL+FL+WL(+X)	770.6	-112.1	2053.0	-0.2	632.2	2197.7	-112.1	2053.0	-0.2	632.2
208 DL+EQDL+FL+WL(-X)	770.4	112.1	-1545.8	0.0	631.0	2197.5	112.1	-1545.8	0.0	631.0
209 DL+EQDL+FL+WL(+Z)	770.9	-0.5	253.4	-148.2	-1669.7	2198.0	-0.5	253.4	-148.2	-1669.7
210 DL+EQDL+FL+WL(-Z)	770.1	0.5	253.7	148.0	2933.1	2197.2	0.5	253.7	148.0	2933.1
211 DL+EQDL+FL+0.7SL(+X)	761.4	-126.7	3123.9	-0.3	560.9	2188.5	-126.7	3123.9	-0.3	560.9
212 DL+EQDL+FL+0.7SL(-X)	779.4	126.7	-2616.8	0.1	702.3	2206.5	126.7	-2616.8	0.1	702.3
213 DL+EQDL+FL+0.7SL(+Z)	788.1	-0.5	193.5	-127.5	-2070.4	2215.2	-0.5	193.5	-127.5	-2070.4
214 DL+EQDL+FL+0.7SL(-Z)	752.6	0.4	313.4	127.3	3333.3	2179.7	0.4	313.4	127.3	3333.3
215 DL+EQDL+FL+0.75WL(+X)+0.7	1442.2	-84.1	2378.4	-0.1	432.0	2869.3	-84.1	2378.4	-0.1	432.0
216 DL+EQDL+FL+0.75WL(-X)+0.7	1442.0	84.1	-320.6	0.0	431.2	2869.1	84.1	-320.6	0.0	431.2
217 DL+EQDL+FL+0.75WL(+Z)+0.7	1442.4	-0.4	1028.4	-111.1	-1293.7	2869.5	-0.4	1028.4	-111.1	-1293.7
218 DL+EQDL+FL+0.75WL(-Z)+0.7	1441.8	0.3	1028.5	111.1	2157.7	2868.9	0.3	1028.5	111.1	2157.7
219 DL+EQDL+FL+0.525SL(+X)+0.7	1435.3	-95.1	3181.3	-0.1	378.8	2862.4	-95.1	3181.3	-0.1	378.8
220 DL+EQDL+FL+0.525SL(-X)+0.7	1448.9	95.0	-1124.4	0.0	485.1	2876.0	95.0	-1124.4	0.0	485.1
221 DL+EQDL+FL+0.525SL(+Z)+0.7	1455.5	-0.4	983.1	-95.6	-1594.5	2882.6	-0.4	983.1	-95.6	-1594.5
222 DL+EQDL+FL+0.525SL(-Z)+0.7	1428.8	0.4	1073.3	95.5	2458.1	2855.9	0.4	1073.3	95.5	2458.1
223 0.6(DL+EQDL)+WL(+X)	365.8	-112.1	1889.5	0.0	225.4	1792.9	-112.1	1889.5	0.0	225.4
224 0.6(DL+EQDL)+WL(-X)	365.6	112.1	-1709.4	0.2	224.3	1792.7	112.1	-1709.4	0.2	224.3
225 0.6(DL+EQDL)+WL(+Z)	366.1	-0.5	90.0	-148.0	-2076.6	1793.2	-0.5	90.0	-148.0	-2076.6
226 0.6(DL+EQDL)+WL(-Z)	365.3	0.5	90.1	148.2	2526.2	1792.4	0.5	90.1	148.2	2526.2
227 0.6(DL+EQDL)+0.7SL(+X)	356.6	-126.7	2960.4	-0.2	154.0	1783.7	-126.7	2960.4	-0.2	154.0
228 0.6(DL+EQDL)+0.7SL(-X)	374.7	126.6	-2779.9	0.2	295.8	1801.8	126.6	-2779.9	0.2	295.8
229 0.6(DL+EQDL)+0.7SL(+Z)	383.5	-0.4	29.9	-127.4	-2477.2	1810.6	-0.4	29.9	-127.4	-2477.2
230 0.6(DL+EQDL)+0.7SL(-Z)	347.9	0.5	150.2	127.4	2926.9	1775.0	0.5	150.2	127.4	2926.9

L/C	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201 DL+EQDL+FL	2197.6	0.1	253.5	-0.1	631.7	33.4	12.1	8790.3	8790.3
202 DL+EQDL+FL+LL+EQLL+TL	3094.2	0.3	1283.2	0.0	372.4	9.3	28.9	4125.6	NA
203 DL+EQDL+FL+0.75(LL+EQI	2869.9	0.1	1025.7	-0.1	437.3	10.8	22.8	11479.5	11479.5
204 DL+EQDL+FL+PL	2199.0	-31.8	926.9	-31.9	35.9	9.1	213.1	27.7	27.6
205 DL+EQDL+FL+PL+LL+EQLL	3095.6	-31.6	1956.6	-31.8	295.1	6.1	36.5	39.2	38.9
206 DL+EQDL+FL+PL+0.75(LL+	2871.3	-31.7	1699.0	-31.9	230.3	6.5	43.3	36.2	36.0
207 DL+EQDL+FL+WL(+X)	2197.7	-112.1	2053.0	-0.2	632.2	4.1	12.1	7.8	4395.4
208 DL+EQDL+FL+WL(-X)	2197.5	112.1	-1545.8	0.0	631.0	5.5	12.1	7.8	NA
209 DL+EQDL+FL+WL(+Z)	2198.0	-0.5	253.4	-148.2	1669.7	33.4	4.6	1758.4	5.9
210 DL+EQDL+FL+WL(-Z)	2197.2	0.5	253.7	148.0	2933.1	33.3	2.6	1757.7	5.9
211 DL+EQDL+FL+0.7SL(+X)	2188.5	-126.7	3123.9	-0.3	560.9	2.7	13.6	6.9	2918.0
212 DL+EQDL+FL+0.7SL(-X)	2206.5	126.7	-2616.8	0.1	702.3	3.2	10.9	7.0	8825.9
213 DL+EQDL+FL+0.7SL(+Z)	2215.2	-0.5	193.5	-127.5	2070.4	44.1	3.7	1772.1	6.9
214 DL+EQDL+FL+0.7SL(-Z)	2179.7	0.4	313.4	127.3	3333.3	26.8	2.3	2179.7	6.8
215 DL+EQDL+FL+0.75WL(+X)+	2869.3	-84.1	2378.4	-0.1	432.0	4.6	23.1	13.6	11477.1
216 DL+EQDL+FL+0.75WL(-X)+(	2869.1	84.1	-320.6	0.0	431.2	34.5	23.1	13.6	NA
217 DL+EQDL+FL+0.75WL(+Z)+	2869.5	-0.4	1028.4	-111.1	1293.7	10.7	7.7	2869.5	10.3
218 DL+EQDL+FL+0.75WL(-Z)+(	2868.9	0.3	1028.5	111.1	2157.7	10.7	4.6	3825.2	10.3
219 DL+EQDL+FL+0.525SL(+X)-	2862.4	-95.1	3181.3	-0.1	378.8	3.5	26.3	12.0	11449.5
220 DL+EQDL+FL+0.525SL(-X)+	2876.0	95.0	-1124.4	0.0	485.1	9.8	20.6	12.1	NA
221 DL+EQDL+FL+0.525SL(+Z)-	2882.6	-0.4	983.1	-95.6	1594.5	11.3	6.3	2882.6	12.1
222 DL+EQDL+FL+0.525SL(-Z)+	2855.9	0.4	1073.3	95.5	2458.1	10.2	4.0	2855.9	12.0
223 0.6(DL+EQDL)+WL(+X)	1792.9	-112.1	1889.5	0.0	225.4	3.7	27.6	6.4	NA
224 0.6(DL+EQDL)+WL(-X)	1792.7	112.1	-1709.4	0.2	224.3	4.0	27.8	6.4	3585.4
225 0.6(DL+EQDL)+WL(+Z)	1793.2	-0.5	90.0	-148.0	2076.6	76.7	3.0	1434.5	4.8
226 0.6(DL+EQDL)+WL(-Z)	1792.4	0.5	90.1	148.2	2526.2	76.6	2.5	1433.9	4.8
227 0.6(DL+EQDL)+0.7SL(+X)	1783.7	-126.7	2960.4	-0.2	154.0	2.3	40.3	5.6	3567.4
228 0.6(DL+EQDL)+0.7SL(-X)	1801.8	126.6	-2779.9	0.2	295.8	2.5	21.2	5.7	3603.6
229 0.6(DL+EQDL)+0.7SL(+Z)	1810.6	-0.4	29.9	-127.4	2477.2	232.9	2.5	1810.6	5.7
230 0.6(DL+EQDL)+0.7SL(-Z)	1775.0	0.5	150.2	127.4	2926.9	45.5	2.1	1420.0	5.6
					Min.=	**2.7**	**2.3**	**6.9**	**5.9**

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Base pressure check of footing F7 :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	46	201	0	37.101	0
Min Px	46	201	0	37.101	0
Max Py	326	219	0	136.097	0
Min Py	46	210	0	0	0
Max Pz	46	201	0	37.101	0
Min Pz	46	201	0	37.101	0

Gross Maximum Base pressure, p_{max} = 136.1 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2 (Bore hole 27)
Depth of soil above footing = 0.45 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 458.1 KN/m^2
> 136.1 KN/m^2
Hence OK

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Design of Foundation Raft F7 :

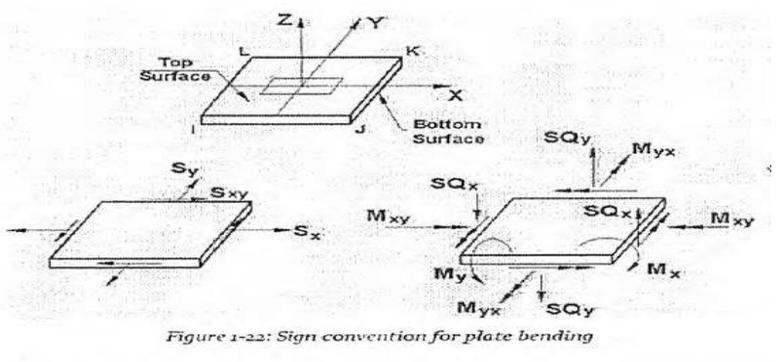
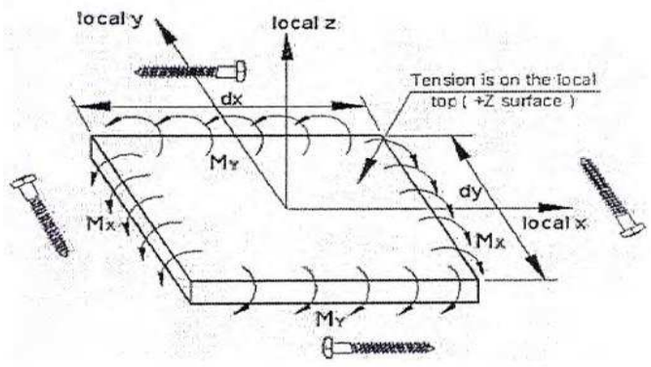


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; : FDN_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	225	0.961	0.001	0	0	0	82.914	3.625	-58.316	141.23	61.941	118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
Min Qx	114	-1.56	0.009	0	0	0	114.99	10.106	0	114.987	10.106	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Max Qy	377	-0.049	1.267	0	0	0	7.253	213.832	-69.335	76.588	283.167	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Qy	461	0.012	-1.3	0	0	0	7.154	61.533	87.689	94.843	149.222	115 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
Max Sx	1111	-0.147	-0.063	0	0	0	66.861	7.906	91.841	158.702	99.747	101 1.4(DL+EQDL+FL)
Min Sx	1111	-0.147	-0.063	0	0	0	66.861	7.906	91.841	158.702	99.747	101 1.4(DL+EQDL+FL)
Max Sy	1111	-0.147	-0.063	0	0	0	66.861	7.906	91.841	158.702	99.747	101 1.4(DL+EQDL+FL)
Min Sy	1111	-0.147	-0.063	0	0	0	66.861	7.906	91.841	158.702	99.747	101 1.4(DL+EQDL+FL)
Max Sxy	1111	-0.147	-0.063	0	0	0	66.861	7.906	91.841	158.702	99.747	101 1.4(DL+EQDL+FL)
Min Sxy	1111	-0.147	-0.063	0	0	0	66.861	7.906	91.841	158.702	99.747	101 1.4(DL+EQDL+FL)
Max Mx	151	-0.154	0.055	0	0	0	290.19	17.471	-75.961	366.146	93.432	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Mx	172	0.059	-0.002	0	0	0	-125.7	37.065	13.816	-139.547	50.881	123 0.9(DL+EQDL)+SL(+X)
Max My	411	-0.058	0.055	0	0	0	20.51	339.127	-25.856	46.366	364.983	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min My	367	-0.02	-0.071	0	0	0	9.394	-111.347	24.103	33.497	-135.45	121 0.9(DL+EQDL)+1.6WL(+Z)
Max Mxy	1084	-0.065	-0.089	0	0	0	157.37	166.667	263.377	420.747	430.044	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Mxy	517	-0.047	0.106	0	0	0	150.87	201.113	-289.301	440.17	490.414	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)

PLATE CENTER STRESS ; FDN_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	176	0.331	-0.059	0	0	0	155.65	16.988	-110.915	266.56	127.903	118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
Min Qx	163	-0.489	0.127	0	0	0	228.81	32.378	0	228.808	32.378	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Max Qy	383	-0.079	0.399	0	0	0	25.562	288.29	-151.948	177.51	440.238	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Qy	359	0.114	-0.446	0	0	0	25.276	83.518	-125.652	150.928	209.17	116 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
Max Sx	123	0.07	0.013	0	0	0	52.995	2.989	15.04	68.035	18.029	101 1.4(DL+EQDL+FL)
Min Sx	123	0.07	0.013	0	0	0	52.995	2.989	15.04	68.035	18.029	101 1.4(DL+EQDL+FL)
Max Sy	123	0.07	0.013	0	0	0	52.995	2.989	15.04	68.035	18.029	101 1.4(DL+EQDL+FL)
Min Sy	123	0.07	0.013	0	0	0	52.995	2.989	15.04	68.035	18.029	101 1.4(DL+EQDL+FL)
Max Sxy	123	0.07	0.013	0	0	0	52.995	2.989	15.04	68.035	18.029	101 1.4(DL+EQDL+FL)
Min Sxy	123	0.07	0.013	0	0	0	52.995	2.989	15.04	68.035	18.029	101 1.4(DL+EQDL+FL)
Max Mx	151	-0.154	0.055	0	0	0	290.19	17.471	-75.961	366.146	93.432	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Mx	176	0.031	-0.026	0	0	0	-71.36	3.441	19.437	-90.794	22.878	123 0.9(DL+EQDL)+SL(+X)
Max My	411	-0.058	0.055	0	0	0	20.51	339.127	-25.856	46.366	364.983	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min My	291	-0.004	-0.053	0	0	0	-3.145	-62.267	-50.56	-53.705	-112.827	126 0.9(DL+EQDL)+SL(-Z)
Max Mxy	1082	-0.071	-0.096	0	0	0	204.23	105.169	185.239	389.465	290.408	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Mxy	515	-0.077	0.118	0	0	0	203.23	123.565	-208.139	411.37	331.704	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

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Reinforcement Along X:

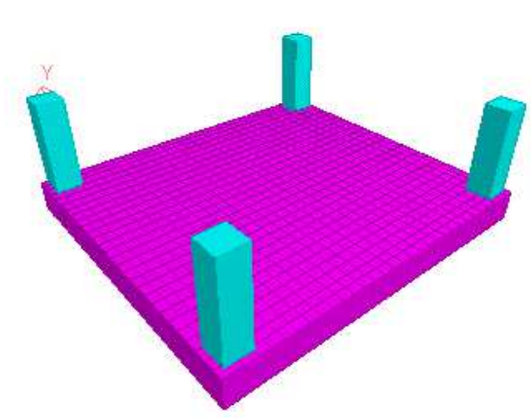
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	366.15	top face	50	688	366.15	1.03	0.0025	1721.1	1350	25	285.32	200	2455	34.62	43.28	0.045	tension control
Min Mx	-139.55	bottom face	50	688	139.55	0.39	0.0009	648.7		25	363.76	200	2455.4	34.62	43.28	0.045	tension control
Max Mxy	420.75	bottom face	50	688	420.75	1.19	0.0029	1983.2		25	247.61	200	2455	34.62	43.28	0.045	tension control
Min Mxy	440.17	bottom face	50	688	440.17	1.24	0.0030	2076.8		25	236.45	200	2455	34.62	43.28	0.045	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.331	1	248.25	266.56	0.004	0.6875	677.495	692.511	1179.52	677.5	2521	> 248.25	ok	508.1	φVc>Vu	not required
	Min Qx	0.489	1	366.75	228.81			696.767			692.51	2533	> 366.75	ok	519.4	φVc>Vu	not required

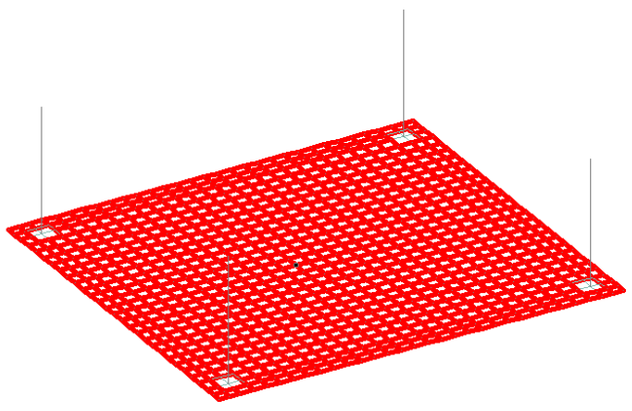
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	364.98	top face	50	688	364.98	1.03	0.0025	1715.6	1350	25	286.24	200	2455.4	34.62	43.28	0.045	tension control
Min My	-135.45	bottom face	50	688	135.45	0.38	0.0009	629.5		25	363.76	200	2455.4	34.62	43.28	0.045	tension control
Max Mxy	430.04	top face	50	688	430.04	1.21	0.0029	2028.0		25	242.15	200	2455.4	34.62	43.28	0.045	tension control
Min Mxy	490.41	top face	50	688	490.41	1.38	0.0034	2319.8		25	211.69	200	2455.4	34.62	43.28	0.045	tension control

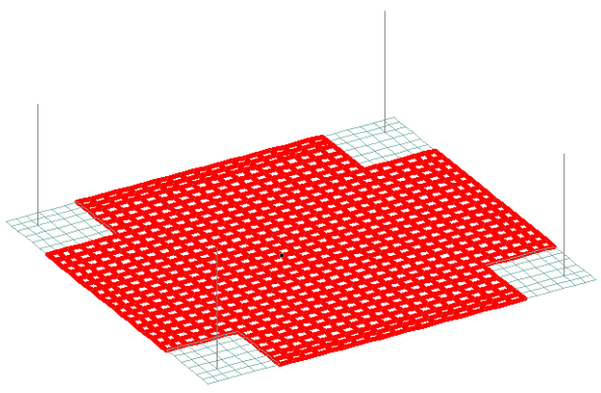
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.399	1	299.25	440.24	0.004	0.6875	fc'670.3	fc'692.5	fc'1180	670.28	704	> 299.25	ok	502.7	φVc>Vu	not required
	Min Qy	0.446	1	334.5	209.17			696.661			692.51	721	> 334.5	ok	519.4	φVc>Vu	not required



3D Rendered View of foundation



Highlighted plate for moment (at column face)



Highlighted plate for shear (at d distance from column face)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	25ϕT	**200**	c/c
At Top	25ϕT	**200**	c/c
Along Z Direction :			
At Bottom	25ϕT	**200**	c/c
At Top	25ϕT	**200**	c/c

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5.8 DESIGN OF CALCULATION OF FOOTING F8

STAAD Node - 933

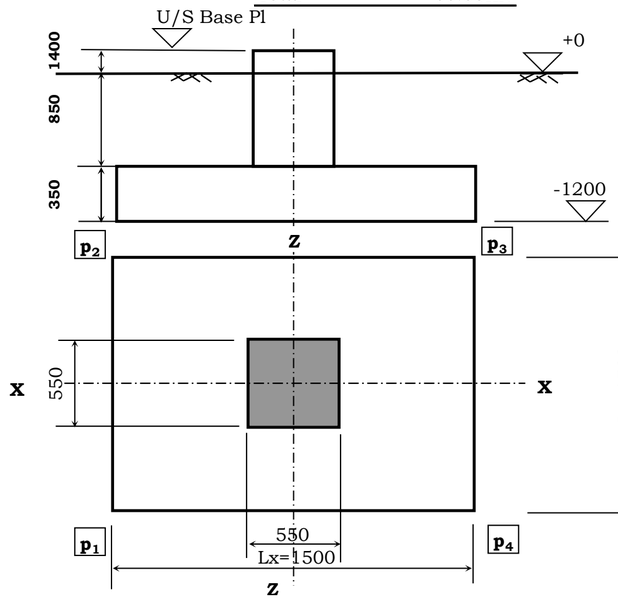
	Along-X	Along-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.20 m	
U/S level Base Plate :	1.40 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

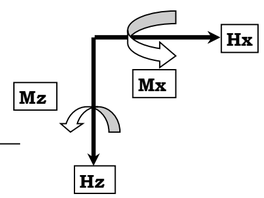
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(1.5*1.5*0.35*24)+(0.55*0.55*2.25*24)	35.24KN
Weight of soil =	((1.5*1.5)-(0.55*0.55))*0.85*18 =	29.80KN
	Total =	65.03KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.475m

Clear projection along X direction = 0.475m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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CHECK FOR OVERTURNING AND SLIDING OF FOOTING F8

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	DL+EQDL+FL	153.0	0.0	0.0	0.0	0.0	218.03	0.00	0.00	0.00	0.00
202	DL+EQDL+FL+LL+EQLL+TL	152.3	-0.1	0.3	0.1	0.3	217.33	-0.10	0.30	0.10	0.30
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	152.4	-0.1	0.2	0.1	0.2	217.43	-0.10	0.20	0.10	0.20
204	DL+EQDL+FL+PL	151.6	0.1	-0.3	0.0	0.0	216.63	0.10	-0.30	0.00	0.00
205	DL+EQDL+FL+PL+LL+EQLL+TL	150.9	0.0	0.0	0.1	0.3	215.93	0.00	0.00	0.10	0.30
206	DL+EQDL+FL+PL+0.75(LL+EQLL)+0.7	151.0	0.0	-0.1	0.1	0.2	216.03	0.00	-0.10	0.10	0.20
207	DL+EQDL+FL+WL(+X)	152.8	0.0	0.0	0.0	0.0	217.83	0.00	0.00	0.00	0.00
208	DL+EQDL+FL+WL(-X)	153.2	0.0	0.0	0.0	0.0	218.23	0.00	0.00	0.00	0.00
209	DL+EQDL+FL+WL(+Z)	152.6	0.4	-1.0	-0.3	-0.8	217.63	0.40	-1.00	-0.30	-0.80
210	DL+EQDL+FL+WL(-Z)	153.4	-0.4	1.0	0.3	0.8	218.43	-0.40	1.00	0.30	0.80
211	DL+EQDL+FL+0.7SL(+X)	162.0	-1.2	3.1	0.1	0.4	227.03	-1.20	3.10	0.10	0.40
212	DL+EQDL+FL+0.7SL(-X)	143.9	1.2	-3.1	-0.1	-0.4	208.93	1.20	-3.10	-0.10	-0.40
213	DL+EQDL+FL+0.7SL(+Z)	135.2	0.5	-1.3	-0.6	-1.5	200.23	0.50	-1.30	-0.60	-1.50
214	DL+EQDL+FL+0.7SL(-Z)	170.7	-0.5	1.3	0.6	1.5	235.73	-0.50	1.30	0.60	1.50
215	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EC	153.0	0.1	-0.2	0.0	0.0	218.03	0.10	-0.20	0.00	0.00
216	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EC	153.3	0.1	-0.2	0.0	0.0	218.33	0.10	-0.20	0.00	0.00
217	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EC	152.8	0.4	-1.0	-0.2	-0.6	217.83	0.40	-1.00	-0.20	-0.60
218	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EC	153.4	-0.2	0.6	0.2	0.6	218.43	-0.20	0.60	0.20	0.60
219	DL+EQDL+FL+0.525SL(+X)+0.75(LL+E	159.9	-0.8	2.1	0.1	0.3	224.93	-0.80	2.10	0.10	0.30
220	DL+EQDL+FL+0.525SL(-X)+0.75(LL+E	146.3	1.0	-2.5	-0.1	-0.3	211.33	1.00	-2.50	-0.10	-0.30
221	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+E	139.8	0.4	-1.2	-0.4	-1.1	204.83	0.40	-1.20	-0.40	-1.10
222	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+E	166.4	-0.3	0.8	0.4	1.1	231.43	-0.30	0.80	0.40	1.10
223	0.6(DL+EQDL)+WL(+X)	55.2	0.0	0.0	0.0	0.0	120.23	0.00	0.00	0.00	0.00
224	0.6(DL+EQDL)+WL(-X)	55.6	0.0	0.0	0.0	0.0	120.63	0.00	0.00	0.00	0.00
225	0.6(DL+EQDL)+WL(+Z)	55.0	0.4	-1.0	-0.3	-0.8	120.03	0.40	-1.00	-0.30	-0.80
226	0.6(DL+EQDL)+WL(-Z)	55.8	-0.4	1.0	0.3	0.8	120.83	-0.40	1.00	0.30	0.80
227	0.6(DL+EQDL)+0.7SL(+X)	64.4	-1.2	3.1	0.1	0.4	129.43	-1.20	3.10	0.10	0.40
228	0.6(DL+EQDL)+0.7SL(-X)	46.3	1.2	-3.1	-0.1	-0.4	111.33	1.20	-3.10	-0.10	-0.40
229	0.6(DL+EQDL)+0.7SL(+Z)	37.6	0.5	-1.3	-0.6	-1.5	102.63	0.50	-1.30	-0.60	-1.50
230	0.6(DL+EQDL)+0.7SL(-Z)	73.1	-0.5	1.3	0.6	1.5	138.13	-0.50	1.30	0.60	1.50
Ultimate Conditions											
101	1.4(DL+EQDL+FL)	214.1	0.0	0.0	0.0	0.0	279.13	0.00	0.00	0.00	0.00
102	1.2(DL+EQDL+FL)+1.6(LL+EQLL)	183.9	0.2	-0.4	0.0	0.0	248.93	0.20	-0.40	0.00	0.00
103	1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)	182.8	-0.1	0.2	0.1	0.3	247.83	-0.10	0.20	0.10	0.30
104	1.4(DL+EQDL+FL+PL)	212.2	0.1	-0.4	0.0	0.0	277.23	0.10	-0.40	0.00	0.00
105	1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)	182.2	0.3	-0.7	0.0	0.0	247.23	0.30	-0.70	0.00	0.00
106	1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQL	181.1	0.0	-0.1	0.1	0.3	246.13	0.00	-0.10	0.10	0.30
107	1.2(DL+EQDL)+0.8WL(+X)	110.6	0.0	0.0	0.0	0.0	175.63	0.00	0.00	0.00	0.00
108	1.2(DL+EQDL)+0.8WL(-X)	110.9	0.0	0.0	0.0	0.0	175.93	0.00	0.00	0.00	0.00
109	1.2(DL+EQDL)+0.8WL(+Z)	110.4	0.3	-0.8	-0.2	-0.6	175.43	0.30	-0.80	-0.20	-0.60
110	1.2(DL+EQDL)+0.8WL(-Z)	111.0	-0.3	0.8	0.2	0.6	176.03	-0.30	0.80	0.20	0.60
111	1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQL	110.5	0.1	-0.1	0.0	0.0	175.53	0.10	-0.10	0.00	0.00
112	1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQL	111.1	0.1	0.0	0.0	0.0	176.13	0.10	0.26	0.00	0.00
113	1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQL	110.2	0.7	0.0	-0.5	0.0	175.23	0.70	1.82	-0.50	-1.30
114	1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQL	111.5	-0.6	0.0	0.5	0.0	176.53	-0.60	-1.56	0.50	1.30
115	1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+(	154.1	-1.7	0.0	0.2	0.0	219.13	-1.70	-4.42	0.20	0.52
116	1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+(	128.3	1.8	0.0	-0.2	0.0	193.33	1.80	4.68	-0.20	-0.52
117	1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+(	115.8	0.8	0.0	-0.8	0.0	180.83	0.80	2.08	-0.80	-2.08
118	1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+(	166.6	-0.7	0.0	0.8	0.0	231.63	-0.70	-1.82	0.80	2.08
119	0.9(DL+EQDL)+1.6WL(+X)	82.7	0.0	0.0	0.0	0.0	147.73	0.00	0.00	0.00	0.00
120	0.9(DL+EQDL)+1.6WL(-X)	83.4	0.0	0.0	0.0	0.0	148.43	0.00	0.00	0.00	0.00
121	0.9(DL+EQDL)+1.6WL(+Z)	82.4	0.6	0.0	-0.5	0.0	147.43	0.60	1.56	-0.50	-1.30
122	0.9(DL+EQDL)+1.6WL(-Z)	83.7	-0.6	0.0	0.5	0.0	148.73	-0.60	-1.56	0.50	1.30
123	0.9(DL+EQDL)+SL(+X)	95.9	-1.7	0.0	0.2	0.0	160.93	-1.70	-4.42	0.20	0.52
124	0.9 1.4 (D + F1)	70.1	1.7	0.0	-0.2	0.0	135.13	1.70	4.42	-0.20	-0.52
125	0.9 1.4 (D + F1)	57.6	0.7	1.0	-0.8	1.0	122.63	0.70	2.82	-0.80	-1.08
126	0.9 1.4 (D + F1)	108.4	-0.7	2.0	0.8	2.0	173.43	-0.70	0.18	0.80	4.08

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Check for overturning and sliding:

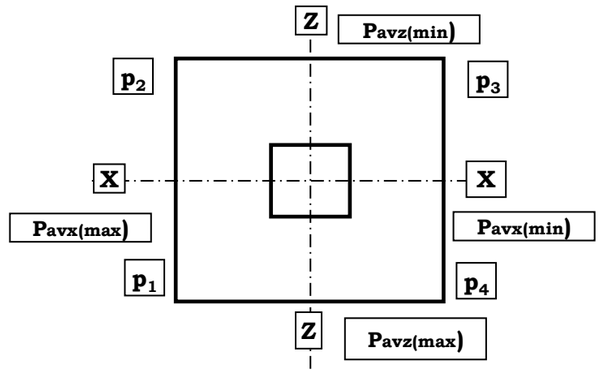
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	DL+EQDL+FL	218.0	0.0	0.0	0.0	0.0	NA	NA	NA	NA
202	DL+EQDL+FL+LL+EQLL+TL	217.3	-0.1	0.3	0.1	0.3	543.3	543.3	869.3	869.3
203	DL+EQDL+FL+0.75(LL+EQLL)+0.75TL	217.4	-0.1	0.2	0.1	0.2	815.4	815.4	869.7	869.7
204	DL+EQDL+FL+PL	216.6	0.1	-0.3	0.0	0.0	541.6	NA	866.5	NA
205	DL+EQDL+FL+PL+LL+EQLL+TL	215.9	0.0	0.0	0.1	0.3	NA	539.8	NA	863.7
206	DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL	216.0	0.0	-0.1	0.1	0.2	1620.2	810.1	NA	864.1
207	DL+EQDL+FL+WL(+X)	217.8	0.0	0.0	0.0	0.0	NA	NA	NA	NA
208	DL+EQDL+FL+WL(-X)	218.2	0.0	0.0	0.0	0.0	NA	NA	NA	NA
209	DL+EQDL+FL+WL(+Z)	217.6	0.4	-1.0	-0.3	-0.8	163.2	204.0	217.6	290.2
210	DL+EQDL+FL+WL(-Z)	218.4	-0.4	1.0	0.3	0.8	163.8	204.8	218.4	291.2
211	DL+EQDL+FL+0.7SL(+X)	227.0	-1.2	3.1	0.1	0.4	54.9	425.7	75.7	908.1
212	DL+EQDL+FL+0.7SL(-X)	208.9	1.2	-3.1	-0.1	-0.4	50.5	391.7	69.6	835.7
213	DL+EQDL+FL+0.7SL(+Z)	200.2	0.5	-1.3	-0.6	-1.5	115.5	100.1	160.2	133.5
214	DL+EQDL+FL+0.7SL(-Z)	235.7	-0.5	1.3	0.6	1.5	136.0	117.9	188.6	157.2
215	DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)	218.0	0.1	-0.2	0.0	0.0	817.6	NA	872.1	NA
216	DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)	218.3	0.1	-0.2	0.0	0.0	818.7	NA	873.3	NA
217	DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)	217.8	0.4	-1.0	-0.2	-0.6	163.4	272.3	217.8	435.7
218	DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)	218.4	-0.2	0.6	0.2	0.6	273.0	273.0	436.9	436.9
219	DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL	224.9	-0.8	2.1	0.1	0.3	80.3	562.3	112.5	899.7
220	DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)	211.3	1.0	-2.5	-0.1	-0.3	63.4	528.3	84.5	845.3
221	DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL	204.8	0.4	-1.2	-0.4	-1.1	128.0	139.7	204.8	204.8
222	DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)	231.4	-0.3	0.8	0.4	1.1	217.0	157.8	308.6	231.4
223	0.6(DL+EQDL)+WL(+X)	120.2	0.0	0.0	0.0	0.0	NA	NA	NA	NA
224	0.6(DL+EQDL)+WL(-X)	120.6	0.0	0.0	0.0	0.0	NA	NA	NA	NA
225	0.6(DL+EQDL)+WL(+Z)	120.0	0.4	-1.0	-0.3	-0.8	90.0	112.5	120.0	160.0
226	0.6(DL+EQDL)+WL(-Z)	120.8	-0.4	1.0	0.3	0.8	90.6	113.3	120.8	161.1
227	0.6(DL+EQDL)+0.7SL(+X)	129.4	-1.2	3.1	0.1	0.4	31.3	242.7	43.1	517.7
228	0.6(DL+EQDL)+0.7SL(-X)	111.3	1.2	-3.1	-0.1	-0.4	26.9	208.7	37.1	445.3
229	0.6(DL+EQDL)+0.7SL(+Z)	102.6	0.5	-1.3	-0.6	-1.5	59.2	51.3	82.1	68.4
230	0.6(DL+EQDL)+0.7SL(-Z)	138.1	-0.5	1.3	0.6	1.5	79.7	69.1	110.5	92.1
						Min.=	**26.9**	**51.3**	**37.1**	**68.4**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F8



Load Combination & Calculation of Base pressure

Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 471.6 kN/m^2 For WL/SL case = 584.1 kN/m^2

Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	Base Pressure (kN/m^2) V/(Lx.Lz)	6.Mz/(Lx)^2.Lz	6.Mx/Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	DL+EQDL+FL	218.03	0.00	0.00	96.90	0.00	0.00	96.90	96.90	96.90	96.90	75.30	96.90	96.90	96.90	96.90	O.K
202	DL+EQDL+FL+LL+EQLL+TL	217.33	0.30	0.30	96.59	0.53	0.53	97.66	96.59	96.59	95.52	76.06	97.13	96.06	97.13	96.06	O.K
203	DL+EQDL+FL+0.75(LL+EQLL)+0	217.43	0.20	0.20	96.64	0.36	0.36	97.35	96.64	96.64	95.92	75.75	97.00	96.28	97.00	96.28	O.K
204	DL+EQDL+FL+PL	216.63	-0.30	0.00	96.28	0.53	0.00	96.81	96.81	95.75	95.75	75.21	96.81	95.75	96.28	96.28	O.K
205	DL+EQDL+FL+PL+LL+EQLL+TL	215.93	0.00	0.30	95.97	0.00	0.53	96.50	95.44	96.50	95.44	74.90	95.97	95.97	96.50	95.44	O.K
206	DL+EQDL+FL+PL+0.75(LL+EQLI	216.03	-0.10	0.20	96.01	0.18	0.36	96.55	95.84	96.19	95.48	74.95	96.20	95.84	96.37	95.66	O.K
207	DL+EQDL+FL+WL(+X)	217.83	0.00	0.00	96.81	0.00	0.00	96.81	96.81	96.81	96.81	75.21	96.81	96.81	96.81	96.81	O.K
208	DL+EQDL+FL+WL(-X)	218.23	0.00	0.00	96.99	0.00	0.00	96.99	96.99	96.99	96.99	75.39	96.99	96.99	96.99	96.99	O.K
209	DL+EQDL+FL+WL(+Z)	217.63	-1.00	-0.80	96.72	1.78	1.42	99.92	97.08	96.37	93.52	78.32	98.50	94.95	98.15	95.30	O.K
210	DL+EQDL+FL+WL(-Z)	218.43	1.00	0.80	97.08	1.78	1.42	100.28	97.44	96.72	93.88	78.68	98.86	95.30	98.50	95.66	O.K
211	DL+EQDL+FL+0.7SL(+X)	227.03	3.10	0.40	100.90	5.51	0.71	107.12	105.70	96.10	94.68	85.52	106.41	95.39	101.61	100.19	O.K
212	DL+EQDL+FL+0.7SL(-X)	208.93	-3.10	-0.40	92.86	5.51	0.71	99.08	97.66	88.06	86.64	77.48	98.37	87.35	93.57	92.15	O.K
213	DL+EQDL+FL+0.7SL(+Z)	200.23	-1.30	-1.50	88.99	2.31	2.67	93.97	88.64	89.35	84.01	72.37	91.31	86.68	91.66	86.33	O.K
214	DL+EQDL+FL+0.7SL(-Z)	235.73	1.30	1.50	104.77	2.31	2.67	109.75	104.41	105.12	99.79	88.15	107.08	102.46	107.44	102.10	O.K

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215 DL+EQDL+FL+0.75WL(+X)+0.75	218.03	-0.20	0.00	96.90	0.36	0.00	97.26	97.26	96.55	96.55	75.66	97.26	96.55	96.91	96.91	O.K
216 DL+EQDL+FL+0.75WL(-X)+0.75(	218.33	-0.20	0.00	97.04	0.36	0.00	97.39	97.39	96.68	96.68	75.79	97.39	96.68	97.04	97.04	O.K
217 DL+EQDL+FL+0.75WL(+Z)+0.75	217.83	-1.00	-0.60	96.81	1.78	1.07	99.66	97.52	96.10	93.97	78.06	98.59	95.04	97.88	95.75	O.K
218 DL+EQDL+FL+0.75WL(-Z)+0.75(	218.43	0.60	0.60	97.08	1.07	1.07	99.21	97.08	97.08	94.95	77.61	98.15	96.02	98.15	96.02	O.K
219 DL+EQDL+FL+0.525SL(+X)+0.75	224.93	2.10	0.30	99.97	3.73	0.53	104.24	103.17	96.77	95.70	82.64	103.71	96.24	100.51	99.44	O.K
220 DL+EQDL+FL+0.525SL(-X)+0.75	211.33	-2.50	-0.30	93.92	4.44	0.53	98.90	97.84	90.01	88.95	77.30	98.37	89.48	94.46	93.40	O.K
221 DL+EQDL+FL+0.525SL(+Z)+0.75	204.83	-1.20	-1.10	91.04	2.13	1.96	95.12	91.21	90.86	86.95	73.52	93.17	88.91	92.99	89.08	O.K
222 DL+EQDL+FL+0.525SL(-Z)+0.75	231.43	0.80	1.10	102.86	1.42	1.96	106.24	102.32	103.39	99.48	84.64	104.28	101.44	104.82	100.90	O.K
223 0.6(DL+EQDL)+WL(+X)	120.23	0.00	0.00	53.44	0.00	0.00	53.44	53.44	53.44	53.44	31.84	53.44	53.44	53.44	53.44	O.K
224 0.6(DL+EQDL)+WL(-X)	120.63	0.00	0.00	53.61	0.00	0.00	53.61	53.61	53.61	53.61	32.01	53.61	53.61	53.61	53.61	O.K
225 0.6(DL+EQDL)+WL(+Z)	120.03	-1.00	-0.80	53.35	1.78	1.42	56.55	53.70	52.99	50.15	34.95	55.13	51.57	54.77	51.93	O.K
226 0.6(DL+EQDL)+WL(-Z)	120.83	1.00	0.80	53.70	1.78	1.42	56.90	54.06	53.35	50.50	35.30	55.48	51.93	55.13	52.28	O.K
227 0.6(DL+EQDL)+0.7SL(+X)	129.43	3.10	0.40	57.52	5.51	0.71	63.75	62.32	52.72	51.30	42.15	63.04	52.01	58.24	56.81	O.K
228 0.6(DL+EQDL)+0.7SL(-X)	111.33	-3.10	-0.40	49.48	5.51	0.71	55.70	54.28	44.68	43.26	34.10	54.99	43.97	50.19	48.77	O.K
229 0.6(DL+EQDL)+0.7SL(+Z)	102.63	-1.30	-1.50	45.61	2.31	2.67	50.59	45.26	45.97	40.64	28.99	47.93	43.31	48.28	42.95	O.K
230 0.6(DL+EQDL)+0.7SL(-Z)	138.13	1.30	1.50	61.39	2.31	2.67	66.37	61.04	61.75	56.41	44.77	63.71	59.08	64.06	58.73	O.K

Ultimate Conditions																
101 1.4(DL+EQDL+FL)	279.13	0.00	0.00	124.06	0.00	0.00	124.06	124.06	124.06	124.06	102.46	124.06	124.06	124.06	124.06	
102 1.2(DL+EQDL+FL)+1.6(LL+EQLL	248.93	-0.40	0.00	110.64	0.71	0.00	111.35	111.35	109.92	109.92	89.75	111.35	109.92	110.64	110.64	
103 1.2(DL+EQDL+FL+TL)+1.6(LL+E	247.83	0.20	0.30	110.15	0.36	0.53	111.04	109.97	110.32	109.26	89.44	110.51	109.79	110.68	109.62	
104 1.4(DL+EQDL+FL+PL)	277.23	-0.40	0.00	123.21	0.71	0.00	123.92	123.92	122.50	122.50	102.32	123.92	122.50	123.21	123.21	
105 1.2(DL+EQDL+FL+PL)+1.6(LL+E	247.23	-0.70	0.00	109.88	1.24	0.00	111.12	111.12	108.64	108.64	89.52	111.12	108.64	109.88	109.88	
106 1.2(DL+EQDL+FL+PL+TL)+1.6(LI	246.13	-0.10	0.30	109.39	0.18	0.53	110.10	109.04	109.75	108.68	88.50	109.57	109.22	109.93	108.86	
107 1.2(DL+EQDL)+0.8WL(+X)	175.63	0.00	0.00	78.06	0.00	0.00	78.06	78.06	78.06	78.06	56.46	78.06	78.06	78.06	78.06	
108 1.2(DL+EQDL)+0.8WL(-X)	175.93	0.00	0.00	78.19	0.00	0.00	78.19	78.19	78.19	78.19	56.59	78.19	78.19	78.19	78.19	

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109 1.2(DL+EQDL)+0.8WL(+Z)	175.43	-0.80	-0.60	77.97	1.42	1.07	80.46	78.32	77.61	75.48	58.86	79.39	76.55	79.04	76.90	
110 1.2(DL+EQDL)+0.8WL(-Z)	176.03	0.80	0.60	78.24	1.42	1.07	80.72	78.59	77.88	75.75	59.12	79.66	76.82	79.30	77.17	
111 1.2(DL+EQDL)+1.6WL(+X)+0.5(L	175.53	-0.10	0.00	78.01	0.18	0.00	78.19	78.19	77.84	77.84	56.59	78.19	77.84	78.02	78.02	
112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LI	176.13	0.26	0.00	78.28	0.46	0.00	78.74	78.74	77.82	77.82	57.14	78.74	77.82	78.28	78.28	
113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(L	175.23	1.82	-1.30	77.88	3.24	2.31	83.43	78.80	76.96	72.33	61.83	81.12	74.65	80.20	75.57	
114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LI	176.53	-1.56	1.30	78.46	2.77	2.31	83.54	78.92	78.00	73.37	61.94	81.23	75.69	80.77	76.15	
115 1.2(DL+EQDL)+SL(+X)+0.5(LL+E	219.13	-4.42	0.52	97.39	7.86	0.92	106.17	104.32	90.46	88.61	84.57	105.25	89.54	98.32	96.47	
116 1.2(DL+EQDL)+SL(-X)+0.5(LL+E	193.33	4.68	-0.52	85.92	8.32	0.92	95.17	93.32	78.53	76.68	73.57	94.25	77.61	86.85	85.00	
117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+E	180.83	2.08	-2.08	80.37	3.70	3.70	87.76	80.37	80.37	72.97	66.16	84.07	76.67	84.07	76.67	
118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+E	231.63	-1.82	2.08	102.95	3.24	3.70	109.88	102.48	103.41	96.01	88.28	106.18	99.71	106.65	99.25	
119 0.9(DL+EQDL)+1.6WL(+X)	147.73	0.00	0.00	65.66	0.00	0.00	65.66	65.66	65.66	65.66	44.06	65.66	65.66	65.66	65.66	
120 0.9(DL+EQDL)+1.6WL(-X)	148.43	0.00	0.00	65.97	0.00	0.00	65.97	65.97	65.97	65.97	44.37	65.97	65.97	65.97	65.97	
121 0.9(DL+EQDL)+1.6WL(+Z)	147.43	1.56	-1.30	65.52	2.77	2.31	70.61	65.99	65.06	60.44	49.01	68.30	62.75	67.84	63.22	
122 0.9(DL+EQDL)+1.6WL(-Z)	148.73	-1.56	1.30	66.10	2.77	2.31	71.19	66.56	65.64	61.02	49.59	68.88	63.33	68.42	63.79	
123 0.9(DL+EQDL)+SL(+X)	160.93	-4.42	0.52	71.52	7.86	0.92	80.31	78.46	64.59	62.74	58.71	79.39	63.67	72.45	70.60	
124 0.9(DL+EQDL)+SL(-X)	135.13	4.42	-0.52	60.06	7.86	0.92	68.84	66.99	53.12	51.28	47.24	67.92	52.20	60.98	59.14	
125 0.9(DL+EQDL)+SL(+Z)	122.63	2.82	-1.08	54.50	5.01	1.92	61.44	57.60	51.41	47.57	39.84	59.52	49.49	56.43	52.59	
126 0.9(DL+EQDL)+SL(-Z)	173.43	0.18	4.08	77.08	0.32	7.25	84.65	70.15	84.01	69.51	63.05	77.40	76.76	84.33	69.83	
Max :	279	5	2													
Min :	103	-4	-2													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 75.30 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 85.30 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 88.15 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 98.15 kN/m²

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Sample calculation of values of base pressure:

Load case : 202 DL+EQDL+FL+LL+EQLL+TL

Maximum pressure, p_1 (pmax) = 96.59+0.53+0.53= 97.66 kN/m^2

p_2= 96.59+0.53-0.53= 96.59 kN/m^2

p_3= 96.59-(0.53+0.53)= 95.52 kN/m^2

p_4= 96.59-0.53+0.53= 96.59 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 97.66-1.2*18= 76.06 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (97.66+96.59)/2= 97.12 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (95.52+96.59)/2= 96.06 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (97.66+96.59)/2= 97.12 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (95.52+96.59)/2= 96.06 kN/m^2

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DESIGN OF ISOLATED FOOTING F8

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

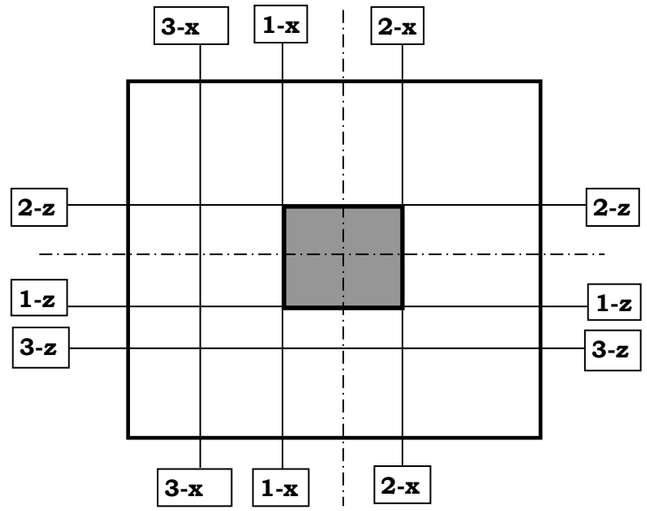
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.55+2*0.5*0.282)+(0.55+2*0.5*0.282))*2
= 3.328 m

Punching Area= ((0.55+2*0.5*0.282)*(0.55+2*0.5*0.282))
= 0.69 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+0.85*18)
= 23.70 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+EQDL+FL)	124.06	124.06	124.06	11.32	11.32	18.17	156.56
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	110.90	110.37	111.18	9.87	9.74	15.85	135.62
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	110.28	110.02	110.42	9.78	9.72	15.70	134.86
104	1.4(DL+EQDL+FL+PL)	123.47	122.95	123.75	11.29	11.16	18.12	155.24
105	1.2(DL+EQDL+FL+PL)+1.6(LL+	110.33	109.43	110.82	9.83	9.61	15.80	134.44
106	1.2(DL+EQDL+FL+PL+TL)+1.6(	109.46	109.33	109.53	9.68	9.65	15.54	133.68
107	1.2(DL+EQDL)+0.8WL(+X)	78.06	78.06	78.06	6.13	6.13	9.84	84.80
108	1.2(DL+EQDL)+0.8WL(-X)	78.19	78.19	78.19	6.15	6.15	9.86	85.01

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109 1.2(DL+EQDL)+0.8WL(+Z)	78.49	77.45	79.05	6.25	6.00	10.05	84.66
110 1.2(DL+EQDL)+0.8WL(-Z)	78.76	77.72	79.32	6.28	6.03	10.10	85.08
111 1.2(DL+EQDL)+1.6WL(+X)+0.5(	78.08	77.95	78.15	6.14	6.11	9.86	84.73
112 1.2(DL+EQDL)+1.6WL(-X)+0.5(	78.45	78.11	78.63	6.20	6.12	9.95	85.14
113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(	79.07	76.70	80.34	6.40	5.82	10.32	84.52
114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(	79.48	77.44	80.56	6.42	5.93	10.35	85.42
115 1.2(DL+EQDL)+SL(+X)+0.5(LL+	100.28	94.51	103.35	9.01	7.61	14.59	114.96
116 1.2(DL+EQDL)+SL(-X)+0.5(LL+	88.98	82.88	92.24	7.76	6.28	12.59	97.07
117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+	81.73	79.01	83.18	6.72	6.06	10.85	88.40
118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+	104.13	101.76	105.40	9.23	8.65	14.86	123.62
119 0.9(DL+EQDL)+1.6WL(+X)	65.66	65.66	65.66	4.73	4.73	7.59	65.45
120 0.9(DL+EQDL)+1.6WL(-X)	65.97	65.97	65.97	4.77	4.77	7.65	65.94
121 0.9(DL+EQDL)+1.6WL(+Z)	66.54	64.51	67.63	4.97	4.47	8.01	65.25
122 0.9(DL+EQDL)+1.6WL(-Z)	67.12	65.09	68.21	5.03	4.54	8.12	66.15
123 0.9(DL+EQDL)+SL(+X)	74.41	68.65	77.49	6.10	4.70	9.91	74.61
124 0.9(DL+EQDL)+SL(-X)	62.94	57.18	66.02	4.80	3.40	7.83	56.72
125 0.9(DL+EQDL)+SL(+Z)	56.34	52.67	58.31	3.92	3.03	6.37	48.05
126 0.9(DL+EQDL)+SL(-Z)	77.20	76.96	77.32	6.05	5.99	9.71	83.27

Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1x} = 124.06-(124.06-124.06)*0.475/1.5 = 124.06 kN/m^2 Overhang = 0.475 m

p_{2x} = 124.06+(124.06-124.06)*0.475/1.5 = 124.06 kN/m^2 Effective depth = 0.294 m

p_{3x} = 124.06-(124.06-124.06)*(0.475-0.294)/1.5 = 124.06 kN/m^2

BM_{1x} = (124.06-23.7)*0.475*0.475/2+0.5*(124.06-124.06)*0.475*0.475*0.67 = 11.32 kN-m/m

BM_{2x} = (124.06-23.7)*0.475*0.475/2-0.5*(-124.06+124.06)*0.475*0.475*0.67 = 11.32 kN-m/m

SF_{3x} = (124.06-23.7)*(0.475-0.294)+0.5*(124.06-124.06)*(0.475-0.294) = 18.17 kN/m

$S.F_{(punching)}$ = (2.25-0.69)*(124.06-23.7) = 156.56 kN

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	11.32	0.13	0.17	0.000	122.61	529.20	12ϕT	213.71	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 18.17 kN/m

Shear stress , τ_v= (18.17*1000)/(1000*294)

= 0.06 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 156.56 kN

Strength of Concrete in shear Vc1 = 2894.24 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1817.02 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1929.49 kN **Table 22.6.5.2_ (c)**

Design Vc= 1817.02 kN

Allowable shear stress, (ϕ Vc)/bd = 1.39 N/mm^2

Design punching shear stress, τ = (156.56*1000)/(3328*294)

= 0.16 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+EQDL+FL)	124.06	124.06	124.06	11.32	11.32	19.37
102	1.2(DL+EQDL+FL)+1.6(LL+EQL	110.64	110.64	110.64	9.81	9.81	16.78
103	1.2(DL+EQDL+FL+TL)+1.6(LL+	110.34	109.96	110.54	9.80	9.71	16.77
104	1.4(DL+EQDL+FL+PL)	123.21	123.21	123.21	11.23	11.23	19.21
105	1.2(DL+EQDL+FL+PL)+1.6(LL+	109.88	109.88	109.88	9.72	9.72	16.63
106	1.2(DL+EQDL+FL+PL+TL)+1.6(	109.59	109.20	109.79	9.72	9.62	16.63
107	1.2(DL+EQDL)+0.8WL(+X)	78.06	78.06	78.06	6.13	6.13	10.49
108	1.2(DL+EQDL)+0.8WL(-X)	78.19	78.19	78.19	6.15	6.15	10.52
109	1.2(DL+EQDL)+0.8WL(+Z)	78.36	77.58	78.76	6.22	6.03	10.65
110	1.2(DL+EQDL)+0.8WL(-Z)	78.63	77.84	79.03	6.25	6.06	10.70
111	1.2(DL+EQDL)+1.6WL(+X)+0.5(	78.02	78.02	78.02	6.13	6.13	10.48
112	1.2(DL+EQDL)+1.6WL(-X)+0.5(	78.28	78.28	78.28	6.16	6.16	10.53
113	1.2(DL+EQDL)+1.6WL(+Z)+0.5(	78.73	77.04	79.60	6.32	5.91	10.85
114	1.2(DL+EQDL)+1.6WL(-Z)+0.5(	79.31	77.61	80.18	6.38	5.97	10.96

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115 1.2(DL+EQDL)+SL(+X)+0.5(LL+	97.73	97.06	98.08	8.40	8.23	14.38
116 1.2(DL+EQDL)+SL(-X)+0.5(LL+	86.26	85.59	86.61	7.10	6.94	12.16
117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+	81.73	79.01	83.12	6.72	6.06	11.56
118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+	104.31	101.59	105.70	9.27	8.61	15.92
119 0.9(DL+EQDL)+1.6WL(+X)	65.66	65.66	65.66	4.73	4.73	8.10
120 0.9(DL+EQDL)+1.6WL(-X)	65.97	65.97	65.97	4.77	4.77	8.16
121 0.9(DL+EQDL)+1.6WL(+Z)	66.38	60.66	67.25	4.93	3.54	8.46
122 0.9(DL+EQDL)+1.6WL(-Z)	66.95	62.03	67.82	4.99	3.93	8.57
123 0.9(DL+EQDL)+SL(+X)	71.86	64.14	72.21	5.48	3.40	9.39
124 0.9(DL+EQDL)+SL(-X)	60.40	40.41	60.74	4.18	-1.17	7.17
125 0.9(DL+EQDL)+SL(+Z)	55.21	35.94	55.94	3.65	-1.34	6.27
126 0.9(DL+EQDL)+SL(-Z)	79.74	87.00	82.46	6.67	9.94	11.52

Sample calculation of BM & SF

Load case : 101 1.4(DL+EQDL+FL)

p_{1z} = 124.06-(124.06-124.06)*0.475/1.5= 124.06 kN/m²

p_{2z} = 124.06+(124.06-124.06)*0.475/1.5= 124.06 kN/m²

p_{3z} = 124.06-(124.06-124.06)*(0.475-0.282)/1.5= 124.06 kN/m²

Overhang = 0.475 m

Effective depth = 0.282 m

BM_{1z} = (124.06-23.7)*0.475*0.475/2+0.5*(124.06-124.06)*0.475*0.475*0.67 = 11.32 kN-m/m

BM_{2z} = (124.06-23.7)*0.475*0.475/2-0.5*(-124.06+124.06)*0.475*0.475*0.67 = 11.32 kN-m/m

p_{1z} = 109.93-(109.93-108.86)*0.475/1.5= 109.59 kN/m²

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.	Ast (mm^2)
At bottom	11.32	0.142	0.190	0.0005	133.31	507.60	12ϕT	222.81	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 19.37 kN/m

Shear stress , τ_v= (19.37*1000)/(1000*282)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **200** c/c

Along Z Direction :

At Bottom **12ϕT** **200** c/c

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5.9 DESIGN OF COLUMN C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.9 % i.e. ρ_g = 0.019

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.71

Y_b = (b-d')/b = 0.71

Unsupported length of column parallel to X-X axis, l_{ux} = 2.1 m kl_{ux}/r_x = 16.8 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.1 m kl_{uz}/r_z = 16.8

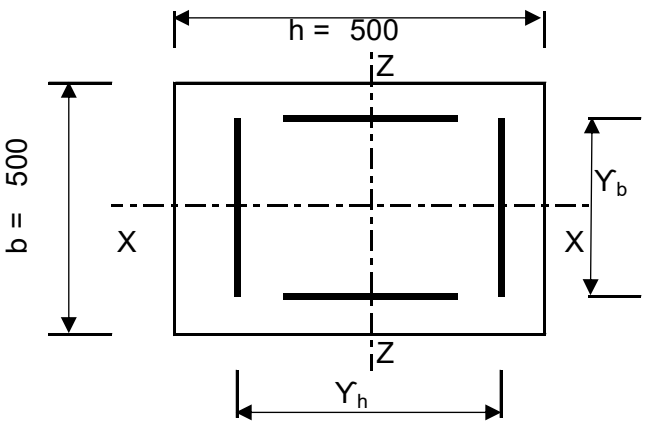
Radius of gyration r_x= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 7789.79 KN safe

Radius of gyration r_z= 150 mm

L/C	Factored Force Value P_u (KN)	M_{ux} (KN-m)	M_{uz} (KN-m)	Design force Value $P_n = P_u/φ$ (KN)	$M_{nx}=M_{ux}/φ$ (KN-m)	$M_{nz}=M_{uz}/φ$ (KN-m)	$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis: $R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1} (KN-m)	M_{nx}/M_{nx1}	about ZZ axis: $R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1} (KN-m)	M_{nz}/M_{nz1}	Check for Interaction Ratio: α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
102	585.6	108.1	111.9	901	166.3	172.2	0.1030	0.120	525	0.32	0.120	525	0.33	1.609	0.324	safe
116	167.4	0.0	0.0	258	0.0	0.0	0.0294	0.095	415.625	0.00	0.095	415.6	0.00		0	safe
116	15.4	2.3	123.8	24	3.5	190.5	0.0027	0.095	415.625	0.01	0.095	415.6	0.46		0.286	safe
103	576.0	192.7	193.6	886	296.5	297.8	0.1013	0.120	525	0.56	0.120	525	0.57		0.801	safe
103	576.0	192.7	193.6	886	296.5	297.8	0.1013	0.120	525	0.56	0.120	525	0.57		0.801	safe
103	423.6	221.8	45.1	652	341.2	69.4	0.0745	0.110	481.25	0.71	0.110	481.3	0.14		0.62	safe
101	111.6	23.0	6.6	172	35.4	10.2	0.0196	0.095	415.625	0.09	0.095	415.6	0.02		0.022	safe
101	111.6	23.0	6.6	172	35.4	10.2	0.0196	0.095	415.625	0.09	0.095	415.6	0.02		0.022	safe
103	423.6	221.8	45.1	652	341.2	69.4	0.0745	0.110	481.25	0.71	0.110	481.3	0.14		0.62	safe
103	576.0	192.7	193.6	886	296.5	297.8	0.1013	0.120	525	0.56	0.120	525	0.57		0.801	safe
116	15.4	2.3	123.8	24	3.5	190.5	0.0027	0.095	415.625	0.01	0.095	415.6	0.46		0.286	safe
103	576.0	192.7	193.6	886	296.5	297.8	0.1013	0.120	525	0.56	0.120	525	0.57		0.801	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
4750	4	25	+	8	25	5893	2.36	10	400	250



P_n : Nominal axial load

M_n : Nominal moment

P_u : Factored design axial load

M_u : Factored design moment

Use interaction diagram R35-420.8

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5.10 DESIGN OF COLUMN C2

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

$Y_h = (h-d')/h$ = 0.72

$Y_b = (b-d')/b$ = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 2.2 m kl_{ux}/r_x = 17.6 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.2 m kl_{uz}/r_z = 17.6

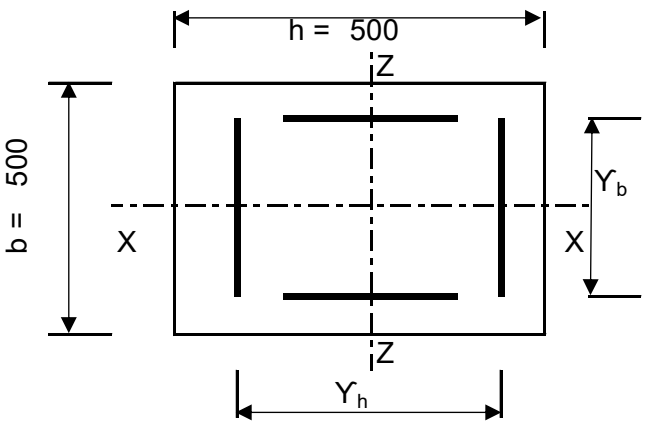
Radius of gyration r_x= 150 mm

Radius of gyration r_z= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845.08 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx}=M_{ux}/φ$	$M_{nz}=M_{uz}/φ$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$α=log0.5/logβ$	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	294.4	16.0	89.3	453	24.6	137.4	0.0518	0.070	306.25	0.08	0.070	306.3	0.45	1.609	0.293	safe
117	12.9	0.0	0.0	20	0.0	0.0	0.0023	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
116	118.5	0.0	0.0	182	0.0	0.0	0.0208	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
103	274.4	0.0	0.0	422	0.0	0.0	0.0482	0.060	262.5	0.00	0.060	262.5	0.00		0	safe
103	274.4	0.0	0.0	422	0.0	0.0	0.0482	0.060	262.5	0.00	0.060	262.5	0.00		0	safe
119	24.1	0.0	0.0	37	0.0	0.0	0.0042	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
101	190.4	0.0	0.0	293	0.0	0.0	0.0335	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
101	190.4	0.0	0.0	293	0.0	0.0	0.0335	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
103	292.0	18.8	173.9	449	28.9	267.5	0.0513	0.070	306.25	0.09	0.070	306.3	0.87		0.828	safe
119	37.3	3.3	15.7	57	5.1	24.2	0.0066	0.050	218.75	0.02	0.050	218.8	0.11		0.032	safe
103	292.0	18.8	173.9	449	28.9	267.5	0.0513	0.070	306.25	0.09	0.070	306.3	0.87		0.828	safe
116	131.7	8.0	3.1	203	12.3	4.8	0.0232	0.050	218.75	0.06	0.050	218.8	0.02		0.012	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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5.11 DESIGN OF COLUMN C3

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 2.25 % i.e. ρ_g = 0.0225

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 650 mm

Depth parallel to Z-Z axis, b = 650 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.78

Y_b = (b-d')/b = 0.78

Unsupported length of column parallel to X-X axis, l_{ux} = 2 m kl_{ux}/r_x = 12.31 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2 m kl_{uz}/r_z = 12.31

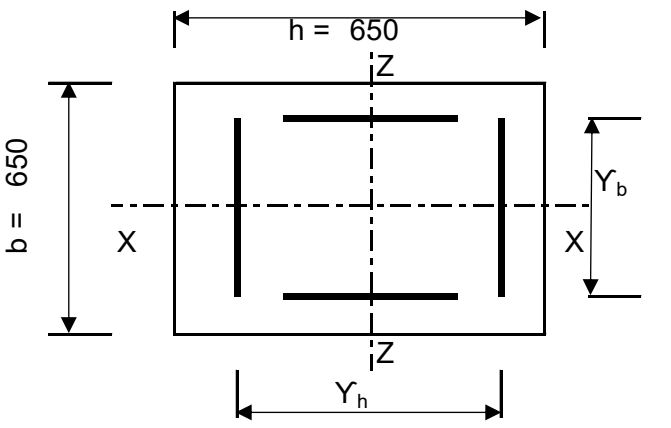
Radius of gyration r_x= 195 mm

Radius of gyration r_z= 195 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 13121.9 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx}=M_{ux}/φ$	$M_{nz}=M_{uz}/φ$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$α=log0.5/logβ$	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
103	747.0	630.7	228.4	1149	970.3	351.4	0.0777	0.125	1201.48	0.81	0.125	1201	0.29	1.609	0.848	safe
117	47.3	0.0	0.0	73	0.0	0.0	0.0049	0.100	961.188	0.00	0.100	961.2	0.00		0	safe
103	717.2	0.0	0.0	1103	0.0	0.0	0.0746	0.120	1153.43	0.00	0.120	1153	0.00		0	safe
117	47.3	0.0	0.0	73	0.0	0.0	0.0049	0.100	961.188	0.00	0.100	961.2	0.00		0	safe
103	535.6	0.0	0.0	824	0.0	0.0	0.0557	0.110	1057.31	0.00	0.110	1057	0.00		0	safe
103	717.2	0.0	0.0	1103	0.0	0.0	0.0746	0.120	1153.43	0.00	0.120	1153	0.00		0	safe
101	392.0	0.0	0.0	603	0.0	0.0	0.0408	0.110	1057.31	0.00	0.110	1057	0.00		0	safe
101	392.0	0.0	0.0	603	0.0	0.0	0.0408	0.110	1057.31	0.00	0.110	1057	0.00		0	safe
103	565.4	598.8	173.9	870	921.2	267.5	0.0588	0.115	1105.37	0.83	0.115	1105	0.24		0.848	safe
103	747.0	630.7	228.4	1149	970.3	351.4	0.0777	0.125	1201.48	0.81	0.125	1201	0.29		0.848	safe
117	25.0	32.9	134.7	38	50.6	207.2	0.0026	0.100	961.188	0.05	0.100	961.2	0.22		0.094	safe
103	747.0	630.7	228.4	1149	970.3	351.4	0.0777	0.125	1201.48	0.81	0.125	1201	0.29		0.848	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
9506	4	25	+	16	25	9822	2.32	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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5.12 DESIGN OF COLUMN C4

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 2.55 m kl_{ux}/r_x = 20.4 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.55 m kl_{uz}/r_z = 20.4

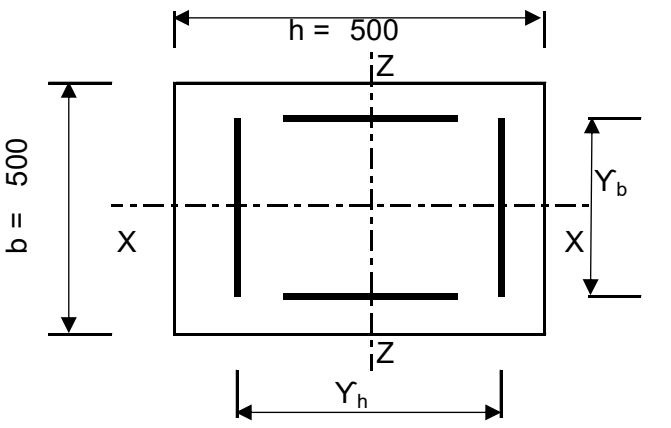
Radius of gyration r_x= 150 mm

Radius of gyration r_z= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845.08 KN safe

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx}=M_{ux}/φ$	$M_{nz}=M_{uz}/φ$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
124	210.1	0.0	0.0	323	0.0	0.0	0.0369	0.050	218.75	0.00	0.050	218.8	0.00	1.609	0	safe
116	18.9	0.0	0.0	29	0.0	0.0	0.0033	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
112	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe
113	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe
115	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
114	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
106	44.3	10.8	0.0	68	16.6	0.0	0.0078	0.050	218.75	0.08	0.050	218.8	0.00		0.016	safe
106	44.3	10.8	0.0	68	16.6	0.0	0.0078	0.050	218.75	0.08	0.050	218.8	0.00		0.016	safe
114	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
115	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
112	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe
113	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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5.13 DESIGN OF COLUMN C5

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 2.1 m kl_{ux}/r_x = 16.8 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.1 m kl_{uz}/r_z = 16.8

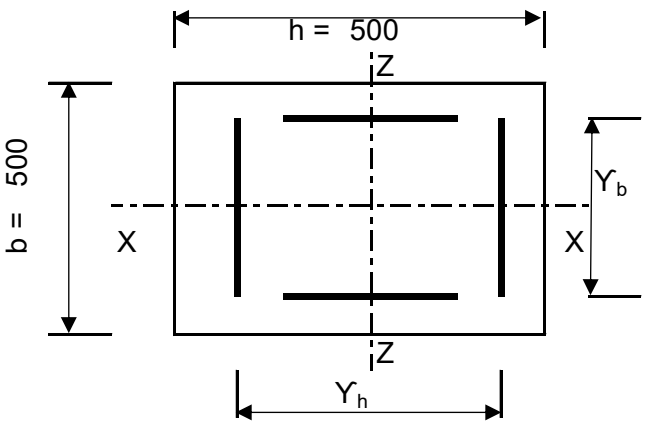
Radius of gyration r_x= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845.08 KN safe

Radius of gyration r_z= 150 mm

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx}=M_{ux}/φ$	$M_{nz}=M_{uz}/φ$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
101	289.7	0.0	0.0	446	0.0	0.0	0.0509	0.050	218.75	0.00	0.050	218.8	0.00	1.609	0	safe
116	18.9	0.0	0.0	29	0.0	0.0	0.0033	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
112	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe
113	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe
115	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
114	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
101	289.7	0.0	0.0	446	0.0	0.0	0.0509	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
101	289.7	0.0	0.0	446	0.0	0.0	0.0509	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
114	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
115	129.3	25.5	0.0	199	39.2	0.0	0.0227	0.050	218.75	0.18	0.050	218.8	0.00		0.063	safe
112	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe
113	129.3	0.0	25.5	199	0.0	39.2	0.0227	0.050	218.75	0.00	0.050	218.8	0.18		0.063	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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5.14 DESIGN OF COLUMN C6

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 600 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.77

Y_b = (b-d')/b = 0.77

Unsupported length of column parallel to X-X axis, l_{ux} = 2.6 m kl_{ux}/r_x = 17.33 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.6 m kl_{uz}/r_z = 17.33

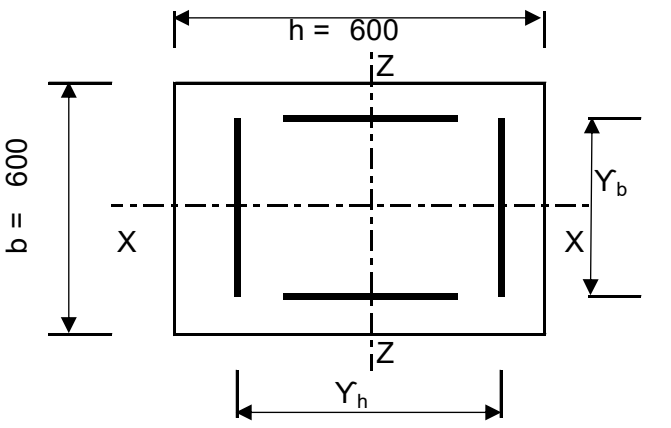
Radius of gyration r_x= 180 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 9745.62 KN safe

Radius of gyration r_z= 180 mm

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx}=M_{ux}/φ$	$M_{nz}=M_{uz}/φ$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$α=log0.5/logβ$	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
101	194.4	0.0	0.0	299	0.0	0.0	0.0237	0.050	378	0.00	0.050	378	0.00	1.609	0	safe
116	32.4	0.0	0.0	50	0.0	0.0	0.0040	0.050	378	0.00	0.050	378	0.00		0	safe
108	71.8	0.0	25.4	110	0.0	39.1	0.0088	0.050	378	0.00	0.050	378	0.10		0.026	safe
109	71.8	0.0	25.4	110	0.0	39.1	0.0088	0.050	378	0.00	0.050	378	0.10		0.026	safe
111	71.8	25.4	0.0	110	39.1	0.0	0.0088	0.050	378	0.10	0.050	378	0.00		0.026	safe
110	71.8	25.4	0.0	110	39.1	0.0	0.0088	0.050	378	0.10	0.050	378	0.00		0.026	safe
101	194.4	0.0	0.0	299	0.0	0.0	0.0237	0.050	378	0.00	0.050	378	0.00		0	safe
101	194.4	0.0	0.0	299	0.0	0.0	0.0237	0.050	378	0.00	0.050	378	0.00		0	safe
110	71.8	25.4	0.0	110	39.1	0.0	0.0088	0.050	378	0.10	0.050	378	0.00		0.026	safe
111	71.8	25.4	0.0	110	39.1	0.0	0.0088	0.050	378	0.10	0.050	378	0.00		0.026	safe
108	71.8	0.0	25.4	110	0.0	39.1	0.0088	0.050	378	0.00	0.050	378	0.10		0.026	safe
109	71.8	0.0	25.4	110	0.0	39.1	0.0088	0.050	378	0.00	0.050	378	0.10		0.026	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3600	4	20	+	8	20	3772	1.05	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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5.15 DESIGN OF COLUMN C7

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.75 % i.e. ρ_g = 0.0175

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 550 mm

Depth parallel to Z-Z axis, b = 550 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.74

Y_b = (b-d')/b = 0.74

Unsupported length of column parallel to X-X axis, l_{ux} = 2.6 m kl_{ux}/r_x = 18.91 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.6 m kl_{uz}/r_z = 18.91

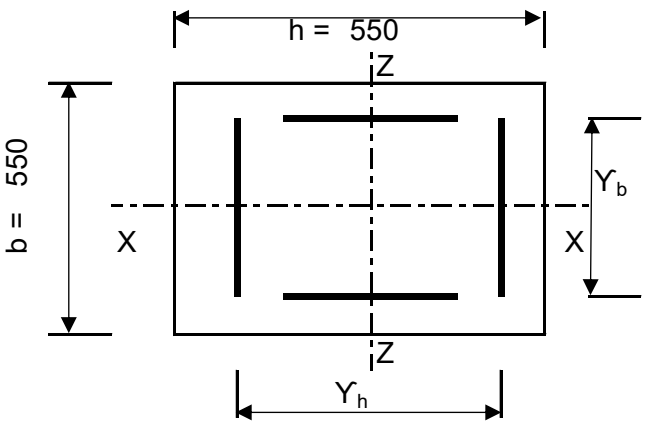
Radius of gyration r_x= 165 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 9039.29 KN safe

Radius of gyration r_z= 165 mm

L/C	Factored Force Value P_u (KN)	M_{ux} (KN-m)	M_{uz} (KN-m)	Design force Value $P_n = P_u/φ$ (KN)	$M_{nx}=M_{ux}/φ$ (KN-m)	$M_{nz}=M_{uz}/φ$ (KN-m)	$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis: $R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1} (KN-m)	M_{nx}/M_{nx1}	Uniaxial moment capacity about ZZ axis: $R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1} (KN-m)	M_{nz}/M_{nz1}	Check for Interaction Ratio: α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
106	811.6	270.2	159.8	1249	415.7	245.8	0.1179	0.110	640.544	0.65	0.110	640.5	0.38	1.609	0.713	safe
126	216.8	0.0	0.0	334	0.0	0.0	0.0315	0.090	524.081	0.00	0.090	524.1	0.00		0	safe
115	670.0	88.1	222.3	1031	135.5	342.0	0.0974	0.100	582.313	0.23	0.100	582.3	0.59		0.521	safe
116	606.1	49.8	231.8	932	76.6	356.6	0.0881	0.100	582.313	0.13	0.100	582.3	0.61		0.493	safe
114	630.5	219.1	84.5	970	337.1	130.0	0.0916	0.100	582.313	0.58	0.100	582.3	0.22		0.505	safe
106	811.6	270.2	159.8	1249	415.7	245.8	0.1179	0.110	640.544	0.65	0.110	640.5	0.38		0.713	safe
101	329.0	1.9	61.1	506	2.9	94.0	0.0478	0.090	524.081	0.01	0.090	524.1	0.18		0.064	safe
101	329.0	1.9	61.1	506	2.9	94.0	0.0478	0.090	524.081	0.01	0.090	524.1	0.18		0.064	safe
106	811.6	270.2	159.8	1249	415.7	245.8	0.1179	0.110	640.544	0.65	0.110	640.5	0.38		0.713	safe
114	630.5	219.1	84.5	970	337.1	130.0	0.0916	0.100	582.313	0.58	0.100	582.3	0.22		0.505	safe
115	670.0	88.1	222.3	1031	135.5	342.0	0.0974	0.100	582.313	0.23	0.100	582.3	0.59		0.521	safe
116	606.1	49.8	231.8	932	76.6	356.6	0.0881	0.100	582.313	0.13	0.100	582.3	0.61		0.493	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
5294	4	25	+	8	25	5893	1.95	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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5.16 DESIGN OF COLUMN C8

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 550 mm

Depth parallel to Z-Z axis, b = 550 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.75

Y_b = (b-d')/b = 0.75

Unsupported length of column parallel to X-X axis, l_{ux} = 2.6 m kl_{ux}/r_x = 18.91 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.6 m kl_{uz}/r_z = 18.91

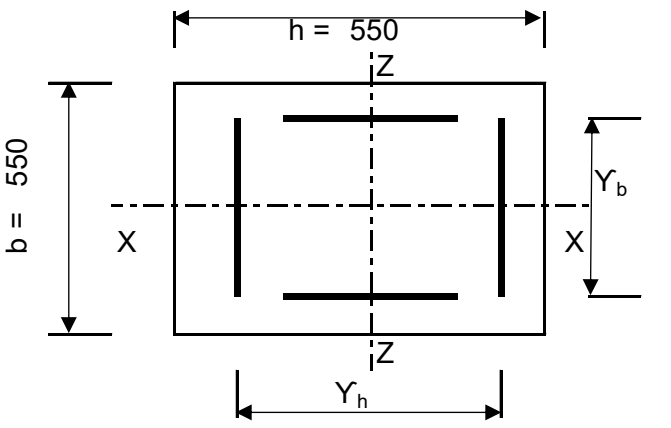
Radius of gyration r_x= 165 mm

$P_{n, Max}$ = 0.8[0.85*f_c'(A_g-A_{st})+$f_y A_{st}$] = 8377.12 KN safe

Radius of gyration r_z= 165 mm

Factored Force Value				Design force Value			$K_n = P_n/f_c bh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx}=M_{ux}/\varphi$	$M_{nz}=M_{uz}/\varphi$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
101	214.1	0.0	0.0	329	0.0	0.0	0.0311	0.050	291.156	0.00	0.050	291.2	0.00	1.609	0	safe
125	39.9	0.0	0.0	61	0.0	0.0	0.0058	0.050	291.156	0.00	0.050	291.2	0.00		0	safe
123	95.9	0.5	4.4	148	0.8	6.8	0.0139	0.050	291.156	0.00	0.050	291.2	0.02		0.003	safe
116	128.3	0.5	4.6	197	0.8	7.1	0.0186	0.050	291.156	0.00	0.050	291.2	0.02		0.003	safe
118	166.6	2.1	1.7	256	3.2	2.6	0.0242	0.050	291.156	0.01	0.050	291.2	0.01		0.002	safe
117	115.8	2.1	2.0	178	3.2	3.1	0.0168	0.050	291.156	0.01	0.050	291.2	0.01		0.002	safe
101	214.1	0.0	0.0	329	0.0	0.0	0.0311	0.050	291.156	0.00	0.050	291.2	0.00		0	safe
101	214.1	0.0	0.0	329	0.0	0.0	0.0311	0.050	291.156	0.00	0.050	291.2	0.00		0	safe
117	115.8	2.1	2.0	178	3.2	3.1	0.0168	0.050	291.156	0.01	0.050	291.2	0.01		0.002	safe
118	166.6	2.1	1.7	256	3.2	2.6	0.0242	0.050	291.156	0.01	0.050	291.2	0.01		0.002	safe
123	95.9	0.5	4.4	148	0.8	6.8	0.0139	0.050	291.156	0.00	0.050	291.2	0.02		0.003	safe
116	128.3	0.5	4.6	197	0.8	7.1	0.0186	0.050	291.156	0.00	0.050	291.2	0.02		0.003	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3025	4	20	+	8	20	3772	1.25	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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ATTACHMENT 1 STAAD INPUT & OUTPUT FILE FOR F1 DESIGN

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*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
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*      USER ID: LT Construction                    *
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     3. ENGINEER DATE 24-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
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425. ELEMENT
426. _RAFT_MOM 31 35 39 50 54 58 68 72 76 87 91 95 129 TO 170 174 176 178 180 182 -
427. 184 186 188 190 192 194 196 199 TO 210 212 TO 253 284 286 288 294 296 298 -
428. 303 305 307 313 315 317 335 TO 376 378 TO 389 392 TO 403 405 TO 446 468 470 -
429. 472 474 476 478 480 482 487 489 491 493 495 497 499 501 508 510 512 514 516 -
430. 518 520 522 527 529 531 533 535 537 539 541 704 TO 799 824 TO 1015
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431. _RAFT_SHEAR 31 35 54 58 68 72 91 95 129 TO 156 159 TO 168 176 178 180 182 -
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433. 303 305 315 317 335 TO 362 365 TO 374 379 TO 388 393 TO 402 407 TO 416 419 -
434. 420 TO 446 468 470 480 482 487 489 499 501 508 510 520 522 527 529 539 541 -
435. 704 TO 799 824 TO 1015
436. _RAFT 31 35 39 42 46 50 54 58 68 72 76 80 83 87 91 95 129 TO 170 172 174 176 -
437. 178 180 182 184 186 188 190 192 194 196 TO 253 284 286 288 290 292 294 296 -
438. 298 303 305 307 309 311 313 315 317 335 TO 446 468 470 472 474 476 478 480 -
439. 482 487 489 491 493 495 497 499 501 508 510 512 514 516 518 520 522 527 529 -
440. 531 533 535 537 539 541 704 TO 799 824 TO 1015
441. MEMBER
442. _C1 461 TO 464
443. END GROUP DEFINITION
444. ELEMENT PROPERTY
445. _RAFT THICKNESS 0.5
446. DEFINE MATERIAL START
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448. E 2.17185E+007
449. POISSON 0.17
450. DENSITY 25
451. ALPHA 1E-005
452. DAMP 0.05
453. TYPE CONCRETE
454. STRENGTH FCU 27579
455. END DEFINE MATERIAL
456. MEMBER PROPERTY AMERICAN
457. 1 TO 4 461 TO 464 PRIS YD 0.5 ZD 0.5
458. CONSTANTS
459. MATERIAL CONCRETE ALL
460. SUPPORTS
461. _RAFT PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
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841	0.061620
842	0.061620
843	0.061623
844	0.061620
845	0.061620
846	0.061620
847	0.061496
848	0.061623
849	0.061620
850	0.061620
851	0.061620
852	0.061620
853	0.061623
854	0.061626
855	0.061626
856	0.061629
857	0.061626
858	0.061626
859	0.061626
860	0.061502
861	0.061629
862	0.061626
863	0.061626
864	0.061626
865	0.061626
866	0.061629
867	0.063629
868	0.063622
869	0.063502
870	0.063495
871	0.063622
872	0.063629
873	0.063495
874	0.063502
875	0.063002
876	0.031503
877	0.063002
878	0.062996
879	0.031500
880	0.062996
881	0.063126
882	0.063126
883	0.031500
884	0.063120
885	0.063120

```
       886          0.031497
       887          0.062996
       888          0.031500
       889          0.062996
       890          0.063002
       891          0.031503
       892          0.063002
       893          0.063120
       894          0.063120
       895          0.031497
       896          0.063126
       897          0.063126
       898          0.031500
    TOTAL          35.999737
462. ********************************************************************************
463. LOAD 1 DEAD LOAD (DL)
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
464. ********************************************************************************
465. JOINT LOAD
466. 1 FX 1 FY -55 FZ -5
467. 2 FX -6 FY -73 FZ -6
468. 4 FX -2 FY -51 FZ 5
469. 5 FX -1 FY -56 FZ 8
470. ELEMENT LOAD
471. _RAFT PR -12.6
472. SELFWEIGHT Y -1
473. ********************************************************************************
474. LOAD 2 LIVE LOAD (LL)
475. ********************************************************************************
476. JOINT LOAD
477. 1 FX 1 FY -133 FZ -8
478. 2 FX -9 FY -159 FZ -11
479. 4 FX -3 FY -101 FZ 8
480. 5 FX -2 FY -115 FZ 15
481. ********************************************************************************
482. LOAD 3 OPERATING FLUID LOAD (FL)
483. ********************************************************************************
484. JOINT LOAD
485. 1 FX 1 FY -7 FZ -1.
486. 2 FX -8 FY -70 FZ -7
487. 4 FX -1 FY -5 FZ 1
488. 5 FX -1 FY -57 FZ 8
489. ********************************************************************************
490. LOAD 4 EQP DL (EQDL)
491. ********************************************************************************
492. JOINT LOAD
493. 1 FX 0 FY -3 FZ -1.
494. 2 FX -3 FY -29 FZ -4
495. 4 FX -1 FY -3 FZ 1
496. 5 FX -1 FY -28 FZ 4
497. ********************************************************************************
498. LOAD 5 EQP LL (EQLL)
499. ********************************************************************************
500. JOINT LOAD
501. 1 FX 1 FY -7 FZ -1.
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502. 2 FX -8 FY -67 FZ -5
503. 4 FX -1 FY -1 FZ 1
504. 5 FX -1 FY -36 FZ 5
505. ******************************************************************************
506. LOAD 6 WIND LOAD (+X) WL(+X)
507. ******************************************************************************
508. JOINT LOAD
509. 1 FX 2 FY 35 FZ 5
510. 2 FX 38 FY 56 FZ 5
511. 4 FX 32 FY 135 FZ -3
512. 5 FX 22 FY -97 FZ 7
513. ******************************************************************************
514. LOAD 7 WIND LOAD (-X) WL(-X)
515. ******************************************************************************
516. JOINT LOAD
517. 1 FX -2 FY -35 FZ -5
518. 2 FX -38 FY -56 FZ -5
519. 4 FX -32 FY -135 FZ 3
520. 5 FX -22 FY 97 FZ -7
521. ******************************************************************************
522. LOAD 8 WIND LOAD (+Z) WL(+Z)
523. ******************************************************************************
524. JOINT LOAD
525. 1 FX -1 FY 42 FZ 15
526. 2 FX 1 FY 48 FZ 29
527. 4 FX 2 FY -33 FZ 15
528. 5 FX 2 FY -36 FZ 24
529. ******************************************************************************
530. LOAD 9 WIND LOAD (-Z) WL(-Z)
531. ******************************************************************************
532. JOINT LOAD
533. 1 FX 1 FY -42 FZ -15
534. 2 FX -1 FY -48 FZ -29
535. 4 FX -2 FY 33 FZ -15
536. 5 FX -2 FY 36 FZ -24
537. ******************************************************************************
538. LOAD 10 SEISMIC LOAD SL(+X)
539. ******************************************************************************
540. JOINT LOAD
541. 1 FX 1 FY 17 FZ 3
542. 2 FX 23 FY 48 FZ 6
543. 4 FX 17 FY 76 FZ -2
544. 5 FX 14 FY -57 FZ 6
545. ******************************************************************************
546. LOAD 11 SEISMIC LOAD SL(-X)
547. ******************************************************************************
548. JOINT LOAD
549. 1 FX -1 FY -17 FZ -3
550. 2 FX -23 FY -48 FZ -6
551. 4 FX -17 FY -76 FZ 2
552. 5 FX -14 FY 57 FZ -6
553. ******************************************************************************
554. LOAD 12 SEISMIC LOAD SL(+Z)
555. ******************************************************************************
556. JOINT LOAD
557. 1 FX -1 FY 14 FZ 4
```

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558. 2 FX 2 FY 35 FZ 19
559. 4 FX 1 FY -11 FZ 5
560. 5 FX 1 FY -21 FZ 15
561. ********************************************************************************
562. LOAD 13 SEISMIC LOAD SL(-Z)
563. ********************************************************************************
564. JOINT LOAD
565. 1 FX 1 FY -14 FZ -4
566. 2 FX -2 FY -35 FZ -19
567. 4 FX -1 FY 11 FZ -5
568. 5 FX -1 FY 21 FZ -15
569. ********************************************************************************
570. LOAD 14 TEMPERATURE LOAD (TL)
571. JOINT LOAD
572. 1 FX -2 FY 1 FZ -24
573. 2 FX -29 FY 8 FZ -30
574. 4 FX -26 FY -10 FZ 24
575. 5 FX 23 FY 4 FZ 32
576. ********************************************************************************
577. ********************************************************************************
578. LOAD 15 EQP TEST LOAD (EQTL)
579. ********************************************************************************
580. JOINT LOAD
581. 1 FX 0 FY -2 FZ -1.
582. 2 FX -2 FY -22 FZ -4
583. 4 FX -1 FY -6 FZ 1
584. 5 FX -1 FY -36 FZ 6
585. ********************************************************************************
586. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
587. ********************************************************************************
588. LOAD 101 1.4(DL+EQDL+FL)
589. REPEAT LOAD
590. 1 1.4 4 1.4 3 1.4
591. LOAD 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
592. REPEAT LOAD
593. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
594. LOAD 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
595. REPEAT LOAD
596. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
597. LOAD 104 1.2(DL+EQDL)+0.8WL(+X)
598. REPEAT LOAD
599. 1 1.2 4 1.2 6 0.8
600. LOAD 105 1.2(DL+EQDL)+0.8WL(-X)
601. REPEAT LOAD
602. 1 1.2 4 1.2 7 0.8
603. LOAD 106 1.2(DL+EQDL)+0.8WL(+Z)
604. REPEAT LOAD
605. 1 1.2 4 1.2 8 0.8
606. LOAD 107 1.2(DL+EQDL)+0.8WL(-Z)
607. REPEAT LOAD
608. 1 1.2 4 1.2 9 0.8
609. LOAD 108 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
610. REPEAT LOAD
611. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
612. LOAD 109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
613. REPEAT LOAD
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614. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
615. LOAD 110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
616. REPEAT LOAD
617. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
618. LOAD 111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
619. REPEAT LOAD
620. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
621. LOAD 112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
622. REPEAT LOAD
623. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
624. LOAD 113 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
625. REPEAT LOAD
626. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
627. LOAD 114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
628. REPEAT LOAD
629. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
630. LOAD 115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
631. REPEAT LOAD
632. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
633. LOAD 116 0.9(DL+EQDL)+1.6WL(+X)
634. REPEAT LOAD
635. 1 0.9 4 0.9 6 1.6
636. LOAD 117 0.9(DL+EQDL)+1.6WL(-X)
637. REPEAT LOAD
638. 1 0.9 4 0.9 7 1.6
639. LOAD 118 0.9(DL+EQDL)+1.6WL(+Z)
640. REPEAT LOAD
641. 1 0.9 4 0.9 8 1.6
642. LOAD 119 0.9(DL+EQDL)+1.6WL(-Z)
643. REPEAT LOAD
644. 1 0.9 4 0.9 9 1.6
645. LOAD 120 0.9(DL+EQDL)+SL(+X)
646. REPEAT LOAD
647. 1 0.9 4 0.9 10 1.0
648. LOAD 121 0.9(DL+EQDL)+SL(-X)
649. REPEAT LOAD
650. 1 0.9 4 0.9 11 1.0
651. LOAD 122 0.9(DL+EQDL)+SL(+Z)
652. REPEAT LOAD
653. 1 0.9 4 0.9 12 1.0
654. LOAD 123 0.9(DL+EQDL)+SL(-Z)
655. REPEAT LOAD
656. 1 0.9 4 0.9 13 1.0
657. LOAD 124 1.2DL+EQTL
658. REPEAT LOAD
659. 1 1.2 15 1.0
660. ****************************************
661. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 658 NUMBER OF MEMBERS 8
NUMBER OF PLATES 576 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 650

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 624/ 25/ 78 DOF
TOTAL PRIMARY LOAD CASES = 39, TOTAL DEGREES OF FREEDOM = 1998
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 156 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 17.7/ 99949.7 MB

662. CHANGE
663. SUPPORTS
664. _RAFT PLATE MAT DIRECT Y SUBGRADE 40500 PRINT COMPRESSION

JOINT	INFLUENCE AREA
161	0.063502
162	0.063502
163	0.063495
164	0.063495
165	0.031500
166	0.031497
176	0.064875
177	0.031503
178	0.063502
179	0.064869
180	0.031500
181	0.063495
182	0.064875
183	0.031503
184	0.063502
185	0.064875
186	0.031503
187	0.063502
188	0.064869
189	0.064875
190	0.031503
191	0.063502
192	0.064875
193	0.031503
194	0.063502
195	0.064875
196	0.031503
197	0.063502

198	0.064875
199	0.031503
200	0.063502
203	0.064751
204	0.063502
205	0.031500
206	0.064745
207	0.063495
208	0.031497
209	0.064751
210	0.063502
211	0.031500
212	0.064751
213	0.063502
214	0.031500
215	0.064745
216	0.064751
217	0.063502
218	0.031500
219	0.064751
220	0.063502
221	0.031500
222	0.064751
223	0.063502
224	0.031500
225	0.064751
226	0.063502
227	0.031500
230	0.061626
232	0.061626
234	0.061629
236	0.061626
238	0.061626
240	0.061626
242	0.061502
244	0.061629
246	0.061626
248	0.061626
250	0.061626
252	0.061626
254	0.061629
255	0.061620
256	0.061620
257	0.061623
258	0.061620
259	0.061620
260	0.061620
261	0.061496
262	0.061623
263	0.061620
264	0.061620
265	0.061620
266	0.061620
267	0.061623
268	0.061626
269	0.061626

270	0.061629
271	0.061626
272	0.061626
273	0.061626
274	0.061502
275	0.061629
276	0.061626
277	0.061626
278	0.061626
279	0.061626
280	0.061629
281	0.061626
282	0.061626
283	0.061629
284	0.061626
285	0.061626
286	0.061626
287	0.061502
288	0.061629
289	0.061626
290	0.061626
291	0.061626
292	0.061626
293	0.061629
294	0.061620
295	0.061620
296	0.061623
297	0.061620
298	0.061620
299	0.061620
300	0.061496
301	0.061623
302	0.061620
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305	0.061620
306	0.061623
307	0.061626
308	0.061626
309	0.061629
310	0.061626
311	0.061626
312	0.061626
313	0.061502
314	0.061629
315	0.061626
316	0.061626
317	0.061626
318	0.061626
319	0.061629
320	0.061626
321	0.061626
322	0.061629
323	0.061626
324	0.061626
325	0.061626

326	0.061502
327	0.061629
328	0.061626
329	0.061626
330	0.061626
331	0.061626
332	0.061629
333	0.061626
334	0.061626
335	0.061629
336	0.061626
337	0.061626
338	0.061626
339	0.061502
340	0.061629
341	0.061626
342	0.061626
343	0.061626
344	0.061626
345	0.061629
346	0.061626
347	0.061626
348	0.061629
349	0.061626
350	0.061626
351	0.061626
352	0.061502
353	0.061629
354	0.061626
355	0.061626
356	0.061626
357	0.061626
358	0.061629
374	0.064875
375	0.031503
377	0.063502
378	0.064875
379	0.031503
380	0.063502
381	0.064869
382	0.031500
383	0.063495
384	0.064875
385	0.031503
386	0.063502
387	0.064875
388	0.031503
389	0.064869
390	0.031500
391	0.063495
392	0.064875
393	0.031503
394	0.063502
395	0.064875
396	0.031503
397	0.063502

398	0.064875
399	0.031503
400	0.063502
406	0.064751
407	0.063502
409	0.031500
410	0.064751
411	0.063502
412	0.031500
413	0.064745
414	0.063495
415	0.031497
416	0.064751
417	0.063502
418	0.031500
419	0.064751
420	0.031500
421	0.064745
422	0.063495
423	0.031497
424	0.064751
425	0.063502
426	0.031500
427	0.064751
428	0.063502
429	0.031500
430	0.064751
431	0.063502
432	0.031500
437	0.061626
439	0.061626
441	0.061629
443	0.061626
445	0.061626
447	0.061626
449	0.061502
451	0.061629
453	0.061626
455	0.061626
457	0.061626
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462	0.061626
463	0.061626
464	0.061629
465	0.061626
466	0.061626
467	0.061626
468	0.061502
469	0.061629
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471	0.061626
472	0.061626
473	0.061626
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475	0.061620

476	0.061620
477	0.061623
478	0.061620
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481	0.061496
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490	0.061629
491	0.061626
492	0.061626
493	0.061626
494	0.061502
495	0.061629
496	0.061626
497	0.061626
498	0.061626
499	0.061626
500	0.061629
501	0.061626
502	0.061626
503	0.061629
504	0.061626
505	0.061626
506	0.061626
507	0.061502
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510	0.061626
511	0.061626
512	0.061626
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515	0.061620
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519	0.061620
520	0.061496
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531	0.061626

532	0.061626
533	0.061502
534	0.061629
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540	0.061626
541	0.061626
542	0.061629
543	0.061626
544	0.061626
545	0.061626
546	0.061502
547	0.061629
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551	0.061626
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555	0.061629
556	0.061626
557	0.061626
558	0.061626
559	0.061502
560	0.061629
561	0.061626
562	0.061626
563	0.061626
564	0.061626
565	0.061629
584	0.063629
585	0.063622
586	0.063629
587	0.063629
588	0.063622
589	0.063629
590	0.063629
591	0.063629
592	0.063629
595	0.063502
596	0.063495
597	0.063502
598	0.063502
599	0.063495
600	0.063502
601	0.063502
602	0.063502
603	0.063502
606	0.063629
607	0.063629
608	0.063622
609	0.063629

610	0.063629
611	0.063622
612	0.063629
613	0.063629
614	0.063629
617	0.063502
618	0.063502
619	0.063495
620	0.063502
621	0.063502
622	0.063495
623	0.063502
624	0.063502
625	0.063502
627	0.031748
628	0.032434
629	0.031748
630	0.032434
631	0.032372
632	0.031748
633	0.032372
634	0.031748
635	0.030810
636	0.030810
637	0.030811
638	0.030810
639	0.030810
640	0.030810
641	0.030748
642	0.030811
643	0.030810
644	0.030810
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649	0.030810
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651	0.030810
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653	0.030810
654	0.030748
655	0.030811
656	0.030810
657	0.030810
658	0.030810
659	0.030810
660	0.030811
661	0.031748
662	0.032434
663	0.031748
664	0.032434
665	0.032372
666	0.031748
667	0.032372
668	0.031748

669	0.030810
670	0.030810
671	0.030811
672	0.030810
673	0.030810
674	0.030810
675	0.030748
676	0.030811
677	0.030810
678	0.030810
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685	0.030810
686	0.030810
687	0.030810
688	0.030748
689	0.030811
690	0.030810
691	0.030810
692	0.030810
693	0.030810
694	0.030811
695	0.031811
696	0.015750
697	0.031811
698	0.015750
699	0.031748
700	0.015748
701	0.031748
702	0.015748
703	0.031811
704	0.015750
705	0.031811
706	0.015750
707	0.031748
708	0.015748
709	0.031748
710	0.015748
711	0.063002
712	0.062996
713	0.063002
714	0.062996
715	0.063002
716	0.063002
717	0.063002
718	0.063002
719	0.062996
720	0.062996
721	0.063002
722	0.063002
723	0.063002
724	0.063002

725	0.063002
726	0.063002
727	0.063002
728	0.063002
729	0.063126
730	0.063120
731	0.063126
732	0.063120
733	0.063126
734	0.063126
735	0.063126
736	0.063126
737	0.063120
738	0.063120
739	0.063126
740	0.063126
741	0.063126
742	0.063126
743	0.063126
744	0.063126
745	0.063126
746	0.063126
747	0.063002
748	0.063002
749	0.063002
750	0.063002
751	0.062996
752	0.062996
753	0.063002
754	0.063002
755	0.063002
756	0.063002
757	0.062996
758	0.062996
759	0.063002
760	0.063002
761	0.063002
762	0.063002
763	0.063002
764	0.063002
765	0.063126
766	0.063126
767	0.063126
768	0.063126
769	0.063120
770	0.063120
771	0.063126
772	0.063126
773	0.063126
774	0.063126
775	0.063120
776	0.063120
777	0.063126
778	0.063126
779	0.063126
780	0.063126

781	0.063126
782	0.063126
783	0.031498
784	0.031498
785	0.031498
786	0.031498
787	0.031560
788	0.031560
789	0.031560
790	0.031560
791	0.031498
792	0.031498
793	0.031498
794	0.031498
795	0.031560
796	0.031560
797	0.031560
798	0.031560
799	0.063502
800	0.064875
801	0.063495
802	0.064869
803	0.064751
804	0.063502
805	0.064745
806	0.063495
807	0.061626
808	0.061626
809	0.061629
810	0.061626
811	0.061626
812	0.061626
813	0.061502
814	0.061629
815	0.061626
816	0.061626
817	0.061626
818	0.061626
819	0.061629
820	0.061620
821	0.061620
822	0.061623
823	0.061620
824	0.061620
825	0.061620
826	0.061496
827	0.061623
828	0.061620
829	0.061620
830	0.061620
831	0.061620
832	0.061623
833	0.063495
834	0.064869
835	0.063502
836	0.064875

837	0.064745
838	0.063495
839	0.064751
840	0.063502
841	0.061620
842	0.061620
843	0.061623
844	0.061620
845	0.061620
846	0.061620
847	0.061496
848	0.061623
849	0.061620
850	0.061620
851	0.061620
852	0.061620
853	0.061623
854	0.061626
855	0.061626
856	0.061629
857	0.061626
858	0.061626
859	0.061626
860	0.061502
861	0.061629
862	0.061626
863	0.061626
864	0.061626
865	0.061626
866	0.061629
867	0.063629
868	0.063622
869	0.063502
870	0.063495
871	0.063622
872	0.063629
873	0.063495
874	0.063502
875	0.063002
876	0.031503
877	0.063002
878	0.062996
879	0.031500
880	0.062996
881	0.063126
882	0.063126
883	0.031500
884	0.063120
885	0.063120
886	0.031497
887	0.062996
888	0.031500
889	0.062996
890	0.063002
891	0.031503
892	0.063002

```
       893          0.063120
       894          0.063120
       895          0.031497
       896          0.063126
       897          0.063126
       898          0.031500
    TOTAL          35.999737
665. ******************************************************
666. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
667. ******************************************************
668. LOAD 201 DL+EQDL+FL
669. REPEAT LOAD
670. 1 1.0 4 1.0 3 1.0
671. LOAD 202 DL+EQDL+FL+LL+EQLL+TL
672. REPEAT LOAD
673. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
674. LOAD 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
675. REPEAT LOAD
676. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
677. LOAD 204 DL+EQDL+FL+WL(+X)
678. REPEAT LOAD
679. 1 1.0 4 1.0 3 1.0 6 1.0
680. LOAD 205 DL+EQDL+FL+WL(-X)
681. REPEAT LOAD
682. 1 1.0 4 1.0 3 1.0 7 1.0
683. LOAD 206 DL+EQDL+FL+WL(+Z)
684. REPEAT LOAD
685. 1 1.0 4 1.0 3 1.0 8 1.0
686. LOAD 207 DL+EQDL+FL+WL(-Z)
687. REPEAT LOAD
688. 1 1.0 4 1.0 3 1.0 9 1.0
689. LOAD 208 DL+EQDL+FL+0.7SL(+X)
690. REPEAT LOAD
691. 1 1.0 4 1.0 3 1.0 10 0.7
692. LOAD 209 DL+EQDL+FL+0.7SL(-X)
693. REPEAT LOAD
694. 1 1.0 4 1.0 3 1.0 11 0.7
695. LOAD 210 DL+EQDL+FL+0.7SL(+Z)
696. REPEAT LOAD
697. 1 1.0 4 1.0 3 1.0 12 0.7
698. LOAD 211 DL+EQDL+FL+0.7SL(-Z)
699. REPEAT LOAD
700. 1 1.0 4 1.0 3 1.0 13 0.7
701. LOAD 212 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
702. REPEAT LOAD
703. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
704. LOAD 213 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
705. REPEAT LOAD
706. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
707. LOAD 214 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
708. REPEAT LOAD
709. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
710. LOAD 215 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
711. REPEAT LOAD
712. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
713. LOAD 216 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
```

```
 714. REPEAT LOAD
 715. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
 716. LOAD 217 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
 717. REPEAT LOAD
 718. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
 719. LOAD 218 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
 720. REPEAT LOAD
 721. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
 722. LOAD 219 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
 723. REPEAT LOAD
 724. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
 725. LOAD 220 0.6(DL+EQDL)+WL(+X)
 726. REPEAT LOAD
 727. 1 0.6 4 0.6 6 1.0
 728. LOAD 221 0.6(DL+EQDL)+WL(-X)
 729. REPEAT LOAD
 730. 1 0.6 4 0.6 7 1.0
 731. LOAD 222 0.6(DL+EQDL)+WL(+Z)
 732. REPEAT LOAD
 733. 1 0.6 4 0.6 8 1.0
 734. LOAD 223 0.6(DL+EQDL)+WL(-Z)
 735. REPEAT LOAD
 736. 1 0.6 4 0.6 9 1.0
 737. LOAD 224 0.6(DL+EQDL)+0.7SL(+X)
 738. REPEAT LOAD
 739. 1 0.6 4 0.6 10 0.7
 740. LOAD 225 0.6(DL+EQDL)+0.7SL(-X)
 741. REPEAT LOAD
 742. 1 0.6 4 0.6 11 0.7
 743. LOAD 226 0.6(DL+EQDL)+0.7SL(+Z)
 744. REPEAT LOAD
 745. 1 0.6 4 0.6 12 0.7
 746. LOAD 227 0.6(DL+EQDL)+0.7SL(-Z)
 747. REPEAT LOAD
 748. 1 0.6 4 0.6 13 0.7
 749. LOAD 228 DL+EQTL
 750. REPEAT LOAD
 751. 1 1.0 15 1.0
 752. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201


**NOTE-Tension/Compression converged after  1 iterations, Case=  202


**NOTE-Tension/Compression converged after  1 iterations, Case=  203


**START ITERATION NO.          2
```

**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 220

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 221

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

**NOTE-Tension/Compression converged after 1 iterations, Case= 223

**NOTE-Tension/Compression converged after 1 iterations, Case= 224

**NOTE-Tension/Compression converged after 1 iterations, Case= 225

**NOTE-Tension/Compression converged after 1 iterations, Case= 226

**NOTE-Tension/Compression converged after 1 iterations, Case= 227

**NOTE-Tension/Compression converged after 1 iterations, Case= 228

753. **********
754. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 15:13:54 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 2 STAAD INPUT & OUTPUT FILE FOR F1 STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    15:11:55                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F1 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 24-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 -212.95 913.6 -435.9; 2 -209.5 913.6 -435.9; 4 -212.95 913.6 -429
    10. 5 -209.5 913.6 -429; 161 -212.95 911 -435.9; 162 -209.5 911 -435.9
    11. 163 -212.95 911 -429; 164 -209.5 911 -429
    12. MEMBER INCIDENCES
    13. 1 161 1; 2 162 2; 3 163 4; 4 164 5
    14. DEFINE MATERIAL START
    15. ISOTROPIC CONCRETE
    16. E 2.17185E+007
    17. POISSON 0.17
    18. DENSITY 25
    19. ALPHA 1E-005
    20. DAMP 0.05
    21. TYPE CONCRETE
    22. STRENGTH FCU 27579
    23. END DEFINE MATERIAL
    24. MEMBER PROPERTY AMERICAN
    25. 1 TO 4 PRIS YD 0.5 ZD 0.5
    26. CONSTANTS
    27. MATERIAL CONCRETE ALL
    28. SUPPORTS
    29. 161 TO 164 FIXED
    30. ******************************************************************************
    31. LOAD 1 DEAD LOAD (DL)
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    32. ******************************************************************************
    33. JOINT LOAD
    34. 1 FX 1 FY -55 FZ -5
    35. 2 FX -6 FY -73 FZ -6
    36. 4 FX -2 FY -51 FZ 5
```

```
37. 5 FX -1 FY -56 FZ 8
38. SELFWEIGHT Y -1 LIST ALL
39. ********************************************************************************
40. LOAD 2 LIVE LOAD (LL)
41. ********************************************************************************
42. JOINT LOAD
43. 1 FX 1 FY -133 FZ -8
44. 2 FX -9 FY -159 FZ -11
45. 4 FX -3 FY -101 FZ 8
46. 5 FX -2 FY -115 FZ 15
47. ********************************************************************************
48. LOAD 3 OPERATING FLUID LOAD (FL)
49. ********************************************************************************
50. JOINT LOAD
51. 1 FX 1 FY -7 FZ -1.
52. 2 FX -8 FY -70 FZ -7
53. 4 FX -1 FY -5 FZ 1
54. 5 FX -1 FY -57 FZ 8
55. ********************************************************************************
56. LOAD 4 EQP DL (EQDL)
57. ********************************************************************************
58. JOINT LOAD
59. 1 FX 0 FY -3 FZ -1.
60. 2 FX -3 FY -29 FZ -4
61. 4 FX -1 FY -3 FZ 1
62. 5 FX -1 FY -28 FZ 4
63. ********************************************************************************
64. LOAD 5 EQP LL (EQLL)
65. ********************************************************************************
66. JOINT LOAD
67. 1 FX 1 FY -7 FZ -1.
68. 2 FX -8 FY -67 FZ -5
69. 4 FX -1 FY -1 FZ 1
70. 5 FX -1 FY -36 FZ 5
71. ********************************************************************************
72. LOAD 6 WIND LOAD (+X) WL(+X)
73. ********************************************************************************
74. JOINT LOAD
75. 1 FX 2 FY 35 FZ 5
76. 2 FX 38 FY 56 FZ 5
77. 4 FX 32 FY 135 FZ -3
78. 5 FX 22 FY -97 FZ 7
79. ********************************************************************************
80. LOAD 7 WIND LOAD (-X) WL(-X)
81. ********************************************************************************
82. JOINT LOAD
83. 1 FX -2 FY -35 FZ -5
84. 2 FX -38 FY -56 FZ -5
85. 4 FX -32 FY -135 FZ 3
86. 5 FX -22 FY 97 FZ -7
87. ********************************************************************************
88. LOAD 8 WIND LOAD (+Z) WL(+Z)
89. ********************************************************************************
90. JOINT LOAD
91. 1 FX -1 FY 42 FZ 15
92. 2 FX 1 FY 48 FZ 29
```

```
 93. 4 FX 2 FY -33 FZ 15
 94. 5 FX 2 FY -36 FZ 24
 95. ********************************************************************************
 96. LOAD 9 WIND LOAD (-Z) WL(-Z)
 97. ********************************************************************************
 98. JOINT LOAD
 99. 1 FX 1 FY -42 FZ -15
100. 2 FX -1 FY -48 FZ -29
101. 4 FX -2 FY 33 FZ -15
102. 5 FX -2 FY 36 FZ -24
103. ********************************************************************************
104. LOAD 10 SEISMIC LOAD SL(+X)
105. ********************************************************************************
106. JOINT LOAD
107. 1 FX 1 FY 17 FZ 3
108. 2 FX 23 FY 48 FZ 6
109. 4 FX 17 FY 76 FZ -2
110. 5 FX 14 FY -57 FZ 6
111. ********************************************************************************
112. LOAD 11 SEISMIC LOAD SL(-X)
113. ********************************************************************************
114. JOINT LOAD
115. 1 FX -1 FY -17 FZ -3
116. 2 FX -23 FY -48 FZ -6
117. 4 FX -17 FY -76 FZ 2
118. 5 FX -14 FY 57 FZ -6
119. ********************************************************************************
120. LOAD 12 SEISMIC LOAD SL(+Z)
121. ********************************************************************************
122. JOINT LOAD
123. 1 FX -1 FY 14 FZ 4
124. 2 FX 2 FY 35 FZ 19
125. 4 FX 1 FY -11 FZ 5
126. 5 FX 1 FY -21 FZ 15
127. ********************************************************************************
128. LOAD 13 SEISMIC LOAD SL(-Z)
129. ********************************************************************************
130. JOINT LOAD
131. 1 FX 1 FY -14 FZ -4
132. 2 FX -2 FY -35 FZ -19
133. 4 FX -1 FY 11 FZ -5
134. 5 FX -1 FY 21 FZ -15
135. ********************************************************************************
136. LOAD 14 TEMPERATURE LOAD (TL)
137. JOINT LOAD
138. 1 FX -2 FY 1 FZ -24
139. 2 FX -29 FY 8 FZ -30
140. 4 FX -26 FY -10 FZ 24
141. 5 FX 23 FY 4 FZ 32
142. ********************************************************************************
143. ********************************************************************************
144. LOAD 15 EQP TEST LOAD (EQTL)
145. ********************************************************************************
146. JOINT LOAD
147. 1 FX 0 FY -2 FZ -1.
148. 2 FX -2 FY -22 FZ -4
```

```
149. 4 FX -1 FY -6 FZ 1
150. 5 FX -1 FY -36 FZ 6
151. ********************************************************************************
152. ****************************************************
153. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
154. ****************************************************
155. LOAD COMB 201 DL+EQDL+FL
156. 1 1.0 4 1.0 3 1.0
157. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
158. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
159. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
160. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
161. LOAD COMB 204 DL+EQDL+FL+WL(+X)
162. 1 1.0 4 1.0 3 1.0 6 1.0
163. LOAD COMB 205 DL+EQDL+FL+WL(-X)
164. 1 1.0 4 1.0 3 1.0 7 1.0
165. LOAD COMB 206 DL+EQDL+FL+WL(+Z)
166. 1 1.0 4 1.0 3 1.0 8 1.0
167. LOAD COMB 207 DL+EQDL+FL+WL(-Z)
168. 1 1.0 4 1.0 3 1.0 9 1.0
169. LOAD COMB 208 DL+EQDL+FL+0.7SL(+X)
170. 1 1.0 4 1.0 3 1.0 10 0.7
171. LOAD COMB 209 DL+EQDL+FL+0.7SL(-X)
172. 1 1.0 4 1.0 3 1.0 11 0.7
173. LOAD COMB 210 DL+EQDL+FL+0.7SL(+Z)
174. 1 1.0 4 1.0 3 1.0 12 0.7
175. LOAD COMB 211 DL+EQDL+FL+0.7SL(-Z)
176. 1 1.0 4 1.0 3 1.0 13 0.7
177. LOAD COMB 212 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
178. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
179. LOAD COMB 213 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
180. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
181. LOAD COMB 214 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
182. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
183. LOAD COMB 215 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
184. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
185. LOAD COMB 216 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
186. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
187. LOAD COMB 217 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
188. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
189. LOAD COMB 218 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
190. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
191. LOAD COMB 219 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
192. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
193. LOAD COMB 220 0.6(DL+EQDL)+WL(+X)
194. 1 0.6 4 0.6 6 1.0
195. LOAD COMB 221 0.6(DL+EQDL)+WL(-X)
196. 1 0.6 4 0.6 7 1.0
197. LOAD COMB 222 0.6(DL+EQDL)+WL(+Z)
198. 1 0.6 4 0.6 8 1.0
199. LOAD COMB 223 0.6(DL+EQDL)+WL(-Z)
200. 1 0.6 4 0.6 9 1.0
201. LOAD COMB 224 0.6(DL+EQDL)+0.7SL(+X)
202. 1 0.6 4 0.6 10 0.7
203. LOAD COMB 225 0.6(DL+EQDL)+0.7SL(-X)
204. 1 0.6 4 0.6 11 0.7
```

```
205. LOAD COMB 226 0.6(DL+EQDL)+0.7SL(+Z)
206. 1 0.6 4 0.6 12 0.7
207. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(-Z)
208. 1 0.6 4 0.6 13 0.7
209. LOAD COMB 228 DL+EQTL
210. 1 1.0 15 1.0
211. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**8**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 4/ 4/ 6 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 28 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 99949.8 MB

```
212. ********************************************************************************
213. FINISH

           *********** END OF THE STAAD.Pro RUN ***********

             **** DATE= MAY 21,2020   TIME= 15:12: 1 ****
```

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE FOR F2 & F3 DESIGN

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    15:12:37                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2 _F3 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 24-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. *SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 3 -202.5 913.6 -435.9; 6 -209.5 913.6 -426.2; 7 -202.5 913.6 -426.2
    10. 161 -202.5 911 -435.9; 162 -209.5 911 -426.2; 163 -202.5 911 -426.2
    11. 164 -202.5 911.25 -435.9; 165 -209.5 911.25 -426.2; 166 -202.5 911.25 -426.2
    12. MEMBER INCIDENCES
    13. 357 3 164; 358 6 165; 359 7 166; 360 164 161; 361 165 162; 362 166 163
    14. START GROUP DEFINITION
    15. MEMBER
    16. _C2 358
    17. _C3 357 359
    18. END GROUP DEFINITION
    19. DEFINE MATERIAL START
    20. ISOTROPIC CONCRETE
    21. E 2.17185E+007
    22. POISSON 0.17
    23. DENSITY 25
    24. ALPHA 1E-005
    25. DAMP 0.05
    26. TYPE CONCRETE
    27. STRENGTH FCU 27579
    28. END DEFINE MATERIAL
    29. MEMBER PROPERTY AMERICAN
    30. 358 361 PRIS YD 0.5 ZD 0.5
    31. MEMBER PROPERTY
    32. 357 359 360 362 PRIS YD 0.65 ZD 0.65
    33. CONSTANTS
    34. MATERIAL CONCRETE ALL
    35. SUPPORTS
    36. 161 TO 163 FIXED
    37. ******************************************************************************
    38. LOAD 1 DEAD LOAD (DL)
```

**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
MASTER/SLAVE OR IF UNCONNECTED JOINTS.

```
39. ********************************************************************************
40. JOINT LOAD
41. 3 FX 7 FY -60 FZ -19
42. 6 FX -5 FY -35 FZ 1
43. 7 FX 5 FY -56 FZ 18
44. SELFWEIGHT Y -1
45. ********************************************************************************
46. LOAD 2 LIVE LOAD (LL)
47. ********************************************************************************
48. JOINT LOAD
49. 3 FX 13 FY -102 FZ -29
50. 6 FX -6 FY -63 FZ 1
51. 7 FX 6 FY -70 FZ 25
52. ********************************************************************************
53. LOAD 3 OPERATING FLUID LOAD (FL)
54. ********************************************************************************
55. JOINT LOAD
56. 3 FX 9 FY -153 FZ -49
57. 6 FX -10 FY -63 FZ 1
58. 7 FX 10 FY -151 FZ 48
59. ********************************************************************************
60. LOAD 4 EQP DL (EQDL)
61. ********************************************************************************
62. JOINT LOAD
63. 3 FX 4 FY -67 FZ -25
64. 6 FX -6 FY -38 FZ 1
65. 7 FX 6 FY -85 FZ 24
66. ********************************************************************************
67. LOAD 5 EQP LL (EQLL)
68. ********************************************************************************
69. JOINT LOAD
70. 3 FX 8 FY -131 FZ -30
71. 6 FX -2 FY -8 FZ 1
72. 7 FX 2 FY -48 FZ 30
73. ********************************************************************************
74. LOAD 6 WIND LOAD (+X) WL(+X)
75. ********************************************************************************
76. JOINT LOAD
77. 3 FX 42 FY -101 FZ -16
78. 6 FX 7 FY -33 FZ 1
79. 7 FX 10 FY 6 FZ 2
80. ********************************************************************************
81. LOAD 7 WIND LOAD (-X) WL(-X)
82. ********************************************************************************
83. JOINT LOAD
84. 3 FX -42 FY 101 FZ 16
85. 6 FX -7 FY 33 FZ -1.
86. 7 FX -10 FY -6 FZ -2
87. ********************************************************************************
88. LOAD 8 WIND LOAD (+Z) WL(+Z)
89. ********************************************************************************
90. JOINT LOAD
91. 3 FX -4 FY 28 FZ 18
92. 6 FX -2 FY -26 FZ 2
```

```
 93. 7 FX 2 FY -23 FZ 16
 94. ************************************************************************
 95. LOAD 9 WIND LOAD (-Z) WL(-Z)
 96. ************************************************************************
 97. JOINT LOAD
 98. 3 FX 4 FY -28 FZ -18
 99. 6 FX 2 FY 26 FZ -2
100. 7 FX -2 FY 23 FZ -16
101. ************************************************************************
102. LOAD 10 SEISMIC LOAD SL(+X)
103. ************************************************************************
104. JOINT LOAD
105. 3 FX 24 FY -70 FZ -12
106. 6 FX 2 FY -22 FZ 1
107. 7 FX 4 FY 8 FZ -1.
108. ************************************************************************
109. LOAD 11 SEISMIC LOAD SL(-X)
110. ************************************************************************
111. JOINT LOAD
112. 3 FX -24 FY 70 FZ 12
113. 6 FX -2 FY 22 FZ -1.
114. 7 FX -4 FY -8 FZ 1
115. ************************************************************************
116. LOAD 12 SEISMIC LOAD SL(+Z)
117. ************************************************************************
118. JOINT LOAD
119. 3 FX -3 FY 39 FZ 22
120. 6 FX -3 FY -20 FZ 1
121. 7 FX 3 FY -37 FZ 20
122. ************************************************************************
123. LOAD 13 SEISMIC LOAD SL(-Z)
124. ************************************************************************
125. JOINT LOAD
126. 3 FX 3 FY -39 FZ -22
127. 6 FX 3 FY 20 FZ -1.
128. 7 FX -3 FY 37 FZ -20
129. ************************************************************************
130. LOAD 14 TEMPERATURE LOAD (TL)
131. JOINT LOAD
132. 3 FX 33 FY -7 FZ -52
133. 6 FX -30 FY 2 FZ 1
134. 7 FX 30 FY 3 FZ 49
135. ************************************************************************
136. ************************************************************************
137. LOAD 15 EQP TEST LOAD (EQTL)
138. ************************************************************************
139. JOINT LOAD
140. 3 FX 4 FY -63 FZ -32
141. 6 FX -11 FY -71 FZ 1
142. 7 FX 11 FY -143 FZ 31
143. ************************************************************************
144. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
145. ************************************************************************
146. LOAD COMB 101 1.4(DL+EQDL+FL)
147. 1 1.4 4 1.4 3 1.4
148. LOAD COMB 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
```

```
149. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
150. LOAD COMB 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
151. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
152. LOAD COMB 104 1.2(DL+EQDL)+0.8WL(+X)
153. 1 1.2 4 1.2 6 0.8
154. LOAD COMB 105 1.2(DL+EQDL)+0.8WL(-X)
155. 1 1.2 4 1.2 7 0.8
156. LOAD COMB 106 1.2(DL+EQDL)+0.8WL(+Z)
157. 1 1.2 4 1.2 8 0.8
158. LOAD COMB 107 1.2(DL+EQDL)+0.8WL(-Z)
159. 1 1.2 4 1.2 9 0.8
160. LOAD COMB 108 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
161. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
162. LOAD COMB 109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
163. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
164. LOAD COMB 110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
165. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
166. LOAD COMB 111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
167. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
168. LOAD COMB 112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
169. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
170. LOAD COMB 113 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
171. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
172. LOAD COMB 114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
173. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
174. LOAD COMB 115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
175. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
176. LOAD COMB 116 0.9(DL+EQDL)+1.6WL(+X)
177. 1 0.9 4 0.9 6 1.6
178. LOAD COMB 117 0.9(DL+EQDL)+1.6WL(-X)
179. 1 0.9 4 0.9 7 1.6
180. LOAD COMB 118 0.9(DL+EQDL)+1.6WL(+Z)
181. 1 0.9 4 0.9 8 1.6
182. LOAD COMB 119 0.9(DL+EQDL)+1.6WL(-Z)
183. 1 0.9 4 0.9 9 1.6
184. LOAD COMB 120 0.9(DL+EQDL)+SL(+X)
185. 1 0.9 4 0.9 10 1.0
186. LOAD COMB 121 0.9(DL+EQDL)+SL(-X)
187. 1 0.9 4 0.9 11 1.0
188. LOAD COMB 122 0.9(DL+EQDL)+SL(+Z)
189. 1 0.9 4 0.9 12 1.0
190. LOAD COMB 123 0.9(DL+EQDL)+SL(-Z)
191. 1 0.9 4 0.9 13 1.0
192. LOAD COMB 124 1.2DL+EQTL
193. 1 1.2 15 1.0
194. ******************************************************
195. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
196. ******************************************************
197. LOAD COMB 201 DL+EQDL+FL
198. 1 1.0 4 1.0 3 1.0
199. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
200. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
201. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
202. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
203. LOAD COMB 204 DL+EQDL+FL+WL(+X)
204. 1 1.0 4 1.0 3 1.0 6 1.0
```

```
205. LOAD COMB 205 DL+EQDL+FL+WL(-X)
206. 1 1.0 4 1.0 3 1.0 7 1.0
207. LOAD COMB 206 DL+EQDL+FL+WL(+Z)
208. 1 1.0 4 1.0 3 1.0 8 1.0
209. LOAD COMB 207 DL+EQDL+FL+WL(-Z)
210. 1 1.0 4 1.0 3 1.0 9 1.0
211. LOAD COMB 208 DL+EQDL+FL+0.7SL(+X)
212. 1 1.0 4 1.0 3 1.0 10 0.7
213. LOAD COMB 209 DL+EQDL+FL+0.7SL(-X)
214. 1 1.0 4 1.0 3 1.0 11 0.7
215. LOAD COMB 210 DL+EQDL+FL+0.7SL(+Z)
216. 1 1.0 4 1.0 3 1.0 12 0.7
217. LOAD COMB 211 DL+EQDL+FL+0.7SL(-Z)
218. 1 1.0 4 1.0 3 1.0 13 0.7
219. LOAD COMB 212 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
220. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
221. LOAD COMB 213 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
222. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
223. LOAD COMB 214 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
224. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
225. LOAD COMB 215 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
226. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
227. LOAD COMB 216 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
228. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
229. LOAD COMB 217 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
230. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
231. LOAD COMB 218 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
232. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
233. LOAD COMB 219 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
234. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
235. LOAD COMB 220 0.6(DL+EQDL)+WL(+X)
236. 1 0.6 4 0.6 6 1.0
237. LOAD COMB 221 0.6(DL+EQDL)+WL(-X)
238. 1 0.6 4 0.6 7 1.0
239. LOAD COMB 222 0.6(DL+EQDL)+WL(+Z)
240. 1 0.6 4 0.6 8 1.0
241. LOAD COMB 223 0.6(DL+EQDL)+WL(-Z)
242. 1 0.6 4 0.6 9 1.0
243. LOAD COMB 224 0.6(DL+EQDL)+0.7SL(+X)
244. 1 0.6 4 0.6 10 0.7
245. LOAD COMB 225 0.6(DL+EQDL)+0.7SL(-X)
246. 1 0.6 4 0.6 11 0.7
247. LOAD COMB 226 0.6(DL+EQDL)+0.7SL(+Z)
248. 1 0.6 4 0.6 12 0.7
249. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(-Z)
250. 1 0.6 4 0.6 13 0.7
251. LOAD COMB 228 DL+EQTL
252. 1 1.0 15 1.0
253. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 9 NUMBER OF MEMBERS 6
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 3

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 6/ 1/ 12 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 36
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 99947.1 MB

254. **
255. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 15:12:47 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 4 STAAD INPUT & OUTPUT FILE FOR F4 DESIGN

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    14:37: 6                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F4 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 29-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 2 -209.5 913.7 -435.9; 162 -209.5 911 -435.9; 580 -209.5 911.15 -435.9
    10. MEMBER INCIDENCES
    11. 2 162 580; 462 580 2
    12. START GROUP DEFINITION
    13. MEMBER
    14. _C5 462
    15. _C4 462
    16. ELEMENT
    17. END GROUP DEFINITION
    18. DEFINE MATERIAL START
    19. ISOTROPIC CONCRETE
    20. E 2.17185E+007
    21. POISSON 0.17
    22. DENSITY 25
    23. ALPHA 1E-005
    24. DAMP 0.05
    25. TYPE CONCRETE
    26. STRENGTH FCU 27579
    27. END DEFINE MATERIAL
    28. MEMBER PROPERTY AMERICAN
    29. 2 462 PRIS YD 0.5 ZD 0.5
    30. CONSTANTS
    31. MATERIAL CONCRETE ALL
    32. ******************************************************************************
    33. SUPPORTS
    34. 162 FIXED
    35. LOAD 1 DEAD LOAD (DL)
    36. ******************************************************************************
    37. JOINT LOAD
    38. 2 FX 0 FY 0 FZ 0
```

```
39. SELFWEIGHT Y -1
40. ********************************************************************************
41. LOAD 2 LIVE LOAD (LL)
42. ********************************************************************************
43. JOINT LOAD
44. 2 FX 0 FY 0 FZ 0
45. ********************************************************************************
46. LOAD 3 OPERATING FLUID LOAD (FL)
47. ********************************************************************************
48. JOINT LOAD
49. 2 FY -170
50. ********************************************************************************
51. LOAD 4 EQP DL (EQDL)
52. ********************************************************************************
53. JOINT LOAD
54. 2 FY -21
55. ********************************************************************************
56. LOAD 5 EQP LL (EQLL)
57. ********************************************************************************
58. JOINT LOAD
59. 2 FX 0 FY 0 FZ 0
60. ********************************************************************************
61. LOAD 6 WIND LOAD (+X) WL(+X)
62. ********************************************************************************
63. JOINT LOAD
64. 2 FX 5.3
65. ********************************************************************************
66. LOAD 7 WIND LOAD (-X) WL(-X)
67. ********************************************************************************
68. JOINT LOAD
69. 2 FX -5.3
70. ********************************************************************************
71. LOAD 8 WIND LOAD (+Z) WL(+Z)
72. ********************************************************************************
73. JOINT LOAD
74. 2 FZ 5.3
75. ********************************************************************************
76. LOAD 9 WIND LOAD (-Z) WL(-Z)
77. ********************************************************************************
78. JOINT LOAD
79. 2 FZ -5.3
80. ********************************************************************************
81. LOAD 10 SEISMIC LOAD SL(+X)
82. ********************************************************************************
83. JOINT LOAD
84. 2 FX 10
85. ********************************************************************************
86. LOAD 11 SEISMIC LOAD SL(-X)
87. ********************************************************************************
88. JOINT LOAD
89. 2 FX -10
90. ********************************************************************************
91. LOAD 12 SEISMIC LOAD SL(+Z)
92. ********************************************************************************
93. JOINT LOAD
94. 2 FZ 10
```

```
 95. ********************************************************************************
 96. LOAD 13 SEISMIC LOAD SL(-Z)
 97. ********************************************************************************
 98. JOINT LOAD
 99. 2 FZ -10
100. ********************************************************************************
101. LOAD 14 TEMPERATURE LOAD (TL)
102. JOINT LOAD
103. 2 FX 0 FY 0 FZ 0
104. ********************************************************************************
105. ********************************************************************************
106. LOAD 15 EQP TEST LOAD (EQTL)
107. ********************************************************************************
108. JOINT LOAD
109. 2 FY -191
110. ********************************************************************************
111. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
112. ********************************************************************************
113. LOAD COMB 101 1.4(DL+EQDL+FL)
114. 1 1.4 4 1.4 3 1.4
115. LOAD COMB 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
116. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
117. LOAD COMB 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
118. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
119. LOAD COMB 104 1.2(DL+EQDL)+0.8WL(+X)
120. 1 1.2 4 1.2 6 0.8
121. LOAD COMB 105 1.2(DL+EQDL)+0.8WL(-X)
122. 1 1.2 4 1.2 7 0.8
123. LOAD COMB 106 1.2(DL+EQDL)+0.8WL(+Z)
124. 1 1.2 4 1.2 8 0.8
125. LOAD COMB 107 1.2(DL+EQDL)+0.8WL(-Z)
126. 1 1.2 4 1.2 9 0.8
127. LOAD COMB 108 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
128. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
129. LOAD COMB 109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
130. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
131. LOAD COMB 110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
132. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
133. LOAD COMB 111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
134. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
135. LOAD COMB 112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
136. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
137. LOAD COMB 113 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
138. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
139. LOAD COMB 114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
140. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
141. LOAD COMB 115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
142. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
143. LOAD COMB 116 0.9(DL+EQDL)+1.6WL(+X)
144. 1 0.9 4 0.9 6 1.6
145. LOAD COMB 117 0.9(DL+EQDL)+1.6WL(-X)
146. 1 0.9 4 0.9 7 1.6
147. LOAD COMB 118 0.9(DL+EQDL)+1.6WL(+Z)
148. 1 0.9 4 0.9 8 1.6
149. LOAD COMB 119 0.9(DL+EQDL)+1.6WL(-Z)
150. 1 0.9 4 0.9 9 1.6
```

```
151. LOAD COMB 120 0.9(DL+EQDL)+SL(+X)
152. 1 0.9 4 0.9 10 1.0
153. LOAD COMB 121 0.9(DL+EQDL)+SL(-X)
154. 1 0.9 4 0.9 11 1.0
155. LOAD COMB 122 0.9(DL+EQDL)+SL(+Z)
156. 1 0.9 4 0.9 12 1.0
157. LOAD COMB 123 0.9(DL+EQDL)+SL(-Z)
158. 1 0.9 4 0.9 13 1.0
159. LOAD COMB 124 1.2DL+EQTL
160. 1 1.2 15 1.0
161. ******************************************************
162. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
163. ******************************************************
164. LOAD COMB 201 DL+EQDL+FL
165. 1 1.0 4 1.0 3 1.0
166. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
167. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
168. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
169. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
170. LOAD COMB 204 DL+EQDL+FL+WL(+X)
171. 1 1.0 4 1.0 3 1.0 6 1.0
172. LOAD COMB 205 DL+EQDL+FL+WL(-X)
173. 1 1.0 4 1.0 3 1.0 7 1.0
174. LOAD COMB 206 DL+EQDL+FL+WL(+Z)
175. 1 1.0 4 1.0 3 1.0 8 1.0
176. LOAD COMB 207 DL+EQDL+FL+WL(-Z)
177. 1 1.0 4 1.0 3 1.0 9 1.0
178. LOAD COMB 208 DL+EQDL+FL+0.7SL(+X)
179. 1 1.0 4 1.0 3 1.0 10 0.7
180. LOAD COMB 209 DL+EQDL+FL+0.7SL(-X)
181. 1 1.0 4 1.0 3 1.0 11 0.7
182. LOAD COMB 210 DL+EQDL+FL+0.7SL(+Z)
183. 1 1.0 4 1.0 3 1.0 12 0.7
184. LOAD COMB 211 DL+EQDL+FL+0.7SL(-Z)
185. 1 1.0 4 1.0 3 1.0 13 0.7
186. LOAD COMB 212 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
187. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
188. LOAD COMB 213 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
189. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
190. LOAD COMB 214 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
191. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
192. LOAD COMB 215 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
193. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
194. LOAD COMB 216 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
195. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
196. LOAD COMB 217 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
197. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
198. LOAD COMB 218 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
199. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
200. LOAD COMB 219 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
201. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
202. LOAD COMB 220 0.6(DL+EQDL)+WL(+X)
203. 1 0.6 4 0.6 6 1.0
204. LOAD COMB 221 0.6(DL+EQDL)+WL(-X)
205. 1 0.6 4 0.6 7 1.0
206. LOAD COMB 222 0.6(DL+EQDL)+WL(+Z)
```

```
207. 1 0.6 4 0.6 8 1.0
208. LOAD COMB 223 0.6(DL+EQDL)+WL(-Z)
209. 1 0.6 4 0.6 9 1.0
210. LOAD COMB 224 0.6(DL+EQDL)+0.7SL(+X)
211. 1 0.6 4 0.6 10 0.7
212. LOAD COMB 225 0.6(DL+EQDL)+0.7SL(-X)
213. 1 0.6 4 0.6 11 0.7
214. LOAD COMB 226 0.6(DL+EQDL)+0.7SL(+Z)
215. 1 0.6 4 0.6 12 0.7
216. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(-Z)
217. 1 0.6 4 0.6 13 0.7
218. LOAD COMB 228 DL+EQTL
219. 1 1.0 15 1.0
220. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**3**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**1**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 12 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 99979.9 MB

```
221. ********************************************************************************
222. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 14:37:13 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 5 STAAD INPUT & OUTPUT FILE FOR F5 DESIGN

PROPRIETARY & CONFIDENTIAL

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    14:28:11                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F5 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 29-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 2 -209.5 913.7 -435.9; 5 -209.5 913.7 -432.4; 162 -209.5 911 -435.9
    10. 164 -209.5 911 -432.4; 580 -209.5 911.15 -435.9; 582 -209.5 911.15 -432.4
    11. 583 -209.5 911 -436.4; 584 -209.5 911 -431.9; 585 -210.25 911 -435.9
    12. 586 -210.25 911 -432.4; 587 -210.25 911 -436.4; 588 -210.25 911 -431.9
    13. 589 -208.75 911 -435.9; 590 -208.75 911 -432.4; 591 -208.75 911 -436.4
    14. 592 -208.75 911 -431.9; 593 -210 911 -431.9; 594 -210 911 -432.15
    15. 595 -210.25 911 -432.15; 596 -209.75 911 -431.9; 597 -209.75 911 -432.15
    16. 598 -209.5 911 -432.15; 599 -210 911 -432.4; 600 -209.75 911 -432.4
    17. 601 -209.25 911 -431.9; 602 -209.25 911 -432.15; 603 -209 911 -431.9
    18. 604 -209 911 -432.15; 605 -208.75 911 -432.15; 606 -209.25 911 -432.4
    19. 607 -209 911 -432.4; 608 -210 911 -435.9; 609 -210 911 -436.15
    20. 610 -210.25 911 -436.15; 611 -209.75 911 -435.9; 612 -209.75 911 -436.15
    21. 613 -209.5 911 -436.15; 614 -210 911 -436.4; 615 -209.75 911 -436.4
    22. 616 -209.25 911 -435.9; 617 -209.25 911 -436.15; 618 -209 911 -435.9
    23. 619 -209 911 -436.15; 620 -208.75 911 -436.15; 621 -209.25 911 -436.4
    24. 622 -209 911 -436.4; 623 -210 911 -432.65; 624 -210.25 911 -432.65
    25. 625 -209.75 911 -432.65; 626 -209.5 911 -432.65; 627 -210 911 -432.9
    26. 628 -210.25 911 -432.9; 629 -209.75 911 -432.9; 630 -209.5 911 -432.9
    27. 631 -210 911 -433.15; 632 -210.25 911 -433.15; 633 -209.75 911 -433.15
    28. 634 -209.5 911 -433.15; 635 -210 911 -433.4; 636 -210.25 911 -433.4
    29. 637 -209.75 911 -433.4; 638 -209.5 911 -433.4; 639 -210 911 -433.65
    30. 640 -210.25 911 -433.65; 641 -209.75 911 -433.65; 642 -209.5 911 -433.65
    31. 643 -210 911 -433.9; 644 -210.25 911 -433.9; 645 -209.75 911 -433.9
    32. 646 -209.5 911 -433.9; 647 -210 911 -434.15; 648 -210.25 911 -434.15
    33. 649 -209.75 911 -434.15; 650 -209.5 911 -434.15; 651 -210 911 -434.4
    34. 652 -210.25 911 -434.4; 653 -209.75 911 -434.4; 654 -209.5 911 -434.4
    35. 655 -210 911 -434.65; 656 -210.25 911 -434.65; 657 -209.75 911 -434.65
    36. 658 -209.5 911 -434.65; 659 -210 911 -434.9; 660 -210.25 911 -434.9
    37. 661 -209.75 911 -434.9; 662 -209.5 911 -434.9; 663 -210 911 -435.15
    38. 664 -210.25 911 -435.15; 665 -209.75 911 -435.15; 666 -209.5 911 -435.15
```

```
39. 667 -210 911 -435.4; 668 -210.25 911 -435.4; 669 -209.75 911 -435.4
40. 670 -209.5 911 -435.4; 671 -210 911 -435.65; 672 -210.25 911 -435.65
41. 673 -209.75 911 -435.65; 674 -209.5 911 -435.65; 675 -209.25 911 -432.65
42. 676 -209 911 -432.65; 677 -208.75 911 -432.65; 678 -209.25 911 -432.9
43. 679 -209 911 -432.9; 680 -208.75 911 -432.9; 681 -209.25 911 -433.15
44. 682 -209 911 -433.15; 683 -208.75 911 -433.15; 684 -209.25 911 -433.4
45. 685 -209 911 -433.4; 686 -208.75 911 -433.4; 687 -209.25 911 -433.65
46. 688 -209 911 -433.65; 689 -208.75 911 -433.65; 690 -209.25 911 -433.9
47. 691 -209 911 -433.9; 692 -208.75 911 -433.9; 693 -209.25 911 -434.15
48. 694 -209 911 -434.15; 695 -208.75 911 -434.15; 696 -209.25 911 -434.4
49. 697 -209 911 -434.4; 698 -208.75 911 -434.4; 699 -209.25 911 -434.65
50. 700 -209 911 -434.65; 701 -208.75 911 -434.65; 702 -209.25 911 -434.9
51. 703 -209 911 -434.9; 704 -208.75 911 -434.9; 705 -209.25 911 -435.15
52. 706 -209 911 -435.15; 707 -208.75 911 -435.15; 708 -209.25 911 -435.4
53. 709 -209 911 -435.4; 710 -208.75 911 -435.4; 711 -209.25 911 -435.65
54. 712 -209 911 -435.65; 713 -208.75 911 -435.65
55. MEMBER INCIDENCES
56. 2 162 580; 4 164 582; 462 580 2; 464 582 5
57. ELEMENT INCIDENCES SHELL
58. 485 588 593 594 595; 487 593 596 597 594; 489 596 584 598 597
59. 491 595 594 599 586; 493 594 597 600 599; 494 597 598 164 600
60. 496 584 601 602 598; 498 601 603 604 602; 500 603 592 605 604
61. 502 598 602 606 164; 504 602 604 607 606; 505 604 605 590 607
62. 508 585 608 609 610; 510 608 611 612 609; 512 611 162 613 612
63. 514 610 609 614 587; 516 609 612 615 614; 517 612 613 583 615
64. 519 162 616 617 613; 521 616 618 619 617; 523 618 589 620 619
65. 525 613 617 621 583; 527 617 619 622 621; 528 619 620 591 622
66. 531 586 599 623 624; 532 599 600 625 623; 534 600 164 626 625
67. 536 624 623 627 628; 537 623 625 629 627; 539 625 626 630 629
68. 541 628 627 631 632; 542 627 629 633 631; 544 629 630 634 633
69. 546 632 631 635 636; 547 631 633 637 635; 549 633 634 638 637
70. 551 636 635 639 640; 552 635 637 641 639; 554 637 638 642 641
71. 556 640 639 643 644; 557 639 641 645 643; 559 641 642 646 645
72. 561 644 643 647 648; 562 643 645 649 647; 564 645 646 650 649
73. 566 648 647 651 652; 567 647 649 653 651; 569 649 650 654 653
74. 571 652 651 655 656; 572 651 653 657 655; 574 653 654 658 657
75. 576 656 655 659 660; 577 655 657 661 659; 579 657 658 662 661
76. 581 660 659 663 664; 582 659 661 665 663; 584 661 662 666 665
77. 586 664 663 667 668; 587 663 665 669 667; 589 665 666 670 669
78. 591 668 667 671 672; 592 667 669 673 671; 594 669 670 674 673
79. 595 672 671 608 585; 596 671 673 611 608; 597 673 674 162 611
80. 598 164 606 675 626; 599 606 607 676 675; 601 607 590 677 676
81. 602 626 675 678 630; 603 675 676 679 678; 605 676 677 680 679
82. 606 630 678 681 634; 607 678 679 682 681; 609 679 680 683 682
83. 610 634 681 684 638; 611 681 682 685 684; 613 682 683 686 685
84. 614 638 684 687 642; 615 684 685 688 687; 617 685 686 689 688
85. 618 642 687 690 646; 619 687 688 691 690; 621 688 689 692 691
86. 622 646 690 693 650; 623 690 691 694 693; 625 691 692 695 694
87. 626 650 693 696 654; 627 693 694 697 696; 629 694 695 698 697
88. 630 654 696 699 658; 631 696 697 700 699; 633 697 698 701 700
89. 634 658 699 702 662; 635 699 700 703 702; 637 700 701 704 703
90. 638 662 702 705 666; 639 702 703 706 705; 641 703 704 707 706
91. 642 666 705 708 670; 643 705 706 709 708; 645 706 707 710 709
92. 646 670 708 711 674; 647 708 709 712 711; 649 709 710 713 712
93. 650 674 711 616 162; 651 711 712 618 616; 652 712 713 589 618
94. START GROUP DEFINITION
```

```
 95. MEMBER
 96. _C5 462 464
 97. ELEMENT
 98. _RAFT 485 487 489 491 493 494 496 498 500 502 504 505 508 510 512 514 516 -
 99. 517 519 521 523 525 527 528 531 532 534 536 537 539 541 542 544 546 547 549 -
100. 551 552 554 556 557 559 561 562 564 566 567 569 571 572 574 576 577 579 581 -
101. 582 584 586 587 589 591 592 594 TO 599 601 TO 603 605 TO 607 609 TO 611 613 -
102. 614 TO 615 617 TO 619 621 TO 623 625 TO 627 629 TO 631 633 TO 635 637 TO 639 -
103. 641 TO 643 645 TO 647 649 TO 652
104. _RAFT_MOM 485 487 489 491 493 496 498 500 504 505 508 510 514 516 517 521 -
105. 523 525 527 528 531 532 536 537 539 541 542 544 546 547 549 551 552 554 556 -
106. 557 559 561 562 564 566 567 569 571 572 574 576 577 579 581 582 584 586 587 -
107. 589 591 592 594 TO 596 599 601 TO 603 605 TO 607 609 TO 611 613 TO 615 617 -
108. 618 TO 619 621 TO 623 625 TO 627 629 TO 631 633 TO 635 637 TO 639 641 TO 643 -
109. 645 TO 647 649 651 652
110. _RAFT_SHEAR 485 491 500 505 508 514 523 528 531 536 541 542 544 546 547 549 -
111. 551 552 554 556 557 559 561 562 564 566 567 569 571 572 574 576 577 579 581 -
112. 582 584 586 587 589 591 595 601 605 TO 607 609 TO 611 613 TO 615 617 TO 619 -
113. 621 TO 623 625 TO 627 629 TO 631 633 TO 635 637 TO 639 641 TO 643 645 649 -
114. 652
115. END GROUP DEFINITION
116. ELEMENT PROPERTY
117. 485 487 489 491 493 494 496 498 500 502 504 505 508 510 512 514 516 517 519 -
118. 521 523 525 527 528 531 532 534 536 537 539 541 542 544 546 547 549 551 552 -
119. 554 556 557 559 561 562 564 566 567 569 571 572 574 576 577 579 581 582 584 -
120. 586 587 589 591 592 594 TO 599 601 TO 603 605 TO 607 609 TO 611 613 TO 615 -
121. 617 TO 619 621 TO 623 625 TO 627 629 TO 631 633 TO 635 637 TO 639 -
122. 641 TO 643 645 TO 647 649 TO 652 THICKNESS 0.35
123. DEFINE MATERIAL START
124. ISOTROPIC CONCRETE
125. E 2.17185E+007
126. POISSON 0.17
127. DENSITY 25
128. ALPHA 1E-005
129. DAMP 0.05
130. TYPE CONCRETE
131. STRENGTH FCU 27579
132. END DEFINE MATERIAL
133. MEMBER PROPERTY AMERICAN
134. 2 4 462 464 PRIS YD 0.5 ZD 0.5
135. CONSTANTS
136. MATERIAL CONCRETE ALL
137. ******************************************************************************
138. SUPPORTS
139. 485 487 489 491 493 494 496 498 500 502 504 505 508 510 512 514 516 517 519 -
140. 521 523 525 527 528 531 532 534 536 537 539 541 542 544 546 547 549 551 552 -
141. 554 556 557 559 561 562 564 566 567 569 571 572 574 576 577 579 581 582 584 -
142. 586 587 589 591 592 594 TO 599 601 TO 603 605 TO 607 609 TO 611 613 TO 615 -
143. 617 TO 619 621 TO 623 625 TO 627 629 TO 631 633 TO 635 637 TO 639 -
144. 641 TO 643 645 TO 647 649 TO 652 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
```

JOINT	INFLUENCE AREA
162	0.062500
164	0.062500

583	0.031247
584	0.031247
585	0.031252
586	0.031252
587	0.015624
588	0.015624
589	0.031252
590	0.031252
591	0.015624
592	0.015624
593	0.031249
594	0.062503
595	0.031252
596	0.031249
597	0.062503
598	0.062500
599	0.062503
600	0.062503
601	0.031247
602	0.062500
603	0.031249
604	0.062503
605	0.031252
606	0.062500
607	0.062503
608	0.062503
609	0.062497
610	0.031249
611	0.062503
612	0.062497
613	0.062494
614	0.031249
615	0.031249
616	0.062500
617	0.062494
618	0.062503
619	0.062497
620	0.031249
621	0.031247
622	0.031249
623	0.062497
624	0.031249
625	0.062497
626	0.062494
627	0.062503
628	0.031252
629	0.062503
630	0.062500
631	0.062503
632	0.031252
633	0.062503
634	0.062500
635	0.062497
636	0.031249
637	0.062497
638	0.062494

639	0.062503
640	0.031252
641	0.062503
642	0.062500
643	0.062503
644	0.031252
645	0.062503
646	0.062500
647	0.062503
648	0.031252
649	0.062503
650	0.062500
651	0.062503
652	0.031252
653	0.062503
654	0.062500
655	0.062497
656	0.031249
657	0.062497
658	0.062494
659	0.062503
660	0.031252
661	0.062503
662	0.062500
663	0.062503
664	0.031252
665	0.062503
666	0.062500
667	0.062497
668	0.031249
669	0.062497
670	0.062494
671	0.062503
672	0.031252
673	0.062503
674	0.062500
675	0.062494
676	0.062497
677	0.031249
678	0.062500
679	0.062503
680	0.031252
681	0.062500
682	0.062503
683	0.031252
684	0.062494
685	0.062497
686	0.031249
687	0.062500
688	0.062503
689	0.031252
690	0.062500
691	0.062503
692	0.031252
693	0.062500
694	0.062503

```
       695          0.031252
       696          0.062500
       697          0.062503
       698          0.031252
       699          0.062494
       700          0.062497
       701          0.031249
       702          0.062500
       703          0.062503
       704          0.031252
       705          0.062500
       706          0.062503
       707          0.031252
       708          0.062494
       709          0.062497
       710          0.031249
       711          0.062500
       712          0.062503
       713          0.031252
    TOTAL           6.750013
145. LOAD 1 DEAD LOAD (DL)
146. ******************************************************************************
147. JOINT LOAD
148. 2 FX 0 FY 0 FZ 0
149. 5 FX 0 FY 0 FZ 0
150. SELFWEIGHT Y -1
151. ELEMENT LOAD
152. 485 487 489 491 493 494 496 498 500 502 504 505 508 510 512 514 516 517 519 -
153. 521 523 525 527 528 531 532 534 536 537 539 541 542 544 546 547 549 551 552 -
154. 554 556 557 559 561 562 564 566 567 569 571 572 574 576 577 579 581 582 584 -
155. 586 587 589 591 592 594 TO 599 601 TO 603 605 TO 607 609 TO 611 613 TO 615 -
156. 617 TO 619 621 TO 623 625 TO 627 629 TO 631 633 TO 635 637 TO 639 -
157. 641 TO 643 645 TO 647 649 TO 652 PR -15.3
158. ******************************************************************************
159. LOAD 2 LIVE LOAD (LL)
160. ******************************************************************************
161. JOINT LOAD
162. 2 FX 0 FY 0 FZ 0
163. 5 FX 0 FY 0 FZ 0
164. ******************************************************************************
165. LOAD 3 OPERATING FLUID LOAD (FL)
166. ******************************************************************************
167. JOINT LOAD
168. 2 FY -170
169. 5 FY -170
170. ******************************************************************************
171. LOAD 4 EQP DL (EQDL)
172. ******************************************************************************
173. JOINT LOAD
174. 2 FY -21
175. 5 FY -21
176. ******************************************************************************
177. LOAD 5 EQP LL (EQLL)
178. ******************************************************************************
179. JOINT LOAD
180. 2 FX 0 FY 0 FZ 0
```

```
5 FX 0 FY 0 FZ 0
********************************************************************************
LOAD 6 WIND LOAD (+X) WL(+X)
********************************************************************************
JOINT LOAD
2 FX 5.3
5 FX 5.3
********************************************************************************
LOAD 7 WIND LOAD (-X) WL(-X)
********************************************************************************
JOINT LOAD
2 FX -5.3
5 FX -5.3
********************************************************************************
LOAD 8 WIND LOAD (+Z) WL(+Z)
********************************************************************************
JOINT LOAD
2 FZ 5.3
5 FZ 5.3
********************************************************************************
LOAD 9 WIND LOAD (-Z) WL(-Z)
********************************************************************************
JOINT LOAD
2 FZ -5.3
5 FZ -5.3
********************************************************************************
LOAD 10 SEISMIC LOAD SL(+X)
********************************************************************************
JOINT LOAD
2 FX 10
5 FX 10
********************************************************************************
LOAD 11 SEISMIC LOAD SL(-X)
********************************************************************************
JOINT LOAD
2 FX -10
5 FX -10
********************************************************************************
LOAD 12 SEISMIC LOAD SL(+Z)
********************************************************************************
JOINT LOAD
2 FZ 10
5 FZ 10
********************************************************************************
LOAD 13 SEISMIC LOAD SL(-Z)
********************************************************************************
JOINT LOAD
2 FZ -10
5 FZ -10
********************************************************************************
LOAD 14 TEMPERATURE LOAD (TL)
JOINT LOAD
2 FX 0 FY 0 FZ 0
5 FX 0 FY 0 FZ 0
********************************************************************************
********************************************************************************
```

```
237. LOAD 15 EQP TEST LOAD (EQTL)
238. ********************************************************************************
239. JOINT LOAD
240. 2 FY -191
241. 5 FY -191
242. ********************************************************************************
243. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
244. ********************************************************************************
245. LOAD COMB 101 1.4(DL+EQDL+FL)
246. 1 1.4 4 1.4 3 1.4
247. LOAD COMB 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
248. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
249. LOAD COMB 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
250. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
251. LOAD COMB 104 1.2(DL+EQDL)+0.8WL(+X)
252. 1 1.2 4 1.2 6 0.8
253. LOAD COMB 105 1.2(DL+EQDL)+0.8WL(-X)
254. 1 1.2 4 1.2 7 0.8
255. LOAD COMB 106 1.2(DL+EQDL)+0.8WL(+Z)
256. 1 1.2 4 1.2 8 0.8
257. LOAD COMB 107 1.2(DL+EQDL)+0.8WL(-Z)
258. 1 1.2 4 1.2 9 0.8
259. LOAD COMB 108 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
260. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
261. LOAD COMB 109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
262. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
263. LOAD COMB 110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
264. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
265. LOAD COMB 111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
266. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
267. LOAD COMB 112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
268. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
269. LOAD COMB 113 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
270. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
271. LOAD COMB 114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
272. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
273. LOAD COMB 115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
274. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
275. LOAD COMB 116 0.9(DL+EQDL)+1.6WL(+X)
276. 1 0.9 4 0.9 6 1.6
277. LOAD COMB 117 0.9(DL+EQDL)+1.6WL(-X)
278. 1 0.9 4 0.9 7 1.6
279. LOAD COMB 118 0.9(DL+EQDL)+1.6WL(+Z)
280. 1 0.9 4 0.9 8 1.6
281. LOAD COMB 119 0.9(DL+EQDL)+1.6WL(-Z)
282. 1 0.9 4 0.9 9 1.6
283. LOAD COMB 120 0.9(DL+EQDL)+SL(+X)
284. 1 0.9 4 0.9 10 1.0
285. LOAD COMB 121 0.9(DL+EQDL)+SL(-X)
286. 1 0.9 4 0.9 11 1.0
287. LOAD COMB 122 0.9(DL+EQDL)+SL(+Z)
288. 1 0.9 4 0.9 12 1.0
289. LOAD COMB 123 0.9(DL+EQDL)+SL(-Z)
290. 1 0.9 4 0.9 13 1.0
291. LOAD COMB 124 1.2DL+EQTL
292. 1 1.2 15 1.0
```

```
293. ******************************************************
294. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
295. ******************************************************
296. LOAD COMB 201 DL+EQDL+FL
297. 1 1.0 4 1.0 3 1.0
298. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
299. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
300. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
301. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
302. LOAD COMB 204 DL+EQDL+FL+WL(+X)
303. 1 1.0 4 1.0 3 1.0 6 1.0
304. LOAD COMB 205 DL+EQDL+FL+WL(-X)
305. 1 1.0 4 1.0 3 1.0 7 1.0
306. LOAD COMB 206 DL+EQDL+FL+WL(+Z)
307. 1 1.0 4 1.0 3 1.0 8 1.0
308. LOAD COMB 207 DL+EQDL+FL+WL(-Z)
309. 1 1.0 4 1.0 3 1.0 9 1.0
310. LOAD COMB 208 DL+EQDL+FL+0.7SL(+X)
311. 1 1.0 4 1.0 3 1.0 10 0.7
312. LOAD COMB 209 DL+EQDL+FL+0.7SL(-X)
313. 1 1.0 4 1.0 3 1.0 11 0.7
314. LOAD COMB 210 DL+EQDL+FL+0.7SL(+Z)
315. 1 1.0 4 1.0 3 1.0 12 0.7
316. LOAD COMB 211 DL+EQDL+FL+0.7SL(-Z)
317. 1 1.0 4 1.0 3 1.0 13 0.7
318. LOAD COMB 212 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
319. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
320. LOAD COMB 213 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
321. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
322. LOAD COMB 214 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
323. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
324. LOAD COMB 215 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
325. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
326. LOAD COMB 216 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
327. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
328. LOAD COMB 217 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
329. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
330. LOAD COMB 218 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
331. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
332. LOAD COMB 219 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
333. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
334. LOAD COMB 220 0.6(DL+EQDL)+WL(+X)
335. 1 0.6 4 0.6 6 1.0
336. LOAD COMB 221 0.6(DL+EQDL)+WL(-X)
337. 1 0.6 4 0.6 7 1.0
338. LOAD COMB 222 0.6(DL+EQDL)+WL(+Z)
339. 1 0.6 4 0.6 8 1.0
340. LOAD COMB 223 0.6(DL+EQDL)+WL(-Z)
341. 1 0.6 4 0.6 9 1.0
342. LOAD COMB 224 0.6(DL+EQDL)+0.7SL(+X)
343. 1 0.6 4 0.6 10 0.7
344. LOAD COMB 225 0.6(DL+EQDL)+0.7SL(-X)
345. 1 0.6 4 0.6 11 0.7
346. LOAD COMB 226 0.6(DL+EQDL)+0.7SL(+Z)
347. 1 0.6 4 0.6 12 0.7
348. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(-Z)
```

```
349. 1 0.6 4 0.6 13 0.7
350. LOAD COMB 228 DL+EQTL
351. 1 1.0 15 1.0
352. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**137**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**108**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**133**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 132/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 423
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 20 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.9/ 99983.9 MB

```
353. ********************************************************************************
354. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 14:28:23 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 6 STAAD INPUT & OUTPUT FILE FOR F5 STABILITY

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    15: 8:25                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F5 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 29-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 2 -209.5 913.7 -435.9; 5 -209.5 913.7 -432.4; 162 -209.5 911 -435.9
    10. 164 -209.5 911 -432.4; 580 -209.5 911.15 -435.9; 582 -209.5 911.15 -432.4
    11. MEMBER INCIDENCES
    12. 2 162 580; 4 164 582; 462 580 2; 464 582 5
    13. DEFINE MATERIAL START
    14. ISOTROPIC CONCRETE
    15. E 2.17185E+007
    16. POISSON 0.17
    17. DENSITY 25
    18. ALPHA 1E-005
    19. DAMP 0.05
    20. TYPE CONCRETE
    21. STRENGTH FCU 27579
    22. END DEFINE MATERIAL
    23. MEMBER PROPERTY AMERICAN
    24. 2 4 462 464 PRIS YD 0.5 ZD 0.5
    25. CONSTANTS
    26. MATERIAL CONCRETE ALL
    27. ******************************************************************************
    28. SUPPORTS
    29. 162 164 FIXED
    30. LOAD 1 DEAD LOAD (DL)
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    31. ******************************************************************************
    32. JOINT LOAD
    33. 2 FX 0 FY 0 FZ 0
    34. 5 FX 0 FY 0 FZ 0
    35. SELFWEIGHT Y -1
    36. ******************************************************************************
```

```
37. LOAD 2 LIVE LOAD (LL)
38. ********************************************************************************
39. JOINT LOAD
40. 2 FX 0 FY 0 FZ 0
41. 5 FX 0 FY 0 FZ 0
42. ********************************************************************************
43. LOAD 3 OPERATING FLUID LOAD (FL)
44. ********************************************************************************
45. JOINT LOAD
46. 2 FY -170
47. 5 FY -170
48. ********************************************************************************
49. LOAD 4 EQP DL (EQDL)
50. ********************************************************************************
51. JOINT LOAD
52. 2 FY -21
53. 5 FY -21
54. ********************************************************************************
55. LOAD 5 EQP LL (EQLL)
56. ********************************************************************************
57. JOINT LOAD
58. 2 FX 0 FY 0 FZ 0
59. 5 FX 0 FY 0 FZ 0
60. ********************************************************************************
61. LOAD 6 WIND LOAD (+X) WL(+X)
62. ********************************************************************************
63. JOINT LOAD
64. 2 FX 5.3
65. 5 FX 5.3
66. ********************************************************************************
67. LOAD 7 WIND LOAD (-X) WL(-X)
68. ********************************************************************************
69. JOINT LOAD
70. 2 FX -5.3
71. 5 FX -5.3
72. ********************************************************************************
73. LOAD 8 WIND LOAD (+Z) WL(+Z)
74. ********************************************************************************
75. JOINT LOAD
76. 2 FZ 5.3
77. 5 FZ 5.3
78. ********************************************************************************
79. LOAD 9 WIND LOAD (-Z) WL(-Z)
80. ********************************************************************************
81. JOINT LOAD
82. 2 FZ -5.3
83. 5 FZ -5.3
84. ********************************************************************************
85. LOAD 10 SEISMIC LOAD SL(+X)
86. ********************************************************************************
87. JOINT LOAD
88. 2 FX 10
89. 5 FX 10
90. ********************************************************************************
91. LOAD 11 SEISMIC LOAD SL(-X)
92. ********************************************************************************
```

```
 93. JOINT LOAD
 94. 2 FX -10
 95. 5 FX -10
 96. ********************************************************************************
 97. LOAD 12 SEISMIC LOAD SL(+Z)
 98. ********************************************************************************
 99. JOINT LOAD
100. 2 FZ 10
101. 5 FZ 10
102. ********************************************************************************
103. LOAD 13 SEISMIC LOAD SL(-Z)
104. ********************************************************************************
105. JOINT LOAD
106. 2 FZ -10
107. 5 FZ -10
108. ********************************************************************************
109. LOAD 14 TEMPERATURE LOAD (TL)
110. JOINT LOAD
111. 2 FX 0 FY 0 FZ 0
112. 5 FX 0 FY 0 FZ 0
113. ********************************************************************************
114. ********************************************************************************
115. LOAD 15 EQP TEST LOAD (EQTL)
116. ********************************************************************************
117. JOINT LOAD
118. 2 FY -191
119. 5 FY -191
120. ********************************************************************************
121. ******************************************************
122. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
123. ******************************************************
124. LOAD COMB 201 DL+EQDL+FL
125. 1 1.0 4 1.0 3 1.0
126. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
127. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
128. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
129. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
130. LOAD COMB 204 DL+EQDL+FL+WL(+X)
131. 1 1.0 4 1.0 3 1.0 6 1.0
132. LOAD COMB 205 DL+EQDL+FL+WL(-X)
133. 1 1.0 4 1.0 3 1.0 7 1.0
134. LOAD COMB 206 DL+EQDL+FL+WL(+Z)
135. 1 1.0 4 1.0 3 1.0 8 1.0
136. LOAD COMB 207 DL+EQDL+FL+WL(-Z)
137. 1 1.0 4 1.0 3 1.0 9 1.0
138. LOAD COMB 208 DL+EQDL+FL+0.7SL(+X)
139. 1 1.0 4 1.0 3 1.0 10 0.7
140. LOAD COMB 209 DL+EQDL+FL+0.7SL(-X)
141. 1 1.0 4 1.0 3 1.0 11 0.7
142. LOAD COMB 210 DL+EQDL+FL+0.7SL(+Z)
143. 1 1.0 4 1.0 3 1.0 12 0.7
144. LOAD COMB 211 DL+EQDL+FL+0.7SL(-Z)
145. 1 1.0 4 1.0 3 1.0 13 0.7
146. LOAD COMB 212 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
147. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
148. LOAD COMB 213 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
```

```
149. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
150. LOAD COMB 214 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
151. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
152. LOAD COMB 215 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
153. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
154. LOAD COMB 216 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
155. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
156. LOAD COMB 217 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
157. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
158. LOAD COMB 218 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
159. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
160. LOAD COMB 219 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
161. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
162. LOAD COMB 220 0.6(DL+EQDL)+WL(+X)
163. 1 0.6 4 0.6 6 1.0
164. LOAD COMB 221 0.6(DL+EQDL)+WL(-X)
165. 1 0.6 4 0.6 7 1.0
166. LOAD COMB 222 0.6(DL+EQDL)+WL(+Z)
167. 1 0.6 4 0.6 8 1.0
168. LOAD COMB 223 0.6(DL+EQDL)+WL(-Z)
169. 1 0.6 4 0.6 9 1.0
170. LOAD COMB 224 0.6(DL+EQDL)+0.7SL(+X)
171. 1 0.6 4 0.6 10 0.7
172. LOAD COMB 225 0.6(DL+EQDL)+0.7SL(-X)
173. 1 0.6 4 0.6 11 0.7
174. LOAD COMB 226 0.6(DL+EQDL)+0.7SL(+Z)
175. 1 0.6 4 0.6 12 0.7
176. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(-Z)
177. 1 0.6 4 0.6 13 0.7
178. LOAD COMB 228 DL+EQTL
179. 1 1.0 15 1.0
180. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 4/ 4/ 18 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 28 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 99944.5 MB

```
181. ********************************************************************************
182. FINISH


          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= MAY 21,2020   TIME= 15: 8:31 ****

       ************************************************************
       *   For technical assistance on STAAD.Pro, please visit    *
       *   http://selectservices.bentley.com/en-US/               *
       *                                                          *
       *   Details about additional assistance from               *
       *   Bentley and Partners can be found at program menu      *
       *   Help->Technical Support                                *
       *                                                          *
       *      Copyright (c) 1997-2014 Bentley Systems, Inc.
       *                http://www.bentley.com                    *
       ************************************************************
```

ATTACHMENT 7 STAAD INPUT & OUTPUT FILE FOR F6 DESIGN

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    14:45:53                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F6 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 24-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 5 -209.5 913.6 -429; 164 -209.5 911 -429; 582 -209.5 911.25 -429
    10. MEMBER INCIDENCES
    11. 4 164 582; 464 582 5
    12. START GROUP DEFINITION
    13. MEMBER
    14. _C6 464
    15. END GROUP DEFINITION
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 4 464 PRIS YD 0.6 ZD 0.6
    28. CONSTANTS
    29. MATERIAL CONCRETE ALL
    30. SUPPORTS
    31. 164 FIXED
    32. ******************************************************************************
    33. LOAD 1 DEAD LOAD (DL)
    34. ******************************************************************************
    35. JOINT LOAD
    36. 5 FX 0 FY 0 FZ 0
    37. SELFWEIGHT Y -1
    38. ******************************************************************************
```

```
39. LOAD 2 LIVE LOAD (LL)
40. ********************************************************************************
41. JOINT LOAD
42. 5 FX 0 FY 0 FZ 0
43. ********************************************************************************
44. LOAD 3 OPERATING FLUID LOAD (FL)
45. ********************************************************************************
46. JOINT LOAD
47. 5 FX 0 FY -79 FZ 0
48. ********************************************************************************
49. LOAD 4 EQP DL (EQDL)
50. ********************************************************************************
51. JOINT LOAD
52. 5 FX 0 FY -36 FZ 0
53. ********************************************************************************
54. LOAD 5 EQP LL (EQLL)
55. ********************************************************************************
56. JOINT LOAD
57. 5 FX 0 FY 0 FZ 0
58. ********************************************************************************
59. LOAD 6 WIND LOAD (+X) WL(+X)
60. ********************************************************************************
61. JOINT LOAD
62. 5 FX 6 FY 0 FZ 0
63. ********************************************************************************
64. LOAD 7 WIND LOAD (-X) WL(-X)
65. ********************************************************************************
66. JOINT LOAD
67. 5 FX -6 FY 0 FZ 0
68. ********************************************************************************
69. LOAD 8 WIND LOAD (+Z) WL(+Z)
70. ********************************************************************************
71. JOINT LOAD
72. 5 FX 0 FY 0 FZ 6
73. ********************************************************************************
74. LOAD 9 WIND LOAD (-Z) WL(-Z)
75. ********************************************************************************
76. JOINT LOAD
77. 5 FX 0 FY 0 FZ -6
78. ********************************************************************************
79. LOAD 10 SEISMIC LOAD SL(+X)
80. ********************************************************************************
81. JOINT LOAD
82. 5 FX 3 FY 0 FZ 0
83. ********************************************************************************
84. LOAD 11 SEISMIC LOAD SL(-X)
85. ********************************************************************************
86. JOINT LOAD
87. 5 FX -3 FY 0 FZ 0
88. ********************************************************************************
89. LOAD 12 SEISMIC LOAD SL(+Z)
90. ********************************************************************************
91. JOINT LOAD
92. 5 FX 0 FY 0 FZ 3
93. ********************************************************************************
94. LOAD 13 SEISMIC LOAD SL(-Z)
```

```
 95. ********************************************************************************
 96. JOINT LOAD
 97. 5 FX 0 FY 0 FZ -3
 98. ********************************************************************************
 99. LOAD 14 TEMPERATURE LOAD (TL)
100. JOINT LOAD
101. 5 FX 0 FY 0 FZ 0
102. ********************************************************************************
103. ********************************************************************************
104. LOAD 15 EQP TEST LOAD (EQTL)
105. ********************************************************************************
106. JOINT LOAD
107. 5 FX 0 FY -115 FZ 0
108. ********************************************************************************
109. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
110. ********************************************************************************
111. LOAD COMB 101 1.4(DL+EQDL+FL)
112. 1 1.4 4 1.4 3 1.4
113. LOAD COMB 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
114. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
115. LOAD COMB 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
116. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
117. LOAD COMB 104 1.2(DL+EQDL)+0.8WL(+X)
118. 1 1.2 4 1.2 6 0.8
119. LOAD COMB 105 1.2(DL+EQDL)+0.8WL(-X)
120. 1 1.2 4 1.2 7 0.8
121. LOAD COMB 106 1.2(DL+EQDL)+0.8WL(+Z)
122. 1 1.2 4 1.2 8 0.8
123. LOAD COMB 107 1.2(DL+EQDL)+0.8WL(-Z)
124. 1 1.2 4 1.2 9 0.8
125. LOAD COMB 108 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
126. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
127. LOAD COMB 109 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
128. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
129. LOAD COMB 110 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
130. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
131. LOAD COMB 111 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
132. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
133. LOAD COMB 112 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
134. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
135. LOAD COMB 113 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
136. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
137. LOAD COMB 114 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
138. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
139. LOAD COMB 115 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
140. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
141. LOAD COMB 116 0.9(DL+EQDL)+1.6WL(+X)
142. 1 0.9 4 0.9 6 1.6
143. LOAD COMB 117 0.9(DL+EQDL)+1.6WL(-X)
144. 1 0.9 4 0.9 7 1.6
145. LOAD COMB 118 0.9(DL+EQDL)+1.6WL(+Z)
146. 1 0.9 4 0.9 8 1.6
147. LOAD COMB 119 0.9(DL+EQDL)+1.6WL(-Z)
148. 1 0.9 4 0.9 9 1.6
149. LOAD COMB 120 0.9(DL+EQDL)+SL(+X)
150. 1 0.9 4 0.9 10 1.0
```

```
151. LOAD COMB 121 0.9(DL+EQDL)+SL(-X)
152. 1 0.9 4 0.9 11 1.0
153. LOAD COMB 122 0.9(DL+EQDL)+SL(+Z)
154. 1 0.9 4 0.9 12 1.0
155. LOAD COMB 123 0.9(DL+EQDL)+SL(-Z)
156. 1 0.9 4 0.9 13 1.0
157. LOAD COMB 124 1.2DL+EQTL
158. 1 1.2 15 1.0
159. ******************************************************
160. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
161. ******************************************************
162. LOAD COMB 201 DL+EQDL+FL
163. 1 1.0 4 1.0 3 1.0
164. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
165. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
166. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
167. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
168. LOAD COMB 204 DL+EQDL+FL+WL(+X)
169. 1 1.0 4 1.0 3 1.0 6 1.0
170. LOAD COMB 205 DL+EQDL+FL+WL(-X)
171. 1 1.0 4 1.0 3 1.0 7 1.0
172. LOAD COMB 206 DL+EQDL+FL+WL(+Z)
173. 1 1.0 4 1.0 3 1.0 8 1.0
174. LOAD COMB 207 DL+EQDL+FL+WL(-Z)
175. 1 1.0 4 1.0 3 1.0 9 1.0
176. LOAD COMB 208 DL+EQDL+FL+0.7SL(+X)
177. 1 1.0 4 1.0 3 1.0 10 0.7
178. LOAD COMB 209 DL+EQDL+FL+0.7SL(-X)
179. 1 1.0 4 1.0 3 1.0 11 0.7
180. LOAD COMB 210 DL+EQDL+FL+0.7SL(+Z)
181. 1 1.0 4 1.0 3 1.0 12 0.7
182. LOAD COMB 211 DL+EQDL+FL+0.7SL(-Z)
183. 1 1.0 4 1.0 3 1.0 13 0.7
184. LOAD COMB 212 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
185. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
186. LOAD COMB 213 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
187. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
188. LOAD COMB 214 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
189. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
190. LOAD COMB 215 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
191. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
192. LOAD COMB 216 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
193. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
194. LOAD COMB 217 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
195. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
196. LOAD COMB 218 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
197. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
198. LOAD COMB 219 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
199. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
200. LOAD COMB 220 0.6(DL+EQDL)+WL(+X)
201. 1 0.6 4 0.6 6 1.0
202. LOAD COMB 221 0.6(DL+EQDL)+WL(-X)
203. 1 0.6 4 0.6 7 1.0
204. LOAD COMB 222 0.6(DL+EQDL)+WL(+Z)
205. 1 0.6 4 0.6 8 1.0
206. LOAD COMB 223 0.6(DL+EQDL)+WL(-Z)
```

```
207. 1 0.6 4 0.6 9 1.0
208. LOAD COMB 224 0.6(DL+EQDL)+0.7SL(+X)
209. 1 0.6 4 0.6 10 0.7
210. LOAD COMB 225 0.6(DL+EQDL)+0.7SL(-X)
211. 1 0.6 4 0.6 11 0.7
212. LOAD COMB 226 0.6(DL+EQDL)+0.7SL(+Z)
213. 1 0.6 4 0.6 12 0.7
214. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(-Z)
215. 1 0.6 4 0.6 13 0.7
216. LOAD COMB 228 DL+EQTL
217. 1 1.0 15 1.0
218. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**3**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**1**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2/ 2/ 12 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 52 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 99979.7 MB

```
219. ********************************************************************************
220. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 14:46: 0 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 8 STAAD INPUT & OUTPUT FILE FOR F7 DESIGN

PROPRIETARY & CONFIDENTIAL

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    15: 5:34                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F7 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 6.9 0 0; 3 0 0 6.15; 4 6.9 0 6.15; 5 0 2.6 0; 6 6.9 2.6 0
    10. 7 0 2.6 6.15; 8 6.9 2.6 6.15; 9 0 0 -0.4; 10 6.9 0 -0.4; 11 0 0 6.55
    11. 12 6.9 0 6.55; 13 -0.4 0 0; 14 -0.4 0 6.15; 15 -0.4 0 -0.4; 16 -0.4 0 6.55
    12. 17 7.3 0 0; 18 7.3 0 6.15; 19 7.3 0 -0.4; 20 7.3 0 6.55; 21 -0.2 0 6.55
    13. 22 -0.2 0 6.35; 23 -0.4 0 6.35; 24 0 0 6.35; 25 -0.2 0 6.15; 26 -0.2 0 0
    14. 27 -0.2 0 -0.2; 28 -0.4 0 -0.2; 29 0 0 -0.2; 30 -0.2 0 -0.4; 31 7.1 0 0
    15. 32 7.1 0 -0.2; 33 6.9 0 -0.2; 34 7.3 0 -0.2; 35 7.1 0 -0.4; 36 7.1 0 6.55
    16. 37 7.1 0 6.35; 38 6.9 0 6.35; 39 7.3 0 6.35; 40 7.1 0 6.15; 41 0.255556 0 6.55
    17. 42 0.255555 0 6.35; 43 0.511111 0 6.55; 44 0.511111 0 6.35; 45 0.766667 0 6.55
    18. 46 0.766667 0 6.35; 47 1.02222 0 6.55; 48 1.02222 0 6.35; 49 1.27778 0 6.55
    19. 50 1.27778 0 6.35; 51 1.53333 0 6.55; 52 1.53333 0 6.35; 53 1.78889 0 6.55
    20. 54 1.78889 0 6.35; 55 2.04444 0 6.55; 56 2.04444 0 6.35; 57 2.3 0 6.55
    21. 58 2.3 0 6.35; 59 2.55556 0 6.55; 60 2.55556 0 6.35; 61 2.81111 0 6.55
    22. 62 2.81111 0 6.35; 63 3.06667 0 6.55; 64 3.06667 0 6.35; 65 3.32222 0 6.55
    23. 66 3.32222 0 6.35; 67 3.57778 0 6.55; 68 3.57778 0 6.35; 69 3.83333 0 6.55
    24. 70 3.83333 0 6.35; 71 4.08889 0 6.55; 72 4.08889 0 6.35; 73 4.34444 0 6.55
    25. 74 4.34444 0 6.35; 75 4.6 0 6.55; 76 4.6 0 6.35; 77 4.85556 0 6.55
    26. 78 4.85556 0 6.35; 79 5.11111 0 6.55; 80 5.11111 0 6.35; 81 5.36667 0 6.55
    27. 82 5.36667 0 6.35; 83 5.62222 0 6.55; 84 5.62222 0 6.35; 85 5.87778 0 6.55
    28. 86 5.87778 0 6.35; 87 6.13333 0 6.55; 88 6.13333 0 6.35; 89 6.38889 0 6.55
    29. 90 6.38889 0 6.35; 91 6.64444 0 6.55; 92 6.64444 0 6.35; 93 0.255555 0 6.15
    30. 94 0.511111 0 6.15; 95 0.766667 0 6.15; 96 1.02222 0 6.15; 97 1.27778 0 6.15
    31. 98 1.53333 0 6.15; 99 1.78889 0 6.15; 100 2.04444 0 6.15; 101 2.3 0 6.15
    32. 102 2.55556 0 6.15; 103 2.81111 0 6.15; 104 3.06667 0 6.15; 105 3.32222 0 6.15
    33. 106 3.57778 0 6.15; 107 3.83333 0 6.15; 108 4.08889 0 6.15; 109 4.34444 0 6.15
    34. 110 4.6 0 6.15; 111 4.85556 0 6.15; 112 5.11111 0 6.15; 113 5.36667 0 6.15
    35. 114 5.62222 0 6.15; 115 5.87778 0 6.15; 116 6.13333 0 6.15; 117 6.38889 0 6.15
    36. 118 6.64444 0 6.15; 119 0.255556 0 0; 120 0.255555 0 -0.2; 121 0.511111 0 0
    37. 122 0.511111 0 -0.2; 123 0.766667 0 0; 124 0.766667 0 -0.2; 125 1.02222 0 0
    38. 126 1.02222 0 -0.2; 127 1.27778 0 0; 128 1.27778 0 -0.2; 129 1.53333 0 0
```

39. 130 1.53333 0 -0.2; 131 1.78889 0 0; 132 1.78889 0 -0.2; 133 2.04444 0 0
40. 134 2.04444 0 -0.2; 135 2.3 0 0; 136 2.3 0 -0.2; 137 2.55556 0 0
41. 138 2.55556 0 -0.2; 139 2.81111 0 0; 140 2.81111 0 -0.2; 141 3.06667 0 0
42. 142 3.06667 0 -0.2; 143 3.32222 0 0; 144 3.32222 0 -0.2; 145 3.57778 0 0
43. 146 3.57778 0 -0.2; 147 3.83333 0 0; 148 3.83333 0 -0.2; 149 4.08889 0 0
44. 150 4.08889 0 -0.2; 151 4.34444 0 0; 152 4.34444 0 -0.2; 153 4.6 0 0
45. 154 4.6 0 -0.2; 155 4.85556 0 0; 156 4.85556 0 -0.2; 157 5.11111 0 0
46. 158 5.11111 0 -0.2; 159 5.36667 0 0; 160 5.36667 0 -0.2; 161 5.62222 0 0
47. 162 5.62222 0 -0.2; 163 5.87778 0 0; 164 5.87778 0 -0.2; 165 6.13333 0 0
48. 166 6.13333 0 -0.2; 167 6.38889 0 0; 168 6.38889 0 -0.2; 169 6.64444 0 0
49. 170 6.64444 0 -0.2; 171 0.255555 0 -0.4; 172 0.511111 0 -0.4
50. 173 0.766667 0 -0.4; 174 1.02222 0 -0.4; 175 1.27778 0 -0.4
51. 176 1.53333 0 -0.4; 177 1.78889 0 -0.4; 178 2.04444 0 -0.4; 179 2.3 0 -0.4
52. 180 2.55556 0 -0.4; 181 2.81111 0 -0.4; 182 3.06667 0 -0.4; 183 3.32222 0 -0.4
53. 184 3.57778 0 -0.4; 185 3.83333 0 -0.4; 186 4.08889 0 -0.4; 187 4.34444 0 -0.4
54. 188 4.6 0 -0.4; 189 4.85556 0 -0.4; 190 5.11111 0 -0.4; 191 5.36667 0 -0.4
55. 192 5.62222 0 -0.4; 193 5.87778 0 -0.4; 194 6.13333 0 -0.4; 195 6.38889 0 -0.4
56. 196 6.64444 0 -0.4; 197 -0.2 0 5.89375; 198 -0.4 0 5.89375; 199 0 0 5.89375
57. 200 -0.2 0 5.6375; 201 -0.4 0 5.6375; 202 0 0 5.6375; 203 -0.2 0 5.38125
58. 204 -0.4 0 5.38125; 205 0 0 5.38125; 206 -0.2 0 5.125; 207 -0.4 0 5.125
59. 208 0 0 5.125; 209 -0.2 0 4.86875; 210 -0.4 0 4.86875; 211 0 0 4.86875
60. 212 -0.2 0 4.6125; 213 -0.4 0 4.6125; 214 0 0 4.6125; 215 -0.2 0 4.35625
61. 216 -0.4 0 4.35625; 217 0 0 4.35625; 218 -0.2 0 4.1; 219 -0.4 0 4.1
62. 220 0 0 4.1; 221 -0.2 0 3.84375; 222 -0.4 0 3.84375; 223 0 0 3.84375
63. 224 -0.2 0 3.5875; 225 -0.4 0 3.5875; 226 0 0 3.5875; 227 -0.2 0 3.33125
64. 228 -0.4 0 3.33125; 229 0 0 3.33125; 230 -0.2 0 3.075; 231 -0.4 0 3.075
65. 232 0 0 3.075; 233 -0.2 0 2.81875; 234 -0.4 0 2.81875; 235 0 0 2.81875
66. 236 -0.2 0 2.5625; 237 -0.4 0 2.5625; 238 0 0 2.5625; 239 -0.2 0 2.30625
67. 240 -0.4 0 2.30625; 241 0 0 2.30625; 242 -0.2 0 2.05; 243 -0.4 0 2.05
68. 244 0 0 2.05; 245 -0.2 0 1.79375; 246 -0.4 0 1.79375; 247 0 0 1.79375
69. 248 -0.2 0 1.5375; 249 -0.4 0 1.5375; 250 0 0 1.5375; 251 -0.2 0 1.28125
70. 252 -0.4 0 1.28125; 253 0 0 1.28125; 254 -0.2 0 1.025; 255 -0.4 0 1.025
71. 256 0 0 1.025; 257 -0.2 0 0.768749; 258 -0.4 0 0.768749; 259 0 0 0.768749
72. 260 -0.2 0 0.512499; 261 -0.4 0 0.512499; 262 0 0 0.512499
73. 263 -0.2 0 0.256249; 264 -0.4 0 0.256249; 265 0 0 0.256249; 266 7.1 0 5.89375
74. 267 6.9 0 5.89375; 268 7.3 0 5.89375; 269 7.1 0 5.6375; 270 6.9 0 5.6375
75. 271 7.3 0 5.6375; 272 7.1 0 5.38125; 273 6.9 0 5.38125; 274 7.3 0 5.38125
76. 275 7.1 0 5.125; 276 6.9 0 5.125; 277 7.3 0 5.125; 278 7.1 0 4.86875
77. 279 6.9 0 4.86875; 280 7.3 0 4.86875; 281 7.1 0 4.6125; 282 6.9 0 4.6125
78. 283 7.3 0 4.6125; 284 7.1 0 4.35625; 285 6.9 0 4.35625; 286 7.3 0 4.35625
79. 287 7.1 0 4.1; 288 6.9 0 4.1; 289 7.3 0 4.1; 290 7.1 0 3.84375
80. 291 6.9 0 3.84375; 292 7.3 0 3.84375; 293 7.1 0 3.5875; 294 6.9 0 3.5875
81. 295 7.3 0 3.5875; 296 7.1 0 3.33125; 297 6.9 0 3.33125; 298 7.3 0 3.33125
82. 299 7.1 0 3.075; 300 6.9 0 3.075; 301 7.3 0 3.075; 302 7.1 0 2.81875
83. 303 6.9 0 2.81875; 304 7.3 0 2.81875; 305 7.1 0 2.5625; 306 6.9 0 2.5625
84. 307 7.3 0 2.5625; 308 7.1 0 2.30625; 309 6.9 0 2.30625; 310 7.3 0 2.30625
85. 311 7.1 0 2.05; 312 6.9 0 2.05; 313 7.3 0 2.05; 314 7.1 0 1.79375
86. 315 6.9 0 1.79375; 316 7.3 0 1.79375; 317 7.1 0 1.5375; 318 6.9 0 1.5375
87. 319 7.3 0 1.5375; 320 7.1 0 1.28125; 321 6.9 0 1.28125; 322 7.3 0 1.28125
88. 323 7.1 0 1.025; 324 6.9 0 1.025; 325 7.3 0 1.025; 326 7.1 0 0.768749
89. 327 6.9 0 0.768749; 328 7.3 0 0.768749; 329 7.1 0 0.512499; 330 6.9 0 0.512499
90. 331 7.3 0 0.512499; 332 7.1 0 0.256249; 333 6.9 0 0.256249; 334 7.3 0 0.256249
91. 335 0.255556 0 5.89375; 336 0.511111 0 5.89375; 337 0.766667 0 5.89375
92. 338 1.02222 0 5.89375; 339 1.27778 0 5.89375; 340 1.53333 0 5.89375
93. 341 1.78889 0 5.89375; 342 2.04444 0 5.89375; 343 2.3 0 5.89375
94. 344 2.55556 0 5.89375; 345 2.81111 0 5.89375; 346 3.06667 0 5.89375

95. 347 3.32222 0 5.89375; 348 3.57778 0 5.89375; 349 3.83333 0 5.89375
96. 350 4.08889 0 5.89375; 351 4.34444 0 5.89375; 352 4.6 0 5.89375
97. 353 4.85556 0 5.89375; 354 5.11111 0 5.89375; 355 5.36667 0 5.89375
98. 356 5.62222 0 5.89375; 357 5.87778 0 5.89375; 358 6.13333 0 5.89375
99. 359 6.38889 0 5.89375; 360 6.64444 0 5.89375; 361 0.255556 0 5.6375
100. 362 0.511111 0 5.6375; 363 0.766667 0 5.6375; 364 1.02222 0 5.6375
101. 365 1.27778 0 5.6375; 366 1.53333 0 5.6375; 367 1.78889 0 5.6375
102. 368 2.04444 0 5.6375; 369 2.3 0 5.6375; 370 2.55556 0 5.6375
103. 371 2.81111 0 5.6375; 372 3.06667 0 5.6375; 373 3.32222 0 5.6375
104. 374 3.57778 0 5.6375; 375 3.83333 0 5.6375; 376 4.08889 0 5.6375
105. 377 4.34444 0 5.6375; 378 4.6 0 5.6375; 379 4.85556 0 5.6375
106. 380 5.11111 0 5.6375; 381 5.36667 0 5.6375; 382 5.62222 0 5.6375
107. 383 5.87778 0 5.6375; 384 6.13333 0 5.6375; 385 6.38889 0 5.6375
108. 386 6.64444 0 5.6375; 387 0.255556 0 5.38125; 388 0.511111 0 5.38125
109. 389 0.766667 0 5.38125; 390 1.02222 0 5.38125; 391 1.27778 0 5.38125
110. 392 1.53333 0 5.38125; 393 1.78889 0 5.38125; 394 2.04444 0 5.38125
111. 395 2.3 0 5.38125; 396 2.55556 0 5.38125; 397 2.81111 0 5.38125
112. 398 3.06667 0 5.38125; 399 3.32222 0 5.38125; 400 3.57778 0 5.38125
113. 401 3.83333 0 5.38125; 402 4.08889 0 5.38125; 403 4.34444 0 5.38125
114. 404 4.6 0 5.38125; 405 4.85556 0 5.38125; 406 5.11111 0 5.38125
115. 407 5.36667 0 5.38125; 408 5.62222 0 5.38125; 409 5.87778 0 5.38125
116. 410 6.13333 0 5.38125; 411 6.38889 0 5.38125; 412 6.64444 0 5.38125
117. 413 0.255556 0 5.125; 414 0.511111 0 5.125; 415 0.766667 0 5.125
118. 416 1.02222 0 5.125; 417 1.27778 0 5.125; 418 1.53333 0 5.125
119. 419 1.78889 0 5.125; 420 2.04444 0 5.125; 421 2.3 0 5.125; 422 2.55556 0 5.125
120. 423 2.81111 0 5.125; 424 3.06667 0 5.125; 425 3.32222 0 5.125
121. 426 3.57778 0 5.125; 427 3.83333 0 5.125; 428 4.08889 0 5.125
122. 429 4.34444 0 5.125; 430 4.6 0 5.125; 431 4.85556 0 5.125; 432 5.11111 0 5.125
123. 433 5.36667 0 5.125; 434 5.62222 0 5.125; 435 5.87778 0 5.125
124. 436 6.13333 0 5.125; 437 6.38889 0 5.125; 438 6.64444 0 5.125
125. 439 0.255556 0 4.86875; 440 0.511111 0 4.86875; 441 0.766667 0 4.86875
126. 442 1.02222 0 4.86875; 443 1.27778 0 4.86875; 444 1.53333 0 4.86875
127. 445 1.78889 0 4.86875; 446 2.04444 0 4.86875; 447 2.3 0 4.86875
128. 448 2.55556 0 4.86875; 449 2.81111 0 4.86875; 450 3.06667 0 4.86875
129. 451 3.32222 0 4.86875; 452 3.57778 0 4.86875; 453 3.83333 0 4.86875
130. 454 4.08889 0 4.86875; 455 4.34444 0 4.86875; 456 4.6 0 4.86875
131. 457 4.85556 0 4.86875; 458 5.11111 0 4.86875; 459 5.36667 0 4.86875
132. 460 5.62222 0 4.86875; 461 5.87778 0 4.86875; 462 6.13333 0 4.86875
133. 463 6.38889 0 4.86875; 464 6.64444 0 4.86875; 465 0.255555 0 4.6125
134. 466 0.511111 0 4.6125; 467 0.766667 0 4.6125; 468 1.02222 0 4.6125
135. 469 1.27778 0 4.6125; 470 1.53333 0 4.6125; 471 1.78889 0 4.6125
136. 472 2.04444 0 4.6125; 473 2.3 0 4.6125; 474 2.55556 0 4.6125
137. 475 2.81111 0 4.6125; 476 3.06667 0 4.6125; 477 3.32222 0 4.6125
138. 478 3.57778 0 4.6125; 479 3.83333 0 4.6125; 480 4.08889 0 4.6125
139. 481 4.34444 0 4.6125; 482 4.6 0 4.6125; 483 4.85556 0 4.6125
140. 484 5.11111 0 4.6125; 485 5.36667 0 4.6125; 486 5.62222 0 4.6125
141. 487 5.87778 0 4.6125; 488 6.13333 0 4.6125; 489 6.38889 0 4.6125
142. 490 6.64444 0 4.6125; 491 0.255555 0 4.35625; 492 0.511111 0 4.35625
143. 493 0.766667 0 4.35625; 494 1.02222 0 4.35625; 495 1.27778 0 4.35625
144. 496 1.53333 0 4.35625; 497 1.78889 0 4.35625; 498 2.04444 0 4.35625
145. 499 2.3 0 4.35625; 500 2.55556 0 4.35625; 501 2.81111 0 4.35625
146. 502 3.06667 0 4.35625; 503 3.32222 0 4.35625; 504 3.57778 0 4.35625
147. 505 3.83333 0 4.35625; 506 4.08889 0 4.35625; 507 4.34444 0 4.35625
148. 508 4.6 0 4.35625; 509 4.85556 0 4.35625; 510 5.11111 0 4.35625
149. 511 5.36667 0 4.35625; 512 5.62222 0 4.35625; 513 5.87778 0 4.35625
150. 514 6.13333 0 4.35625; 515 6.38889 0 4.35625; 516 6.64444 0 4.35625

151. 517 0.255555 0 4.1; 518 0.511111 0 4.1; 519 0.766667 0 4.1; 520 1.02222 0 4.1
152. 521 1.27778 0 4.1; 522 1.53333 0 4.1; 523 1.78889 0 4.1; 524 2.04444 0 4.1
153. 525 2.3 0 4.1; 526 2.55556 0 4.1; 527 2.81111 0 4.1; 528 3.06667 0 4.1
154. 529 3.32222 0 4.1; 530 3.57778 0 4.1; 531 3.83333 0 4.1; 532 4.08889 0 4.1
155. 533 4.34444 0 4.1; 534 4.6 0 4.1; 535 4.85556 0 4.1; 536 5.11111 0 4.1
156. 537 5.36667 0 4.1; 538 5.62222 0 4.1; 539 5.87778 0 4.1; 540 6.13333 0 4.1
157. 541 6.38889 0 4.1; 542 6.64444 0 4.1; 543 0.255555 0 3.84375
158. 544 0.511111 0 3.84375; 545 0.766667 0 3.84375; 546 1.02222 0 3.84375
159. 547 1.27778 0 3.84375; 548 1.53333 0 3.84375; 549 1.78889 0 3.84375
160. 550 2.04444 0 3.84375; 551 2.3 0 3.84375; 552 2.55556 0 3.84375
161. 553 2.81111 0 3.84375; 554 3.06667 0 3.84375; 555 3.32222 0 3.84375
162. 556 3.57778 0 3.84375; 557 3.83333 0 3.84375; 558 4.08889 0 3.84375
163. 559 4.34444 0 3.84375; 560 4.6 0 3.84375; 561 4.85556 0 3.84375
164. 562 5.11111 0 3.84375; 563 5.36667 0 3.84375; 564 5.62222 0 3.84375
165. 565 5.87778 0 3.84375; 566 6.13333 0 3.84375; 567 6.38889 0 3.84375
166. 568 6.64444 0 3.84375; 569 0.255555 0 3.5875; 570 0.511111 0 3.5875
167. 571 0.766667 0 3.5875; 572 1.02222 0 3.5875; 573 1.27778 0 3.5875
168. 574 1.53333 0 3.5875; 575 1.78889 0 3.5875; 576 2.04444 0 3.5875
169. 577 2.3 0 3.5875; 578 2.55556 0 3.5875; 579 2.81111 0 3.5875
170. 580 3.06667 0 3.5875; 581 3.32222 0 3.5875; 582 3.57778 0 3.5875
171. 583 3.83333 0 3.5875; 584 4.08889 0 3.5875; 585 4.34444 0 3.5875
172. 586 4.6 0 3.5875; 587 4.85556 0 3.5875; 588 5.11111 0 3.5875
173. 589 5.36667 0 3.5875; 590 5.62222 0 3.5875; 591 5.87778 0 3.5875
174. 592 6.13333 0 3.5875; 593 6.38889 0 3.5875; 594 6.64444 0 3.5875
175. 595 0.255555 0 3.33125; 596 0.511111 0 3.33125; 597 0.766667 0 3.33125
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286. MEMBER INCIDENCES
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570. START GROUP DEFINITION
571. ELEMENT
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583. 457 459 461 TO 463 465 TO 1112
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587. 165 167 172 174 176 178 180 182 184 186 188 190 192 194 196 198 200 202 204 -
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595. 1087 TO 1111
596. _RAFT_SHEAR 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100 102 104 106 108 -
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598. 178 180 182 184 186 188 190 192 194 196 198 200 202 204 206 208 210 212 214 -
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602. 399 401 403 405 407 409 411 413 415 417 419 421 423 425 427 429 431 433 435 -
603. 437 439 441 443 445 447 449 469 TO 487 496 TO 514 523 TO 541 550 TO 568 573 -
604. 574 TO 1004 1009 TO 1027 1036 TO 1054 1063 TO 1081 1090 TO 1108
605. MEMBER
606. _C7 1113 TO 1116
607. END GROUP DEFINITION
608. ELEMENT PROPERTY
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611. 123 125 127 129 131 133 135 137 139 141 143 145 147 149 151 153 155 157 159 -
612. 161 163 165 167 168 170 172 174 176 178 180 182 184 186 188 190 192 194 196 -
613. 198 200 202 204 206 208 210 212 214 216 218 220 221 223 225 227 229 231 233 -
614. 235 237 239 241 243 245 247 249 251 253 255 257 259 261 263 265 267 269 271 -
615. 273 274 277 279 281 283 285 287 289 291 293 295 297 299 301 303 305 307 309 -
616. 311 313 315 317 319 321 323 325 327 329 331 333 335 337 339 341 343 345 347 -
617. 349 351 353 355 357 359 361 363 365 367 TO 369 371 373 375 377 379 381 383 -
618. 385 387 389 391 393 395 397 399 401 403 405 407 409 411 413 415 417 419 421 -
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620. 461 TO 463 465 TO 1112 THICKNESS 0.75
621. DEFINE MATERIAL START
622. ISOTROPIC CONCRETE
623. E 2.17185E+007
624. POISSON 0.17
625. DENSITY 25
626. ALPHA 1E-005
627. DAMP 0.05
628. TYPE CONCRETE
629. STRENGTH FCU 27579
630. END DEFINE MATERIAL
631. MEMBER PROPERTY AMERICAN
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633. CONSTANTS
634. MATERIAL CONCRETE ALL
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636. SUPPORTS
637. _RAFT PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
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TOTAL	53.515366

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638. ************************************************************
639. LOAD 1 DEAD LOAD (DL)
640. JOINT LOAD
641. 5 FX -12.8 FY -154.2 FZ -0.6
642. 6 FX 11.5 FY -132.5 FZ -12.2
643. 7 FX -5.2 FY -78.5 FZ -0.5
644. 8 FX 6.5 FY -85.7 FZ 13.4
645. SELFWEIGHT Y -1
646. ELEMENT LOAD
647. _RAFT PR -8.1
648. ********************************************************************************
649. ********************************************************************************
650. LOAD 2 LIVE LOAD (LL)
651. JOINT LOAD
652. 5 FX -14.1 FY -106 FZ -2
653. 6 FX 10 FY -193 FZ -24.4
654. 7 FX -10.8 FY -156.4 FZ -1.7
655. 8 FX 14.9 FY -232.6 FZ 28
656. ********************************************************************************
657. ********************************************************************************
658. LOAD 3 OPERATING FLUID LOAD (FL)
659. JOINT LOAD
660. 5 FX -5.1 FY -58.2 FZ 0
661. 6 FX 5.1 FY -64.1 FZ -4.8
662. 7 FX -1.4 FY -7.3 FZ 0
663. 8 FX 1.4 FY -31.4 FZ 4.9
664. ********************************************************************************
665. ********************************************************************************
666. LOAD 4 EQP DL (EQDL)
667. JOINT LOAD
668. 5 FX -1.7 FY -5.8 FZ 0
669. 6 FX 1.8 FY -33.8 FZ -3.6
670. 7 FX -1.6 FY -5.1 FZ 0
671. 8 FX 1.5 FY -35.1 FZ 3.5
672. ********************************************************************************
673. ********************************************************************************
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674. LOAD 5 EQP LL (EQLL)
675. JOINT LOAD
676. 5 FX -4.6 FY -20.1 FZ 0
677. 6 FX 4.6 FY -85.4 FZ -8.9
678. 7 FX -3.9 FY -15.6 FZ 0
679. 8 FX 3.9 FY -86.6 FZ 8.9
680. ******************************************************************************
681. ******************************************************************************
682. LOAD 6 WIND LOAD (+X) WL(+X)
683. JOINT LOAD
684. 5 FX 27.8 FY 114.8 FZ 0.4
685. 6 FX 29.1 FY -115 FZ -9.8
686. 7 FX 27.2 FY 103.7 FZ 0.3
687. 8 FX 28 FY -103.6 FZ 9.2
688. ******************************************************************************
689. ******************************************************************************
690. LOAD 7 WIND LOAD (-X) WL(-X)
691. JOINT LOAD
692. 5 FX -27.8 FY -114.8 FZ -0.4
693. 6 FX -29.1 FY 115 FZ 9.8
694. 7 FX -27.2 FY -103.7 FZ -0.3
695. 8 FX -28 FY 103.6 FZ -9.2
696. ******************************************************************************
697. ******************************************************************************
698. LOAD 8 WIND LOAD (+Z) WL(+Z)
699. JOINT LOAD
700. 5 FX 10.5 FY 141.7 FZ 36.4
701. 6 FX -9.2 FY 169.7 FZ 39.3
702. 7 FX -11.4 FY -142.1 FZ 33
703. 8 FX 10.6 FY -169.7 FZ 39.4
704. ******************************************************************************
705. ******************************************************************************
706. LOAD 9 WIND LOAD (-Z) WL(-Z)
707. JOINT LOAD
708. 5 FX -10.5 FY -141.7 FZ -36.4
709. 6 FX 9.2 FY -169.7 FZ -39.3
710. 7 FX 11.4 FY 142.1 FZ -33
711. 8 FX -10.6 FY 169.7 FZ -39.4
712. ******************************************************************************
713. ******************************************************************************
714. LOAD 10 SEISMIC LOAD SL(+X)
715. JOINT LOAD
716. 5 FX 74.9 FY 301.8 FZ -24.1
717. 6 FX 74.1 FY -279 FZ -1.6
718. 7 FX 15.8 FY 230.7 FZ -21.9
719. 8 FX 16.2 FY -240.6 FZ 47.8
720. ******************************************************************************
721. ******************************************************************************
722. LOAD 11 SEISMIC LOAD SL(-X)
723. JOINT LOAD
724. 5 FX -74.9 FY -301.8 FZ 24.1
725. 6 FX -74.1 FY 279 FZ 1.6
726. 7 FX -15.8 FY -230.7 FZ 21.9
727. 8 FX -16.2 FY 240.6 FZ -47.8
728. ******************************************************************************
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730. LOAD 12 SEISMIC LOAD SL(+Z)
731. JOINT LOAD
732. 5 FX 19 FY 266.6 FZ 56.2
733. 6 FX -17.5 FY 271.4 FZ 37
734. 7 FX -21.1 FY -292 FZ 51.7
735. 8 FX 20.3 FY -271.4 FZ 37.1
736. ******************************************************************************
737. ******************************************************************************
738. LOAD 13 SEISMIC LOAD SL(-Z)
739. JOINT LOAD
740. 5 FX -19 FY -266.6 FZ -56.2
741. 6 FX 17.5 FY -271.4 FZ -37
742. 7 FX 21.1 FY 292 FZ -51.7
743. 8 FX -20.3 FY 271.4 FZ -37.1
744. ******************************************************************************
745. ******************************************************************************
746. LOAD 14 TEMPERATURE LOAD (TL)
747. JOINT LOAD
748. 5 FX -22.1 FY -2.3 FZ -11.5
749. 6 FX 21.8 FY 0.7 FZ -16.8
750. 7 FX -18.9 FY 1.4 FZ 11.5
751. 8 FX 19 FY -0.7 FZ 16.8
752. ******************************************************************************
753. ******************************************************************************
754. LOAD 15 PIPE LATERAL LOAD (PL)
755. JOINT LOAD
756. 5 FX 16 FY 99.7 FZ 6.8
757. 6 FX 10.1 FY -5.3 FZ 5.4
758. 7 FX -0.7 FY -14.8 FZ 6.2
759. 8 FX 6.4 FY -81 FZ 13.4
760. ******************************************************************************
761. ******************************************************************************
762. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
763. ******************************************************************************
764. LOAD 101 1.4(DL+EQDL+FL)
765. REPEAT LOAD
766. 1 1.4 4 1.4 3 1.4
767. LOAD 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
768. REPEAT LOAD
769. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
770. LOAD 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
771. REPEAT LOAD
772. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
773. LOAD 104 1.4(DL+EQDL+FL+PL)
774. REPEAT LOAD
775. 1 1.4 4 1.4 3 1.4 15 1.4
776. LOAD 105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
777. REPEAT LOAD
778. 1 1.2 4 1.2 3 1.2 15 1.2 2 1.6 5 1.6
779. LOAD 106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
780. REPEAT LOAD
781. 1 1.2 4 1.2 3 1.2 15 1.2 14 1.2 2 1.6 5 1.6
782. LOAD 107 1.2(DL+EQDL)+0.8WL(+X)
783. REPEAT LOAD
784. 1 1.2 4 1.2 6 0.8
785. LOAD 108 1.2(DL+EQDL)+0.8WL(-X)
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786. REPEAT LOAD
787. 1 1.2 4 1.2 7 0.8
788. LOAD 109 1.2(DL+EQDL)+0.8WL(+Z)
789. REPEAT LOAD
790. 1 1.2 4 1.2 8 0.8
791. LOAD 110 1.2(DL+EQDL)+0.8WL(-Z)
792. REPEAT LOAD
793. 1 1.2 4 1.2 9 0.8
794. LOAD 111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
795. REPEAT LOAD
796. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
797. LOAD 112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
798. REPEAT LOAD
799. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
800. LOAD 113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
801. REPEAT LOAD
802. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
803. LOAD 114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
804. REPEAT LOAD
805. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
806. LOAD 115 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
807. REPEAT LOAD
808. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
809. LOAD 116 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
810. REPEAT LOAD
811. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
812. LOAD 117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
813. REPEAT LOAD
814. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
815. LOAD 118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
816. REPEAT LOAD
817. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
818. LOAD 119 0.9(DL+EQDL)+1.6WL(+X)
819. REPEAT LOAD
820. 1 0.9 4 0.9 6 1.6
821. LOAD 120 0.9(DL+EQDL)+1.6WL(-X)
822. REPEAT LOAD
823. 1 0.9 4 0.9 7 1.6
824. LOAD 121 0.9(DL+EQDL)+1.6WL(+Z)
825. REPEAT LOAD
826. 1 0.9 4 0.9 8 1.6
827. LOAD 122 0.9(DL+EQDL)+1.6WL(-Z)
828. REPEAT LOAD
829. 1 0.9 4 0.9 9 1.6
830. LOAD 123 0.9(DL+EQDL)+SL(+X)
831. REPEAT LOAD
832. 1 0.9 4 0.9 10 1.0
833. LOAD 124 0.9(DL+EQDL)+SL(-X)
834. REPEAT LOAD
835. 1 0.9 4 0.9 11 1.0
836. LOAD 125 0.9(DL+EQDL)+SL(+Z)
837. REPEAT LOAD
838. 1 0.9 4 0.9 12 1.0
839. LOAD 126 0.9(DL+EQDL)+SL(-Z)
840. REPEAT LOAD
841. 1 0.9 4 0.9 13 1.0
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842. ************************************
843. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**936**	**NUMBER OF MEMBERS**	**8**
NUMBER OF PLATES	**868**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**928**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 932/ 48/ 150 DOF
TOTAL PRIMARY LOAD CASES = 41, TOTAL DEGREES OF FREEDOM = 2832
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 425 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 22.7/ 99979.3 MB

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844. CHANGE
845. SUPPORTS
846. _RAFT PLATE MAT DIRECT Y SUBGRADE 40500 PRINT COMPRESSION
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JOINT	INFLUENCE AREA
1	0.051962
2	0.051962
3	0.051962
4	0.051962
9	0.022778
10	0.022778
11	0.022778
12	0.022778
13	0.022812
14	0.022813
15	0.010000
16	0.010000
17	0.022812
18	0.022813
19	0.010000
20	0.010000
21	0.020000
22	0.040000
23	0.020000
24	0.045556
25	0.045625
26	0.045625
27	0.040000

28	0.020000
29	0.045556
30	0.020000
31	0.045625
32	0.040000
33	0.045556
34	0.020000
35	0.020000
36	0.020000
37	0.040000
38	0.045556
39	0.020000
40	0.045625
41	0.025556
42	0.051111
43	0.025556
44	0.051111
45	0.025555
46	0.051111
47	0.025556
48	0.051111
49	0.025556
50	0.051111
51	0.025556
52	0.051111
53	0.025556
54	0.051111
55	0.025556
56	0.051111
57	0.025556
58	0.051112
59	0.025556
60	0.051111
61	0.025556
62	0.051111
63	0.025556
64	0.051111
65	0.025556
66	0.051111
67	0.025556
68	0.051111
69	0.025556
70	0.051111
71	0.025556
72	0.051111
73	0.025556
74	0.051111
75	0.025556
76	0.051112
77	0.025556
78	0.051111
79	0.025556
80	0.051111
81	0.025556
82	0.051111
83	0.025556

84	0.051111
85	0.025556
86	0.051111
87	0.025556
88	0.051111
89	0.025556
90	0.051111
91	0.025556
92	0.051111
93	0.058299
94	0.058299
95	0.058298
96	0.058299
97	0.058299
98	0.058298
99	0.058298
100	0.058298
101	0.058300
102	0.058299
103	0.058298
104	0.058299
105	0.058299
106	0.058298
107	0.058298
108	0.058298
109	0.058298
110	0.058300
111	0.058299
112	0.058298
113	0.058298
114	0.058298
115	0.058298
116	0.058298
117	0.058299
118	0.058298
119	0.058298
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121	0.058298
122	0.051111
123	0.058298
124	0.051111
125	0.058299
126	0.051111
127	0.058298
128	0.051111
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130	0.051111
131	0.058298
132	0.051111
133	0.058298
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135	0.058300
136	0.051112
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202	0.058368
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       924          0.065487
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       929          0.065486
       930          0.065486
       931          0.065486
       932          0.065486
    TOTAL          53.515366
847. ********************************************************
848. *** UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
849. ********************************************************
850. LOAD 201 DL+EQDL+FL
851. REPEAT LOAD
852. 1 1.0 4 1.0 3 1.0
853. LOAD 202 DL+EQDL+FL+LL+EQLL+TL
854. REPEAT LOAD
855. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
856. LOAD 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
857. REPEAT LOAD
858. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
859. LOAD 204 DL+EQDL+FL+PL
860. REPEAT LOAD
861. 1 1.0 4 1.0 3 1.0 15 1.0
862. LOAD 205 DL+EQDL+FL+PL+LL+EQLL+TL
863. REPEAT LOAD
864. 1 1.0 4 1.0 3 1.0 15 1.0 2 1.0 5 1.0 14 1.0
865. LOAD 206 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL
866. REPEAT LOAD
867. 1 1.0 4 1.0 3 1.0 15 1.0 2 0.75 5 0.75 14 0.75
868. LOAD 207 DL+EQDL+FL+WL(+X)
869. REPEAT LOAD
870. 1 1.0 4 1.0 3 1.0 6 1.0
871. LOAD 208 DL+EQDL+FL+WL(-X)
872. REPEAT LOAD
873. 1 1.0 4 1.0 3 1.0 7 1.0
874. LOAD 209 DL+EQDL+FL+WL(+Z)
875. REPEAT LOAD
876. 1 1.0 4 1.0 3 1.0 8 1.0
877. LOAD 210 DL+EQDL+FL+WL(-Z)
878. REPEAT LOAD
879. 1 1.0 4 1.0 3 1.0 9 1.0
880. LOAD 211 DL+EQDL+FL+0.7SL(+X)
881. REPEAT LOAD
882. 1 1.0 4 1.0 3 1.0 10 0.7
883. LOAD 212 DL+EQDL+FL+0.7SL(-X)
884. REPEAT LOAD
885. 1 1.0 4 1.0 3 1.0 11 0.7
886. LOAD 213 DL+EQDL+FL+0.7SL(+Z)
887. REPEAT LOAD
888. 1 1.0 4 1.0 3 1.0 12 0.7
889. LOAD 214 DL+EQDL+FL+0.7SL(-Z)
890. REPEAT LOAD
891. 1 1.0 4 1.0 3 1.0 13 0.7
892. LOAD 215 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
```

```
893. REPEAT LOAD
894. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
895. LOAD 216 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
896. REPEAT LOAD
897. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
898. LOAD 217 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
899. REPEAT LOAD
900. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
901. LOAD 218 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
902. REPEAT LOAD
903. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
904. LOAD 219 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
905. REPEAT LOAD
906. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
907. LOAD 220 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
908. REPEAT LOAD
909. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
910. LOAD 221 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
911. REPEAT LOAD
912. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
913. LOAD 222 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
914. REPEAT LOAD
915. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
916. LOAD 223 0.6(DL+EQDL)+WL(+X)
917. REPEAT LOAD
918. 1 0.6 4 0.6 6 1.0
919. LOAD 224 0.6(DL+EQDL)+WL(-X)
920. REPEAT LOAD
921. 1 0.6 4 0.6 7 1.0
922. LOAD 225 0.6(DL+EQDL)+WL(+Z)
923. REPEAT LOAD
924. 1 0.6 4 0.6 8 1.0
925. LOAD 226 0.6(DL+EQDL)+WL(-Z)
926. REPEAT LOAD
927. 1 0.6 4 0.6 9 1.0
928. LOAD 227 0.6(DL+EQDL)+0.7SL(+X)
929. REPEAT LOAD
930. 1 0.6 4 0.6 10 0.7
931. LOAD 228 0.6(DL+EQDL)+0.7SL(-X)
932. REPEAT LOAD
933. 1 0.6 4 0.6 11 0.7
934. LOAD 229 0.6(DL+EQDL)+0.7SL(+Z)
935. REPEAT LOAD
936. 1 0.6 4 0.6 12 0.7
937. LOAD 230 0.6(DL+EQDL)+0.7SL(-Z)
938. REPEAT LOAD
939. 1 0.6 4 0.6 13 0.7
940. ************************************
941. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 210

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 223

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 224

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 225

```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  226



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  227



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  228



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  229



**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5
**START ITERATION NO.          6

**NOTE-Tension/Compression converged after  6 iterations, Case=  230



942. ******************************************************************************
943. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 15: 8:26 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 9 STAAD INPUT & OUTPUT FILE FOR F7 STABILITY

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 21, 2020                *
*           Time=    15: 6:28                    *
*                                                *
*      USER ID: LT Construction                 *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F7 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 6.9 0 0; 3 0 0 6.15; 4 6.9 0 6.15; 5 0 2.6 0; 6 6.9 2.6 0
    10. 7 0 2.6 6.15; 8 6.9 2.6 6.15; 933 0 0.3 0; 934 6.9 0.3 0; 935 0 0.3 6.15
    11. 936 6.9 0.3 6.15
    12. MEMBER INCIDENCES
    13. 5 1 933; 6 2 934; 7 3 935; 8 4 936; 1113 933 5; 1114 934 6; 1115 935 7
    14. 1116 936 8
    15. START GROUP DEFINITION
    17. MEMBER
    18. _C7 1113 TO 1116
    19. END GROUP DEFINITION
    20. DEFINE MATERIAL START
    21. ISOTROPIC CONCRETE
    22. E 2.17185E+007
    23. POISSON 0.17
    24. DENSITY 25
    25. ALPHA 1E-005
    26. DAMP 0.05
    27. TYPE CONCRETE
    28. STRENGTH FCU 27579
    29. END DEFINE MATERIAL
    30. MEMBER PROPERTY AMERICAN
    31. 5 TO 8 1113 TO 1116 PRIS YD 0.55 ZD 0.55
    32. CONSTANTS
    33. MATERIAL CONCRETE ALL
    34. ******************************************************************************
    35. SUPPORTS
    36. 1 TO 4 FIXED
    37. ************************************************************************
    38. LOAD 1 DEAD LOAD (DL)
```

**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
MASTER/SLAVE OR IF UNCONNECTED JOINTS.

```
39. JOINT LOAD
40. 5 FX -12.8 FY -154.2 FZ -0.6
41. 6 FX 11.5 FY -132.5 FZ -12.2
42. 7 FX -5.2 FY -78.5 FZ -0.5
43. 8 FX 6.5 FY -85.7 FZ 13.4
44. SELFWEIGHT Y -1 LIST ALL
45. ********************************************************************************
46. ********************************************************************************
47. LOAD 2 LIVE LOAD (LL)
48. JOINT LOAD
49. 5 FX -14.1 FY -106 FZ -2
50. 6 FX 10 FY -193 FZ -24.4
51. 7 FX -10.8 FY -156.4 FZ -1.7
52. 8 FX 14.9 FY -232.6 FZ 28
53. ********************************************************************************
54. ********************************************************************************
55. LOAD 3 OPERATING FLUID LOAD (FL)
56. JOINT LOAD
57. 5 FX -5.1 FY -58.2 FZ 0
58. 6 FX 5.1 FY -64.1 FZ -4.8
59. 7 FX -1.4 FY -7.3 FZ 0
60. 8 FX 1.4 FY -31.4 FZ 4.9
61. ********************************************************************************
62. ********************************************************************************
63. LOAD 4 EQP DL (EQDL)
64. JOINT LOAD
65. 5 FX -1.7 FY -5.8 FZ 0
66. 6 FX 1.8 FY -33.8 FZ -3.6
67. 7 FX -1.6 FY -5.1 FZ 0
68. 8 FX 1.5 FY -35.1 FZ 3.5
69. ********************************************************************************
70. ********************************************************************************
71. LOAD 5 EQP LL (EQLL)
72. JOINT LOAD
73. 5 FX -4.6 FY -20.1 FZ 0
74. 6 FX 4.6 FY -85.4 FZ -8.9
75. 7 FX -3.9 FY -15.6 FZ 0
76. 8 FX 3.9 FY -86.6 FZ 8.9
77. ********************************************************************************
78. ********************************************************************************
79. LOAD 6 WIND LOAD (+X) WL(+X)
80. JOINT LOAD
81. 5 FX 27.8 FY 114.8 FZ 0.4
82. 6 FX 29.1 FY -115 FZ -9.8
83. 7 FX 27.2 FY 103.7 FZ 0.3
84. 8 FX 28 FY -103.6 FZ 9.2
85. ********************************************************************************
86. ********************************************************************************
87. LOAD 7 WIND LOAD (-X) WL(-X)
88. JOINT LOAD
89. 5 FX -27.8 FY -114.8 FZ -0.4
90. 6 FX -29.1 FY 115 FZ 9.8
91. 7 FX -27.2 FY -103.7 FZ -0.3
92. 8 FX -28 FY 103.6 FZ -9.2
```

```
 93. ********************************************************************************
 94. ********************************************************************************
 95. LOAD 8 WIND LOAD (+Z) WL(+Z)
 96. JOINT LOAD
 97. 5 FX 10.5 FY 141.7 FZ 36.4
 98. 6 FX -9.2 FY 169.7 FZ 39.3
 99. 7 FX -11.4 FY -142.1 FZ 33
100. 8 FX 10.6 FY -169.7 FZ 39.4
101. ********************************************************************************
102. ********************************************************************************
103. LOAD 9 WIND LOAD (-Z) WL(-Z)
104. JOINT LOAD
105. 5 FX -10.5 FY -141.7 FZ -36.4
106. 6 FX 9.2 FY -169.7 FZ -39.3
107. 7 FX 11.4 FY 142.1 FZ -33
108. 8 FX -10.6 FY 169.7 FZ -39.4
109. ********************************************************************************
110. ********************************************************************************
111. LOAD 10 SEISMIC LOAD SL(+X)
112. JOINT LOAD
113. 5 FX 74.9 FY 301.8 FZ -24.1
114. 6 FX 74.1 FY -279 FZ -1.6
115. 7 FX 15.8 FY 230.7 FZ -21.9
116. 8 FX 16.2 FY -240.6 FZ 47.8
117. ********************************************************************************
118. ********************************************************************************
119. LOAD 11 SEISMIC LOAD SL(-X)
120. JOINT LOAD
121. 5 FX -74.9 FY -301.8 FZ 24.1
122. 6 FX -74.1 FY 279 FZ 1.6
123. 7 FX -15.8 FY -230.7 FZ 21.9
124. 8 FX -16.2 FY 240.6 FZ -47.8
125. ********************************************************************************
126. ********************************************************************************
127. LOAD 12 SEISMIC LOAD SL(+Z)
128. JOINT LOAD
129. 5 FX 19 FY 266.6 FZ 56.2
130. 6 FX -17.5 FY 271.4 FZ 37
131. 7 FX -21.1 FY -292 FZ 51.7
132. 8 FX 20.3 FY -271.4 FZ 37.1
133. ********************************************************************************
134. ********************************************************************************
135. LOAD 13 SEISMIC LOAD SL(-Z)
136. JOINT LOAD
137. 5 FX -19 FY -266.6 FZ -56.2
138. 6 FX 17.5 FY -271.4 FZ -37
139. 7 FX 21.1 FY 292 FZ -51.7
140. 8 FX -20.3 FY 271.4 FZ -37.1
141. ********************************************************************************
142. ********************************************************************************
143. LOAD 14 TEMPERATURE LOAD (TL)
144. JOINT LOAD
145. 5 FX -22.1 FY -2.3 FZ -11.5
146. 6 FX 21.8 FY 0.7 FZ -16.8
147. 7 FX -18.9 FY 1.4 FZ 11.5
148. 8 FX 19 FY -0.7 FZ 16.8
```

```
149. ********************************************************************************
150. ********************************************************************************
151. LOAD 15 PIPE LATERAL LOAD (PL)
152. JOINT LOAD
153. 5 FX 16 FY 99.7 FZ 6.8
154. 6 FX 10.1 FY -5.3 FZ 5.4
155. 7 FX -0.7 FY -14.8 FZ 6.2
156. 8 FX 6.4 FY -81 FZ 13.4
157. ********************************************************************************
158. ********************************************************************************
159. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
160. ********************************************************************************
161. LOAD COMB 101 1.4(DL+EQDL+FL)
162. 1 1.4 4 1.4 3 1.4
163. LOAD COMB 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
164. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
165. LOAD COMB 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
166. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
167. LOAD COMB 104 1.4(DL+EQDL+FL+PL)
168. 1 1.4 4 1.4 3 1.4 15 1.4
169. LOAD COMB 105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
170. 1 1.2 4 1.2 3 1.2 15 1.2 2 1.6 5 1.6
171. LOAD COMB 106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
172. 1 1.2 4 1.2 3 1.2 15 1.2 14 1.2 2 1.6 5 1.6
173. LOAD COMB 107 1.2(DL+EQDL)+0.8WL(+X)
174. 1 1.2 4 1.2 6 0.8
175. LOAD COMB 108 1.2(DL+EQDL)+0.8WL(-X)
176. 1 1.2 4 1.2 7 0.8
177. LOAD COMB 109 1.2(DL+EQDL)+0.8WL(+Z)
178. 1 1.2 4 1.2 8 0.8
179. LOAD COMB 110 1.2(DL+EQDL)+0.8WL(-Z)
180. 1 1.2 4 1.2 9 0.8
181. LOAD COMB 111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
182. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
183. LOAD COMB 112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
184. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
185. LOAD COMB 113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
186. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
187. LOAD COMB 114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
188. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
189. LOAD COMB 115 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
190. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
191. LOAD COMB 116 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
192. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
193. LOAD COMB 117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
194. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
195. LOAD COMB 118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
196. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
197. LOAD COMB 119 0.9(DL+EQDL)+1.6WL(+X)
198. 1 0.9 4 0.9 6 1.6
199. LOAD COMB 120 0.9(DL+EQDL)+1.6WL(-X)
200. 1 0.9 4 0.9 7 1.6
201. LOAD COMB 121 0.9(DL+EQDL)+1.6WL(+Z)
202. 1 0.9 4 0.9 8 1.6
203. LOAD COMB 122 0.9(DL+EQDL)+1.6WL(-Z)
204. 1 0.9 4 0.9 9 1.6
```

```
205. LOAD COMB 123 0.9(DL+EQDL)+SL(+X)
206. 1 0.9 4 0.9 10 1.0
207. LOAD COMB 124 0.9(DL+EQDL)+SL(-X)
208. 1 0.9 4 0.9 11 1.0
209. LOAD COMB 125 0.9(DL+EQDL)+SL(+Z)
210. 1 0.9 4 0.9 12 1.0
211. LOAD COMB 126 0.9(DL+EQDL)+SL(-Z)
212. 1 0.9 4 0.9 13 1.0
213. ******************************************************
214. *** UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
215. ******************************************************
216. LOAD COMB 201 DL+EQDL+FL
217. 1 1.0 4 1.0 3 1.0
218. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
219. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
220. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
221. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
222. LOAD COMB 204 DL+EQDL+FL+PL
223. 1 1.0 4 1.0 3 1.0 15 1.0
224. LOAD COMB 205 DL+EQDL+FL+PL+LL+EQLL+TL
225. 1 1.0 4 1.0 3 1.0 15 1.0 2 1.0 5 1.0 14 1.0
226. LOAD COMB 206 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL
227. 1 1.0 4 1.0 3 1.0 15 1.0 2 0.75 5 0.75 14 0.75
228. LOAD COMB 207 DL+EQDL+FL+WL(+X)
229. 1 1.0 4 1.0 3 1.0 6 1.0
230. LOAD COMB 208 DL+EQDL+FL+WL(-X)
231. 1 1.0 4 1.0 3 1.0 7 1.0
232. LOAD COMB 209 DL+EQDL+FL+WL(+Z)
233. 1 1.0 4 1.0 3 1.0 8 1.0
234. LOAD COMB 210 DL+EQDL+FL+WL(-Z)
235. 1 1.0 4 1.0 3 1.0 9 1.0
236. LOAD COMB 211 DL+EQDL+FL+0.7SL(+X)
237. 1 1.0 4 1.0 3 1.0 10 0.7
238. LOAD COMB 212 DL+EQDL+FL+0.7SL(-X)
239. 1 1.0 4 1.0 3 1.0 11 0.7
240. LOAD COMB 213 DL+EQDL+FL+0.7SL(+Z)
241. 1 1.0 4 1.0 3 1.0 12 0.7
242. LOAD COMB 214 DL+EQDL+FL+0.7SL(-Z)
243. 1 1.0 4 1.0 3 1.0 13 0.7
244. LOAD COMB 215 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
245. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
246. LOAD COMB 216 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
247. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
248. LOAD COMB 217 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
249. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
250. LOAD COMB 218 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
251. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
252. LOAD COMB 219 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
253. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
254. LOAD COMB 220 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
255. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
256. LOAD COMB 221 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
257. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
258. LOAD COMB 222 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
259. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
260. LOAD COMB 223 0.6(DL+EQDL)+WL(+X)
```

```
261. 1 0.6 4 0.6 6 1.0
262. LOAD COMB 224 0.6(DL+EQDL)+WL(-X)
263. 1 0.6 4 0.6 7 1.0
264. LOAD COMB 225 0.6(DL+EQDL)+WL(+Z)
265. 1 0.6 4 0.6 8 1.0
266. LOAD COMB 226 0.6(DL+EQDL)+WL(-Z)
267. 1 0.6 4 0.6 9 1.0
268. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(+X)
269. 1 0.6 4 0.6 10 0.7
270. LOAD COMB 228 0.6(DL+EQDL)+0.7SL(-X)
271. 1 0.6 4 0.6 11 0.7
272. LOAD COMB 229 0.6(DL+EQDL)+0.7SL(+Z)
273. 1 0.6 4 0.6 12 0.7
274. LOAD COMB 230 0.6(DL+EQDL)+0.7SL(-Z)
275. 1 0.6 4 0.6 13 0.7
276. ************************************
277. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**12**	**NUMBER OF MEMBERS**	**8**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```
  ORIGINAL/FINAL BAND-WIDTH=     8/     1/     12 DOF
TOTAL      PRIMARY LOAD CASES =    15, TOTAL DEGREES OF FREEDOM =      48
TOTAL LOAD COMBINATION  CASES =    56  SO FAR.
  SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/  99974.3 MB
```

```
278. ********************************************************************************
279. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 15: 6:36 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 10 STAAD INPUT & OUTPUT FILE FOR F8 DESIGN

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*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 21, 2020                   *
*           Time=    14:49:34                       *
*                                                   *
*      USER ID: LT Construction                    *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F8 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 933 0 0 -4.75; 934 0 2.6 -4.75
    10. MEMBER INCIDENCES
    11. 1113 933 934
    12. DEFINE MATERIAL START
    13. ISOTROPIC CONCRETE
    14. E 2.17185E+007
    15. POISSON 0.17
    16. DENSITY 25
    17. ALPHA 1E-005
    18. DAMP 0.05
    19. TYPE CONCRETE
    20. STRENGTH FCU 27579
    21. END DEFINE MATERIAL
    22. MEMBER PROPERTY AMERICAN
    23. 1113 PRIS YD 0.55 ZD 0.55
    24. CONSTANTS
    25. MATERIAL CONCRETE ALL
    26. ******************************************************************************
    27. SUPPORTS
    28. 933 FIXED
    29. LOAD 1 DEAD LOAD (DL)
    30. JOINT LOAD
    31. 934 FX 0 FY -72.6 FZ 0
    32. SELFWEIGHT Y -1
    33. ******************************************************************************
    34. ******************************************************************************
    35. LOAD 2 LIVE LOAD (LL)
    36. JOINT LOAD
    37. 934 FX -0.1 FY -0.2 FZ 0
    38. ******************************************************************************
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39. ********************************************************************************
40. LOAD 3 OPERATING FLUID LOAD (FL)
41. JOINT LOAD
42. 934 FX 0 FY -60.7 FZ 0
43. ********************************************************************************
44. ********************************************************************************
45. LOAD 4 EQP DL (EQDL)
46. JOINT LOAD
47. 934 FX 0 FY 0 FZ 0
48. ********************************************************************************
49. ********************************************************************************
50. LOAD 5 EQP LL (EQLL)
51. JOINT LOAD
52. 934 FX 0 FY 0 FZ 0
53. ********************************************************************************
54. ********************************************************************************
55. LOAD 6 WIND LOAD (+X) WL(+X)
56. JOINT LOAD
57. 934 FX 0 FY 0.2 FZ 0
58. ********************************************************************************
59. ********************************************************************************
60. LOAD 7 WIND LOAD (-X) WL(-X)
61. JOINT LOAD
62. 934 FX 0 FY -0.2 FZ 0
63. ********************************************************************************
64. ********************************************************************************
65. LOAD 8 WIND LOAD (+Z) WL(+Z)
66. JOINT LOAD
67. 934 FX -0.4 FY 0.4 FZ 0.3
68. ********************************************************************************
69. ********************************************************************************
70. LOAD 9 WIND LOAD (-Z) WL(-Z)
71. JOINT LOAD
72. 934 FX 0.4 FY -0.4 FZ -0.3
73. ********************************************************************************
74. ********************************************************************************
75. LOAD 10 SEISMIC LOAD SL(+X)
76. JOINT LOAD
77. 934 FX 1.7 FY -12.9 FZ -0.2
78. ********************************************************************************
79. ********************************************************************************
80. LOAD 11 SEISMIC LOAD SL(-X)
81. JOINT LOAD
82. 934 FX -1.7 FY 12.9 FZ 0.2
83. ********************************************************************************
84. ********************************************************************************
85. LOAD 12 SEISMIC LOAD SL(+Z)
86. JOINT LOAD
87. 934 FX -0.7 FY 25.4 FZ 0.8
88. ********************************************************************************
89. ********************************************************************************
90. LOAD 13 SEISMIC LOAD SL(-Z)
91. JOINT LOAD
92. 934 FX 0.7 FY -25.4 FZ -0.8
93. ********************************************************************************
94. ********************************************************************************
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 95. LOAD 14 TEMPERATURE LOAD (TL)
 96. JOINT LOAD
 97. 934 FX 0.2 FY 0.9 FZ -0.1
 98. ********************************************************************************
 99. ********************************************************************************
100. LOAD 15 PIPE LATERAL LOAD (PL)
101. JOINT LOAD
102. 934 FX -0.1 FY 1.4 FZ 0
103. ********************************************************************************
104. ********************************************************************************
105. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
106. ********************************************************************************
107. LOAD COMB 101 1.4(DL+EQDL+FL)
108. 1 1.4 4 1.4 3 1.4
109. LOAD COMB 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
110. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
111. LOAD COMB 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
112. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
113. LOAD COMB 104 1.4(DL+EQDL+FL+PL)
114. 1 1.4 4 1.4 3 1.4 15 1.4
115. LOAD COMB 105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
116. 1 1.2 4 1.2 3 1.2 15 1.2 2 1.6 5 1.6
117. LOAD COMB 106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
118. 1 1.2 4 1.2 3 1.2 15 1.2 14 1.2 2 1.6 5 1.6
119. LOAD COMB 107 1.2(DL+EQDL)+0.8WL(+X)
120. 1 1.2 4 1.2 6 0.8
121. LOAD COMB 108 1.2(DL+EQDL)+0.8WL(-X)
122. 1 1.2 4 1.2 7 0.8
123. LOAD COMB 109 1.2(DL+EQDL)+0.8WL(+Z)
124. 1 1.2 4 1.2 8 0.8
125. LOAD COMB 110 1.2(DL+EQDL)+0.8WL(-Z)
126. 1 1.2 4 1.2 9 0.8
127. LOAD COMB 111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
128. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
129. LOAD COMB 112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
130. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
131. LOAD COMB 113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
132. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
133. LOAD COMB 114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
134. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
135. LOAD COMB 115 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
136. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
137. LOAD COMB 116 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
138. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
139. LOAD COMB 117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
140. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
141. LOAD COMB 118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
142. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
143. LOAD COMB 119 0.9(DL+EQDL)+1.6WL(+X)
144. 1 0.9 4 0.9 6 1.6
145. LOAD COMB 120 0.9(DL+EQDL)+1.6WL(-X)
146. 1 0.9 4 0.9 7 1.6
147. LOAD COMB 121 0.9(DL+EQDL)+1.6WL(+Z)
148. 1 0.9 4 0.9 8 1.6
149. LOAD COMB 122 0.9(DL+EQDL)+1.6WL(-Z)
150. 1 0.9 4 0.9 9 1.6
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151. LOAD COMB 123 0.9(DL+EQDL)+SL(+X)
152. 1 0.9 4 0.9 10 1.0
153. LOAD COMB 124 0.9(DL+EQDL)+SL(-X)
154. 1 0.9 4 0.9 11 1.0
155. LOAD COMB 125 0.9(DL+EQDL)+SL(+Z)
156. 1 0.9 4 0.9 12 1.0
157. LOAD COMB 126 0.9(DL+EQDL)+SL(-Z)
158. 1 0.9 4 0.9 13 1.0
159. ******************************************************
160. *** UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
161. ******************************************************
162. LOAD COMB 201 DL+EQDL+FL
163. 1 1.0 4 1.0 3 1.0
164. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
165. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
166. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
167. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
168. LOAD COMB 204 DL+EQDL+FL+PL
169. 1 1.0 4 1.0 3 1.0 15 1.0
170. LOAD COMB 205 DL+EQDL+FL+PL+LL+EQLL+TL
171. 1 1.0 4 1.0 3 1.0 15 1.0 2 1.0 5 1.0 14 1.0
172. LOAD COMB 206 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL
173. 1 1.0 4 1.0 3 1.0 15 1.0 2 0.75 5 0.75 14 0.75
174. LOAD COMB 207 DL+EQDL+FL+WL(+X)
175. 1 1.0 4 1.0 3 1.0 6 1.0
176. LOAD COMB 208 DL+EQDL+FL+WL(-X)
177. 1 1.0 4 1.0 3 1.0 7 1.0
178. LOAD COMB 209 DL+EQDL+FL+WL(+Z)
179. 1 1.0 4 1.0 3 1.0 8 1.0
180. LOAD COMB 210 DL+EQDL+FL+WL(-Z)
181. 1 1.0 4 1.0 3 1.0 9 1.0
182. LOAD COMB 211 DL+EQDL+FL+0.7SL(+X)
183. 1 1.0 4 1.0 3 1.0 10 0.7
184. LOAD COMB 212 DL+EQDL+FL+0.7SL(-X)
185. 1 1.0 4 1.0 3 1.0 11 0.7
186. LOAD COMB 213 DL+EQDL+FL+0.7SL(+Z)
187. 1 1.0 4 1.0 3 1.0 12 0.7
188. LOAD COMB 214 DL+EQDL+FL+0.7SL(-Z)
189. 1 1.0 4 1.0 3 1.0 13 0.7
190. LOAD COMB 215 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
191. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
192. LOAD COMB 216 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
193. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
194. LOAD COMB 217 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
195. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
196. LOAD COMB 218 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
197. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
198. LOAD COMB 219 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
199. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
200. LOAD COMB 220 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
201. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
202. LOAD COMB 221 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
203. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
204. LOAD COMB 222 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
205. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
206. LOAD COMB 223 0.6(DL+EQDL)+WL(+X)
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207. 1 0.6 4 0.6 6 1.0
208. LOAD COMB 224 0.6(DL+EQDL)+WL(-X)
209. 1 0.6 4 0.6 7 1.0
210. LOAD COMB 225 0.6(DL+EQDL)+WL(+Z)
211. 1 0.6 4 0.6 8 1.0
212. LOAD COMB 226 0.6(DL+EQDL)+WL(-Z)
213. 1 0.6 4 0.6 9 1.0
214. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(+X)
215. 1 0.6 4 0.6 10 0.7
216. LOAD COMB 228 0.6(DL+EQDL)+0.7SL(-X)
217. 1 0.6 4 0.6 11 0.7
218. LOAD COMB 229 0.6(DL+EQDL)+0.7SL(+Z)
219. 1 0.6 4 0.6 12 0.7
220. LOAD COMB 230 0.6(DL+EQDL)+0.7SL(-Z)
221. 1 0.6 4 0.6 13 0.7
222. ************************************
223. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**2**	**NUMBER OF MEMBERS**	**1**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**1**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 56 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 99979.6 MB

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224. ********************************************************************************
225. FINISH
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********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 21,2020 TIME= 14:49:40 ****

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